

Multiplicative Number Theory III

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Notation

Some symbols are used in more than one way. The intended interpretation should be clear from the context in which it arises.

Symbol	Meaning
$\mathbb{C}$	The set of complex numbers. See page 1.
$\mathbb{Q}$	The set of rational numbers.
$\mathbb{R}$	The set of real numbers.
$\mathbb{T}$	$\mathbb{R}/\mathbb{Z}$, i.e., the real numbers modulo 1. See page xx.
$\mathbb{Z}$	The set of rational integers.
A^*	The adjoint of the matrix A . See page ??.
$\ \mathbf{x}\ $	Norm of the vector $\mathbf{x}$.
$\ A\ $	The operator norm of the matrix A . See page ??.
$\ \alpha\ $	The distance from α to the nearest integer. See page ??.
$E_0(\chi)$	$= 1$ if $\chi = \chi_0$, $= 0$ otherwise. See page ??
$s(x)$	The sawtooth function. See page ??.
$\text{si}(x)$	The sine integral. See page ??.
$\deg P$	The degree of the polynomial P .
$\Delta(s)$	$\zeta(s) = \Delta(s)\zeta(1-s)$. See page 81.
$\sum_{\chi}^*$	A sum over primitive characters modulo q . See page ??.

23

Probabilistic number theory

C:ProbNoThy

We say that an arithmetic function f is *additive* if

$$f(mn) = f(m) + f(n) \quad (23.1) \quad \text{E: defaddfcn}$$

whenever $(m, n) = 1$. The values of an additive function are determined by its values on prime-powers, since

$$f(n) = \sum_{p^k \parallel n} f(p^k). \quad (23.2) \quad \text{E: formaddfcn}$$

If the identity (23.1) holds for all pairs m, n , then we say that f is *totally* (or *completely*) *additive*. If f is additive and $f(p^k) = f(p)$ for all p and all $k \geq 1$, then we say that f is *strongly additive*. For example, $\log n$ and $\Omega(n)$ are totally additive functions, $\log(n/\varphi(n))$ and $\omega(n)$ are strongly additive, while $\Omega(n) - \omega(n)$ and $\log d(n)$ are additive but neither totally additive nor strongly additive.

In our study of sieves in Chapter 3 we saw that things do not always work out as one would expect on probabilistic grounds. However, we find that the distribution of the values of an additive function follow the natural probabilistic model very closely. Suppose that f is an additive function. The asymptotic density of integers n for which $p^k \parallel n$ is $p^{-k}(1 - 1/p)$. It is with this ‘probability’ that the term $f(p^k)$ is one of the terms in the sum. Accordingly, for each prime number p we define a random variable X_p that has the distribution

$$\begin{aligned} \mathbf{P}(X_p = f(p^k)) &= \frac{1 - 1/p}{p^k} \quad (k = 1, 2, \dots), \\ \mathbf{P}(X_p = 0) &= 1 - 1/p. \end{aligned} \quad (23.3) \quad \text{E: DistrvX_p}$$

If p and q are distinct primes, then by the Chinese remainder theorem we see that the asymptotic density of the integers n for which both

$p^k \parallel n$ and $q^\ell \parallel n$ is $p^{-k}(1-1/p)q^{-\ell}(1-1/q)$. Hence the two events $p^k \parallel n$ and $q^\ell \parallel n$ are asymptotically independent. Thus it is natural to take the random variables X_p to be independent, and we set

$$X = \sum_p X_p. \quad (23.4) \quad \boxed{\text{E: defrvX}}$$

This sum either converges almost always or almost nowhere. We shall find that when it converges almost always, the values of f have a limiting distribution that is the same as the distribution of X , and that when it converges almost nowhere, f does not have a limiting distribution.

We have already established a scattering of results concerning a few additive functions. In §2.3 we estimated the mean of $\omega(n)$, and also its variance about its mean. In §2.4 we determined the distribution of the additive function $\Omega(n) - \omega(n)$ by calculating the mean value of the multiplicative function $z^{\Omega(n) - \omega(n)}$. In §7.4 we put

$$\alpha_n = \frac{\Omega(n) - \log \log n}{\sqrt{\log \log n}},$$

and found that the distribution of α_n is asymptotically normal with mean 0 and variance 1. In this chapter we are more concerned with developing a general theory than with special examples.

23.1 The Turán–Kubilius inequality

S:TKIneq

PT34 Turán (1934) showed (cf Theorem 2.12) that

$$\sum_{n \leq x} (\omega(n) - \log \log x)^2 \ll x \log \log x,$$

and JK56 Kubilius (1956) generalized this to arbitrary additive functions.

Suppose that f is an additive function. Then

$$\begin{aligned}
 \sum_{n \leq x} f(n) &= \sum_{n \leq x} \sum_{p^k \parallel n} f(p^k) \\
 &= \sum_{p^k \leq x} f(p^k) \sum_{\substack{n \leq x \\ p^k \parallel n}} 1 \\
 &= \sum_{p^k \leq x} f(p^k) \left(\left\lfloor \frac{x}{p^k} \right\rfloor - \left\lfloor \frac{x}{p^{k+1}} \right\rfloor \right) \\
 &= x \sum_{p^k \leq x} \frac{f(p^k)}{p^k} \left(1 - \frac{1}{p} \right) + O \left(\sum_{p^k \leq x} |f(p^k)| \right). \tag{23.5}
 \end{aligned}$$

E:meanaddfcn

For ease of reference, we set

$$A(x) = A(f, x) = \sum_{p^k \leq x} \frac{f(p^k)}{p^k} \left(1 - \frac{1}{p} \right). \tag{23.6}$$

E:defA

We anticipate that the variance of f about its mean should not be much more than

$$B(x) = B(f, x) = \sum_{p^k \leq x} \frac{|f(p^k)|^2}{p^k}. \tag{23.7}$$

E:defB

By Cauchy's inequality,

$$\sum_{p^k \leq x} |f(p^k)| \leq B(x)^{1/2} \left(\sum_{p^k \leq x} p^k \right)^{1/2} \ll B(x)^{1/2} \frac{x}{\sqrt{\log x}}.$$

Thus from (23.5) we see that

$$\sum_{n \leq x} f(n) = A(x)x + O(B(x)^{1/2}x(\log x)^{-1/2}). \tag{23.8}$$

E:meanfest1

As concerns the potential size of $A(x)$ relative to $B(x)$, we note by Cauchy's inequality that

$$|A(x)|^2 \leq \left(\sum_{p^k \leq x} \frac{|f(p^k)|^2}{p^k} \right) \left(\sum_{p^k \leq x} \frac{1}{p^k} \right) \ll B(x) \log \log x. \tag{23.9}$$

E:ArelBest

We now show that $B(x)$ is within a constant factor of being an upper bound for the variance of the values of f about its mean.

T:TKIneq

Theorem 23.1 (The Turán–Kubilius inequality). *Let f be an additive function, with $A(x)$ and $B(x)$ defined as in (23.6) and (23.7). Then*

$$\sum_{n \leq x} (f(n) - A(x))^2 \ll xB(x). \tag{23.10}$$

E:TKIneq

The implicit constant in the above is absolute: It is independent of both f and x . From (23.9) we see that

$$\sum_{n \leq x} |A(x)|^2 \ll xB(x) \log \log x,$$

and by (23.10) it follows that also

$$\sum_{n \leq x} |f(n)|^2 \ll xB(x) \log \log x.$$

Thus we see that the estimate (23.10) is never more than a factor of $\log \log x$ from being trivial. Despite this lack of quantitative depth, the Turán–Kubilius inequality turns out to be a quite useful result.

Proof We expand the sum on the left hand side, and obtain three terms. The simplest is

$$\begin{aligned} T_0 &= \sum_{n \leq x} |A(x)|^2 = |A(x)|^2 [x] = |A(x)|^2 x + O(|A(x)|^2) \\ &= x|A(x)|^2 + O(B(x) \log \log x) \end{aligned} \quad (23.11) \quad \boxed{\text{E:T_0Est}}$$

by (23.9). The intermediate term is $T_1 = -2 \operatorname{Re} \overline{A(x)} \sum_{n \leq x} f(n)$. Thus by (23.8),

$$T_1 = -2|A(x)|^2 x + O(|A(x)|B(x)^{1/2}x(\log x)^{-1/2}),$$

and by (23.9) this is

$$= -2|A(x)|^2 x + O(B(x)x(\log x)^{-1/2}(\log \log x)^{1/2}). \quad (23.12) \quad \boxed{\text{E:T_1Est}}$$

Finally,

$$\begin{aligned} T_2 &= \sum_{n \leq x} |f(n)|^2 = \sum_{n \leq x} \sum_{p^k \parallel n} f(p^k) \sum_{q^\ell \parallel n} \bar{f}(q^\ell) \\ &= \sum_{p^k \leq x} \sum_{q^\ell \leq x} f(p^k) \bar{f}(q^\ell) \sum_{\substack{n \leq x \\ p^k \parallel n \\ q^\ell \parallel n}} 1 \end{aligned}$$

where q denotes a prime number. The contribution of those terms for which $p = q$ is

$$T_2' = \sum_{p^k \leq x} |f(p^k)|^2 \left(\left[\frac{x}{p^k} \right] - \left[\frac{x}{p^{k+1}} \right] \right) \leq xB(x). \quad (23.13) \quad \boxed{\text{E:T2diag}}$$

The remaining terms contribute

$$\begin{aligned}
 T_2'' &= \sum_{\substack{p^k q^\ell \leq x \\ p \neq q}} f(p^k) \bar{f}(q^\ell) \left(\left\lfloor \frac{x}{p^k q^\ell} \right\rfloor - \left\lfloor \frac{x}{p^{k+1} q^\ell} \right\rfloor - \left\lfloor \frac{x}{p^k q^{\ell+1}} \right\rfloor + \left\lfloor \frac{x}{p^{k+1} q^{\ell+1}} \right\rfloor \right) \\
 &= x \sum_{p^k q^\ell \leq x} \frac{f(p^k)}{p^k} \frac{\bar{f}(q^\ell)}{q^\ell} \left(1 - \frac{1}{p}\right) \left(1 - \frac{1}{q}\right) \quad (23.14) \quad \boxed{\text{E:T2nondiag1}}
 \end{aligned}$$

$$+ O\left(\sum_{p^k q^\ell \leq x} |f(p^k) \bar{f}(q^\ell)| \right) + O\left(x \sum_{p^{k+\ell} \leq x} \frac{|f(p^k) \bar{f}(p^\ell)|}{p^{k+\ell}} \right). \quad (23.15) \quad \boxed{\text{E:T2nondiag2}}$$

By Cauchy's inequality the first error term is

$$\ll \left(\sum_{p^k q^\ell \leq x} \frac{|f(p^k)|^2 |f(q^\ell)|^2}{p^k q^\ell} \right)^{1/2} \left(\sum_{p^k q^\ell \leq x} p^k q^\ell \right)^{1/2}.$$

Here the first sum is $\leq B(x)^2$, and the second sum is

$$\leq x \sum_{\substack{n \leq x \\ \omega(n) \leq 2}} 1 \ll x^2 (\log x)^{-1} \log \log x$$

by (7.54). By Cauchy's inequality the second error term in (23.15) is

$$\begin{aligned}
 &\ll x \left(\sum_{p^{k+\ell} \leq x} \frac{|f(p^k)|^2}{p^{k+\ell}} \right)^{1/2} \left(\sum_{p^{k+\ell} \leq x} \frac{|f(p^\ell)|^2}{p^{k+\ell}} \right)^{1/2} \\
 &= x \sum_{p^k \leq x} \frac{|f(p^k)|^2}{p^k} \sum_{\substack{\ell \\ p^\ell \leq x/p^k}} \frac{1}{p^\ell} \ll x B(x).
 \end{aligned}$$

The expression (23.14) is

$$= x |A(x)|^2 - x \sum_{\substack{p^k \leq x \\ q^\ell \leq x \\ p^k q^\ell > x}} \frac{f(p^k)}{p^k} \frac{\bar{f}(q^\ell)}{q^\ell} \left(1 - \frac{1}{p}\right) \left(1 - \frac{1}{q}\right).$$

By Cauchy's inequality the sum on the right has absolute value

$$\leq \left(\sum_{\substack{p^k \leq x \\ q^\ell \leq x \\ p^k q^\ell > x}} \frac{|f(p^k)|^2}{p^k} \frac{|f(q^\ell)|^2}{q^\ell} \right)^{1/2} \left(\sum_{\substack{p^k \leq x \\ q^\ell \leq x \\ p^k q^\ell > x}} \frac{1}{p^k q^\ell} \right)^{1/2}.$$

Here the first sum is $\leq B(x)^2$, and the second sum is

$$\begin{aligned}
 &= 2 \sum_{p^k \leq x^{1/2}} \frac{1}{p^k} \sum_{x/p^k < q^\ell \leq x} \frac{1}{q^\ell} + \left(\sum_{x^{1/2} < p^k \leq x} \frac{1}{p^k} \right)^2 \\
 &\ll \sum_{p^k \leq x^{1/2}} \frac{k \log p}{p^k \log x} + 1 \ll 1.
 \end{aligned}$$

On assembling our estimates we deduce that

$$T_2'' = x|A(x)|^2 + O(x(B(x))).$$

The stated result now follows by combining this with (23.11)–(23.13). $\square$

23.1.1 Exercises

1. Show that almost all integers n have $(1/2 + o(1)) \log \log n$ prime factors $\equiv 1 \pmod{4}$.
2. Let k be a fixed positive integer. Show that $d_k(n) = (\log n)^{(1+o(1)) \log k}$ for almost all integers.
3. Show that

$$\sum_{n \leq x} \Omega(n) \Omega(n+k) = x(\log \log x)^2 + cx \log \log x + O(x)$$

where

$$c =$$

4. Show that $\sum_{n \leq x} (\omega(n^2 + 1) - \log \log n)^2 \ll x \log \log x$.
5. Show that $\sum_{p \leq x} (\omega(p+1) - \log \log p)^2 \ll x \log \log x$.
6. Suppose that f is an additive function, and let $A(x)$ and $B(x)$ be defined as in (23.6) and (23.7).

(a) Show that if $n \leq x$, then

$$|A(n) - A(x)|^2 \leq B(x) \sum_{n < p^k \leq x} \frac{1}{p^k}.$$

(b) Show that

$$\sum_{n \leq x} |A(n) - A(x)|^2 \ll B(x)x / \log x.$$

(c) Conclude that

$$\sum_{n \leq x} |f(n) - A(n)|^2 \ll xB(x).$$

7. Suppose that f is an additive function and that $B(x)$ is defined as in (23.7).

(a) Show that

$$\left| \sum_{p^k \leq x} \frac{f(p^k)}{p^{k+1}} \right|^2 \ll B(x).$$

(b) Show that

$$\left| \sum_{\substack{p^k \leq x \\ k > 1}} \frac{f(p^k)}{p^k} \right|^2 \ll B(x).$$

(c) Put $A'(x) = \sum_{p \leq x} f(p)/p$. Show that if f is an additive function, then

$$\sum_{n \leq x} |f(n) - A'(x)|^2 \ll xB(x).$$

8. The *Kubilius class* $\mathcal{H}$ consists of those additive functions f with the two properties

(i) $B(f, x) \rightarrow \infty$ as $x \rightarrow \infty$;

(ii) $\sum_{x^{1/2} < p^k \leq x} \frac{|f(p^k)|^2}{p^k} = o(B(x)) \quad (x \rightarrow \infty).$

Show that if $f \in \mathcal{H}$, then

$$\sum_{n \leq x} |f(n) - A(x)|^2 = (1 + o(1))xB(x)$$

as $x \rightarrow \infty$.

9. Let $f(n) = \log n$.

(a) Show that $A(x) = \log x + O(1)$.

(b) Show that $B(x) = \frac{1}{2}(\log x)^2 + O(\log x)$.

(c) Deduce that $f \notin \mathcal{H}$.

(d) Show that $\sum_{n \leq x} |f(n) - A(x)|^2 \ll x$.

10. Suppose that f is strongly additive, so that $f(n) = \sum_{p|n} f(p)$ for all n . Consider the bilinear form inequality

$$\sum_{n \leq x} \left| \sum_{p|n} f(p) - \sum_{p \leq x} \frac{f(p)}{p} \right|^2 \leq \Delta \sum_{p \leq x} \frac{|f(p)|^2}{p} \quad (23.16) \quad \boxed{\text{E:biform1}}$$

in the variables $f(p)$.

- (a) By the change of variables $g(p) = f(p)/\sqrt{p}$, show that the above is equivalent to the bilinear form inequality

$$\sum_{n \leq x} \left| \sum_{p|n} g(p)p^{1/2} - \sum_{p \leq x} \frac{g(p)}{p^{1/2}} \right|^2 \leq \Delta \sum_{p \leq x} |g(p)|^2. \quad (23.17) \quad \boxed{\text{E:biform2}}$$

- (b) Use Theorem ?? to show that the above is equivalent to the bilinear form inequality

$$\sum_{p \leq x} p \left| \sum_{\substack{n \leq x \\ p|n}} h(n) - \frac{1}{p} \sum_{n \leq x} h(n) \right|^2 \leq \Delta \sum_{n \leq x} |h(n)|^2 \quad (23.18) \quad \boxed{\text{E:biform3}}$$

in the variables $h(n)$.

- (c) Apply the large sieve, as discussed in §xx.x to show that

$$\sum_{p \leq x^{1/2}} p \left| \sum_{\substack{n \leq x \\ p|n}} h(n) - \frac{1}{p} \sum_{n \leq x} h(n) \right|^2 \ll x \sum_{n \leq x} |h(n)|^2.$$

- (d) Show that if $x^{1/2} < p \leq x$, then

$$\left| \sum_{\substack{n \leq x \\ p|n}} h(n) \right|^2 \ll \frac{x}{p} \sum_{\substack{n \leq x \\ p|n}} |h(n)|^2.$$

- (e) Show that

$$\sum_{x^{1/2} < p \leq x} p \left| \sum_{\substack{n \leq x \\ p|n}} h(n) \right|^2 \ll x \sum_{n \leq x} |h(n)|^2.$$

- (f) Show that

$$\sum_{x^{1/2} < p \leq x} \frac{1}{p} \left| \sum_{n \leq x} h(n) \right|^2 \ll x \sum_{n \leq x} |h(n)|^2.$$

- (g) Deduce that (23.18) and hence also (23.16) hold with $\Delta \ll x$.

23.2 Mean values of multiplicative functions

S:MVMF

Suppose that f is a multiplicative function, which is to say that $f(mn) = f(m)f(n)$ whenever $(m, n) = 1$. Hence $f(n) = \prod_{p^k \parallel n} f(p^k)$. We let $\mathcal{M}_0$ denote the class of those multiplicative functions f for which $|f(n)| \leq 1$ for all n . Our object is to characterize those members of $\mathcal{M}_0$ that have an asymptotic mean value. If f is a real-valued additive function, then $e(tf(n)) \in \mathcal{M}_0$, so an ability to compute mean values of multiplicative functions will help us to determine the Fourier transform of the distribution of additive functions. We begin with several simple results concerning (not necessarily multiplicative) arithmetic functions.

T:MVf=g*1

Theorem 23.2. *If $f(n) = \sum_{d|n} g(d)$, if the series $\sum_{d=1}^{\infty} g(d)/d$ converges, say to a , and if $\sum_{d \leq x} |g(d)| = o(x)$ as $x \rightarrow \infty$, then*

$$S(x) = \sum_{n \leq x} f(n) = ax + o(x). \quad (23.19) \quad \text{E:MVf}$$

Proof Clearly

$$S(x) = \sum_{n \leq x} \sum_{d|n} g(d) = \sum_{d \leq x} g(d) [x/d] = x \sum_{d \leq x} g(d)/d + O\left(\sum_{d \leq x} |g(d)|\right).$$

Thus we have the stated result. $\square$

Co:Wintner

Corollary 23.3 (Wintner). *If $f(n) = \sum_{d|n} g(d)$ and $\sum_{d=1}^{\infty} |g(d)|/d < \infty$, then (23.19) holds with $a = \sum_{d=1}^{\infty} g(d)/d$.*

Proof From the hypothesis that $\sum_{d=1}^{\infty} |g(d)|/d < \infty$, it follows by partial summation that $\sum_{d \leq x} |g(d)| = o(x)$. $\square$

Co:MVmultfcn1

Corollary 23.4. *If f is multiplicative, if*

$$\sum_p \frac{|1 - f(p)|}{p} < \infty, \quad (23.20) \quad \text{E:sum|1-f(p)|/p<infy}$$

and if

$$\sum_{\substack{p^k \\ k > 1}} \frac{|f(p^k)|}{p^k} < \infty,$$

then (23.19) holds with

$$a = \prod_p \left(1 - \frac{1}{p}\right) \left(1 + \frac{f(p)}{p} + \frac{f(p^2)}{p^2} + \cdots\right). \quad (23.21) \quad \text{E:MVaform}$$

Proof Let g be the multiplicative function for which $g(p^k) = f(p^k) - f(p^{k-1})$. Then $f(n) = \sum_{d|n} g(d)$, and

$$\sum_{d=1}^{\infty} \frac{|g(d)|}{d} = \prod_p \left(1 + \frac{|f(p) - 1|}{p} + \frac{|f(p^2) - f(p)|}{p^2} + \dots \right) < \infty,$$

and thus

$$\sum_{d=1}^{\infty} \frac{g(d)}{d} = \prod_p \left(1 + \frac{f(p) - 1}{p} + \frac{f(p^2) - f(p)}{p^2} + \dots \right) = a$$

where a is defined by (23.21). $\square$

In the same vein we have

Theorem 23.5. *If $f(n) = \sum_{d|n} g(d)h(n/d)$, if $\sum_{m=1}^{\infty} |h(m)|/m < \infty$, and if $\sum_{d \leq x} g(d) = bx + o(x)$, then we have (23.19) with $a = b \sum_{m=1}^{\infty} h(m)/m$.*

Here we see that a mean value for g yields one for f , provided that f is near g in the sense that $\sum |h(m)|/m < \infty$. If $h(1) = 1$ and $h(m) = 0$ for all $m > 1$, then $f = g$.

Proof Put

$$r(x) = \sum_{d \leq x} g(d) - bx.$$

Then

$$\begin{aligned} S(x) &= \sum_{n \leq x} \sum_{d|n} g(d)h(n/d) = \sum_{m \leq x} h(m) \sum_{d \leq x/m} g(d) \\ &= bx \sum_{m \leq x} \frac{h(m)}{m} + \sum_{m \leq x} h(m)r(x/m). \end{aligned}$$

There is a constant C (depending on g) such that $|r(x)| \leq Cx$ for all $x \geq 1$, and for every $\varepsilon > 0$ there is a δ such that $|r(x)| \leq \varepsilon x$ for all $x \geq 1/\delta$. Thus the second sum above has absolute value not exceeding

$$\varepsilon x \sum_{m \leq \delta x} |h(m)|/m + Cx \sum_{\delta x < m \leq x} |h(m)|/m.$$

Here the first term is $\ll \varepsilon x$, and the second sum is small since it is part of the tail of a convergent series. Thus we have the stated result. $\square$

In Theorem 23.2 we found a connection between the mean value of f and the convergence of the series $\sum g(d)/d$, but we find it more productive to pursue the line suggested by Corollary 23.4, which we now sharpen.

T:Delange

Theorem 23.6 (Delange). *Suppose that $f \in \mathcal{M}_0$, and that the series*

$$\sum_p \frac{1 - f(p)}{p} \quad (23.22) \quad \boxed{\text{E:sum1-f(p)/pconv}}$$

converges. Then (23.19) holds with a given by (23.21).

Since $\operatorname{Re} f(p) \leq |f(p)| \leq 1$, we see that the convergence of the series (23.22) implies that the sum of the real parts is absolutely convergent, just as it was in Corollary 23.4. Thus Theorem 23.6 is stronger by virtue of the fact that we are no longer assuming that the sum of the imaginary parts is absolutely convergent. Given the convergence of the product (23.21), we see that $a \neq 0$ unless one of the individual factors vanishes. This happens only in the single case that

$$f(2^k) = -1 \quad (k = 1, 2, 3, \dots). \quad (23.23) \quad \boxed{\text{E:f(2^k)=-1}}$$

Proof We suppose first that in addition to the stated hypotheses, f has the further properties that

$$f(p^k) = f(p)^k \quad (k = 1, 2, \dots), \quad (23.24) \quad \boxed{\text{E:fTotMult}}$$

and that

$$\operatorname{Re} f(p) \geq 1/2 \quad (23.25) \quad \boxed{\text{E:Reflarge}}$$

for all p . Once we have established the theorem for such f , we extend the result to general f by an appeal to Theorem 23.5. Let P be a large parameter, let $\mathcal{P}_1$ denote the set of primes not exceeding P , and let $\mathcal{P}_2$ denote the primes larger than P . Let f_i be multiplicative, $f_i(p^k) = f(p)^k$ for $p \in \mathcal{P}_i$, and $f_i(p^k) = 1$ for $p \notin \mathcal{P}_i$. Thus $f = f_1 f_2$, and by Corollary 23.4,

$$\sum_{n \leq x} f_1(n) = a(P)x + o(x)$$

with

$$a(P) = \prod_{p \leq P} \left(1 - \frac{1}{p}\right) \left(1 - \frac{f(p)}{p}\right)^{-1}.$$

Since $|f(p^k)| \leq 1$, we see by (23.24) and (23.25) that we may write $f_2(n) = e^{g(n)}$ where g is an additive function such that $-\log 2 \leq \operatorname{Re} g(p) \leq 0$ and $|\operatorname{Im} g(p)| \leq \pi/3$. If $\operatorname{Re} z \leq 0$, then

$$|e^z - 1| = \left| \int_0^z e^w dw \right| \leq |z|,$$

so we see that

$$\left| \sum_{n \leq x} f(n) - f_1(n) \right| \leq \sum_{n \leq x} |f_2(n) - 1| \leq \sum_{n \leq x} |g(n)|.$$

Let $A(x) = A(g, x)$ be defined as in (23.6). Then by Cauchy's inequality the above is

$$\leq x|A(x)| + \sum_{n \leq x} |g(n) - A(x)| \leq x|A(x)| + x^{1/2} \left(\sum_{n \leq x} |g(n) - A(x)|^2 \right)^{1/2},$$

and by the Turán–Kubilius inequality (Theorem 23.1), this is

$$\ll x|A(x)| + xB(x)^{1/2}.$$

We now relate $A(x)$ and $B(x)$ to $\sum_p (1 - f(p))/p$. While the imaginary part of this sum is not necessarily absolutely convergent, the real part of each term is nonnegative, and so the sum of the real parts is absolutely convergent. Also, $|1 - f(p)| = 1 - 2 \operatorname{Re} f(p) + |f(p)|^2 \leq 2 - 2 \operatorname{Re} f(p)$, $g(p^k) \ll k$, and $|g(p)| \asymp |1 - f(p)|$, so that

$$\begin{aligned} B(x) &= \sum_{p^k \leq x} \frac{|g(p^k)|}{p^k} \ll \sum_{p > P} \frac{1}{p^2} + \sum_{P < p \leq x} \frac{|1 - f(p)|^2}{p} \\ &\ll \frac{1}{P} + \operatorname{Re} \sum_{P < p \leq x} \frac{1 - f(p)}{p}. \end{aligned}$$

We also observe that $g(p) = f(p) - 1 + O(|1 - f(p)|^2)$, so that

$$\begin{aligned} A(x) &= \sum_{p^k \leq x} \frac{g(p^k)}{p^k} \left(1 - \frac{1}{p}\right) = \sum_{P < p \leq x} \frac{f(p) - 1}{p} \\ &\quad + O\left(\sum_{P < p \leq x} \frac{|1 - f(p)|^2}{p}\right) + O\left(\sum_{P < p \leq x} \frac{1}{p^2}\right) \\ &\ll \frac{1}{P} + \left| \sum_{P < p \leq x} \frac{1 - f(p)}{p} \right|. \end{aligned}$$

On assembling our estimates, we find that

$$S(x) = a(P)x + o(x) + O(x/P^{1/2}) + O\left(x \left| \sum_{P < p \leq x} \frac{1 - f(p)}{p} \right|^{1/2}\right).$$

Since P can be arbitrarily large, this gives the desired result, subject to (23.24) and (23.25).

To complete the proof we now suppose only that $f \in \mathcal{M}_0$ and that

$\sum_p (1 - f(p))/p$ converges. Let $\mathcal{P}$ denote the set of primes p for which $\operatorname{Re} f(p) \leq 1/2$. We note that

$$\sum_{p \in \mathcal{P}} \frac{1}{p} \leq 2 \operatorname{Re} \sum_p \frac{1 - f(p)}{p} < \infty.$$

We define multiplicative functions g and h by the Euler products

$$\begin{aligned} G(s) &= \sum_{n=1}^{\infty} \frac{g(n)}{n^s} = \prod_{p \in \mathcal{P}} \left(1 - \frac{1}{p^s}\right)^{-1} \prod_{p \notin \mathcal{P}} \left(1 - \frac{f(p)}{p^s}\right)^{-1}, \\ H(s) &= \sum_{n=1}^{\infty} \frac{h(n)}{n^s} = \prod_{p \in \mathcal{P}} \left(1 - \frac{1}{p^s}\right) \left(1 + \frac{f(p)}{p^s} + \frac{f(p^2)}{p^{2s}} + \cdots\right) \\ &\quad \times \prod_{p \notin \mathcal{P}} \left(1 - \frac{f(p)}{p^s}\right) \left(1 + \frac{f(p)}{p^s} + \frac{f(p^2)}{p^{2s}} + \cdots\right). \end{aligned}$$

Thus

$$F(s) = \sum_{n=1}^{\infty} \frac{f(n)}{n^s} = \prod_p \left(1 + \frac{f(p)}{p^s} + \frac{f(p^2)}{p^{2s}} + \cdots\right) = G(s)H(s).$$

We observe that $g \in \mathcal{M}_0$, $\sum_p (1 - g(p))/p$ converges, and that g satisfies (23.24) and (23.25). Hence

$$\sum_{n \leq x} g(n) = x \prod_{p \notin \mathcal{P}} \left(1 - \frac{1}{p}\right) \left(1 - \frac{f(p)}{p}\right)^{-1} + o(x),$$

and we obtain the desired result from Theorem 23.5, since

$$\begin{aligned} \sum_{m=1}^{\infty} \frac{|h(m)|}{m} &\leq \prod_{p \in \mathcal{P}} \left(1 + \frac{2}{p} + \frac{2}{p^2} + \cdots\right) \prod_{p \notin \mathcal{P}} \left(1 + \frac{2}{p^2} + \frac{2}{p^3} + \cdots\right) \\ &\ll \exp\left(2 \sum_{p \in \mathcal{P}} \frac{1}{p}\right) < \infty. \end{aligned}$$

□

In Delange's Theorem (Theorem 23.6), the mean value is nonzero unless (23.23) holds. In §23.5 we shall characterize those $f \in \mathcal{M}_0$ with vanishing mean value, in terms of the behaviour of the generating Dirichlet series

$$F(s) = \sum_{n=1}^{\infty} \frac{f(n)}{n^s} \quad (\sigma > 1). \quad (23.26) \quad \boxed{\text{E:DefF}(s)}$$

To prepare for the proof of our next result we establish a variant of the Hardy–Littlewood tauberian theorem (Theorem 5.7).

L:HLTaub4primes

Lemma 23.7. *Suppose that the numbers $c(p)$ are bounded, and that*

$$\lim_{\sigma \rightarrow 1^+} \sum_p \frac{c(p)}{p^\sigma}$$

exists and has the (finite) value c . Then $\sum_p c(p)/p$ converges, and has the value c .

Proof Put

$$a(u) = \sum_{e^{u-1} < p \leq e^u} \frac{c(p)}{p}.$$

Then $a(u) \ll 1/u$ for $u \geq 1$, and

$$I(u) = \int_0^\infty a(u) e^{-\delta u} du = \frac{1 - e^{-\delta}}{\delta} \sum_p \frac{c(p)}{p^{1+\delta}}$$

tends to c as $\delta \rightarrow 0^+$. Thus by the Hardy–Littlewood tauberian theorem (Theorem 5.7 with $\beta = 0$) it follows that $\int_0^U a(u) du$ tends to c as $U \rightarrow \infty$. But

$$\int_0^U a(u) du = \sum_{p \leq e^U} \frac{c(p)}{p} - \sum_{e^{U-1} < p \leq e^U} \frac{c(p)}{p} (U - \log p),$$

and the second sum is $\ll 1/U$, so $\sum_p c(p)/p$ converges to c . $\square$

For members of $\mathcal{M}_0$ with nonzero mean value, we have the following comprehensive result.

T:Delange2

Theorem 23.8 (Delange). *Suppose that $f \in \mathcal{M}_0$, and let $S(x)$, the number a , and the function $F(s)$ be defined as in (23.19), (23.21) and (23.26), respectively. Then the following assertions are equivalent:*

- (a) $S(x) \sim ax$ and $a \neq 0$;
- (b) $\sum_{n \leq x} \frac{f(n)}{n} \sim a \log x$ and $a \neq 0$;
- (c) $F(\sigma) \sim \frac{a}{\sigma - 1}$ as $\sigma \rightarrow 1^+$ and $a \neq 0$;
- (d) $\lim_{\sigma \rightarrow 1^+} \sum_p \frac{1 - f(p)}{p^\sigma}$ exists and (23.23) fails;
- (e) $\sum_p \frac{1 - f(p)}{p}$ converges and (23.23) fails.

Proof We deduce (b) from (a) by partial summation, and similarly deduce (c) from (b). But (c) asserts that $\lim_{\sigma \rightarrow 1^+} F(\sigma)/\zeta(\sigma) = a$, which is to say that

$$\lim_{\sigma \rightarrow 1^+} \prod_p \left(1 - \frac{1}{p^\sigma}\right) \left(1 + \frac{f(p)}{p^\sigma} + \frac{f(p^2)}{p^{2\sigma}} + \cdots\right) = a \neq 0.$$

Each factor of the product has modulus not exceeding 1, so if (23.23) were to hold, then the limit would be 0. Thus (23.23) fails and the product is comparable to

$$\exp \left(\sum_p \frac{1 - f(p)}{p^\sigma} \right).$$

Hence we have (d). That (d) implies (e) is immediate from Lemma 23.7, and that (e) implies (a) follows from Theorem 23.6. $\square$

23.2.1 Exercise

1. Suppose that $\sum_{d=1}^{\infty} g(d)/d$ converges, say to a .
 - (a) Show that $\sum_{d \leq x} g(d) = o(x)$.
 - (b) Suppose also that $\sum_{d \leq x} |g(d)| \ll x$. Use Axer's Theorem (Theorem 8.1) to show that $\sum_{d \leq x} g(d) \{x/d\} = o(x)$.
 - (c) Put $f(n) = \sum_{d|n} g(d)$. Under the above hypotheses, show that $\sum_{n \leq x} f(n) = ax + o(x)$. (Note that this improves upon Theorem 23.2.)

23.3 The distribution of additive functions

S:DistAddFcns

We now employ our understanding of the mean values of multiplicative functions to establish

T:E-WThm

Theorem 23.9 (Erdős–Wintner). *Let f be a real-valued additive function. The following are equivalent:*

- (a) *Each of the following series is convergent:*

$$\sum_{\substack{p \\ |f(p)| \leq 1}} \frac{f(p)}{p}, \quad \sum_{\substack{p \\ |f(p)| \leq 1}} \frac{|f(p)|^2}{p}, \quad \sum_{\substack{p \\ |f(p)| > 1}} \frac{1}{p}. \quad (23.27) \quad \text{E:3series}$$

- (b) *There is an increasing function $F(u)$ such that $\lim_{u \rightarrow -\infty} F(u) = 0$, $\lim_{n \rightarrow +\infty} \frac{1}{N} \text{card}\{n \leq N : f(n) \leq u\} = F(u)$, and such that*

$$\lim_{N \rightarrow \infty} \frac{1}{N} \text{card}\{n \leq N : f(n) \leq u\} = F(u), \quad (23.28) \quad \boxed{\text{E:DistFcnf}}$$

whenever u is not a point of discontinuity of F .

Since F is increasing, the set of its discontinuities is at most countable. Later we shall see that if we define F to be right-continuous, so that $F(u) = F(u^+)$, then (23.28) holds for all values of u . When (b) holds we may say that F is the asymptotic distribution of f . Given F with the above properties, there is a unique probability measure μ such that $F(u) = \int_{-\infty}^u 1 d\mu$. Moreover, we can construct a probability measure μ_N that attaches weight $1/N$ to each of the points $f(n)$ for $1 \leq n \leq N$. Then (23.28) asserts that

$$\lim_{N \rightarrow \infty} \int_{-\infty}^u 1 d\mu = \int_{-\infty}^u 1 d\mu,$$

which is to say that the measures μ_N tend weakly to μ . Our proof of the Erdős–Wintner Theorem depends on our discussion in §?? concerning the weak convergence of measures.

Proof Suppose that (a) holds. By virtue of Theorem I.7, in order to show that (b) holds it suffices to show that

$$\hat{\mu}_N(t) = \int_{\mathbb{R}} e(-tu) d\mu_N(u) = \frac{1}{N} \sum_{n=1}^N e(-tf(n)) \quad (23.29) \quad \boxed{\text{E:muhatN}}$$

has a limit $r(t)$ as $N \rightarrow \infty$, and that r is continuous at $t = 0$. Let $g(n) = g_t(n) = e(-tf(n))$. Then $g \in \mathcal{M}_0$, so by Theorem 23.6 the above tends to

$$r(t) = \prod_p \left(1 - \frac{1}{p}\right) \left(1 + \frac{e(-tf(p))}{p} + \frac{e(-tf(p^2))}{p^2} + \cdots\right), \quad (23.30) \quad \boxed{\text{E:rform}}$$

provided that the sum

$$\sum_p \frac{1 - e(-tf(p))}{p} \quad (23.31) \quad \boxed{\text{E:sumpconv1}}$$

converges. The above is

$$\begin{aligned}
 &= 2\pi it \sum_{|f(p)| \leq 1} \frac{f(p)}{p} + \sum_{|f(p)| \leq 1} \frac{1 - 2\pi it f(p) - e(-tf(p))}{p} \\
 &\quad + \sum_{|f(p)| > 1} \frac{1 - e(-tf(p))}{p}.
 \end{aligned} \tag{23.32}$$

E:sumpexpanded

By the hypotheses (a) we see that the first sum is a constant, and that the third sum is absolutely and uniformly convergent. Since $e(\theta) = 1 + 2\pi i\theta + O(\theta^2)$, the second sum is absolutely convergent, and uniformly so for t in a bounded set. Thus the sum (23.31) converges. Moreover, the expression (23.32) tends to 0 as $t \rightarrow 0$, so $r(t)$ tends to 1, and hence we have (b).

We now show that (b) implies (a). If the μ_N tend weakly to μ , then by Theorem I.6 it follows that $\hat{\mu}_N(t) \rightarrow \hat{\mu}(t)$. Since $\hat{\mu}(t)$ is continuous and $\hat{\mu}(0) = 1$, it follows from (23.29) that the multiplicative function g_t has a non-zero mean value for all t near 0, and that this mean value tends to 1 as $t \rightarrow 0$. Hence by Theorem 23.8 we deduce that the sum (23.31) converges for all small t , and tends to 0 as $t \rightarrow 0$. Let $s(t)$ denote the real part of the series (23.31). Since each term has non-negative real part, the sum of the real parts is absolutely convergent, and uniformly bounded for $|t| \leq \delta$. But then

$$\frac{1}{2\delta} \int_{-\delta}^{\delta} s(t) dt = \sum_p \frac{1}{p} \left(1 - \frac{\sin 2\pi\delta f(p)}{2\pi\delta f(p)} \right),$$

and hence this latter sum is finite. But

$$1 - \frac{\sin \theta}{\theta} \gg \min(1, \theta^2),$$

so the second and third sums in (23.27) are convergent. Hence the second and third sums in (23.32) are convergent, and since the sum (23.31) is convergent, it follows that the first sum in (23.32) is convergent. $\square$

In the next section we shall find that much can be said about the distribution function F of a real-valued additive function f . At this point we content ourselves with the following simple result.

T:DVC **Theorem 23.10.** *Let f be a real-valued additive function with limiting distribution function F . If the series*

$$\sum_{f(p) \neq 0} \frac{1}{p} \tag{23.33}$$

E:sumpconv2

converges, then each value assumed by f is attained on a set of positive density, so that F has jump discontinuities but is otherwise constant (i.e., the associated measure μ is discrete). If the series diverges, then the distribution function F is continuous, and hence any value of f is assumed only on a set of density 0.

Proof Let $\mathcal{P}$ denote the set of primes p for which $f(p) \neq 0$, let $\mathcal{N}_1$ denote the set of positive integers composed entirely of primes $p \in \mathcal{P}$, let $\mathcal{N}_2$ denote the set of integers composed entirely of primes $p \notin \mathcal{P}$ with each prime occurring with multiplicity > 1 , and finally let $\mathcal{N}_3$ denote the set of squarefree integers composed entirely of primes $p \notin \mathcal{P}$. Each n can be written uniquely in the form $n = n_1 n_2 n_3$ with $n_i \in \mathcal{N}_i$ and $(n_2, n_3) = 1$, and $f(n)$ depends only on n_1 and n_2 . The number of $n \leq x$ with prescribed n_1 and n_2 is the number of squarefree $n_3 \leq x/(n_1 n_2)$ such that $n_3 \in \mathcal{N}_3$ and $(n_2, n_3) = 1$. By Corollary 23.4, this is

$$\begin{aligned} & \sim \frac{x}{n_1 n_2} \prod_{p \in \mathcal{P}} \left(1 - \frac{1}{p}\right) \prod_{p|n_2} \left(1 - \frac{1}{p}\right) \prod_{\substack{p \notin \mathcal{P} \\ p \nmid n_2}} \left(1 - \frac{1}{p^2}\right) \\ & = \frac{x}{n_1 n_2} \prod_{p \in \mathcal{P}} \left(1 - \frac{1}{p}\right) \prod_{p|n_2} \left(1 + \frac{1}{p}\right)^{-1} \prod_{p \notin \mathcal{P}} \left(1 - \frac{1}{p^2}\right). \end{aligned}$$

Moreover, these densities sum to 1 as n_1 and n_2 range over $\mathcal{N}_1$ and $\mathcal{N}_2$.

Now suppose that the series (23.33) diverges. By Theorem 23.33 it suffices to show that

$$\int_{-T}^T |\hat{\mu}(t)|^2 dt = o(T) \quad (23.34) \quad \boxed{\text{E:avemusmall}}$$

as $T \rightarrow \infty$. In the case at hand we know that

$$\hat{\mu}(t) = \prod_p \left(1 - \frac{1}{p}\right) \left(1 + \frac{e(-tf(p))}{p} + \frac{e(-tf(p^2))}{p^2} + \dots\right). \quad (23.35) \quad \boxed{\text{E:muhatform}}$$

Let $\mathcal{P}$ be a finite set of primes for which $f(p) \neq 0$, and put $s = \sum_{p \in \mathcal{P}} 1/p$. In the above product, each prime contributes a factor whose absolute value is ≤ 1 . Thus

$$\begin{aligned} |\hat{\mu}(t)| & \leq \prod_{p \in \mathcal{P}} \left| \left(1 - \frac{1}{p}\right) \left(1 + \frac{e(-tf(p))}{p} + \dots\right) \right| \\ & \ll \prod_{p \in \mathcal{P}} \left| 1 - \frac{1}{p} + \frac{e(-tf(p))}{p} \right| \ll \exp \left(-2 \sum_{p \in \mathcal{P}} \frac{\sin^2 \pi t f(p)}{p} \right). \end{aligned}$$

Hence by Hölder's inequality

$$\begin{aligned} \frac{1}{2T} \int_{-T}^T |\widehat{\mu}(t)|^2 dt &\ll \frac{1}{2T} \int_{-T}^T \exp\left(-4 \sum_{p \in \mathcal{P}} \frac{\sin^2 \pi t f(p)}{p}\right) dt \\ &\leq \prod_{p \in \mathcal{P}} \left(\frac{1}{2T} \int_{-T}^T \exp(-4s \sin^2 \pi t f(p)) dt \right)^{1/(sp)}. \end{aligned}$$

Suppose that $f(p) > 0$. The integrand has period $1/f(p)$, and

$$\begin{aligned} f(p) \int_0^{1/f(p)} \exp(-4s \sin^2 \pi t f(p)) dt &= \int_0^1 \exp(-4s \sin^2 \pi t) dt \\ &\leq \int_{-1/2}^{1/2} \exp(-16st^2) dt \leq \int_{-\infty}^{\infty} \exp(-16st^2) dt = \frac{\sqrt{\pi}}{4\sqrt{s}}. \end{aligned}$$

Hence

$$\frac{1}{2T} \int_{-T}^T \exp(-4s \sin^2 \pi t f(p)) dt \leq \frac{1}{\sqrt{s}}$$

for all sufficiently large T , and so

$$\limsup_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T |\widehat{\mu}(t)|^2 dt \leq \frac{1}{\sqrt{s}}.$$

By choosing $\mathcal{P}$ suitably, we may make s as large as we please. Thus we have (23.34), and the proof is complete. $\square$

23.3.1 Exercises

1. (a) Show that $\log \sigma(n)/n$ has a limiting distribution.
 (b) Show that this limiting distribution is continuous.
 (c) Deduce that the set of perfect numbers (i.e., those for which $\sigma(n) = 2n$) is a set of density 0.
2. Show that an integer-valued additive function f has a limiting distribution if and only if

$$\sum_{f(p) \neq 0} \frac{1}{p} < \infty.$$

3. Let f be a multiplicative function that takes only positive real values.

Show that f has a limiting distribution if and only if each of the following four series converges:

$$\sum_{1/2 \leq |f(p)| \leq 2} \frac{1-f(p)}{p}, \quad \sum_{1/2 \leq |f(p)| \leq 2} \frac{|1-f(p)|^2}{p}, \quad \sum_{|f(p)| > 2} \frac{1}{p}, \quad \sum_{|f(p)| < 1/2} \frac{1}{p}.$$

4. Let $f_1, \dots, f_k$ be real-valued additive functions, and put $\mathbf{f}(n) = (f_1(n), \dots, f_k(n))$. Give necessary and sufficient conditions that $\mathbf{f}$ should have a limiting distribution in $\mathbb{R}^k$. Deduce a variant of the Erdős–Wintner Theorem (Theorem 23.9) for complex-valued additive functions.
5. Let f be a real-valued additive function with limiting distribution F , and let μ denote the associated limiting measure. Show that either $\hat{\mu}(t)$ is never 0, or that its zeros form an arithmetic progression of the form $c(2k+1)$ for $k \in \mathbb{Z}$.

23.4 Applications of probability theory

S:AppProbThy

Let f be a real-valued additive function, and for each prime p let X_p denote the random variable defined in (23.3). We take the X_p to be independent, and ask whether the random variable X defined in (23.4) exists. In this connection we quote without proof

Theorem 23.11 (Kolmogorov’s Three Series Theorem). *Let Y_n be independent random variables. If each of the three series*

$$\sum_n \int_{|Y_n| \leq 1} Y_n, \quad \sum_n \int_{|Y_n| \leq 1} |Y_n|^2, \quad \sum_n \int_{|Y_n| > 1} 1$$

converges, then the sum $Y = \sum_n Y_n$ converges almost everywhere. If any one of these series diverges, then the sum $\sum_n Y_n$ diverges almost everywhere.

For our sum (23.4), the conditions of Kolmogorov’s theorem are precisely the conditions of part (a) of the Erdős–Wintner Theorem (Theorem 23.9). Hence the random variable X exists precisely when f has a limiting distribution F . In the context of Kolmogorov’s Three Series Theorem, when Y exists its Fourier transform is

$$\hat{Y}(t) = \int e(-tY) = \int \prod_n e(-tY_n) = \prod_n \int e(-tY_n) = \prod_n \hat{Y}_n(t)$$

by the independence of the Y_n . In the case of the variable X , we find that

$$\begin{aligned}\widehat{X}_p(t) &= \int e(-tX_p) = 1 - \frac{1}{p} + e(-tf(p))\frac{1}{p}\left(1 - \frac{1}{p}\right) + \cdots \\ &= \left(1 - \frac{1}{p}\right)\left(1 + \frac{e(-tf(p))}{p} + \frac{e(-tf(p^2))}{p^2} + \cdots\right),\end{aligned}$$

and hence

$$\widehat{X}(t) = \prod_p \left(1 - \frac{1}{p}\right) \left(1 + \frac{e(-tf(p))}{p} + \frac{e(-tf(p^2))}{p^2} + \cdots\right).$$

But this is the same as $\widehat{\mu}(t)$ given in (23.35), so by the uniqueness of the Fourier transform (Corollary I.5) it follows that F is the distribution function of X . A great deal is known concerning the distribution function of a sum of random variables, so by appealing to this theory we obtain further information concerning F . In particular, we note the Law of Pure Types:

Theorem 23.12 (Jessen–Wintner). *Let Y_n be independent random variables such that $Y = \sum_n Y_n$ converges almost everywhere, and suppose that there is a countable set $\mathcal{C}$ such that $P(Y_n \in \mathcal{C}) = 1$ for all n . Then the distribution of Y is of pure type: Either it is discrete, singular, or absolutely continuous.*

Hence we see that the distribution function F of a real-valued additive function is of pure type. In Theorem 23.10 we characterized the situation in which the distribution is discrete; this can also be obtained by applying a general theorem of Lévy (get reference) concerning sums of independent random variables. We have no similar criterion to distinguish between singular and absolutely continuous distributions, although all three types do occur. In particular, the distribution of $\log \varphi(n)n$ is singular, as we now show.

T:DistPhiSing

Theorem 23.13. *Let μ denote the probability measure such that*

$$\lim_{N \rightarrow \infty} \frac{1}{N} \text{card}\{n \leq N : \varphi(n)/n \leq c\} = \mu((-\infty, c]).$$

Let α be fixed, $1 < \alpha < e - 1$, put $I_k = [\varphi(k)/k - 1/k^\alpha, \varphi(k)/k]$, and set

$$\mathcal{S} = \{x \in [0, 1] : x \in I_k \text{ for infinitely many } k\}.$$

Then $m(\mathcal{S}) = 0$, and $\mu(\mathcal{S}) = 1$.

Here $m(\mathcal{S})$ denotes the Lebesgue measure of $\mathcal{S}$.

Proof The first assertion is clear, since

$$\mathcal{S} \subseteq \bigcup_{k>K} I_k$$

for any K , so that

$$m(\mathcal{S}) \leq \sum_{k>K} k^{-\alpha}.$$

As for the second assertion, we show that for any $\varepsilon > 0$ and any K there is an L such that

$$\mu\left(\bigcup_{K < k \leq L} I_k\right) \geq 1 - \varepsilon. \quad (23.36) \quad \boxed{\text{E:muDist}}$$

The advantage of this finite form of the assertion is that we can estimate the left hand side by considering densities of set of integers: If $\mathcal{N} = \{n : \varphi(n)/n \in \bigcup_{K < k \leq L} I_k\}$, then $d(\mathcal{N}) = \mu(\bigcup_{K < k \leq L} I_k)$. In establishing (23.36), we may assume that K is large, for if the above holds for one value of K , then it also holds for all smaller values of K . For a given number n , write $n = \prod_{i=1}^{\Omega(n)} p_i$ with $p_1 \leq p_2 \leq \dots \leq p_{\Omega(n)}$, and set $d_r = \prod_{i \leq r} p_i$. We shall show that if L is sufficiently large, then most integers n have a divisor d_r , $K < d_r \leq L$, such that

$$\frac{\varphi(n/d_r)}{n/d_r} \geq 1 - d_r^{-\alpha}.$$

In this case $\varphi(n)/n \in I_{d_r}$, since

$$\frac{\varphi(d_r)}{d_r} \geq \frac{\varphi(n)}{n} \geq \frac{\varphi(d_r)}{d_r} \frac{\varphi(n/d_r)}{n/d_r} \geq \frac{\varphi(d_r)}{d_r} (1 - d_r^{-\alpha}) \geq \frac{\varphi(d_r)}{d_r} - d_r^{-\alpha}.$$

Let $\mathcal{N}_0$ denote the complementary set of numbers, i.e., the n for which $\varphi(n/d_r)/(n/d_r) < 1 - d_r^{-\alpha}$ for all $d_r \in (K, L]$. To estimate the size of $\mathcal{N}_0$ we consider various possibilities. Let $\mathcal{N}_1$ be the set of n such that the interval $(K, \log L]$ contains none of the special divisors d_r . Let $\beta = \alpha/2 + (e-1)/2$, so that $1 < \alpha < \beta < e-1$, and let $\mathcal{N}_2$ be the set of numbers n such that $p_{r+1} < d_r^\beta$ whenever $d_r \in (K, L]$. Finally, let $\mathcal{N}_3$ be the set of those n such that there is a $d_r \in (K, L]$ for which $p_{r+1} > d_r^\beta$ and $\varphi(n/d_r)/(n/d_r) < 1 - d_r^{-\alpha}$. The sets $\mathcal{N}_i$ possess asymptotic densities, but for our present purpose it suffices to bound their upper asymptotic densities where the upper asymptotic density of a set $\mathcal{A}$ is

$$\bar{d}(\mathcal{A}) = \limsup_{x \rightarrow \infty} \frac{1}{x} \text{card}\{n \leq x : n \in \mathcal{A}\}.$$

The main estimates to be established are that

$$\bar{d}(\mathcal{N}_1) \ll \frac{\log K}{\log \log L}, \quad (23.37) \quad \boxed{\text{E:DensityN1}}$$

$$\bar{d}(\mathcal{N}_1^c \mathcal{N}_2) \ll \frac{1}{\log \log L}, \quad (23.38) \quad \boxed{\text{E:DensityN2}}$$

$$\bar{d}(\mathcal{N}_3) \ll K^{\alpha-e+1}. \quad (23.39) \quad \boxed{\text{E:DensityN3}}$$

Once these estimates are in place, we argue that $\mathcal{N}_2 = \mathcal{N}_1 \mathcal{N}_2 \cup \mathcal{N}_1 \mathcal{N}_2^c \subseteq \mathcal{N}_1 \cup \mathcal{N}_1^c \mathcal{N}_2$. Since $\bar{d}(\mathcal{A} \cup \mathcal{B}) \leq \bar{d}(\mathcal{A}) + \bar{d}(\mathcal{B})$, it follows from (23.37) and (23.38) that

$$\bar{d}(\mathcal{N}_2) \ll \frac{\log K}{\log \log L}. \quad (23.40) \quad \boxed{\text{E:DensityN2Est}}$$

We also observe that $\mathcal{N}_0 \mathcal{N}_2^c \subseteq \mathcal{N}_3$. Thus $\mathcal{N}_0 = \mathcal{N}_0 \mathcal{N}_2 \cup \mathcal{N}_0 \mathcal{N}_2^c \subseteq \mathcal{N}_2 \cup \mathcal{N}_3$, so from (23.39) and (23.40) we deduce that

$$\bar{d}(\mathcal{N}_0) \leq \bar{d}(\mathcal{N}_2) + \bar{d}(\mathcal{N}_3) \ll K^{\alpha-e+1} + \frac{\log K}{\log \log L}.$$

Thus $\bar{d}(\mathcal{N}_0) < \varepsilon$ if K is sufficiently large and if L is sufficiently large compared with K .

To prove (23.37), we suppose, as we may, that $L > \exp(K^2)$. For $n \in \mathcal{N}_1$, choose r so that $d_r < K$ and $d_{r+1} > \log L$. Thus $n = d_r m$ with m composed entirely of primes $> (\log L)/d_r$. This decomposition is unique, since d_r is composed entirely of primes $< K$, and m is composed entirely of primes $> (\log L)/d_r > K^2/d_r \geq K$. Hence

$$\text{card}\{n \leq x : n \in \mathcal{N}_1\} = \sum_{d < K} \text{card}\{m \leq x/d : p|m \implies p > (\log L)/d\}.$$

By the theorem of Eratosthenes–Legendre (Theorem 3.1), this is

$$\begin{aligned} & \sim x \sum_{d < K} \frac{1}{d} \prod_{p < \frac{\log L}{d}} \left(1 - \frac{1}{p}\right) \leq x \left(\sum_{d < K} \frac{1}{d} \right) \prod_{p < \frac{\log L}{K}} \left(1 - \frac{1}{p}\right) \\ & \ll x \frac{\log K}{\log((\log L)/K)}, \end{aligned}$$

so we have (23.37).

As for (23.38), suppose that $n \in \mathcal{N}_1^c \mathcal{N}_2$ and that $d_r \in (K, L]$. Then $d_{r+1} = d_r p_{r+1} \leq d_r^{1+b}$. Thus by induction, if r_0 is the least r for which $d_r > K$, then

$$d_r \leq d_{r_0}^{(1+b)^{r-r_0}} < (\log L)^{(1+b)^{r-r_0}}$$

provided that this bound is $\leq L$. Set

$$R = \left\lceil \frac{\log \log L - \log \log \log L}{\log(1+b)} \right\rceil.$$

Then $d_r \leq L$ for $r \leq r_0 + R$. Let

$$\Omega_L(n) = \sum_{\substack{p^k \parallel n \\ p \leq L}} 1.$$

Thus $\Omega_L(n) \geq R$ if $n \in \mathcal{N}_2$. By the Turán–Kubilius inequality (Theorem 23.1),

$$\sum_{n \leq x} (\Omega_L(n) - \log \log L)^2 \ll x \log \log L.$$

Let $c = 1/\log(1+b)$. Here $c > 1$, since $1+b < e$, and $R > (c-\varepsilon) \log \log L$ if L is sufficiently large. Thus $\Omega_L(n) - \log \log L \gg \log \log L$ when $n \in \mathcal{N}_2$, and so we have (23.38).

If $n \in \mathcal{N}_3$, then we may write $n = dm$ where $K < d \leq L$, $p|m$ implies $p > d^\beta$, and $\varphi(m)/m < 1 - d^{-\alpha}$. This decomposition may not be unique, but

$$\begin{aligned} \text{card}\{n \leq x : n \in \mathcal{N}_3\} \\ \leq \sum_{K < d \leq L} \text{card}\{m \leq x/d : p|m \implies p > d^\beta, \varphi(m)/m < 1 - d^{-\alpha}\}. \end{aligned}$$

(23.41) E:DensityN3Est

Let

$$f_y(m) = \sum_{\substack{p|m \\ p > y}} \log(1 - 1/p)^{-1}.$$

This is an additive function with

$$A(f_y, z) \leq (1 + o(1))(y \log y)^{-1}, \quad B(f_y, z) \ll y^{-2}(\log y)^{-1}.$$

Thus if $V \geq 2/(y \log y)$, then by the Turán–Kubilius inequality we see that

$$\text{card}\{m \leq z : f_y(m) > V\} \ll \frac{z}{V^2 y^2 \log y}.$$

On taking $z = x/d$, $y = d^\beta$, $V = \log(1 - d^{-\alpha})^{-1} \asymp d^\alpha$, we see that the $m \leq x/d$ for which $f_y(m) > V$ includes the m in (23.41), and hence

$$\text{card}\{n \leq x : n \in \mathcal{N}_3\} \ll x \sum_{K < d \leq L} d^{-1+2\alpha-2\beta} (\log d)^{-1} \ll \frac{x K^{2\alpha-2\beta}}{\log K}.$$

This gives (23.39), in view of the definition of β . Thus the proof is complete. $\square$

23.5 Multiplicative functions with vanishing mean value

S:MFMVO

Suppose that

$$S(x) = \sum_{n \leq x} f(n). \quad (23.42) \quad \text{E:DefS(x)}$$

If $S(x) \ll x$, then by Theorem 1.3 the Dirichlet series

$$F(s) = \sum_{n=1}^{\infty} f(n)n^{-s} \quad (23.43) \quad \text{E:DefGenFcnF}$$

converges for $\sigma > 1$, and

$$F(s) = s \int_1^{\infty} S(x)x^{-s-1} dx$$

for $\sigma > 1$. From this formula it is immediate that if $S(x) = ax + o(x)$, then

$$F(s) = \frac{a}{s-1} + o\left(\frac{\tau}{\sigma-1}\right)$$

as $\sigma \rightarrow 1^+$. This is a simple abelian theorem. In prior discussions of tauberian converses, such as in §5.2, we imposed a bound on the size of $f(n)$ so that $S(x)$ could not change too quickly. In the present context, the hypothesis that $|f(n)| \leq 1$ for all n does not yield a converse (cf Exercise 28.5.1.1), but we find that the hypothesis that $f \in \mathcal{M}_0$ is sufficient. The lesson is that for $f \in \mathcal{M}_0$, the quantity $|S(x)|$ changes more slowly on average than it might under the weaker assumption that $|f(n)| \leq 1$.

T:MFMVO

Theorem 23.14. *Suppose that $f \in \mathcal{M}_0$, let $S(x)$ and $F(s)$ be defined by (23.42) and (23.43), and for $\alpha > 0$ put*

$$M(\alpha) = \left(\sum_{k=-\infty}^{\infty} \frac{1}{k^2 + 1} \max_{\substack{\sigma \geq 1+\alpha \\ |t-k| \leq 1/2}} |F(s)|^2 \right)^{1/2}.$$

Then

$$S(x) \ll \frac{x}{\log x} \int_{1/\log x}^1 \frac{M(\alpha)}{\alpha} d\alpha. \quad (23.44) \quad \text{E:MVEst}$$

From the trivial bound $F(s) \ll 1/(\sigma - 1)$ it follows that $M(\alpha) \ll 1/\alpha$, and when this is inserted in (23.44) we find that $S(x) \ll x$, which is also trivial. However, if for every T we have $F(s) = o(1/(\sigma - 1))$ uniformly for $|t| \leq T$, then $M(\alpha) = o(1/\alpha)$, and hence $S(x) = o(x)$.

We show below that

$$M(\alpha) \gg 1 \quad (23.45) \quad \boxed{\text{E:M}>>1}$$

uniformly for $f \in \mathcal{M}_0$. Thus the right hand side of (23.44) is

$$\gg x \frac{\log \log x}{\log x}.$$

That this should be the limit of the method is not surprising, in view of the example considered in Exercise 27.5.1.2, for which $f \in \mathcal{M}_0$, $M(\alpha) \asymp 1$, and yet there is a large x for which $|S(x)| \gg x(\log x)^{-1} \log \log x$.

To establish (23.45), we write

$$F(s) = (1 + D(s))G(s)H(s) \quad (23.46) \quad \boxed{\text{E:DecompF}}$$

where

$$D(s) = \sum_{k=1}^{\infty} \frac{f(2^k)}{2^{ks}}, \quad G(s) = \prod_{p>2} \left(1 - \frac{f(p)}{p^s}\right)^{-1},$$

$$H(s) = \prod_{p>2} \left(1 - \frac{f(p)}{p^s}\right) \left(1 + \frac{f(p)}{p^s} + \frac{f(p^2)}{p^{2s}} + \cdots\right).$$

Here

$$\begin{aligned} & \left(1 - \frac{f(p)}{p^s}\right) \left(1 + \frac{f(p)}{p^s} + \frac{f(p^2)}{p^{2s}} + \cdots\right) \\ &= 1 + \frac{f(p^2) - f(p)^2}{p^{2s}} + \frac{f(p^3) - f(p)f(p^2)}{p^{3s}} + \cdots, \end{aligned}$$

which is 1 plus an amount not exceeding $2p^{-2\sigma}(1 - p^{-\sigma})^{-1}$ in absolute value. Thus the product $H(s)$ is absolutely and uniformly convergent for $\sigma \geq 2/3$, and so

$$\log H(s) \ll 1 \quad (\sigma \geq 2/3). \quad (23.47) \quad \boxed{\text{E:H}<<1}$$

Choose a real number t_0 with $|t_0| \leq \pi/\log 2$, so that $f(2)/2^{it_0}$ is positive real. Then $\operatorname{Re} f(2)/2^{it} \geq 0$ for $|t - t_0| \leq \pi/(2\log 2)$, and so $|1 + D(s)| \geq$

$1/2$ for $\sigma \geq 1$, $|t - t_0| \leq \pi/(2 \log 2)$. Also,

$$\begin{aligned} \int_{t_0-1}^{t_0+1} \log G(\sigma + it) dt &= \sum_{p>2} \sum_{k=1}^{\infty} \frac{f(p)^k}{k p^{k(1+\alpha)}} \int_{t_0-1}^{t_0+1} p^{-ikt} dt \\ &\ll \sum_p \sum_{k=1}^{\infty} \frac{1}{k^2 p^k \log p} \ll 1. \end{aligned}$$

Thus there is an absolute constant $c > 0$, and a t_1 , $|t_0 - t_1| \leq 1$, such that $|1 + D(1 + \alpha + it_1)| \geq 1/2$ and $|G(1 + \alpha + it_1)| \geq c$, so we have (23.45).

Proof of Theorem 23.14 We shall establish the two main estimates

$$S(x) \ll \frac{x}{\log x} \int_1^x \frac{|S(u)|}{u^2} du + \frac{x \log \log x}{\log x}, \quad (23.48) \quad \boxed{\text{E:S(x)Est1}}$$

$$\int_1^x \frac{|S(u)| \log u}{u^2} du \ll M(2/\log x) \log x. \quad (23.49) \quad \boxed{\text{E:S(x)Est2}}$$

These suffice to give the stated result, since from (23.49) it is evident that

$$\int_{x^{1/2}}^x \frac{|S(u)|}{u^2} du \ll M(2/\log x) \ll \int_{1/\log x}^{2/\log x} \frac{M(\alpha)}{\alpha} d\alpha.$$

We replace x by $x^{1/2^k}$ and sum over k to show that

$$\int_1^x \frac{|S(u)|}{u^2} du \ll \int_{1/\log x}^1 \frac{M(\alpha)}{\alpha} d\alpha.$$

We insert this in (23.48) to obtain the stated result. The second term in (23.48) can be neglected, in view of (23.45).

To establish (23.48) we first observe that

$$\begin{aligned} (\log x) \sum_{n \leq x} f(n) - \sum_{n \leq x} f(n) \log n \\ = \sum_{n \leq x} f(n) \log x/n \ll \sum_{n \leq x} \log x/n \ll x. \end{aligned} \quad (23.50) \quad \boxed{\text{E:f(n)logx/n}}$$

Furthermore,

$$\begin{aligned}
\sum_{n \leq x} f(n) \log n &= \sum_{n \leq x} f(n) \sum_{d|n} \Lambda(d) \\
&= \sum_{d \leq x} \Lambda(d) \sum_{m \leq x/d} f(md) \\
&= \sum_{p \leq x} (\log p) \sum_{m \leq x/p} f(mp) + O\left(x \sum_{\substack{p^k \leq x \\ k > 1}} \frac{\log p}{p^k}\right) \\
&= \sum_{p \leq x} (\log p) f(p) S(x/p) \\
&\quad + O\left(\sum_{p \leq x} (\log p) \sum_{m \leq x/p} |f(mp) - f(m)f(p)|\right) + O(x).
\end{aligned}$$

Since $f(mp) = f(m)f(p)$ unless $p|m$, we see that the sum over $m \leq x/p$ is $\ll x/p^2$, and so the first error term above is $\ll x$. On combining this with (23.50), we deduce that

$$S(x) \log x \ll x + \sum_{p \leq x} |S(x/p)| \log p. \quad (23.51) \quad \boxed{\text{E:S(x)Est3}}$$

Here we have a bound for $|S(x)|$ in terms of S at smaller arguments. The trivial bound for either side is $x \log x$. Thus if it were the case that $S(x)$ were of the order of x , then $S(x/p)$ would have to be of the order of x/p for many primes p . If the primes were exactly uniformly distributed, then the sum over p would be

$$\int_1^x |S(x/v)| dv = \int_1^x \frac{|S(u)|}{u^2} du.$$

Of course the primes are rather irregularly distributed, but as x varies the points x/p also move, so by averaging over x we can pass to a smoother average of $|S|$. Suppose that $X > Y > Z \geq 2$. Since $|S(X) - S(x)| \leq |X - x| + 1$, we see that

$$|S(X)| \log X \ll \frac{1}{Y} \int_X^{X+Y} |S(x)| \log x dx + Y \log X.$$

By (27.50) this is

$$\ll X + Y \log X + \frac{1}{Y} \int_X^{X+Y} \sum_{p \leq x} |S(x/p)| \log p dx.$$

We bound the contribution of the smaller primes trivially:

$$\sum_{p \leq X/Z} |S(x/p)| \log p \ll x \sum_{p \leq X/Z} \frac{\log p}{p} \ll X \log X/Z.$$

As for the contribution of the larger primes, we note that

$$\begin{aligned} \int_X^{X+Y} \sum_{X/Z < p \leq 2X} |S(x/p)| \log p \, dx &= \sum_{X/Z < p \leq 2X} \int_X^{X+Y} |S(x/p)| \, dx \log p \\ &= \sum_{X/Z < p \leq 2X} \int_{X/p}^{(X+Y)/p} |S(u)| \, du \, p \log p \\ &= \int_1^{2Z} |S(u)| \sum_{\substack{X/Z < p \leq 2X \\ X/u < p \leq (X+Y)/u}} p \log p \, du. \end{aligned}$$

Here we can restrict to $u \leq 2Z$ because the two intervals that p must lie in are disjoint if $u > 2Z$. In estimating the above, we now drop the condition $X/Z < p \leq 2X$. The remaining condition stipulates that p must lie in an interval whose length is $Y/u \geq Y/(2Z) \geq 2$ if $Z \leq Y/4$. Thus by the Brun–Titchmarsh inequality (Corollary 3.4) the number of primes in the interval is bounded by the length of the interval divided by the logarithm of its length. Hence the above sum over primes is

$$\ll \frac{XY \log X/u}{u^2 \log Y/u}.$$

Here the quotient of logarithms is an increasing function of u , so the above is uniformly

$$\ll \frac{XY \log X/(2Z)}{u^2 \log Y/(2Z)}.$$

We take $Y = X/\log X$ and $Z = X/(\log X)^2$, and on assembling our estimates discover that

$$S(X) \log X \ll X \int_1^X \frac{|S(u)|}{u^2} \, du + X \log \log X.$$

That is, we have (23.48).

Finally we prove (23.49). Let $S_1(x) = \sum_{n \leq x} f(n) \log n$. By (23.45) and (23.50) it suffices to show that

$$\int_1^x \frac{|S_1(u)|}{u^2} \, du \ll M(2/\log x) \log x. \quad (23.52) \quad \boxed{\text{E:S1Est1}}$$

By the Cauchy–Schwarz inequality,

$$\int_1^x \frac{|S_1(u)|}{u^2} du \leq \left(\int_1^x \frac{|S_1(u)|^2}{u^3} du \right)^{1/2} \left(\int_1^x \frac{1}{u} du \right)^{1/2}.$$

It now suffices to show that

$$\int_1^\infty \frac{|S_1(u)|^2}{u^{3+2\alpha}} du \ll \frac{M(\alpha)^2}{\alpha} \quad (23.53) \quad \boxed{\text{E:S1Est2}}$$

for $0 < \alpha \leq 1$, since we obtain (23.52) by taking $\alpha = 2/\log x$. By Plancherel’s formula as in (5.26), we see that

$$\begin{aligned} \int_1^\infty \frac{|S_1(u)|^2}{u^{3+2\alpha}} du &= \frac{1}{2\pi} \int_{-\infty}^\infty \left| \frac{F'(1+\alpha+it)}{1+\alpha+it} \right| dt \\ &\ll \sum_{k=-\infty}^\infty \frac{1}{k^2+1} \int_{k-1/2}^{k+1/2} |F'(1+\alpha+it)|^2 dt. \end{aligned}$$

We multiply and divide by $|F(1+\alpha+it)|^2$ to see that the above is

$$\leq \sum_{k=-\infty}^\infty \frac{\mu(k)}{k^2+1} \int_{k-1/2}^{k+1/2} \left| \frac{F'}{F}(1+\alpha+it) \right|^2 dt$$

where

$$\mu(k) = \max_{|t-k| \leq 1/2} |F(1+\alpha+it)|^2.$$

Thus to obtain (23.53) it suffices to show that

$$\int_{k-1/2}^{k+1/2} \left| \frac{F'}{F}(1+\alpha+it) \right|^2 dt \ll \frac{1}{\alpha}. \quad (23.54) \quad \boxed{\text{E: Int } |F'/F|^2 \text{ Est}}$$

By (23.46) we see that

$$\frac{F'}{F}(s) = \frac{D'(s)}{1+D(s)} + \frac{G'}{G}(s) + \frac{H'}{H}(s).$$

From (23.47) we deduce that $\frac{H'}{H}(s) \ll 1$ uniformly for $\sigma \geq 1$. By Theorem J.1 we see that

$$\int_{k-1/2}^{k+1/2} \left| \frac{G'}{G}(1+\alpha+it) \right|^2 dt \leq 3 \int_{-1/2}^{1/2} \left| \frac{\zeta'}{\zeta}(1+\alpha+it) \right|^2 dt.$$

If $0 < \alpha \leq 1$, then by Theorem 6.7 this latter integral is

$$\ll \int_{-1/2}^{1/2} |\alpha+it|^{-2} dt \ll \frac{1}{\alpha}.$$

Clearly $D'(s) \ll 1$ for $\sigma \geq 1$. For $\sigma > 1$ we write

$$\frac{1}{1+D(s)} = \sum_{j=0}^{\infty} (-D(s))^j.$$

This is a Dirichlet series whose coefficients do not exceed those of

$$1 + \sum_{j=1}^{\infty} \left(\sum_{k=1}^{\infty} 2^{-ks} \right)^j = 1 + \sum_{j=1}^{\infty} (2^s - 1)^{-j} = \frac{2^s - 1}{2^s - 2}.$$

Hence by Lemma ??,

$$\int_{k-1/2}^{k+1/2} |1 + D(1 + \alpha + it)|^{-2} dt \leq 3 \int_{-1/2}^{1/2} \left| \frac{2^{1+\alpha+it} - 1}{2^{1+\alpha+it} - 2} \right|^2 dt. \quad (23.55) \quad \boxed{\text{E: } |D|^{-2} \text{Est}}$$

Here $|2^{1+\alpha+it} - 1| \asymp 1$ uniformly for $0 \leq \alpha \leq 1$, and

$$2^{1+\alpha+it} - 2 = 2(\log 2) \int_0^{\alpha+it} 2^s ds.$$

This integrand has real part $\geq 1/2$ for $\sigma \geq 0$ and $|t| \leq \pi/(3 \log 2) = 1.5107867 \dots$, so $|2^{1+\alpha+it} - 2| \gg |\alpha + it|$ for $\alpha \geq 0$ and $|t| \leq 1/2$. Thus the right hand side of (23.55) is

$$\ll \int_{-1/2}^{1/2} |\alpha + it|^{-2} dt \ll \frac{1}{\alpha},$$

so we have (23.54), and the proof is complete. $\square$

We comment that the first part of our proof is reminiscent of the elementary proof of the Prime Number Theorem, as found in §8.2. Moreover, the identity on the left hand side of (23.50) is equivalent to integrating by parts in Perron's formula, as

$$\begin{aligned} \frac{\log x}{2\pi i} \int_{c-i\infty}^{c+i\infty} \frac{F(s)x^s}{s} ds &= \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \frac{F(s)x^s}{s^2} ds \\ &\quad - \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \frac{F'(s)x^s}{s} ds. \end{aligned}$$

This is expected to produce a gain, since we expect that $F(s)/s$ is not generally very rapidly changing, while x^s is spinning fairly rapidly. Indeed, suppose we tried to do something as simple as using Perron's formula to show that $[x] \ll x$. Since

$$\int_{-1}^1 |\zeta(1 + \alpha + it)| x^{1+\alpha} dt \asymp x^{1+\alpha} \log \frac{1}{\alpha},$$

we are unable to obtain a bound better than $x \log \log x$. On the other hand, if we were to use Perron's formula to show that $\sum_{n \leq x} \log n \ll x \log x$, we fare better, since

$$\int_{-1}^1 |\zeta'(1 + \alpha + it)| x^{1+\alpha} dt \asymp \frac{x^{1+\alpha}}{\alpha},$$

and we can take $\alpha = 1/\log x$. In both of these approaches, we would still have the problem that the kernel in Perron's formula decays only like an inverse first power. This could be overcome by smoothing, but in the argument just completed we avoided that problem by averaging over x , which allows us to appeal to Plancherel's identity.

The proof just completed depends only on properties of the zeta function in a neighbourhood of $s = 1$, but if we take $f(n) = \mu(n)$, so that $F(s) = 1/\zeta(s)$, then the further information that $\zeta(1 + it) \neq 0$ allows us to deduce that $M(x) = o(x)$.

We now relate the behaviour of $F(s)$ to the values of $f(p)$.

T:HalaszThm

Theorem 23.15 (Halász). *Suppose that $f \in \mathcal{M}_0$, and let $S(x)$ and $F(s)$ be defined as in (23.42) and (23.43). Then the following are equivalent:*

- (a) $S(x) = o(x)$ as $x \rightarrow \infty$;
- (b) For each $T > 0$, $F(s) = o(1/(\sigma - 1))$ as $\sigma \rightarrow 1^+$ uniformly for $|t| \leq T$;
- (c) For each fixed t , $F(\sigma + it) = o(1/(\sigma - 1))$ as $\sigma \rightarrow 1^+$;
- (d) For each t , at least one of the following holds:

$$\begin{aligned} \text{(i)} \quad & \sum_p \frac{1 - \operatorname{Re}(f(p)p^{-it})}{p} = +\infty, \\ \text{(ii)} \quad & f(2^k) = -2^{ikt} \text{ for } k = 1, 2, \dots \end{aligned} \tag{23.56} \quad \text{E:Conditions}$$

Moreover, there is at most one real number t for which (23.56)(i) fails, and at most one real number t for which (23.56)(ii) is true.

Delange's Theorem (Theorem 23.8), when combined with Halász's Theorem (Theorem 23.15) is rather comprehensive, for if (23.56) fails and

$$\sum_p \frac{\operatorname{Im}(f(p)p^{-it})}{p} \tag{23.57} \quad \text{E:ConvSum}$$

converges, then by Theorem 23.8 we have

$$\sum_{n \leq x} f(n)n^{-it} = ax + o(x)$$

where

$$a = \prod_p \left(1 - \frac{1}{p}\right) \left(1 + \frac{f(p)}{p^{1+it}} + \frac{f(p^2)}{p^{2+2it}} + \cdots\right)$$

is nonzero, and by partial summation,

$$S(x) = \frac{a}{1+it} x^{1+it} + o(x).$$

In the one remaining case, in which (23.56) fails for some t , and (23.57) does not hold for that t , then with more work it can be shown that

$$S(x) = \frac{x^{1+it}}{1+it} \prod_{p \leq x} \left(1 - \frac{1}{p}\right) \left(1 + \frac{f(p)}{p^{1+it}} + \frac{f(p^2)}{p^{2+2it}} + \cdots\right) + o(x). \quad (23.58) \quad \boxed{\text{E:AltEstS(x)}}$$

Proof That (a) implies (b) was the subject of the opening remarks of this section. That (b) implies (a) was established in a strong quantitative form in Theorem 23.14. To see that (c) and (d) are equivalent, we recall the decomposition (23.46). By (23.47) we know that $H(s) \asymp 1$ uniformly for $\sigma > 1$. Also, $\lim_{\sigma \rightarrow 1} D(s) = D(1+it)$, and $1 + D(1+it) = 0$ if and only if (23.56)(ii) holds. Finally,

$$|G(s)(\sigma-1)| \asymp \left| \frac{G(s)}{\zeta(s)} \right| \asymp \exp \left(- \sum_p \frac{\operatorname{Re}(1 - f(p)p^{-it})}{p} \right).$$

Here the summands are nonnegative, so the expression is bounded, and tends to 0 if and only if the sum tends to infinity. But since the summands are nonnegative, this is equivalent to (23.56)(i). Thus (c) and (d) are equivalent. Next we show that (d) implies (b). Suppose first that (23.56)(i) holds for all t in an interval $[T_1, T_2]$. We observe that

$$F(s)(\sigma-1) \ll \exp \left(- \sum_p \frac{\operatorname{Re}(1 - f(p)p^{-it})}{p^\sigma} \right).$$

The function on the right hand side decreases to 0 as $\sigma \rightarrow 1^+$. Thus we obtain (b) for the interval $[T_1, T_2]$ by appealing to the following elementary consequence of compactness: If $r(\sigma, t)$ is continuous in t for each fixed $\sigma > 1$, and if for each fixed $t \in [T_1, T_2]$ the function $r(\sigma, t)$ is monotonically decreasing to 0 as $\sigma \rightarrow 1^+$, then $r(\sigma, t)$ tends to 0 as

$\sigma \rightarrow 1^+$ uniformly for $t \in [T_1, T_2]$. Now suppose that (23.56)(i) fails for $t = t_0$, but that (23.56)(ii) holds for $t = t_0$. Then for $|t - t_0| \leq 1$ we have

$$F(s)(\sigma - 1) \ll |s - 1 - it_0| \exp \left(- \sum_p \frac{\operatorname{Re}(1 - f(p)p^{-it})}{p^\sigma} \right).$$

Again the right hand side decreases monotonically to 0 as $\sigma \rightarrow 1^+$, since (23.56)(i) holds for $0 < |t - t_0| \leq 1$. Thus by the compactness principle again, we have (b) uniformly for $|t - t_0| \leq 1$. Thus (d) implies (b). Since (b) clearly implies (c), we have shown that (a)–(d) are equivalent.

As for the last assertion, let $t_1 < t_2$ be fixed real numbers, and let $\mathcal{P}$ be the set of primes p for which $\arg p^{i(t_2 - t_1)} \in [2\pi/3, 4\pi/3] \pmod{2\pi}$. That is, if

$$I_k = [\exp(2\pi(k + 1/3)/(t_2 - t_1)), \exp(2\pi(k + 2/3)/(t_2 - t_1))],$$

then $\mathcal{P}$ consists of those primes such that $p \in I_k$ for some k . By the Prime Number Theorem we see that $\sum_{p \in I_k} 1/p \asymp 1/k$ for all large k . Hence $\sum_{p \in \mathcal{P}} 1/p = +\infty$. If $\operatorname{Re}(1 - f(p)p^{-it}) \leq 1/2$, then $|\arg f(p)p^{-it}| \leq \pi/3$. If this holds for both t_1 and t_2 , then $|\arg p^{i(t_1 - t_2)}| \leq 2\pi/3$. Thus if $p \in \mathcal{P}$, then the inequality $\operatorname{Re}(1 - f(p)p^{-it_j}) \leq 1/2$ fails for at least one value of j , and so

$$\sum_{p \in \mathcal{P}} \frac{1}{p} \leq 2 \sum_p \frac{\operatorname{Re}(1 - f(p)p^{-it_1})}{p} + 2 \sum_p \frac{\operatorname{Re}(1 - f(p)p^{-it_2})}{p}.$$

Consequently at most one of the sums on the right is convergent, and the proof is complete. $\square$

Suppose that $f \in \mathcal{M}_0$. If there is a point of the unit circle $|z| = 1$ that is not a limit point of the numbers $f(p)$, then for any $t \neq 0$ there is a delta such that

$$\sum_{\substack{p \\ \operatorname{Re}(1 - f(p)p^{-it}) > \delta}} \frac{1}{p} = +\infty,$$

so (23.56)(i) holds for all $t \neq 0$. In closing we mention a commonly occurring situation.

Co:MFVSpecCase

Corollary 23.16. *Suppose that $f \in \mathcal{M}_0$, and that there is a constant $c > 0$ such that $|\operatorname{Im} f(p)| \leq c \operatorname{Re}(1 - f(p))$. Then*

$$\lim_{x \rightarrow \infty} \frac{1}{x} \sum_{n \leq x} f(n)$$

exists.

Proof In view of the remark made prior to this Corollary, the condition (23.56)(i) holds for all $t \neq 0$. If (23.56) holds when $t = 0$, then by Theorem 23.15 the mean value tends to 0. Otherwise,

$$\begin{aligned} \sum_p \frac{|1 - f(p)|}{p} &\leq \sum_p \frac{\operatorname{Re}(1 - f(p)) + |\operatorname{Im} f(p)|}{p} \\ &\leq (c + 1) \sum_p \frac{\operatorname{Re}(1 - f(p))}{p} < \infty, \end{aligned}$$

so the mean value exists and is non-zero, by Corollary 23.4. $\square$

23.5.1 Exercises

- Put $f(n) = 1$ for $N < n \leq 2N$, $f(n) = 0$ otherwise, and let $F(s)$ be defined as in (23.26).
 - By Theorem 1.12, or otherwise, show that $F(s) \ll 1 + \tau/N$ uniformly for $\sigma \geq 1$.
 - Note that $S(x) \asymp x$ when $x = 2N$.
- (Montgomery 1978) (a) Let $f_0(n) = i^{\Omega(n)}$. By Theorem 7.18, or otherwise, show that

$$S_0(x) = \sum_{n \leq x} f_0(n) = cx(\log x)^{i-1} + O(x(\log x)^{-2})$$

where

$$c = \frac{1}{\Gamma(i)} \prod_p \left(1 - \frac{i}{p}\right)^{-1} \left(1 - \frac{1}{p}\right)^i.$$

- Let $F_0(s) = \sum_{n=1}^{\infty} f_0(n)n^{-s}$. By means of Theorem 6.7, or otherwise, show that $F_0(s) \ll \log \tau$ uniformly for $\sigma > 1$.
- Let f be a totally multiplicative function with

$$f(p) = \begin{cases} i & (\text{if } p \leq x^{1/2} \text{ or } p > x), \\ e(\theta_p) & (\text{for } x^{1/2} < p \leq x) \end{cases}$$

where the θ_p are to be determined.

- Explain why

$$S(x) = \sum_{\substack{n \leq x \\ p|n \implies p \leq x^{1/2}}} f_0(n) + \sum_{x^{1/2} < p \leq x} e(\theta_p) S_0(x/p).$$

- (e) Deduce that there is a choice of the θ_p such that

$$|S(x)| = \left| \sum_{\substack{n \leq x \\ p|n \implies p \leq x^{1/2}}} f_0(n) \right| + \sum_{x^{1/2} < p \leq x} |S_0(x/p)|.$$

- (f) Show that

$$\sum_{x^{1/2} < p \leq x} |S_0(x/p)| \asymp \frac{x \log \log x}{\log x}.$$

- (g) Show that $M(\alpha) \ll 1$ uniformly for $\alpha > 0$.

3. Recall that the ‘negative binomial theorem’ asserts that

$$(1 - z)^{-r-1} = \sum_{n=0}^{\infty} \binom{n+r}{n} z^n$$

for $|z| < 1$. Here r is any complex number.

- (a) Show that if $r > -1$, then $\binom{n+r}{n} \geq 0$ for all n .
 (b) Suppose that f is totally multiplicative, that $|f(n)| \leq 1$ for all n , and let F be defined as in (23.26). Show that if q is a positive real number, then

$$\int_{T_0-T}^{T_0+T} |F(\sigma + it)|^q dt \leq 3 \int_{-T}^T |\zeta(\sigma + it)|^q dt.$$

- (c) Use (23.46) to show that if $f \in \mathcal{M}_0$ and $q > 0$, then

$$\int_{T_0-T}^{T_0+T} |F(\sigma + it)|^q dt \ll_q \int_{-T}^T |\zeta(\sigma + it)|^q dt.$$

4. (Turán) Let $f(n)$ be an integer-valued additive function, and let $N(x; q, a)$ denote the number of $n \leq x$ such that $f(n) \equiv a \pmod{q}$.

- (a) Show that $\lim_{x \rightarrow \infty} N(x; q, a)/x = n(q, a)$ exists for all a and q .
 (b) Show that $n(q, a) = 1/q$ for all a if and only if both the following hold:

- (i) For each odd prime $p_1 | q$,

$$\sum_{\substack{p \\ p_1 | f(p)}} \frac{1}{p} = +\infty;$$

- (ii) If $2 | q$, then

$$\sum_{\substack{p \\ 2 | f(p)}} \frac{1}{p} = +\infty$$

or both of the following hold: $f(2^k)$ is odd for all $k > 0$, and if $4|q$, then

$$\sum_{\substack{p \\ 4|f(p)}} \frac{1}{p} = +\infty.$$

23.6 Notes

S:NotesProbNoThy

Our use of probabilistic modelling is necessarily a little informal, since the size of sets of integers as measured by asymptotic density do not form a probability space. One of Kolmogorov's fundamental axioms states that if $E_1, E_2, \dots$, are pairwise disjoint events (i.e., sets) in a probability space, then

$$\mathbf{P}\left(\bigcup_{n=1}^{\infty} E_n\right) = \sum_{n=1}^{\infty} \mathbf{P}(E_n).$$

However, if $E_n = \{n\}$, then the asymptotic density of E_n is 0 for all n , while the density of the union of the E_n is 1.

23.7 References

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24

Exponential Sums II: Vinogradov's method

C:ExpSumII

24.1 Vinogradov's mean value theorem

S:Vmvt

In Chapter ??—we derived nontrivial bounds for sums of the sort

$$\sum_{a \leq n \leq b} e(f(n))$$

from information regarding the first few derivatives of f . However, in studying the zeta function near the line $\operatorname{Re} s = 1$ we have to examine sums such as

$$\sum_{a \leq n \leq b} n^{-1-it},$$

and here the situation is much more delicate. To obtain a satisfactory treatment it is necessary to consider derivatives of

$$f(x) = \frac{t}{2\pi} \log x$$

of order r , where r depends on the interrelationship between a and t . It is possible to make use of Theorem ?? in this regard, although the dependence on r in the implicit constant there needs to be made explicit. The amount of saving which can be made in this way is rather poor because the exponent is dropping off like 2^{-r} , but Littlewood was able to show that the zeta function has a zero free region of the form

$$\sigma \geq 1 - c \frac{\log \log \tau}{\log \tau}.$$

We now introduce a much more efficient way of making use of higher derivatives, or rather, what is tantamount to the same thing, polynomial approximations to $f(x)$ of arbitrary degree. The underlying idea is to

relate the individual sums to a mean value. Let

$$\mathbf{n}^{(k)} = (n, n^2, \dots, n^k)$$

and let

$$\begin{aligned}\boldsymbol{\alpha} &= (\alpha_1, \alpha_2, \dots, \alpha_k), \\ S(\boldsymbol{\alpha}, \mathcal{A}) &= \sum_{n \in \mathcal{A}} e(\mathbf{n}^{(k)} \cdot \boldsymbol{\alpha})\end{aligned}$$

where $\mathcal{A}$ is a finite set of integers. We are interested in the mean value

$$J_k(\mathcal{A}, b) = \int_{\mathbb{T}^k} |S(\boldsymbol{\alpha}, \mathcal{A})|^{2b} d\boldsymbol{\alpha}.$$

Now

$$S(\boldsymbol{\alpha}, \mathcal{A})^b = \sum_{\mathbf{m}} r(\mathbf{m}, \mathcal{A}^b) e(\mathbf{m} \cdot \boldsymbol{\alpha})$$

where $r(\mathbf{m}, \mathcal{A}^b)$ denotes the number of solutions of the system

$$\begin{array}{ccccccc} n_1 & + & \cdots & + & n_b & = & m_1 \\ n_1^2 & + & \cdots & + & n_b^2 & = & m_2 \\ \vdots & & & & \vdots & & \vdots \\ n_1^k & + & \cdots & + & n_b^k & = & m_k \end{array} \quad (24.1) \quad \boxed{\text{E:kxbeqns}}$$

with $n_i \in \mathcal{A}$. Thus by Parseval's identity,

$$J_k(\mathcal{A}, b) = \sum_{\mathbf{m}} r(\mathbf{m}, \mathcal{A}^b)^2.$$

When $\mathcal{B}$ and $\mathcal{C}$ are subsets of $\mathbb{R}^b$ containing only finitely many lattice points, let $N(\mathcal{B}, \mathcal{C}, \mathbf{l})$ denote the number of solutions of

$$\begin{array}{ccccccc} m_1 & + & \cdots & + & m_b & = & n_1 + \cdots + n_b + l_1 \\ m_1^2 & + & \cdots & + & m_b^2 & = & n_1^2 + \cdots + n_b^2 + l_2 \\ \vdots & & & & \vdots & & \vdots \\ m_1^k & + & \cdots & + & m_b^k & = & n_1^k + \cdots + n_b^k + l_k \end{array}$$

with $\mathbf{m} \in \mathcal{B}$ and $\mathbf{n} \in \mathcal{C}$, and for brevity write $N(\mathcal{B}, \mathbf{l}) = N(\mathcal{B}, \mathcal{B}, \mathbf{l})$, $N(\mathcal{B}) = N(\mathcal{B}, \mathbf{0})$ and $N(\mathcal{B}, \mathcal{C}) = N(\mathcal{B}, \mathcal{C}, \mathbf{0})$. Then we can define the more general mean

$$J_k(\mathcal{A}, b, \mathbf{l}) = N(\mathcal{A}^b, \mathbf{l}),$$

so that

$$J_k(\mathcal{A}, b) = N(\mathcal{A}^b),$$

concerning which the following elementary observations are useful.

$L:N(B)eqs+ineqs$

Lemma 24.1. *In the above notation,*

- (a) *If $\mathcal{B} \subseteq \mathcal{C}$, then $N(\mathcal{B}, \mathbf{l}) \leq N(\mathcal{C}, \mathbf{l})$,*
- (b) *$N(\mathcal{B}, \mathbf{l}) \leq N(\mathcal{B})$ for all $\mathbf{l}$,*
- (c) *If $\mathcal{C} = \mathcal{B}_1 \cup \dots \cup \mathcal{B}_j$, then $N(\mathcal{C}) \leq j \sum_{i=1}^j N(\mathcal{B}_i)$,*
- (d) *If $\mathbf{d} = (d, d, \dots, d)$, then $N(\mathcal{B} + \mathbf{d}, \mathcal{C} + \mathbf{d}) = N(\mathcal{B}, \mathcal{C})$,*
- (e) *$J_k(\mathcal{A}, b, \mathbf{l}) \leq J_k(\mathcal{A}, b)$.*

Proof (a) If $\mathcal{B} \subseteq \mathcal{C}$, then the solutions counted in $N(\mathcal{B}, \mathcal{B}, \mathbf{l})$ are also counted in $N(\mathcal{C}, \mathcal{C}, \mathbf{l})$.

(b) Let $r(\mathbf{m}, \mathcal{B})$ denote the number of solutions of (24.1) with $\mathbf{n} \in \mathcal{B}$. Then

$$\begin{aligned} N(\mathcal{B}, \mathbf{l}) &= \int_{\mathbb{T}^k} \left| \sum_{\mathbf{m}} r(\mathbf{m}, \mathcal{B}) e(\mathbf{m} \cdot \boldsymbol{\alpha}) \right|^2 e(-\mathbf{l} \cdot \boldsymbol{\alpha}) d\boldsymbol{\alpha} \\ &\leq \int_{\mathbb{T}^k} \left| \sum_{\mathbf{m}} r(\mathbf{m}, \mathcal{B}) e(\mathbf{m} \cdot \boldsymbol{\alpha}) \right|^2 d\boldsymbol{\alpha} = N(\mathcal{B}), \end{aligned}$$

as desired.

(c) In the above notation,

$$r(\mathbf{m}, \mathcal{C}) \leq \sum_{i=1}^j r(\mathbf{m}, \mathcal{B}_i)$$

and so by Cauchy's inequality

$$r(\mathbf{m}, \mathcal{C})^2 \leq j \sum_{i=1}^j r(\mathbf{m}, \mathcal{B}_i)^2.$$

It now suffices to sum this over $\mathbf{m}$, since

$$N(\mathcal{C}) = \sum_{\mathbf{m}} r(\mathbf{m}, \mathcal{C})^2, \quad N(\mathcal{B}_i) = \sum_{\mathbf{m}} r(\mathbf{m}, \mathcal{B}_i)^2.$$

(d) Suppose that

$$\begin{array}{ccccccc} m_1 & + & \cdots & + & m_b & = & n_1 & + & \cdots & + & n_b, \\ m_1^2 & + & \cdots & + & m_b^2 & = & n_1^2 & + & \cdots & + & n_b^2, \\ \vdots & & & & \vdots & & \vdots & & & & \vdots \\ m_1^k & + & \cdots & + & m_b^k & = & n_1^k & + & \cdots & + & n_b^k. \end{array}$$

By the binomial theorem we see that if $1 \leq j \leq k$, then

$$\sum_{i=1}^b (m_i + d)^j = \sum_{l=0}^j \binom{j}{l} d^{j-l} \sum_{i=1}^b m_i^l = \sum_{l=0}^j \binom{j}{l} d^{j-l} \sum_{i=1}^b n_i^l = \sum_{i=1}^b (n_i + d)^j.$$

Thus $N(\mathcal{B}, C) \leq N(\mathcal{B} + \mathbf{d}, C + \mathbf{d})$. Hence also,

$$N(\mathcal{B} + \mathbf{d}, C + \mathbf{d}) \leq N((\mathcal{B} + \mathbf{d}) - \mathbf{d}, (C + \mathbf{d}) - \mathbf{d}) = N(\mathcal{B}, C).$$

(e) is a special case of (b). $\square$

Let $\theta_1, \dots, \theta_q$ denote real or complex numbers and let

$$Q(z; \boldsymbol{\theta}) = \prod_{j=1}^q (z - \theta_j) = \sum_{k=0}^q (-1)^k \sigma_k z^{q-k} \quad (24.2) \quad \boxed{\text{E:DefQ}}$$

where $\sigma_0 = 1$ and $\sigma_1, \sigma_2, \dots, \sigma_q$ are the *elementary symmetric functions* of the θ_j , which is to say that if $Q = \{1, 2, \dots, q\}$, then

$$\sigma_r = \sigma_r(\boldsymbol{\theta}) = \sum_{\substack{S \subseteq Q \\ \text{card } S = r}} \prod_{j \in S} \theta_j.$$

For $m = 1, 2, \dots$ we also form the *power sums* of the θ_j ,

$$s_m = s_m(\boldsymbol{\theta}) = \sum_{j=1}^q \theta_j^m.$$

These are also symmetric polynomials in the θ_j , and are related to the σ_j by means of the *Newton–Girard formulæ*, which assert that

$$\sum_{k=0}^{r-1} (-1)^{r-1-k} \sigma_k s_{r-k} = r \sigma_r \quad (24.3) \quad \boxed{\text{E:NewGir1}}$$

for $1 \leq r \leq q$, and that

$$\sum_{k=0}^q (-1)^k \sigma_k s_{r-k} = 0 \quad (24.4) \quad \boxed{\text{E:NewGir2}}$$

for $r \geq q$. In this second identity, the quantity s_0 arises when $k = r = q$. It is to be understood that $s_0 = q$ even if one or more of the θ_j vanishes. We use the first of these identities (a proof of which is sketched in Exercise 1) to establish our next result.

$\boxed{\text{L:pwrsums}}$

Lemma 24.2. (a) Suppose that $\theta_1, \dots, \theta_q, \phi_1, \dots, \phi_q$ are such that

$$s_r(\boldsymbol{\theta}) = s_r(\boldsymbol{\phi}) \quad (1 \leq r \leq q).$$

Then

$$Q(z; \boldsymbol{\theta}) = Q(z; \boldsymbol{\phi})$$

identically.

(b) Suppose that p is a prime number with $p > q$, that u is a positive integer and that $\theta_1, \dots, \theta_q, \phi_1, \dots, \phi_q$ are integers such that

$$s_r(\boldsymbol{\theta}) \equiv s_r(\boldsymbol{\phi}) \pmod{p^u} \quad (1 \leq r \leq q).$$

Then

$$Q(z; \boldsymbol{\theta}) \equiv Q(z; \boldsymbol{\phi}) \pmod{p^u}$$

for all integers z .

Proof (a) For $r = 1$ we observe that $\sigma_1(\boldsymbol{\theta}) = s_1(\boldsymbol{\theta}) = s_1(\boldsymbol{\phi}) = \sigma_1(\boldsymbol{\phi})$. For $r > 1$ we argue by induction. In (24.3) we see that σ_r is expressed in terms of $s_1, s_2, \dots, s_r$ and $\sigma_0, \sigma_1, \dots, \sigma_{r-1}$. Hence by the inductive hypothesis the left hand side of (24.3) with respect to $\boldsymbol{\theta}$ is equal to the same expression with respect to $\boldsymbol{\phi}$.

(b) As in the preceding case we find that $\sigma_1(\boldsymbol{\theta}) \equiv \sigma_1(\boldsymbol{\phi}) \pmod{p^u}$. In the inductive step we find that $r\sigma_r(\boldsymbol{\theta}) \equiv r\sigma_r(\boldsymbol{\phi}) \pmod{p^u}$. Since $r \leq q < p$, it follows that $(r, p^u) = 1$, and hence that $\sigma_r(\boldsymbol{\theta}) \equiv \sigma_r(\boldsymbol{\phi}) \pmod{p^u}$. $\square$

L:J_kProps

Lemma 24.3. Let $J_k(x, b) = J_k((0, x], b)$. Then

- (a) $J_k(x, b) \leq b!x^b$ when $b \leq k$,
- (b) $J_k(x, b) \leq k!x^{2b-k}$ when $b > k$,
- (c) $J_k(x, b) \geq \lfloor x \rfloor^b$,
- (d) $J_k(x, b) \geq (2b+1)^{-k} \lfloor x \rfloor^{2b-k(k+1)/2}$.

Proof (a) From (24.1) we see that $J_k(x, b)$ is the number of choices of $\mathbf{m}, \mathbf{n}$ in $(0, x]^b$ such that

$$s_r(\mathbf{m}) = s_r(\mathbf{n}) \quad (1 \leq r \leq k). \quad (24.5) \quad \boxed{\mathbf{E: sr(m)=sr(n)}}$$

Hence by Lemma 24.2 with $\boldsymbol{\theta} = \mathbf{m}$, $\boldsymbol{\phi} = \mathbf{n}$ we have

$$Q(z; \mathbf{m}) = Q(z; \mathbf{n})$$

identically. Hence the roots (counting multiplicity) coincide. Thus the n_i are permutations of the m_i .

(b) When $b \geq k$,

$$\begin{aligned} J_k(x, b) &= \int_{\mathbb{T}^k} \left| \sum_{\mathbf{n} \leq x} e(\mathbf{n}^{(k)} \cdot \boldsymbol{\alpha}) \right|^{2b} d\boldsymbol{\alpha} \leq x^{2b-2k} \int_{\mathbb{T}^k} \left| \sum_{\mathbf{n} \leq x} e(\mathbf{n}^{(k)} \cdot \boldsymbol{\alpha}) \right|^{2k} d\boldsymbol{\alpha} \\ &\leq x^{2b-2k} \cdot k!x^k \end{aligned}$$

by part (a).

(c) Since $J_k(x, b)$ is the number of solutions of (24.5), by taking $m_1 = n_1, m_2 = n_2, \dots, m_b = n_b$ we see that there exist at least $\lfloor x \rfloor^b$ solutions.

(d) For brevity put $N = \lfloor x \rfloor$. Then

$$\left| \int_{\mathbb{T}^k} \left| \sum_{n=1}^N e(\mathbf{n}^{(k)} \cdot \boldsymbol{\alpha}) \right|^{2b} e(-\mathbf{l} \cdot \boldsymbol{\alpha}) d\boldsymbol{\alpha} \right| \leq J_k(x, b). \quad (24.6) \quad \boxed{\text{E:IS|2bFC}}$$

The integral on the left hand side is the number of solutions of

$$s_r(\mathbf{m}) - s_r(\mathbf{n}) = l_r \quad (1 \leq r \leq k)$$

with $\mathbf{m}, \mathbf{n}$ in $(0, x]^b$. Since $0 < s_r(\mathbf{m}) \leq bN^r$ there are no solutions unless $\mathbf{l}$ satisfies

$$|l_r| \leq bN^r \quad (1 \leq r \leq k). \quad (24.7) \quad \boxed{\text{E:|l}_r| \leq bN^r}$$

We sum both sides of (24.6) over all such $\mathbf{l}$, and note that on the left we are just counting all possible choices of $\mathbf{m}$ and $\mathbf{n}$, the number of which is N^{2b} . The number of $\mathbf{l}$ satisfying (24.7) is at most

$$(2b+1)^k N^{\frac{1}{2}k(k+1)}.$$

Thus

$$N^{2b} \leq (2b+1)^k N^{\frac{1}{2}k(k+1)} J_k(x, b),$$

which gives the desired conclusion. $\square$

Our treatment of $J_k(x, b)$ when $b > k$ is via a local or “ p -adic” argument, and the following lemma, due originally to Linnik, is the sparking point of the method.

$\boxed{\text{L:LinnikLocal}}$

Lemma 24.4. *Suppose that p is a prime number with $p > k$. Let $A(p, \mathbf{h})$ denote the number of solutions of the simultaneous congruences*

$$\sum_{r=1}^k m_r^j \equiv h_j \pmod{p^j} \quad (1 \leq j \leq k)$$

with $m_r \leq p^k$ and the m_r distinct modulo p . Then

$$A(p, \mathbf{h}) \leq k! p^{\frac{1}{2}k(k-1)}.$$

Proof Let $B(p, \mathbf{g})$ denote the number of solutions of

$$\sum_{r=1}^k m_r^j \equiv g_j \pmod{p^k} \quad (1 \leq j \leq k) \quad (24.8) \quad \boxed{\text{E:summrjmodpk}}$$

with $m_r \leq p^k$ and the m_r distinct modulo p . Then for each $\mathbf{h}$, $A(p, \mathbf{h})$

is the sum of those $B(p, \mathbf{g})$ with $g_j \equiv h_j \pmod{p^j}$ and $1 \leq g_j \leq p^k$ for $1 \leq j \leq k$. The total number of possible choices for $\mathbf{g}$ is $p^{\frac{1}{2}k(k-1)}$. Thus it suffices to show that $B(p, \mathbf{g}) \leq k!$. For a given $\mathbf{g}$ suppose that $m_1, \dots, m_k$ is a solution of (??) with $m_r \leq p^k$ and the m_r distinct modulo p . Suppose that $n_1, \dots, n_k$ is another such solution. Then, by Lemma 24.2(b),

$$Q(z; \mathbf{m}) \equiv Q(z; \mathbf{n}) \pmod{p^k}$$

and so $Q(n_s; \mathbf{m}) \equiv 0 \pmod{p^k}$ whenever $1 \leq s \leq k$. Since

$$Q(z; \mathbf{m}) = \prod_{r=1}^k (z - m_r),$$

for each s there is an r such that $n_s \equiv m_r \pmod{p}$. Also, since the m_r are distinct modulo p it follows that m_r is unique, and so $n_s \equiv m_r \pmod{p^k}$. Thus $n_s = m_r$. Since the n_s are distinct modulo p , and so are distinct, it follows that the $\mathbf{n}$ are a permutation of the $\mathbf{m}$. $\square$

We now have all the machinery we need to establish a useful version of the Vinogradov Mean Value Theorem.

T:VMVT **Theorem 24.5.** *There is a positive number C such that when $k \geq 2$, r is a positive integer and x is a real number with $x \geq 1$ we have*

$$J_k(x, kr) \leq D(k, r) x^{2rk - \frac{1}{2}k(k+1) + \eta(k, r)}$$

where

$$D(k, r) = \exp(Crk^2 \log k)$$

and

$$\eta(k, r) = \frac{1}{2}k^2 \left(1 - \frac{1}{k}\right)^r.$$

Proof We induct on r . The case $r = 1$ is immediate from 24.3(a) and the observations that $2k - \frac{1}{2}k(k+1) + \eta(k, 1) = k$ and that $k! \leq k^k \leq D(k, 1)$ provided that $C \geq 1$.

Suppose now that $r \geq 2$ and that the Theorem holds with r replaced by $r - 1$. Then $1 - \left(1 - \frac{1}{k}\right)^r \leq \frac{r}{k}$, so

$$\frac{1}{2}k(k+1) - \eta(k, r) \leq \min(k^2, rk).$$

Hence if $x \leq \exp(C \max(k, r) \log k)$, then $x^{\frac{1}{2}k(k+1) - \eta(k, r)} \leq D(k, r)$, and

the trivial estimate $J_k(x, kr) \leq x^{2kr}$ gives the Theorem. Thus we can suppose that

$$x > \exp(C \max(k, r) \log k). \quad (24.9) \quad \boxed{\text{E: xLB}}$$

Let

$$y = x^{1/k}$$

and choose z to be the least number such that the number of primes p with $y < p \leq y + z$ is $\frac{1}{2}k(k^2 - 1)$. From (24.9) it follows that $y > k^C$, and so by the prime number theorem

$$z \leq y$$

if C is a sufficiently large absolute constant. For brevity we write b for kr . Let $R_1(\mathbf{h})$ denote the number of solutions to the system

$$\sum_{r=1}^b m_r^j = h_j \quad (1 \leq j \leq k) \quad (24.10) \quad \boxed{\text{E: summrj=hj}}$$

with $m_r \leq x$ and $m_1, \dots, m_k$ distinct, and let $R_2(\mathbf{h})$ denote the number of solutions in which the $m_1, \dots, m_k$ are not distinct. Then

$$J_k(x, b) = \sum_{\mathbf{h}} (R_1(\mathbf{h}) + R_2(\mathbf{h}))^2 \leq 2(S_1 + S_2)$$

where

$$S_i = \sum_{\mathbf{h}} R_i(\mathbf{h})^2.$$

We consider two cases: $S_1 \leq S_2$, and $S_1 > S_2$. We deal first with the easy case, in which $S_2 \geq S_1$. In this case, $J_k(x, b) \leq 4S_2$ and $R_2(\mathbf{h}) \leq \binom{k}{2} R_3(\mathbf{h})$ where $R_3(\mathbf{h})$ is the number of solutions to the system (24.10) with $m_r \leq x$ and $m_1 = m_2$. Let

$$f(\boldsymbol{\beta}) = \sum_{n \leq x} e(\mathbf{n}^{(k)} \cdot \boldsymbol{\beta}).$$

Then

$$S_2 \leq k^4 \int_{\mathbb{T}^k} |f(2\boldsymbol{\alpha})^2 f(\boldsymbol{\alpha})^{2b-4}| d\boldsymbol{\alpha}.$$

Hence by two applications of the simplest form of Hölder's inequality (or by a single application of the extended form found in Exercise 4)

$$S_2 \leq k^4 \left(\int_{\mathbb{T}^k} |f(2\boldsymbol{\alpha})|^{2b} d\boldsymbol{\alpha} \right)^{\frac{1}{b}} \left(\int_{\mathbb{T}^k} |f(\boldsymbol{\alpha})|^{2b} d\boldsymbol{\alpha} \right)^{1-\frac{2}{b}} = k^4 J_k(x, b)^{1-\frac{1}{b}}.$$

Thus the Theorem holds in the first case provided that $C \geq 6$.

We now suppose that $S_2 < S_1$, which implies that

$$J_k(x, b) \leq 4S_1.$$

For a solution $\mathbf{m}$ of (24.10) counted by $R_1(\mathbf{h})$ let

$$P(\mathbf{m}) = \prod_{1 \leq i < j \leq k} (m_i - m_j).$$

Then $0 < |P| \leq x^{\frac{1}{2}k(k-1)}$. Since $y = x^{1/k}$, the number of prime divisors p of P with $p > y$ is at most $\frac{1}{2}k^2(k-1) < \frac{1}{2}k(k^2-1)$. Thus there is a prime p with

$$y < p \leq z$$

such that $p \nmid P$, and so for such a p the $m_1, \dots, m_k$ are distinct modulo p . Hence

$$R_1(\mathbf{h}) \leq \sum_{y < p \leq y+z} R_4(\mathbf{h}, p)$$

where $R_4(\mathbf{h}, p)$ denotes the number of solutions of (24.10) subject to $m_r \leq x$ and $m_1, \dots, m_k$ distinct modulo p . Let

$$I(p) = \sum_{\mathbf{h}} R_4(\mathbf{h}, p)^2.$$

Then $I(p)$ is the number of solutions of

$$s_j(\mathbf{m}) = s_j(\mathbf{n}) \quad (1 \leq j \leq k)$$

with $m_1, \dots, m_b, n_1, \dots, n_b$ in $(0, x]$, $m_1, \dots, m_k$ distinct modulo p and $n_1, \dots, n_k$ distinct modulo p . Thus

$$\begin{aligned} J_k(x, b) &\leq 4 \sum_{\mathbf{h}} R_1(\mathbf{h})^2 \\ &\leq 4 \sum_{\mathbf{h}} \left(\sum_{y < p \leq z} R_4(\mathbf{h}, p) \right)^2 \\ &\leq 4 \sum_{\mathbf{h}} \frac{1}{2} k^3 \sum_{y < p \leq y+z} R_4(\mathbf{h}, p)^2 \\ &\leq 2k^3 \sum_{y < p \leq y+z} I(p) \\ &\leq k^6 \max_{\substack{p \\ y < p \leq y+z}} I(p). \end{aligned}$$

Let

$$g(\boldsymbol{\alpha}, a) = \sum_{\substack{n \leq x \\ n \equiv a \pmod{p}}} e(\mathbf{n}^{(k)} \cdot \boldsymbol{\alpha}).$$

Then

$$I(p) = \int_{\mathbb{T}^k} \left| \sum_{\mathbf{a} \in \mathcal{A}} g(\boldsymbol{\alpha}, a_1) \cdots g(\boldsymbol{\alpha}, a_k) \right|^2 \left| \sum_{a=0}^{p-1} g(\boldsymbol{\alpha}, a) \right|^{2b-2k} d\boldsymbol{\alpha}$$

where $\mathcal{A}$ denotes the set of k -tuples $\mathbf{a} = (a_1, \dots, a_k)$ with $0 \leq a_r < p$ and the a_r distinct. By Hölder's inequality

$$\left| \sum_{a=0}^{p-1} g(\boldsymbol{\alpha}, a) \right|^{2b-2k} \leq p^{2b-2k-1} \sum_{a=0}^{p-1} |g(\boldsymbol{\alpha}, a)|^{2b-2k},$$

and so

$$I(p) \leq p^{2b-2k} \max_{0 \leq a < p} I_1(p, a)$$

where

$$I_1(p, a) = \int_{\mathbb{T}^k} \left| \sum_{\mathbf{a} \in \mathcal{A}} g(\boldsymbol{\alpha}, a_1) \cdots g(\boldsymbol{\alpha}, a_k) \right|^2 |g(\boldsymbol{\alpha}, a)|^{2b-2k} d\boldsymbol{\alpha},$$

and this is the number of solutions of the simultaneous equations

$$\sum_{i=1}^k (m_i^j - n_i^j) = \sum_{r=1}^{b-k} ((pu_r + a)^j - (pv_r + a)^j) \quad (1 \leq j \leq k)$$

with $m_i \leq x$, $n_i \leq x$, $-a/p < u_r \leq (x-a)/p$, $-a/p < v_r \leq (x-a)/p$, $m_1, \dots, m_k$ distinct modulo p and $n_1, \dots, n_k$ distinct modulo p . By Lemma 24.1(d) this is the number of solutions of

$$\sum_{i=1}^k ((m_i - a)^j - (n_i - a)^j) = \sum_{r=1}^{b-k} p^j (u_r^j - v_r^j) \quad (1 \leq j \leq k)$$

under the same conditions. Let $\mathcal{B}(p, a)$ denote the set of $2k$ -tuples

$$(\mathbf{m}, \mathbf{n}) = (m_1, \dots, m_k, n_1, \dots, n_k)$$

such that $m_i \leq x$, $n_i \leq x$, $m_1, \dots, m_k$ are distinct modulo p , $n_1, \dots, n_k$ are distinct modulo p , and

$$\sum_{i=1}^k (m_i - a)^j \equiv \sum_{i=1}^k (n_i - a)^j \pmod{p^j} \quad (1 \leq j \leq k).$$

For each such element $(\mathbf{m}, \mathbf{n})$ put

$$h_j(\mathbf{m}, \mathbf{n}) = p^{-j} \sum_{i=1}^k ((m_i - a)^j - (n_i - a)^j).$$

Then by Lemma 24.1(e),

$$\begin{aligned} I_1(p, a) &= \sum_{(\mathbf{m}, \mathbf{n}) \in \mathcal{B}(p, a)} J_k \left(\left(-\frac{a}{p}, \frac{x-a}{p} \right], b-k, \mathbf{h}(\mathbf{m}, \mathbf{n}) \right) \\ &\leq \text{card } \mathcal{B}(p, a) J_k \left(\left(-\frac{a}{p}, \frac{x-a}{p} \right], b-k \right). \end{aligned}$$

By Lemma 24.4, since $p^k > y^k = x$ we have

$$\text{card } \mathcal{B}(p, a) \leq x^k k! p^{\frac{1}{2}k(k-1)}.$$

Therefore, by Lemma 24.1(d),

$$I_1(p, a) \leq x^k k! p^{\frac{1}{2}k(k-1)} J_k \left(1 + \frac{x}{p}, (r-1)k \right)$$

and so by the inductive hypothesis

$$I_1(p, a) \leq x^k k! p^{\frac{1}{2}k(k-1)} D(k, r-1) \left(1 + \frac{x}{p} \right)^{2(r-1)k - \frac{1}{2}k(k-1) + \eta(r-1, k)}.$$

Hence

$$J_k(x, rk) \leq \max_{y < p \leq y+z} \lambda p^{k^2 - \eta(r-1, k)} x^{2rk - k - \frac{1}{2}k(k+1) + \eta(r-1, k)}$$

where

$$\lambda = k^6 k! \left(1 + \frac{x}{p} \right)^{2rk - 2k - \frac{1}{2}k(k+1) + \eta(r-1, k)} D(k, r-1).$$

Each prime p here satisfies $p \leq y + z \leq 2y = 2x^{1/k}$ and so

$$J_k(x, rk) \leq 2^{k^2} \lambda x^{2rk - \frac{1}{2}k(k+1) + \eta(r, k)}$$

and

$$2^{k^2} \lambda \leq k^6 k! 2^{k^2} (1 + 2x^{1/k-1})^{2rk} \exp(C(r-1)k^2 \log k).$$

Now

$$(1 + 2x^{1/k-1})^{2rk} \leq \exp(4rkx^{-1/2}),$$

and by (24.9) this does not exceed e provided that $C \geq 6$. Thus

$$2^{k^2} \lambda \leq \exp(6 \log k + k \log k + k^2 + 1 + C(r-1)k^2 \log k),$$

and this does not exceed

$$\exp(Crk^2 \log k)$$

provided that $C \geq 6$. This completes the proof of the Theorem. $\square$

24.1.1 Exercises

Exer:NewGir1

1. Let $Q(z) = Q(z; \theta)$, σ_r and s_r be defined as in (24.2) and the subsequent discussion.

- (a) Put $P(z) = P(z; \theta) = z^q Q(1/z)$. Show that

$$P(z) = \prod_{j=1}^q (1 - \theta_j z) = \sum_{r=0}^q (-1)^r \sigma_r z^r.$$

- (b) Deduce that

$$\frac{P'}{P}(z) = - \sum_{j=1}^q \frac{\theta_j}{1 - \theta_j z}.$$

- (c) Let R be determined by the equation $1/R = \max_j |\theta_j|$. Show that the above is

$$= - \sum_{m=1}^{\infty} s_m z^{m-1}$$

for $|z| < R$.

- (d) Explain why

$$- \left(\sum_{k=0}^q (-1)^k \sigma_k z^k \right) \left(\sum_{m=1}^{\infty} s_m z^{m-1} \right) = \sum_{r=1}^q (-1)^r r \sigma_r z^{r-1}$$

for $|z| < R$.

- (e) By considering pairs k, m in the above with $k + m = r$, deduce the first Newton–Girard formula, i.e., equation (24.3).

Exer:NewGir2

2. Let $Q(z)$ be defined as in (24.2). Observe that

$$0 = Q(\theta_j) = \sum_{k=0}^q (-1)^k \sigma_k \theta_j^{q-k}$$

where it is understood that $\theta_j^0 = 1$ even if $\theta_j = 0$. Suppose that $r \geq q$. Multiply both sides of the above by θ_j^{r-q} , and then sum over j to obtain the second Newton–Girard formula, i.e., equation (24.4).

3. We say that a sequence u_n satisfies a linear recurrence of order q if there exist real or complex numbers $a_1, a_2, \dots, a_q$ such that $u_n = a_1 u_{n-1} + a_2 u_{n-2} + \dots + a_q u_{n-q}$ for all $n > q$. Thus (24.4) asserts that the s_m satisfy a linear recurrence of order q . More generally, for arbitrary real or complex constants $c_1, c_2, \dots, c_q$ put

$$u_m = \sum_{j=1}^q c_j \theta_j^m.$$

Show that

$$\sum_{k=0}^q (-1)^k \sigma_k u_{r-k} = 0$$

for all $r \geq q$.

Exer:Holder3

4. Suppose that $p > 1, q > 1, r > 1$ are real numbers such that $\frac{1}{p} + \frac{1}{q} + \frac{1}{r} = 1$.

- (a) Let a_n, b_n, c_n be nonnegative real numbers. By two applications of the simplest form of Hölder's inequality, or otherwise, show that

$$\sum_{n=1}^N a_n b_n c_n \leq \left(\sum_{n=1}^N a_n^p \right)^{1/p} \left(\sum_{n=1}^N b_n^q \right)^{1/q} \left(\sum_{n=1}^N c_n^r \right)^{1/r}.$$

- (b) Let u_n, v_n, w_n be arbitrary real or complex numbers. Show that

$$\left| \sum_{n=1}^N u_n v_n w_n \right| \leq \left(\sum_{n=1}^N |u_n|^p \right)^{1/p} \left(\sum_{n=1}^N |v_n|^q \right)^{1/q} \left(\sum_{n=1}^N |w_n|^r \right)^{1/r}.$$

5. Let

$$S(p, \mathbf{a}) = \sum_{n=1}^p e((a_1 n + a_2 n^2 + \dots + a_k n^k) p^{-1})$$

and

$$M(p) = \sum_{a_1=1}^p \dots \sum_{a_k=1}^p |S(p, \mathbf{a})|^{2k}.$$

- (a) Show that if $p > k$, then $M(p) \leq k! p^{2k}$.
 (b) Suppose that $p > k$ and $p \nmid a_k$. Show that

$$|S(p, \mathbf{a})| \leq (2k)^{\frac{1}{2}} p^{1 - \frac{1}{k}}.$$

6. Show that if $p \nmid a$, then

$$\left| \sum_{n=1}^p e\left(\frac{an^k}{p}\right) \right| \leq ((k, p-1) - 1)p^{\frac{1}{2}}.$$

7. Show that $N(\mathcal{B}, C, \mathbf{h}) \leq N(\mathcal{B})^{\frac{1}{2}} N(C)^{\frac{1}{2}}$.

8. Show that in Linnik's lemma, either $A(p, \mathbf{h}) = 0$ or $A(p, \mathbf{h}) = k!p^{\frac{1}{2}k(k-1)}$. Show that if the condition that the m_j are distinct is omitted, then $A(p, \mathbf{0}) \geq p^{k(k-1)}$.

9. Show that $J_k(hN, b) \leq h^{2b} J_k(N, b)$.

10. Suppose that

$$J_k(q, b) \ll q^{2b - \frac{k(k+1)}{2} + \eta}$$

and for a Dirichlet character χ modulo q

$$W_k(q; \mathbf{a}, \chi) = \sum_{x=1}^q e(a_1 x + \cdots + a_k x^k / q) \chi(x).$$

(a) Show that

$$\sum_{a_1=1}^q \cdots \sum_{a_k=1}^q |W_k(q; \mathbf{a}, \chi)|^{2b} \ll q^{2b+\eta}$$

(b) Show that if $(q, a_1, \dots, a_k) = 1$, then

$$W_k(q; \mathbf{a}, \chi) \ll q^{1 - \frac{1-\eta}{2b} + \epsilon}$$

24.2 Vinogradov's method

S:Vm

Here we are mostly concerned to relate an individual exponential sum to a mean value of exponential sums. A very fruitful way of doing this is to perturb the endpoints of summation. Exercise ?? gives a simple method for doing this. However the situation of particular interest to us is one of very great delicacy and the presence of the logarithmic factor in that exercise means that if the bound we are able to obtain for $S(\beta)$ is close to the trivial bound, then our resulting estimate is worse than trivial. Here we develop a method which has no loss of this kind.

We start by establishing such a connection for a rather special exponential sum, for which we require an auxiliary lemma.

L:TechIneq

Lemma 24.6. Suppose that $X \geq 1$, $Y \geq 1$, and that $|\alpha - a/q| \leq q^{-2}$ with $(q, a) = 1$. Then

$$\sum_{n \leq X} \min \left(Y, \frac{1}{Y \|n\alpha\|^2} \right) \ll \frac{XY}{q} + X + Y + q.$$

Proof Let $\theta = q^2\alpha - aq$, so that $|\theta| \leq 1$. Then for $1 \leq n \leq \frac{1}{2}q$ we have $n\alpha = \frac{na}{q} + \frac{\theta'}{2q}$ where $\theta' = \frac{2n\theta}{q}$, so $|\theta'| \leq 1$ and hence $\|n\alpha\| \geq \frac{1}{2}\|na/q\|$. Now suppose that $n = uq + v$ where u and v are integers with $u \geq 1$ and $|v| \leq \frac{1}{2}q$. Then

$$n\alpha = ua + \frac{va}{q} + \frac{u\theta}{q} + \frac{v\theta}{q^2} = ua + \frac{va + \lfloor u\theta \rfloor}{q} + \frac{3\theta''}{2q}.$$

Thus, given u , we have

$$\|n\alpha\| \gg \left\| \frac{va + \lfloor u\theta \rfloor}{q} \right\|$$

for all but at most 3 values of v . Thus

$$\begin{aligned} \sum_{n \leq X} \min \left(Y, \frac{1}{Y \|n\alpha\|^2} \right) &\ll \sum_{0 \leq u \leq \frac{X}{q} + \frac{1}{2}} \left(Y + \sum_{0 \leq v \leq q} \min \left(Y, \frac{1}{Y \left\| \frac{va + \lfloor u\theta \rfloor}{q} \right\|^2} \right) \right) \\ &\ll \sum_{0 \leq u \leq \frac{X}{q} + \frac{1}{2}} \left(Y + \sum_{1 \leq r \leq \frac{1}{2}q} \min \left(Y, \frac{q^2}{Y r^2} \right) \right), \end{aligned}$$

and for a suitable $R \geq 0$ this is

$$\ll \left(1 + \frac{X}{q} \right) \left(Y + YR + \frac{q^2}{Y(R+1)} \right).$$

The choice $R = \lfloor q/Y \rfloor$ gives the desired conclusion. $\square$

T:VMTechEst

Theorem 24.7. Let $f(n) = \mathbf{n}^{(k)} \cdot \boldsymbol{\alpha}$ and suppose that for $j = 1, 2, \dots, k$ there are a_j, q_j so that $(a_j, q_j) = 1$, $|\alpha_j - a_j/q_j| \leq q_j^{-2}$, and that $1 \leq X \leq X' \leq 2X$, $1 \leq Y \leq Y' \leq 2Y$. Let

$$T = \sum_{X < l \leq X'} \sum_{Y < m \leq Y'} e(f(lm)).$$

Then for each $b \geq k$ we have

$$T \ll XY \left(\frac{J_k(2X, b) J_k(2Y, b)}{(4XY)^{2b - \frac{1}{2}k(k+1)}} \right)^{\frac{1}{4b^2}} \prod_{j=1}^k \left(\frac{1}{q_j} + \frac{1}{X^j} + \frac{1}{Y^j} + \frac{q_j}{X^j Y^j} \right)^{\frac{1}{4b^2}}.$$

Proof The number of integers l such that $X < l \leq 2X$ is $\leq X+1 \leq 2X$, so by Hölder's inequality

$$|T|^{2b} \leq (2X)^{2b-1} \sum_{X < l \leq 2X} \left| \sum_{Y < m \leq Y'} e(f(lm)) \right|^{2b}.$$

Now

$$\left| \sum_{Y < m \leq Y'} e(f(lm)) \right|^{2b} = \sum_{Y < m_1, \dots, m_{2b} \leq Y'} e\left(\sum_{j=1}^k \alpha_j l^j F_j(\mathbf{m})\right)$$

where

$$F_j(\mathbf{m}) = m_1^j + \dots + m_b^j - m_{b+1}^j - \dots - m_{2b}^j.$$

Then

$$|T|^{2b} \leq (2X)^{2b-1} \sum_{Y < m_1, \dots, m_{2b} \leq 2Y} \left| \sum_{X < l \leq 2X} e\left(\sum_{j=1}^k \alpha_j l^j F_j(\mathbf{m})\right) \right|.$$

Now Hölder's inequality, once more, gives

$$|T|^{4b^2} \leq (4XY)^{4b^2-2b} \sum_{Y < m_1, \dots, m_{2b} \leq 2Y} \left| \sum_{X < l \leq 2X} e\left(\sum_{j=1}^k \alpha_j l^j F_j(\mathbf{m})\right) \right|^{2b}.$$

For given $h_1, \dots, h_k$ we collect those $\mathbf{m}$ for which $F_j(\mathbf{m}) = h_j$ for $1 \leq j \leq k$. We note that the contribution is 0 unless $|h_j| \leq b(2Y)^j$ for all j , $1 \leq j \leq k$. Therefore,

$$|T|^{4b^2} \leq (4XY)^{4b^2-2b} \sum_{\substack{h_1, \dots, h_k \\ |h_j| \leq b(2Y)^j}} J_k(2Y, b, \mathbf{h}) \left| \sum_{X < l \leq 2X} e\left(\sum_{j=1}^k \alpha_j h_j l^j\right) \right|^{2b}.$$

Let

$$H_j = 2 \lfloor b(2Y)^j \rfloor.$$

Then

$$\begin{aligned} |T|^{4b^2} &\leq (2X)^{4b^2-2b} J_k(2Y, b) \\ &\quad \times \sum_{\substack{h_1, \dots, h_k \\ |h_j| \leq H_j}} \prod_{j=1}^k 2 \left(1 - \frac{|h_j|}{H_j}\right) \left| \sum_{X < l \leq 2X} e\left(\sum_{j=1}^k \alpha_j h_j l^j\right) \right|^{2b}. \end{aligned}$$

On expanding the power to form a $2b$ -fold sum, and taking the sum over

the h_j inside, we find that the last factor above is

$$= \sum_{X < l_1, \dots, l_{2b} \leq 2X} \prod_{j=1}^k \left(\frac{2}{H_j} \left| \sum_{h=1}^{H_j} e(\alpha_j h F_j(\mathbf{l})) \right|^2 \right).$$

By the estimate (??) for the sum of a geometric progression we see that the above is

$$\begin{aligned} &\leq 2^k \sum_{X < l_1, \dots, l_{2b} \leq 2X} \prod_{j=1}^k \min \left(H_j, \frac{1}{H_j \|\alpha_j F_j(\mathbf{l})\|^2} \right) \\ &\leq 2^k \sum_{\substack{g_1, \dots, g_k \\ |g_j| \leq G_j}} J_k(2X, b, \mathbf{g}) \prod_{j=1}^k \min \left(H_j, \frac{1}{H_j \|\alpha_j g_j\|^2} \right) \end{aligned}$$

where $G_j = b(2X)^j$. By Lemma 24.6 this does not exceed, for some absolute constant C_1 ,

$$\begin{aligned} &C_1^k J_k(2X, b) \prod_{j=1}^k \left(\frac{G_j H_j}{q_j} + G_j + H_j + q_j \right) \\ &\leq C_1^k J_k(2X, b) \prod_{j=1}^k \left(4b^2 (4XY)^j \left(\frac{1}{q_j} + \frac{1}{X^j} + \frac{1}{Y^j} + \frac{q_j}{X^j Y^j} \right) \right). \end{aligned}$$

Therefore

$$|T|^{4b^2} \leq (4XY)^{4b^2 - 2b} J_k(2X, b) J_k(2Y, b) C_1^k (4b^2)^k (4XY)^{\frac{1}{2}k(k+1)} \Delta$$

where

$$\Delta = \prod_{j=1}^k \left(\frac{1}{q_j} + \frac{1}{X^j} + \frac{1}{Y^j} + \frac{q_j}{X^j Y^j} \right).$$

The desired conclusion now follows since $b \geq k$. $\square$

L:CoefSumBnd **Lemma 24.8.** *For integers h let c_h denote arbitrary complex numbers with $\sum_h |c_h| < \infty$, and let*

$$S(\beta) = \sum_h c_h e(h\beta).$$

Further let H be an integer and L and M be natural numbers. Then

$$\left| \sum_{|h-H| \leq M} c_h \right| \leq \left(\frac{2M}{L} + 1 \right) \max_{\beta} |S(\beta)| + \sum_{M < |h-H| < M+L} |c_h|.$$

We note that $S(\beta)$ is uniformly approximated by the continuous functions $s_N(\beta) = \sum_{h=-N}^N c_h e(h\beta)$, so $S(\beta)$ is continuous. Also, $S(\beta)$ has period 1, and the circle group $\mathbb{T}$ is compact, so the supremum of $|S(\beta)|$ is attained, so we are free to refer to $\max_{\beta} |S(\beta)|$.

Proof By replacing $S(\beta)$ by $S(\beta)e(-H\beta)$ we may assume that $H = 0$. We note the basic formulæ that define the Fejér kernel $\Delta_M(\beta)$, namely that

$$\Delta_M(\beta) = \sum_{h=-M+1}^{M-1} \left(1 - \frac{|h|}{M}\right) e(h\beta) = \frac{1}{M} \left(\frac{\sin \pi M \beta}{\sin \pi \beta} \right)^2.$$

We note that $\Delta_M(\beta) \geq 0$ for all β , and hence that

$$\int_0^1 |\Delta_M(\beta)| d\beta = \int_0^1 \Delta_M(\beta) d\beta = \widehat{\Delta}_M(0) = 1.$$

(If these properties are not familiar, see Exercise 1.) Let

$$K(\beta) = \frac{(M+L)\Delta_{M+L}(\beta) - M\Delta_M(\beta)}{L} = \sum_{h=-M-L+1}^{M+L-1} \widehat{K}(h) e(h\beta)$$

where

$$\widehat{K}(h) = \begin{cases} 1 & (|h| \leq M), \\ (M+L-|h|)/L & (M \leq |h| \leq M+L), \\ 0 & (|h| > M+L). \end{cases}$$

Since $|\widehat{K}_M(h)| \leq 1$ for $M < |h| < M+L$ it follows by the triangle inequality that

$$\left| \sum_{h=-M}^M c_h - \sum_{h=-M-L+1}^{M+L-1} c_h \widehat{K}(h) \right| \leq \sum_{M < |h| < M+L} |c_h|.$$

On the other hand,

$$\left| \sum_h c_h \widehat{K}(h) \right| = \left| \int_0^1 S(\beta) K(-\beta) d\beta \right| \leq \int_0^1 |K(-\beta)| d\beta \max_{\beta} |S(\beta)|$$

and $|K(-\beta)| = |K(\beta)| \leq \frac{M+L}{L} \Delta_{M+L}(\beta) + \frac{M}{L} \Delta_M(\beta)$, so

$$\int_0^1 |K(\beta)| d\beta \leq \frac{M+L}{L} + \frac{M}{L},$$

which gives the stated result. $\square$

Note that in Lemma 24.8, if the c_h are bounded, then a good choice for L would be

$$L \asymp (M \sup_{\beta} |S(\beta)|)^{\frac{1}{2}}.$$

However we would need to deal with sums over h on the right under somewhat more general conditions, and the following lemma shows how to do this.

L:SumBizcn **Lemma 24.9.** *Suppose that $c_1, c_2, \dots$ are complex numbers, and that $0 < X \leq N^{\frac{1}{2}}$, $\sum_{n=1}^{\infty} |c_n| < \infty$ and*

$$|c_n| \ll \sum_{\substack{d|n \\ X < d \leq 2X}} 1.$$

Then for each Y, Z with $0 \leq Y \leq Z \leq N$ we have

$$\sum_{Y < n \leq Z} c_n \ll N^{\frac{1}{2}} \left(1 + \max_{\beta} \left| \sum_{n=1}^{\infty} c_n e(n\beta) \right| \right)^{\frac{1}{2}}.$$

Proof Let H be one of the integers closest to $\frac{1}{2}(Y + Z)$ and choose $M \geq 1$ minimally so that $H - M \leq Y$ and $H + M \geq Z$. Note that $M \leq N$. Then $\sum_{Y < n \leq Z} c_n$ differs from $\sum_{|h-H| \leq M} c_h$ by an amount $\ll \max_{n \leq N} d(n) \ll N^{\frac{1}{2}}$. For convenience we put $c_n = 0$ when $n \leq 0$. Now Lemma 24.8 gives

$$\sum_{Y < n \leq Z} c_n \ll (ML^{-1} + 1) \max_{\beta} \left| \sum_{n=1}^{\infty} c_n e(n\beta) \right| + N^{\frac{1}{2}} + \sum_{M < |h-H| < M+L} |c_h|.$$

The last sum here is bounded by two sums of the form

$$\sum_{K < h \leq K+L'} |c_h|$$

where K is a non-negative integer and $L' \leq L$. Such a sum is

$$\ll \sum_{K < h \leq K+L'} \sum_{\substack{d|h \\ X < d \leq 2X}} 1 \leq \sum_{X < d \leq 2X} \left(\frac{L}{d} + 1 \right) \ll L + N^{\frac{1}{2}}.$$

Hence

$$\sum_{Y < n \leq Z} c_n \ll (NL^{-1} + 1) \max_{\beta} \left| \sum_{n=1}^{\infty} c_n e(n\beta) \right| + L + N^{\frac{1}{2}}.$$

Let $S = \max_{\beta} \left| \sum_{n=1}^{\infty} c_n e(n\beta) \right|$. If $S \leq N$, then the choice $L = \lfloor (NS)^{\frac{1}{2}} \rfloor + 1$ gives the lemma. If $S > N$, then the trivial bound

$$\sum_{Y < n \leq Z} c_n \ll \sum_{X < d \leq 2X} \frac{N}{d} \ll N$$

suffices. $\square$

L:SumBizcn2

Lemma 24.10. *Suppose that M, M', N and N' are natural numbers with $M \leq N$ and $N < N' \leq 2N$, that X is a positive real number with $X \leq (2M)^{\frac{1}{2}}$, and that c_m ($m \in \mathbb{N}$) and d_h ($h \in \mathbb{Z}$) are complex numbers such that*

$$c_m \ll \sum_{\substack{d|m \\ X < d \leq 2X}} 1, \quad \sum_{m=1}^{\infty} |c_m| < \infty, \quad |d_h| \leq 1.$$

Then

$$\left| \sum_{M < m \leq 2M} c_m \sum_{N < n \leq N'} d_n \right| \ll NM^{\frac{1}{2}} \left(1 + \max_{n \in (M+N, 2M+2N]} \max_{\beta} \left| \sum_{m=1}^{\infty} c_m d_{n-m} e(m\beta) \right| \right)^{\frac{1}{2}}$$

Proof The product

$$\sum_{M < m \leq 2M} c_m \sum_{N < n \leq N'} d_n$$

can be rearranged to give

$$\sum_{M < m \leq 2M} \sum_{N+m < n \leq N'+m} c_m d_{n-m} = \sum_{M+N < n \leq 2M+N'} \sum_{\substack{M < m \leq 2M \\ n-N' \leq m \leq n-N}} c_m d_{n-m}.$$

Now we apply the preceding lemma, with N replaced by $2M$, to the inner sum. Thus the inner sum is

$$\ll M^{\frac{1}{2}} \left(\max_{\beta} \left| \sum_{m=1}^{\infty} c_m d_{n-m} e(m\beta) \right| \right)^{\frac{1}{2}},$$

from which the conclusion follows. $\square$

T:VT1

Theorem 24.11. *Suppose that $N < N' \leq 2N$, $3 \leq Q \leq N^{\frac{1}{2}}$, $k \geq 2$, $R > 0$, that f is a $k+1$ times continuously differentiable real-valued function defined on the interval $[N-3Q^2, 2N+4Q^2]$, and that*

$$\frac{|f^{(k+1)}(u)|}{(k+1)!} \leq (2Q)^{-2k-2} R^{-1}$$

for all u . Suppose further that $b \geq k$, that

$$0 < R_j \leq Q^j,$$

that for each integer $n \in [N + Q^2, 2N + 2Q^2]$ there are $a_2/q_2, \dots, a_k/q_k$ such that $(a_j, q_j) = 1$,

$$R_j \leq q_j \leq Q^{2j} R_j^{-1},$$

and

$$\left| \frac{f^{(j)}(n)}{j!} - \frac{a_j}{q_j} \right| \leq \frac{1}{q_j^2}.$$

Then the exponential sum

$$S = \sum_{N < n \leq N'} e(f(n))$$

satisfies

$$S \ll N \left(\frac{1}{Q} + \frac{1}{\sqrt{R}} + \left(\frac{J_k(2Q, b)}{(2Q)^{2b - \frac{1}{2}k(k+1)}} \right)^{\frac{1}{4b^2}} \left(\prod_{j=2}^k \frac{1}{R_j} \right)^{\frac{1}{8b^2}} \right).$$

Proof Define

$$c_n = \sum_{\substack{l > Q, m \leq 2Q \\ lm = n}} 1.$$

Let $M = Q^2$. Now

$$\sum_{M < n \leq 2M} c_n$$

is a sum over the ordered pairs of integers l, m with $Q < l \leq 2Q$, $Q < m \leq 2Q$ and $M/m < l \leq 2M/l$, and this certainly counts every pair l, m with $Q < l \leq \sqrt{2}Q$, $Q < m \leq \sqrt{2}Q$ and for $Q \geq 3$ there are always $\gg Q^2$ such pairs. Hence, by the preceding lemma

$$Q^2 S \ll NQ \left(1 + \max_{n \in (Q^2 + N, 2Q^2 + 2N]} \max_{\beta} \left| \sum_{m=1}^{\infty} c_m e(f(n-m) + m\beta) \right| \right)^{\frac{1}{2}}.$$

The innermost sum is

$$\sum_{Q < l \leq 2Q} \sum_{Q < m \leq 2Q} e(f(n-lm) + lm\beta).$$

By Taylor's theorem with Lagrange's form of the remainder,

$$f(n-lm) = \sum_{j=0}^k (-lm)^j \frac{f^{(j)}(n)}{j!} + x^{k+1} \frac{f^{(k+1)}(n-\theta lm)(\theta x)}{(k+1)!}$$

with $0 < \theta < 1$. Let $\alpha_1 = -f'(n) + \beta$ and $\alpha_j = (-1)^j \frac{f^{(j)}(n)}{j!}$ ($2 \leq j \leq k$). Then

$$lm\beta + f(n - lm) = f(n) + \sum_{j=1}^k (lm)^j \alpha_j + O(R^{-1})$$

and so

$$\begin{aligned} \sum_{Q < l \leq 2Q} \sum_{Q < y \leq 2Q} e(f(n - lm) + lm\beta) \\ = e(f(n)) \sum_{Q < l \leq 2Q} \sum_{Q < y \leq 2Q} e\left(\sum_{j=1}^k (lm)^j \alpha_j\right) + O(Q^2 R^{-1}). \end{aligned}$$

For $2 \leq j \leq k$, the hypothesis of Theorem 24.7, with a_j replaced by $(-1)^j a_j$, is satisfied and we may certainly take $q_1 = 1$. Hence, by Theorem 24.7,

$$\begin{aligned} \sum_{Q < l \leq 2Q} \sum_{Q < y \leq 2Q} e(f(n - lm) + lm\beta) \\ \ll \frac{Q^2}{R} + Q^2 \left(\frac{J_k(2Q, b)}{(2Q)^{2b - \frac{1}{2}k(k+1)}} \right)^{\frac{1}{2s^2}} \prod_{j=2}^k \left(\frac{4}{R_j} \right)^{\frac{1}{4b^2}}. \end{aligned}$$

We note that $k \ll 4b^2$. Thus

$$S \ll \frac{N}{Q} \left(1 + \frac{Q^2}{R} + Q^2 \left(\frac{J_k(2Q, b)}{(2Q)^{2b - \frac{1}{2}k(k+1)}} \right)^{\frac{1}{2b^2}} \prod_{j=2}^k \left(\frac{1}{R_j} \right)^{\frac{1}{4b^2}} \right)^{\frac{1}{2}}.$$

□

T:VT2 **Theorem 24.12.** Suppose that $0 < \delta < 1$, $N < N' \leq 2N$, $R > 0$, $3 \leq Q \leq N^{\frac{1}{2}}$, $k \geq 2$, that f is a $k+1$ times continuously differentiable real-valued function on the interval $[N - 3Q^2, 2N + 4Q^2]$, and that

$$\frac{|f^{(k+1)}(u)|}{(k+1)!} \leq (2Q)^{-2k-2} R^{-1}.$$

Suppose further that

$$0 < R_j \leq Q^j,$$

that for each integer $n \in [N + Q^2, 2N + 2Q^2]$ there are $a_2/q_2, \dots, a_k/q_k$ such that $(a_j, q_j) = 1$,

$$R_j \leq q_j \leq Q^{2j} R_j^{-1}$$

and

$$\left| \frac{f(j)(n)}{j!} - \frac{a_j}{q_j} \right| \leq \frac{1}{q_j^2},$$

and that there are $h \geq \delta k$ values of j with $2 \leq j \leq k$ such that $R_j \geq Q^{\delta j}$. Then there is a number $\lambda > 0$ depending only on δ such that the exponential sum

$$S = \sum_{N < n \leq N'} e(f(n))$$

satisfies

$$S \ll NQ^{-\lambda k^{-2}} + NR^{-\frac{1}{2}}.$$

Proof By Theorem 24.5,

$$\left(\frac{J_k(2Q, kr)}{(2Q)^{2kr - \frac{1}{2}k(k+1)}} \right)^{\frac{1}{4k^2r^2}} \leq \exp\left(\frac{C \log k}{4r}\right) Q^{\frac{1}{8r^2}(1-1/k)^r}.$$

The sum of δj over the h values of j for which $R_j \geq Q^{\delta j}$ is bounded below by

$$\sum_{j=2}^h \delta j = \frac{\delta}{2}(h+3)h > \frac{\delta^3 h k^2}{2}.$$

Thus

$$S \ll NQ^{-1} + NR^{-\frac{1}{2}} + \exp\left(\frac{C \log k}{4r}\right) NQ^{\frac{1}{8r^2}((1-1/k)^r - \frac{\delta^3}{2})}.$$

Now we take $r = \varkappa k$ where $\varkappa$ is sufficiently large in terms of δ so that

$$(1 - 1/k)^r < \frac{\delta^3}{4}.$$

□

T:VT3 Theorem 24.13. *There is a positive number C such that if t is a real number and r, N, N' are natural numbers with*

$$t > N^{\frac{1}{9}}, \quad N^{r-1} < t \leq N^r, \quad N < N' \leq 2N,$$

then

$$\sum_{N < n \leq N'} (n + \alpha)^{it} \ll N^{1-Cr^{-2}}$$

uniformly for $0 \leq \alpha \leq 1$.

Proof Let $f(x) = \frac{t}{2\pi} \log(x + \alpha)$. Then, for $j \geq 1$

$$\frac{f^{(j)}(x)}{j!} = (-1)^{j-1} \frac{t}{2\pi j(x + \alpha)^j}.$$

Suppose that

$$Q \leq \frac{1}{2}N^{\frac{1}{2}}.$$

Then for $N - 3Q^2 \leq x \leq 2N + 4Q^2$,

$$\left| \frac{f^{(k+1)}(x)}{(k+1)!} \right| \leq N^r \left(\frac{4}{N} \right)^{k+1}.$$

Also, since $t > N^{r-1} \geq N^{r/9}$, if $N + Q^2 \leq n \leq 2N + 2Q^2$, then $n + \alpha \leq 2N + N/2 + 1 < 4N$ and

$$\frac{N^{r/9}}{(8\pi N)^j} \leq \frac{t}{2\pi j(4N+1)^j} \leq \left| \frac{f^{(j)}(n)}{j!} \right| \leq N^{r-j}. \quad (24.11) \quad \boxed{\text{E:UsefulIneq}}$$

Our aim is to apply Theorem 24.12. Thus we need to introduce a parameter R , and it will be suitable provided it satisfies

$$N^r \left(\frac{4}{N} \right)^{k+1} \leq (2Q)^{-2k-2} R^{-1}. \quad (24.12) \quad \boxed{\text{E:RIIneq}}$$

We also need to find suitable rational approximations a_j/q_j . To this end for each $n \in [N + Q^2, 2N + 2Q^2]$ and each j with $2 \leq j \leq k$ let

$$Q_j = 2 \left| \frac{j!}{f^{(j)}(n)} \right|.$$

Then by Dirichlet's theorem choose q_j, a_j so that $(a_j, q_j) = 1$, $q_j \leq Q$, and

$$\left| \frac{f^{(j)}(n)}{j!} - \frac{a_j}{q_j} \right| \leq \frac{1}{q_j Q_j}.$$

If $a_j = 0$, then we would have

$$\left| \frac{f^{(j)}(n)}{j!} \right| \leq \frac{1}{2} \left| \frac{f^{(j)}(n)}{j!} \right|,$$

which is impossible. Thus

$$1 \leq |a_j| \leq q_j \left| \frac{f^{(j)}(n)}{j!} \right| + \frac{1}{Q_j} = \left(q_j + \frac{1}{2} \right) \left| \frac{f^{(j)}(n)}{j!} \right|,$$

and so

$$\left| \frac{j!}{f^{(j)}(n)} \right| - \frac{1}{2} \leq q_j \leq Q_j = 2 \left| \frac{j!}{f^{(j)}(n)} \right|.$$

From the inequalities (24.11) we now deduce the further inequalities

$$\frac{1}{2}N^{j-r} \leq q_j \leq (16\pi N)^j N^{-r/9}.$$

The upper bound is immediate, as is the lower one provided that $j \geq r$. But $q_j \geq 1$, so the lower bound above is trivial when $j < r$. Now we will be in a position to apply the preceding theorem with $R = N^\delta$, $R_j = R^j$ provided that (24.12) holds and that

$$N^{j\delta} \leq \frac{1}{2} N^{j-r}, \quad (16\pi N)^j N^{-r/9} \leq Q^{2j} N^{-j\delta} \quad (24.13) \quad \boxed{\text{E:MoreIneqs}}$$

hold for an appreciable range of j . To this end we take

$$\delta = \frac{1}{100}, \quad Q = N^{\frac{1}{2}-\delta}.$$

Then (24.12) holds provided that $16^{k+1} N^{r+\delta} \leq N^{\delta(2k+2)}$, and this follows as long as $k\delta > r$ and $N \geq 16^{1/\delta}$. We may certainly suppose the latter of these inequalities since otherwise the conclusion is trivial. Thus for (24.12) it suffices that

$$r \leq k\delta. \quad (24.14) \quad \boxed{\text{E:rlekdelta}}$$

The inequalities (24.13) will follow provided that

$$N^{2j\delta} \leq N^{j-r}, \quad N^{3j\delta} \leq N^{r/9}$$

and $N \geq 56^{1/\delta}$, which again we may certainly suppose. The two inequalities above reduce to

$$\frac{r}{1-2\delta} \leq j \leq \frac{r}{27\delta},$$

which is to say that

$$\frac{50}{49}r \leq j \leq \frac{100}{18}r.$$

It remains to make a suitable choice for k in terms of r , and $k = 100r$ suffices. $\square$

24.2.1 Exercises

Exer:Fejer

1. (a) By expanding the mod-square and collecting terms, show that

$$\left| \sum_{m=0}^{N-1} e(mx) \right|^2 = \sum_{n=-N+1}^{N-1} (N-|n|) e(nx) = N + 2 \sum_{n=1}^{N-1} (N-n) \cos 2\pi nx.$$

- (b) Use the formula for the sum of a geometric series to show that

$$\sum_{m=0}^{N-1} e(mx) = e\left(\frac{N-1}{2}x\right) \frac{\sin \pi Nx}{\sin \pi x}.$$

- (c) Let the Fejér kernel $\Delta_N(x)$ be defined by the formula

$$\Delta_N(x) = \frac{1}{N} \left(\frac{\sin \pi N x}{\sin \pi x} \right)^2.$$

Show that the above is

$$= \sum_{n=-N}^N (1 - |n|/N) e(nx).$$

- (d) Show that

$$0 \leq \Delta_N(x) \leq \min \left(N, \frac{1}{4N\|x\|^2} \right)$$

for all x .

- (e) Let $L^1(\mathbb{T})$ denote the set of functions f with period 1 such that $\int_0^1 |f(x)| dx < \infty$. For $f \in L^1(\mathbb{T})$ let

$$\hat{f}(n) = \int_0^1 f(x) e(-nx) dx$$

denote the Fourier coefficients of f , and set

$$\sigma_N(f; x) = \sum_{n=-N}^N (1 - |n|/N) \hat{f}(n) e(nx).$$

Show that the above is

$$= \int_0^1 \Delta_N(x-u) f(u) du.$$

- (f) Suppose that $f \in L^1(\mathbb{T})$. Show that

$$f(x) - \sigma_N(f; x) = \int_0^1 \Delta_N(x-u) (f(x) - f(u)) du.$$

- (g) Let $C(\mathbb{T})$ denote the set of continuous functions with period 1. Suppose that $f \in C(\mathbb{T})$. Show that for any $\varepsilon > 0$ there is an $N_0 = N_0(\varepsilon, f)$ such that

$$|f(x) - \sigma_N(f; x)| < \varepsilon$$

for all x , if $N > N_0$.

2. Let c_h $h \in \mathbb{Z}$ denote arbitrary complex numbers with $\sum_h |c_h| < \infty$, and let

$$S(\beta) = \sum_h c_h e(h\beta).$$

Show that for any integer H and natural number N we have

$$\sum_{H < h \leq H+N} c_h = \int_0^1 S(\beta) \sum_{H < h \leq H+N} e(-h\beta) d\beta.$$

Hence, or otherwise, show that

$$\left| \sum_{H < h \leq H+N} c_h \right| \leq (\log(eN)) \sup_{\beta} |S(\beta)|.$$

3. Let $S^*(\alpha) = \max_{M \leq N} \left| \sum_{n \leq M} e(\mathbf{n}^{(k)} \cdot \alpha) \right|$.

(a) Prove that

$$\int_{\mathbb{T}^k} S^*(\alpha) d\alpha < (\log 4N)^{2b} J_k(N, b).$$

(b) Suppose that $|\alpha_j - \beta_j| \leq \frac{1}{2kN^j}$ for $j = 1, \dots, k$. Show that

$$e^{-\pi} S^*(\alpha) < S^*(\beta) < e^{\pi} S^*(\alpha).$$

(c) Suppose that $k \geq 3$, that $|\alpha_k - a/q| \leq q^{-2}$, $(a, q) = 1$ and that $N \leq q \leq N^{k-1}$. Show that $S(\alpha) \ll_k N^{1-(4k^2 \log k)^{-1}}$.

(d) Suppose that $k \geq 3$, that $|\alpha_k - a/q| \leq q^{-2}$, $(a, q) = 1$ and that $N^{\frac{1}{4}k} \leq q \leq N^{\frac{3}{4}k}$. Show that there is a positive constant c such that $S(\alpha) \ll_k N^{1-ck^{-2}}$.

24.3 The Korobov-Vinogradov zero-free region

S:KVZFR

The Hurwitz zeta function $\zeta(s, w)$ is defined for $\operatorname{Re} s > 1$ and $0 \leq \operatorname{Re} w \leq 1$ by

$$\zeta(s, w) = \sum_{n=0}^{\infty} \frac{1}{(n+w)^s}. \quad (24.15)$$

The following lemma can be established in exact analogy to Theorem 1.12.

T:Hurcont

Lemma 24.14. *Suppose that $\operatorname{Re} s > 0$, $0 < \alpha \leq 1$ and $x \geq 1$. Then*

$$\begin{aligned} \zeta(s, \alpha) - \frac{1}{s-1} &= \sum_{0 \leq n \leq x} \frac{1}{(n+\alpha)^s} \\ &\quad + \frac{(x+\alpha)^{s-1} - 1}{s-1} + \frac{\{x\} + \alpha}{(x+\alpha)^s} - s \int_x^{\infty} \frac{\{u\}}{(u+\alpha)^{s+1}} du. \end{aligned}$$

T:VZetaEst

Theorem 24.15. *There is a positive number C such that if $0 < \alpha \leq 1$ and σ and t are real numbers with $\sigma \geq \frac{1}{2}$, then*

$$\zeta(s, \alpha) - \frac{1}{s-1} - \alpha^{-s} \ll (\log \tau)^{\frac{2}{3}} \tau^{C\Theta^{\frac{3}{2}}}$$

where $\tau = |t| + 4$ and $\Theta = \max(0, 1 - \sigma)$, and the implicit constant is absolute.

To deduce the following corollary we need only observe that when χ is non-principal modulo q we have

$$L(s, \chi) = q^{-s} \sum_{a=1}^q \chi(a) \zeta(s, a/q)$$

and then

$$\sum_{a=1}^q \chi(a) a^{-s} \ll \frac{q^{\Theta} - 1}{\Theta}$$

with the interpretation that this is $\log q$ when $\Theta = 0$.

T:VinZL

Corollary 24.16. *There is a positive number C such that if σ and t are real numbers with $\sigma \geq \frac{1}{2}$ and $\Theta = \max(0, 1 - \sigma)$, then*

$$\zeta(s) - \frac{1}{s-1} \ll (\log \tau)^{\frac{2}{3}} \tau^{C\Theta^{\frac{3}{2}}} \quad (24.16) \quad \text{E:vinzeta}$$

If in addition χ is a non-principle character modulo q , then

$$L(s, \chi) \ll \frac{q^{\Theta} - 1}{\Theta} + q^{1-\sigma} (\log \tau)^{\frac{2}{3}} \tau^{C\Theta^{\frac{3}{2}}} \quad (24.17) \quad \text{E:VinL}$$

Proof If necessary by taking complex conjugates, we may assume that $t \geq 0$ and by Lemma 24.14 with $x = 1$ the result is immediate when $t \leq 4$. Thus we may suppose that $t \geq 4$. Moreover, the conclusion is trivial when $\sigma \geq 2$.

The lemma with $x = t$ gives

$$\zeta(s, \alpha) - \frac{1}{s-1} - \alpha^{-s} = \sum_{n \leq t} (n + \alpha)^{-s} + O(t^{-\sigma}).$$

Suppose that $N < N' \leq 2N$, $N' \leq t$. Then, by Theorem, 24.13 there is a positive constant c_1 such that

$$\sum_{N < n \leq N'} (n + \alpha)^{-it} \ll N \exp\left(-c_1 \frac{(\log N)^3}{(\log t)^2}\right).$$

Thus, by partial summation, whenever $N < N'' \leq 2N$ and $N'' \leq t$,

$$\sum_{N < n \leq N'} (n + \alpha)^{-\sigma - it} \ll N^\Theta \exp \left(-c_1 \frac{(\log N)^3}{(\log t)^2} \right).$$

We now let N take on the successive values $2^0, 2^1, 2^2, \dots, 2^k$ where k is chosen so that $2^k < t \leq 2^{k+1}$. Thus

$$\sum_{n \leq t} (n + \alpha)^{-s} \ll 1 + \sum_{j=0}^k \exp \left(\Theta j \log 2 - c_1 (\log 2)^3 \frac{j^3}{(\log t)^2} \right).$$

When $j^2 > \frac{2\Theta(\log t)^2}{c_1(\log 2)^3}$ the general term is

$$\ll \exp \left(-\frac{c_1}{2} (\log 2)^3 \frac{j^3}{(\log t)^2} \right)$$

and when $j^2 \leq \frac{2\Theta(\log t)^2}{c_1(\log 2)^3}$ is

$$\exp(c_2 \Theta^{\frac{3}{2}} \log t) \exp \left(-c_1 (\log 2)^3 \frac{j^3}{(\log t)^2} \right).$$

Thus, in general, each term is

$$\ll \exp(c_2 \Theta^{\frac{3}{2}} \log t) \exp \left(-\frac{c_1}{2} (\log 2)^3 \frac{j^3}{(\log t)^2} \right).$$

Hence

$$\sum_{n \leq t} n^{-s} \ll 1 + \exp(c_2 \Theta^{\frac{3}{2}} \log t) \sum_{j=0}^k \exp \left(-c_1 (\log 2)^3 \frac{j^3}{(\log t)^2} \right).$$

By monotonicity

$$\sum_{j=0}^{\infty} \exp(-\delta j^3) \leq 1 + \int_0^{\infty} \exp(-\delta u^3) du \ll \delta^{-1/3}$$

uniformly for $0 < \delta \leq 1$. Thus

$$\sum_{j=0}^k \exp \left(-c_1 (\log 2)^3 \frac{j^3}{(\log t)^2} \right) \ll (\log t)^{\frac{2}{3}}$$

and the theorem follows. $\square$

L:6.3rescaled **Lemma 24.17.** Suppose that $0 < r < R < \Upsilon$ and f is analytic in a

domain containing the closed disc centered at s_0 and of radius Υ . Then for $|s - s_0| \leq r$ we have

$$\left| \frac{f'}{f}(s) - \sum_{k=1}^K \frac{1}{s - s_k} \right| \ll \frac{1}{(R - r) \log \frac{\Upsilon}{R}} \log \frac{M}{|f(s_0)|}$$

where the sum is over all zeros s_k of f for which $|s_k| \leq R$ and M is the maximum modulus of f on the circle centered at s_0 and of radius Υ .

Proof This follows by the proof of Lemma 6.3 re-scaled by a factor of Υ and keeping the constant explicit. $\square$

T:Vzfr,Ests

Theorem 24.18. *There is a positive number C such that whenever*

$$\sigma > 1 - \frac{1}{C(\log \tau)^{\frac{2}{3}}(\log \log \tau)^{\frac{1}{3}}}$$

and $\tau = |t| + 4$ we have

$$\begin{aligned} \zeta(s) &\neq 0, \\ \frac{\zeta'}{\zeta}(s) &\ll (\log \tau)^{\frac{2}{3}}(\log \log \tau)^{\frac{1}{3}}, \\ |\log \zeta(s)| &\leq \frac{2}{3} \log \log \tau + \frac{1}{3} \log \log \log \tau + O(1), \end{aligned}$$

and

$$\frac{1}{\zeta(s)} \ll (\log \tau)^{\frac{2}{3}}(\log \log \tau)^{\frac{1}{3}}.$$

Proof We first establish the zero free region. Let $\beta + i\gamma$ be a zero of ζ with $\gamma \geq 30$. Put $t = \gamma$, $\tau = t + 4$ and let

$$\Upsilon = \frac{(\log \log \tau)^{\frac{2}{3}}}{c_1(\log \tau)^{\frac{2}{3}}}$$

where c_1 is chosen large enough to ensure that $\Upsilon \leq \frac{1}{2}$. Suppose that

$$\sigma_0 = 1 + \frac{1}{4}\Upsilon,$$

and put $s_0 = \sigma_0 + i\gamma$, $s'_0 = \sigma_0 + 2i\gamma$. We have

$$\frac{1}{|\zeta(s_0)|}, \frac{1}{|\zeta(s'_0)|} \leq \frac{\zeta(2\sigma_0)}{\zeta(\sigma_0)} \ll \log \tau$$

and by (24.16),

$$|\zeta(s)| \leq (\log \tau)^{c_2} \quad (|s - s_0| \leq \Upsilon \text{ or } |s - s'_0| \leq \Upsilon).$$

If $1 - \beta \geq \Upsilon/12$, then we are done. Hence we may suppose that $\beta > 1 - \Upsilon/12$. Suppose that $1 < \sigma < \sigma_0$. Then by Lemma 24.17 with $r = \Upsilon/3$, $R = \Upsilon/2$ we find that

$$\begin{aligned} -\operatorname{Re} \frac{\zeta'}{\zeta}(\sigma + 2i\gamma) &< c_3(\log \tau)^{\frac{2}{3}}(\log \log \tau)^{\frac{1}{3}}, \\ -\operatorname{Re} \frac{\zeta'}{\zeta}(\sigma + i\gamma) &< c_3(\log \tau)^{\frac{2}{3}}(\log \log \tau)^{\frac{1}{3}} - \frac{1}{\sigma - \beta}. \end{aligned}$$

Moreover,

$$-\frac{\zeta'}{\zeta}(\sigma) < \frac{1}{\sigma - 1} + C_3.$$

Then, by the inequality

$$0 \leq -3\frac{\zeta'}{\zeta}(\sigma) - 4\operatorname{Re} \frac{\zeta'}{\zeta}(\sigma + i\gamma) - \operatorname{Re} \frac{\zeta'}{\zeta}(\sigma + 2i\gamma)$$

we find that

$$\frac{4}{\sigma - \beta} - \frac{3}{\sigma - 1} \leq 3C_4(\log \tau)^{\frac{2}{3}}(\log \log \tau)^{\frac{1}{3}}.$$

Also $-\Upsilon/4 < 1 + 6(1 - \beta) - \sigma_0 < \Upsilon/4$. Thus we may take $\sigma = 1 + 6(1 - \beta)$. Then the left hand side is

$$= \left(\frac{4}{7} - \frac{1}{2}\right) \frac{1}{1 - \beta} = \frac{1}{14(1 - \beta)},$$

and the result follows.

To bound the logarithmic derivative we follow the essentials of the proof of Theorem 6.7 but with

$$s_1 = 1 + \frac{1}{c_1(\log \tau)^{\frac{2}{3}}(\log \log \tau)^{\frac{1}{3}}} + it,$$

and in place of Lemma 6.4 we use Lemma 24.17 with

$$4r = 2R = \Upsilon = \frac{1}{c_1} \left(\frac{\log \log \tau}{\log \tau} \right)^{\frac{2}{3}}$$

and $s_0 = s_1$. The bounds for $\log \zeta(s)$ and $1/\zeta(s)$ then follow as in the proof of Theorem 6.7. $\square$

T:VzfrL **Theorem 24.19.** *There is a positive number C such that if χ is a Dirichlet character modulo q , then the region*

$$\mathcal{R}_q = \left\{ s : \sigma > 1 - \frac{1}{C(\log q + (\log \tau)^{2/3}(\log \log \tau)^{1/3})} \right\}$$

contains no zero pf $L(s, \chi)$ unless χ is a quadratic character, in which case $L(s, \chi)$ has at most one, necessarily real zero $\beta < 1$ in $\mathcal{R}_q$

Proof If χ is principal, then the conclusion follows at once from Theorem 24.18. Hence we may suppose that $q \geq 3$. If $4 \log q > \log \tau$, then the theorem follows at once by Theorem 11.3. Hence we may suppose also that $4 \log q \leq \log \tau$, and in particular $|t| \geq 50$, so that in the proof we need only consider zeros $\rho = \beta + i\gamma$ with $|\gamma| \geq 50$. Now we follow the proof of Theorem 11.3, but we use (24.17) and Lemma 24.17 in place of Lemma 11.1, following the pattern of the proof of Theorem 24.18.

We suppose first that

$$\log q \leq (\log \tau)^{2/3} (\log \log \tau)^{1/3}.$$

Now when χ is complex we can follow the exact analogue of the proof of the first assertion of Theorem 24.18. Let ρ be a zero with $|\gamma| \geq 50$ and as before let $t = \gamma$, $\tau = 4||t|$. We take

$$\Upsilon = \frac{(\log \log \tau)^{2/3}}{c_1 (\log \tau)^{2/3}}$$

where c_1 is chosen large enough to ensure that $\Upsilon \leq \frac{1}{2}$ and as before in Lemma 24.17 take $s_0 = 1 + \frac{1}{4}\Upsilon$, $r = \Upsilon/3$, $R = \Upsilon/2$. Then for $\beta > 1 - \Upsilon/12$ and $1 < \sigma < 1 + \Upsilon/4$ we have the inequalities

$$\begin{aligned} -\operatorname{Re} \frac{L'}{L}(\sigma + 2i\gamma, \chi^2) &< c_2 (\log \tau)^{2/3} (\log \log \tau)^{1/3}, \\ -\operatorname{Re} \frac{L'}{L}(\sigma + i\gamma, \chi) &< c_2 (\log \tau)^{2/3} (\log \log \tau)^{1/3} - \frac{1}{\sigma - \beta}, \\ -\frac{L'}{L}(\sigma, \chi_0) &< c_2 \log \log q + \frac{1}{\sigma - 1}. \end{aligned}$$

and we can proceed as in the usual way.

When $\chi_2 = \chi_0$, we have to replace the first of the above inequalities by

$$-\operatorname{Re} \frac{L'}{L}(\sigma + 2i\gamma, \chi^2) < \frac{1 - \sigma}{(1 - \sigma)^2 + 4\gamma^2} + c_2 (\log \tau)^{2/3} (\log \log \tau)^{1/3}$$

and again the proof proceeds as that of Theorem 11.3.

That leaves the case when $(\log \tau)^{2/3} (\log \log \tau)^{1/3} < \log q \leq \frac{1}{4} \log \tau$. Now we choose

$$\Upsilon = \frac{(\log q)^2}{c_1 (\log \tau)^2}$$

and we find that the first two inequalities above are replaced by

$$\begin{aligned} -\operatorname{Re} \frac{L'}{L}(\sigma + 2i\gamma, \chi^2) &< c_2 \log q, \\ -\operatorname{Re} \frac{L'}{L}(\sigma + i\gamma, \chi) &< c_2 \log q - \frac{1}{\sigma - \beta}, \end{aligned}$$

in the case when χ is complex and with the same adjustment as before when $\chi^2 = \chi_0$. $\square$

T:VestsL **Theorem 24.20.** *Let χ be a non-principal character modulo q , let C be the constant of Theorem 24.19, and suppose that*

$$\sigma \geq 1 - \frac{1}{2C(\log q + (\log \tau)^{2/3}(\log \log \tau)^{1/3})}.$$

If $L(s, \chi)$ has no exceptional zero, or if β_1 is an exceptional zero of $L(s, \chi)$ but $|s - \beta_1| \geq 1/\log q$, then

$$\frac{L'}{L}(s, \chi) \ll (\log q) + (\log \tau)^{2/3}(\log \log \tau)^{1/3}, \quad (24.18)$$

$$|\log L(s, \chi)| \leq \log(\log q + (\log \tau)^{2/3}(\log \log \tau)^{1/3}) + O(1) \quad (24.19)$$

and

$$\frac{1}{L(s, \chi)} \ll (\log q) + (\log \tau)^{2/3}(\log \log \tau)^{1/3}. \quad (24.20)$$

We do not have anything new to add when there is an exceptional zero β_1 with $|s - \beta_1| < 1/\log q$

24.4 Improvements in the Distribution of Prime Numbers

S:IDP

We now apply the new zero-free regions to the distribution of primes.

T:PET **Theorem 24.21.** *There is a constant $c > 0$ such that*

$$\psi(x) = x + O\left(x \exp\left(-\frac{c(\log x)^{3/5}}{(\log \log x)^{1/5}}\right)\right), \quad (24.21) \quad \text{E:I6.12}$$

$$\vartheta(x) = x + O\left(x \exp\left(-\frac{c(\log x)^{3/5}}{(\log \log x)^{1/5}}\right)\right), \quad (24.22) \quad \text{E:I6.13}$$

and

$$\pi(x) = \text{li}(x) + O\left(x \exp\left(-\frac{c(\log x)^{3/5}}{(\log \log x)^{1/5}}\right)\right). \quad (24.23) \quad \boxed{\text{E:I6.14}}$$

Proof The proof follows that of Theorem 6.9, but using Theorem 24.18 in place of Theorems 6.6 and 6.7 and with T taken to be

$$T = \frac{c_1(\log x)^{3/5}}{(\log \log x)^{1/5}}$$

with an suitable positive constant c_1 . $\square$

For primes in arithmetic progressions there is, as usual, a trade-off between the error term, the range for q , and any possible exceptional zero.

T:PAPET **Theorem 24.22.** *There is a constant c_1 such that if χ is a character modulo q and*

$$q \leq \exp\left(\frac{\log x}{c_1 \log \log x}\right),$$

then

$$\begin{aligned} \psi(x, \chi) &= E_0(\chi)x - E_1(\chi) \frac{x^{\beta_1}}{\beta_1} \\ &+ O\left(x \exp\left(-\frac{\log x}{c_1(\log q + (\log x)^{2/5}(\log \log x)^{1/5})}\right)\right) \end{aligned} \quad (24.24) \quad \boxed{\text{E:I11.23}}$$

where $E_0(\chi) = 0$ unless χ is principal in which case it is 1 and $E_1(\chi) = 0$ unless $L(s, \chi)$ has an exceptional zero β_1 in which case it is 1.

Proof Here one can follow the proof of Theorem 11.16 but with

$$T = \exp\left(\frac{\log x}{c_2(\log q + (\log x)^{2/5}(\log \log x)^{1/5})}\right)$$

for some constant c_2 . $\square$

In applying this to $\psi(x, q, a)$ there is some limitation on the size of q imposed by the requirement to have an error that is small compared with

$$\frac{x}{\phi(q)}.$$

T:VPAGE **Corollary 24.23.** *There is a constant c_1 such that if*

$$q \leq \exp(c_1 \sqrt{\log x})$$

and $(q, a) = 1$, then

$$\psi(x, q, a) = \frac{x}{\phi(q)} - E_1(\chi) \frac{\chi_1(a)x^{\beta_1}}{\phi(q)\beta_1} + O\left(x \exp\left(-\frac{\log x}{c_1(\log q + (\log x)^{2/5}(\log \log x)^{1/5})}\right)\right). \quad (24.25) \quad \boxed{\text{E:I11.29}}$$

When one further applies the consequence, Corollary 11.15, of Siegel's theorem, Theorem 11.14, one has

T:VSiegel1 **Corollary 24.24.** *Let c_1 be the same constant as in Corollary 24.22. For any positive A there is an $x_0(A)$ such that if $q \leq (\log x)^A$, then*

$$\psi(x, \chi) = E_0(\chi)x + O\left(x \exp\left(-\frac{(\log x)^{3/5}}{c_1(\log \log x)^{1/5}}\right)\right). \quad (24.26) \quad \boxed{\text{E:I11.31}}$$

T:VSiegel2 **Corollary 24.25.** *Let c_1 be the same constant as in Corollary 24.22. For any positive A there is an $x_0(A)$ such that if $q \leq (\log x)^A$ and $(q, a) = 1$, then*

$$\psi(x, q, a) = \frac{x}{\phi(q)} + O\left(x \exp\left(-\frac{(\log x)^{3/5}}{c_1(\log \log x)^{1/5}}\right)\right) \quad (24.27) \quad \boxed{\text{E:I11.31a}}$$

and

$$\pi(x, q, a) = \frac{\text{li}(x)}{\phi(q)} + O\left(x \exp\left(-\frac{(\log x)^{3/5}}{c_1(\log \log x)^{1/5}}\right)\right) \quad (24.28) \quad \boxed{\text{E:I11.37}}$$

Note that c_1 is effective. However, in the current state of knowledge, $x_0(A)$ is not. Alternatively one can make the implicit constant in the O -notation ineffective so that the theorem holds for all $x \geq 2$.

Further improvements are dependent on the distribution and density of zeros near the 1-line and are dependent on the main results of Chapter 28.

24.5 Notes

S:ExpSumII Notes

§24.1 The Vinogradov mean value theorem first appeared in IV35, IV36 Vinogradov (1935, 1936). The main motivation at that stage was additive number theory, and especially with the aim of improving the known results in Waring's Problem, at least for larger exponents, and this it did spectacularly. There is some indication that Mordell's LM32 Mordell (1932) work on

complete sums was suggestive. The first application to the Riemann zeta function was not made by Vinogradov, but by ^{NC36}Chudakov (1936). Theorem 24.7 is already in the form given in ????. It is crucial in estimating the kind of sums useful for application to the Riemann zeta-function that the growth of the number $D(k, r)$ should not be too rapid in terms of k and r .

Lemma 24.4 is in ^{YL43}Linnik (1943).

^{RV81}Vaughan (1981) conjectured that

$$J_k(X, b) \sim C_{k,b} \min(X^{2b-k(k+1)/2}, X^b),$$

and a weaker form of this,

$$J_k(X, b) \ll_{k,b,\varepsilon} X^\varepsilon \min(X^{2b-k(k+1)/2}, X^b)$$

was established by ^{JB16}Bourgain, Demeter & Guth (2016). Not quite so strong bounds had been established earlier by ^{TW12}Wooley (2012). However none of these more recent methods give estimates for $D(k, r)$ which are suitable for application here.

§24.2 There is a long succession of papers, ^{NC46}Chudakov (1946), ^{TT50}Tatuzawa (1950), first edition (1951) of ^{ET86}Fitchmarsh (1986), ^{PT53}Turán (1953), ^{LS57}Schoenfeld (1957), each with small improvements on what went before. The best methods we have currently for deducing bounds from the Vinogradov mean value theorem for the sums that are the subject of Theorem 24.13 were developed by ^{NK58}Korobov (1958) and ^{IV58}Vinogradov (1958) independently. However in both these papers claims are made about the error in the prime number theorem, namely that

$$\psi(x) - x \ll x \exp(-c(\log x)^{3/5})$$

which have never been substantiated. There is a very scathing comment by ^{AI64}Ingham (1964) in a review “it is highly desirable that the claim to the stronger and neater result should be substantiated or withdrawn without further delay”. It never has been, although in 1980 Vinogradov does simply write “my method gives” (24.21), tacitly admitting by neglect that this is the best that can be done. There are full accounts of the ^{NK58}Korobov (1958) and ^{IV58}Vinogradov (1958) methods in ^{AW63}Walfisz (1963) (actually written by Richert), and ^{HR67}Richert (1967) makes the application to the Hurwitz zeta function.

The kernel $K(\beta)$ defined in the proof of Lemma 24.8 is an example of a de la Vallée Poussin kernel. The upper bound (??) suffices for our purposes, but ^{HM15}Meha (2015) determined the exact value of this quantity.

§24.4 The application of these methods to primes in arithmetic progressions is not often done, but once one has the bound for the Hurwitz zeta function it is routine and potentially useful. There is an excellent account of the state of play prior to 1958 in ^{KP57}7, Chapter VIII, including earlier versions of the results of this section.

24.6 References

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25

Approximate Functional Equations

C:AFE

We recall that Theorem 10.2 asserts that if s is a complex number, $s \neq 0$, $s \neq 1$, and z is a complex number with $\operatorname{Re} z \geq 0$, then

$$\begin{aligned} \zeta(s)\Gamma(s/2)\pi^{-s/2} &= \pi^{-s/2} \sum_{n=1}^{\infty} n^{-s} \Gamma(s/2, \pi n^2 z) \\ &\quad + \pi^{(s-1)/2} \sum_{n=1}^{\infty} n^{s-1} \Gamma((1-s)/2, \pi n^2/z) \\ &\quad + \frac{z^{(s-1)/2}}{s-1} - \frac{z^{s/2}}{s}. \end{aligned} \quad (25.1) \quad \text{E:zetaFE}$$

Here $\Gamma(s, a)$ is the incomplete gamma function,

$$\Gamma(s, a) = \int_a^{\infty} e^{-z} z^{s-1} dz. \quad (25.2) \quad \text{E:DefIncGamFcn}$$

The integrand is regular at $z = 0$ only when s is a positive integer. To avoid issues of the singularity at 0 we insist that a , and the path joining a to ∞ should lie in the slit complex plane $\mathbb{C} \setminus (-\infty, 0]$. For any fixed such a , $\Gamma(s, a)$ is an entire function of s . When we discussed this formula in Chapter 10, we simply took $z = 1$ (which is essentially what Riemann did). However, the formula (25.1) is not useful when $|t|$ is large and z is real, since then the factors $\Gamma(s/2, \pi n^2 z)$ are much larger than $\Gamma(s/2)$. For example, $|\Gamma(1+it, 1)| \asymp 1/|t|$ while $\Gamma(1+it)$ is exponentially small. More precisely, in Appendix C we used Stirling's formula to show that

$$|\Gamma(s)| \asymp \tau^{\sigma-1/2} e^{-\pi\tau/2} \quad (25.3) \quad \text{E:GamEst}$$

when $|t| \geq 1$ and $|\sigma|$ is uniformly bounded. Fortunately, $\Gamma(s, z) \ll |\Gamma(s)|$ for such s when z is of the form $z = ae^{i\phi}$ where $a > 0$ and

$$\phi = \arctan t. \quad (25.4) \quad \text{E:Defphi}$$

Throughout our discussion, the second argument of the incomplete gamma function will have this argument. We note that from this definition it follows that

$$\cos \phi = \frac{1}{\sqrt{t^2 + 1}}, \quad \sin \phi = \frac{t}{\sqrt{t^2 + 1}}. \quad (25.5) \quad \boxed{\text{E:trigvals}}$$

We note some useful estimates concerning ϕ . It is clear from the above that

$$\frac{1}{|t| + 1} < \cos \phi < \frac{1}{|t|} \quad (25.6) \quad \boxed{\text{E:cosphiEst1}}$$

uniformly for all $t \neq 0$. Since $\tau = |t| + 4$, we also see that

$$\frac{1}{\tau} < \cos \phi \ll \frac{1}{\tau} \quad (25.7) \quad \boxed{\text{E:cosphiEst1.5}}$$

uniformly for all t . From this latter estimate we deduce that

$$|(\cos \phi)^{-s}| = (\cos \phi)^{-\sigma} \asymp \tau^\sigma \quad (25.8) \quad \boxed{\text{E:cosphiEst2}}$$

uniformly for $-A \leq \sigma \leq A$, with the implicit constant depending on A .

Suppose that $t > 0$. Then

$$\phi = \arctan t = \int_0^t \frac{du}{u^2 + 1} = \frac{\pi}{2} - \int_t^\infty \frac{du}{u^2 + 1} > \frac{\pi}{2} - \int_t^\infty \frac{du}{u^2} = \frac{\pi}{2} - \frac{1}{t}.$$

Thus $\frac{\pi}{2}t - 1 < \phi t < \frac{\pi}{2}t$. The function $\phi t = t \arctan t$ is an even function of t , so it follows that $\frac{\pi}{2}|t| - 1 < \phi t < \frac{\pi}{2}|t|$, so

$$\phi t = \frac{\pi}{2}\tau + O(1) \quad (25.9) \quad \boxed{\text{E:phitIneq}}$$

uniformly for all real t . Hence

$$|e^{i\phi s}| = e^{-\phi t} \asymp e^{-\pi\tau/2} \quad (25.10) \quad \boxed{\text{E:e~iphis}}$$

uniformly for all s .

To estimate

$$\Gamma(s, ae^{i\phi}) = \int_{ae^{i\phi}}^\infty e^{-z} z^{s-1} dz$$

we integrate along the ray $z = ue^{i\phi}$ for $a \leq u < \infty$. Thus the above is

$$= e^{i\phi s} \int_a^\infty \exp(-ue^{i\phi}) u^{s-1} du. \quad (25.11) \quad \boxed{\text{E:IncGamForm0}}$$

By the further change of variable $v = u \cos \phi$ we see that the above is

$$\begin{aligned} &= \frac{e^{i\phi s}}{(\cos \phi)^s} \int_{a \cos \phi}^{\infty} e^{-v} v^{\sigma-1} e^{it(\log v - v)} dv \\ &= \frac{e^{i\phi s}}{(\cos \phi)^s} \int_{a \cos \phi}^{\infty} r(v) e^{i\theta(v)} dv \end{aligned} \quad (25.12) \quad \boxed{\text{E:IncGamForm}}$$

where

$$\begin{aligned} r(v) &= e^{-v} v^{\sigma-1}, \\ \theta(v) &= t(\log v - v). \end{aligned} \quad (25.13) \quad \boxed{\text{E:Defr,theta}}$$

On combining (25.3), (25.8), (25.10), and (25.12), we find that

$$|\Gamma(s, ae^{i\phi})| \asymp |\Gamma(s)| \tau^{1/2} \left| \int_{a \cos \phi}^{\infty} r(v) e^{i\theta(v)} dv \right|. \quad (25.14) \quad \boxed{\text{E:MainReduc}}$$

L:IncGamEst **Lemma 25.1.** *Suppose that ϕ is given by (25.4), and that A is a positive constant. Then*

$$\Gamma(s, ae^{i\phi}) = \Gamma(s) \left(\chi_{[0, \tau]}(a) + O\left(e^{-a/\tau} \left(\frac{a}{\tau}\right)^{\sigma} \min\left(1, \frac{\tau^{1/2}}{|a - \tau|}\right)\right) \right) \quad (25.15) \quad \boxed{\text{E:IncGamEst}}$$

uniformly for $a \geq 0$, $0 < \sigma \leq A$, $|t| \geq 1$.

When $|t| \leq 1$ we have available the trivial estimates

$$|\Gamma(s, ae^{i\phi})| \ll \Gamma(\sigma, a \cos \phi) \leq \Gamma(\sigma). \quad (25.16) \quad \boxed{\text{E:IncGamTrivEst}}$$

Proof First suppose that $a \geq \tau + \tau^{1/2}$. Let $r(v)$ and $\theta(v)$ be defined as in (25.13). We note that

$$\frac{r(v)}{\theta'(v)} = \frac{e^{-v} v^{\sigma}}{t(1-v)}.$$

If $V > 1$, then the maximum modulus of the above on the interval $[V, \infty)$, and the its total variation on the same interval are both

$$\ll \frac{e^{-V} V^{\sigma}}{t(V-1)}.$$

By Theorem ??

$$\int_{a \cos \phi}^{\infty} r(v) e^{i\theta(v)} dv \ll \frac{e^{-a \cos \phi} (a \cos \phi)^{\sigma}}{\tau(a \cos \phi - 1)}$$

since $a \cos \phi > a/\tau > 1$. From (25.7) and (25.8) it follows that the above is

$$\ll |\Gamma(s)| e^{-a/\tau} \left(\frac{a}{\tau}\right)^\sigma \frac{\tau^{1/2}}{a - \tau}, \quad (25.17) \quad \boxed{\text{E:Case1Est}}$$

which is the desired bound in this case.

Now suppose that $\tau - \tau^{1/2} \leq a \leq a + \tau^{1/2} = b$. We write the integral in (25.14) as

$$\int_{a \cos \phi}^b r(v) e^{i\theta(v)} dv + \int_b^\infty r(v) e^{i\theta(v)} dv.$$

From (25.17) we know that the second integral above is $\ll \tau^{-1/2}$. From the information that $\tau - \tau^{1/2} \leq a \leq \tau + \tau^{1/2}$, by means of a little calculation we deduce that $1 - \tau^{-1/2} \leq a \cos \phi \leq 1 + 17\tau^{-1/2}$. Now $r(v) \asymp 1$ when $v \asymp 1$, so the first integral above is

$$\ll \int_{a \cos \phi}^b r(v) dv \leq \int_{1 - \tau^{-1/2}}^{1 + 17\tau^{-1/2}} r(v) dv \ll \int_{1 - \tau^{-1/2}}^{1 + 17\tau^{-1/2}} 1 dv \ll \tau^{-1/2}.$$

By (25.14) it follows that $\Gamma(s, ae^{i\phi}) \ll |\Gamma(s)|$ in this case.

Finally, we suppose that $0 \leq a \leq \tau - \tau^{1/2}$. By taking $a = 0$ in (25.12) we obtain a formula for $\Gamma(s)$. On subtracting, we find that

$$\Gamma(s, ae^{i\phi}) = \Gamma(s) - \frac{e^{i\phi s}}{(\cos \phi)^s} \int_0^{a \cos \phi} r(v) e^{i\theta(v)} dv.$$

We treat this last integral as we did in the first case, and thus find that

$$\Gamma(s, ae^{i\phi}) = \Gamma(s) \left(1 + O\left(e^{-a/\tau} \left(\frac{a}{\tau}\right)^\sigma \frac{\tau^{1/2}}{\tau - a} \right) \right),$$

so we have the stated result. $\square$

We now combine the functional equation as expressed by the identity identity (25.1) with Lemma 25.1, to obtain a useful *approximate functional equation*. In what follows, we put

$$\Delta(s) = \frac{\Gamma(\frac{1-s}{2})}{\Gamma(\frac{s}{2})} \pi^{s-\frac{1}{2}} = 2^s \pi^{s-1} \Gamma(1-s) \sin \frac{\pi s}{2}. \quad (25.18) \quad \boxed{\text{E:DefDelta}}$$

Thus the functional equation for the zeta function in its asymmetric form (as in Corollary 10.4) asserts that $\zeta(s) = \Delta(s) \zeta(1-s)$. From (25.3) it follows that

$$|\Delta(s)| \asymp \tau^{1/2-\sigma} \quad (25.19) \quad \boxed{\text{E:DeltaEst}}$$

when $|t| \geq 1$ and σ is uniformly bounded.

T:zetaAFE **Theorem 25.2.** Suppose that $0 < \delta \leq 1/2$ is fixed, that $\delta \leq \sigma \leq 1 - \delta$, and that $2\pi xy = \tau$ where $x \geq 1$, $y \geq 1$. Then

$$\begin{aligned} \zeta(s) = \sum_{n \leq x} n^{-s} + \Delta(s) \sum_{n \leq y} n^{s-1} \\ + O\left((x^{-\sigma} + \tau^{1/2-\sigma} y^{\sigma-1}) \log \tau\right). \end{aligned} \quad (25.20) \quad \text{E:zetaAFE}$$

Note that the last term in the first sum above is of size $x^{-\sigma}$, and that the last term in the second sum is of size $\tau^{1/2-\sigma} y^{\sigma-1}$. Thus the error term is larger than these quantities by only one logarithm.

Proof We divide both sides of (25.1) by $\Gamma(s/2)\pi^{-s/2}$ to see that

$$\begin{aligned} \zeta(s) = \sum_{n=1}^{\infty} n^{-s} \frac{\Gamma(s/2, \pi n^2 z)}{\Gamma(s/2)} + \Delta(s) \sum_{n=1}^{\infty} n^{s-1} \frac{\Gamma((1-s)/2, \pi n^2/z)}{\Gamma((1-s)/2)} \\ + O\left(\frac{|z^{(s-1)/2}| + |z^{s/2}|}{|\Gamma(s/2)| \tau}\right). \end{aligned} \quad (25.21) \quad \text{E:zetaAFE0}$$

We take $z = re^{i\phi}$ where $r = \tau/(2\pi x^2)$ and $\phi = \arctan(t/2)$. The error term above is

$$\ll (x^{-\sigma} + \tau^{1/2-\sigma} y^{\sigma-1}) \tau^{-1/2},$$

which is inconsequential in (25.20). By Lemma 25.1, the first sum above is $\sum_{n \leq x} n^{-s}$ plus an error term that is

$$\begin{aligned} &\ll \sum_{n=1}^{\infty} n^{-\sigma} \left(\frac{n^2 r}{\tau}\right)^{\sigma/2} \min\left(1, \frac{(\tau/2)^{1/2}}{|\pi n^2 r - \tau/2|}\right) e^{-2\pi n^2 r/\tau} \\ &\ll x^{-\sigma} \tau^{-1/2} \sum_{n \leq x-1} \frac{1}{1 - n^2/x^2} + x^{-\sigma} + x^{-\sigma} \tau^{-1/2} \sum_{n \geq x+1} \frac{e^{-n^2/x^2}}{n^2/x^2 - 1} \\ &\ll x^{1-\sigma} \tau^{-1/2} \sum_{n \leq x-1} \frac{1}{x - n} + x^{-\sigma} + x^{1-\sigma} \tau^{-1/2} \sum_{n \geq x+1} \frac{e^{-n^2/x^2}}{n - x} \\ &\ll \tau^{1/2-\sigma} y^{\sigma-1} \log x + x^{-\sigma}. \end{aligned}$$

This is admissible in (25.20). Similarly, the second sum in (25.21) is

$\sum_{n \leq y} y^{s-1}$ plus an amount that is

$$\begin{aligned}
 &\ll \sum_{n=1}^{\infty} n^{\sigma-1} \left(\frac{n^2}{r\tau} \right)^{(1-\sigma)/2} \min \left(1, \frac{(\tau/2)^{1/2}}{|\tau/2 - \pi n^2/r|} \right) e^{-2\pi n^2/(r\tau)} \\
 &\ll y^{\sigma-1} \tau^{-1/2} \sum_{n \leq y-1} \frac{1}{1 - n^2/y^2} + y^{\sigma-1} + y^{\sigma-1} \tau^{-1/2} \sum_{n \geq y+1} \frac{e^{-n^2/y^2}}{n^2/y^2 - 1} \\
 &\ll y^{\sigma} \tau^{-1/2} \sum_{n \leq y-1} \frac{1}{y-n} + y^{\sigma-1} + y^{\sigma} \tau^{-1/2} \sum_{n \geq y+1} \frac{e^{-n^2/y^2}}{n-y} \\
 &\ll y^{\sigma} \tau^{-1/2} \log y + y^{\sigma-1}.
 \end{aligned}$$

We multiply by $\Delta(s)$, and appeal to (25.19) to see that the error term here is

$$\ll (y/\tau)^{\sigma} \log y + y^{\sigma-1} \tau^{1/2-\sigma} \ll x^{-\sigma} \log y + y^{\sigma-1} \tau^{1/2-\sigma}.$$

This is admissible in (25.20), so the proof is complete. $\square$

Despite its elegance, the approximate functional equation is not immediately useful for estimating mean values, since the number of terms in the sums depends on t , while our basic tools concern one fixed sum. To overcome this difficulty we average over the parameters. Set

$$x = \sqrt{2\pi\tau} \cdot r, \quad y = \sqrt{2\pi\tau}/r$$

in the approximate functional equation (25.20), divide both sides by r , integrate both sides from $1/2$ to 2 , and finally divide both sides by $\log 4$. The result is that

$$\begin{aligned}
 \zeta(s) &= \sum_n w(n/\sqrt{\tau}) n^{-s} + \Delta(s) \sum_n w(n/\sqrt{\tau}) n^{s-1} \\
 &\quad + O(\tau^{-\sigma/2} \log \tau)
 \end{aligned} \tag{25.22} \quad \boxed{\text{E:Wtdafe}}$$

where

$$w(u) = \begin{cases} 1 & (0 \leq u \leq \sqrt{\pi/2}), \\ \frac{\log(\sqrt{8\pi}/u)}{\log 4} & (\sqrt{\pi/2} \leq u \leq \sqrt{8\pi}), \\ 0 & (\sqrt{8\pi} \leq u). \end{cases} \tag{25.23} \quad \boxed{\text{E:Defw}}$$

Let

$$A(s, x) = \sum_{n \leq x} n^{-s} \tag{25.24} \quad \boxed{\text{E:DefA(s,x)}}$$

denote a partial sum of the Dirichlet series that defines $\zeta(s)$. Then

$$\sum_n w(n/\sqrt{\tau})n^{-s} = \int_{1-}^{\infty} w(x/\sqrt{\tau}) dA(s, x),$$

which by integration by parts is

$$= \left[w(x/\sqrt{\tau})A(s, x) \right]_{1-}^{\infty} - \int_1^{\infty} A(s, x) dw(x/\sqrt{\tau}).$$

Now $A(s, x) = 0$ for $x < 1$, and $w(x/\sqrt{\tau}) = 0$ for $x > \sqrt{8\pi\tau}$, so the contributions of the endpoints vanishes. The function $w(u)$ is continuous and piecewise differentiable, so the above is

$$= \frac{-1}{\sqrt{\tau}} \int_1^{\infty} A(s, x) w'(x/\sqrt{\tau}) dx.$$

As $w'(u) = -1/(u \log 4)$ for $\sqrt{\pi/2} < u < \sqrt{8\pi}$, and $w'(u) = 0$ otherwise, the above is

$$= \frac{1}{\log 4} \int_{\sqrt{\pi\tau/2}}^{\sqrt{8\pi\tau}} A(s, x) \frac{dx}{x} = \int A(s, x) d\mu_{\tau}(x),$$

say. Here μ_{τ} is a probability measure (i.e., a nonnegative measure with total mass 1), so the identity above expresses the left hand side as a weighted average of the partial sums $A(s, x)$. The support of μ_{τ} depends on t , but by integrating over a longer interval we can obtain an upper bound that holds uniformly for t in an interval. Suppose that k is a positive integer. Then by Hölder's Inequality (with exponents $2k$ and $2k/(2k-1)$, whose reciprocals sum to 1) we find that

$$\begin{aligned} \left| \sum_n w(n/\sqrt{\tau})n^{-s} \right|^{2k} &\leq \left(\int |A(s, x)|^{2k} d\mu_{\tau}(x) \right) \left(\int 1 d\mu_{\tau}(x) \right)^{2k-1} \\ &= \int |A(s, x)|^{2k} d\mu_{\tau}(x). \end{aligned}$$

(25.25) E: Sumub1

Hence the upper bound

$$\left| \sum_n w(n/\sqrt{\tau})n^{-s} \right|^{2k} \leq \int_{T^{1/2}}^{8T^{1/2}} |A(s, x)|^{2k} \frac{dx}{x} \quad (25.26) \quad \text{E: Sumub2}$$

holds uniformly for $T \leq t \leq 2T$. We do not obtain sharp constants by arguing in this way, but sometimes we can obtain useful bounds.

We now turn to L functions. If χ is a primitive character modulo q with $q > 1$, then (by Theorem 10.7) we know that

$$\begin{aligned} L(s, \chi) \Gamma((s + \kappa)/2) (q/\pi)^{(s+\kappa)/2} \\ = (q/\pi)^{(s+\kappa)/2} \sum_{n=1}^{\infty} \chi(n) n^{-s} \Gamma((s + \kappa)/2, \pi n^2 z / q) \\ = \varepsilon(\chi) (q/\pi)^{(1-s+\kappa)/2} \sum_{n=1}^{\infty} \bar{\chi}(n) n^{s-1} \Gamma((1-s + \kappa)/2, \pi n^2 / (qz)) \end{aligned} \quad (25.27) \quad \boxed{\text{E:LfcnFE}}$$

for z with $\operatorname{Re} z > 0$. As usual,

$$\kappa = \kappa(\chi) = \begin{cases} 0 & \text{if } \chi(-1) = 1, \\ 1 & \text{if } \chi(-1) = -1 \end{cases} \quad \varepsilon(\chi) = \frac{\tau(\chi)}{i^\kappa \sqrt{q}}. \quad (25.28) \quad \boxed{\text{E:LfePars}}$$

In general, the incomplete gamma function $\Gamma(s, a)$ undergoes a change of behaviour when $|a|$ passes $|s|$; evidence of this can be seen in Lemma 25.1. The crude estimate (25.15) sufficed to give Theorem 25.2 because the numbers n^2 pass τ so quickly. However, in the formula above n^2 is replaced by n^2/q , which passes τ comparatively slowly if q is large. This would not be a problem if the weight $w_0(u) = w_0(s, u) = \Gamma(s, ue^{i\theta})/\Gamma(s)$ were to move rather smoothly from near 1 to near 0. To estimate a weighted sum $\sum_{n=1}^N w(n) a_n$ in terms of its partial sums $A(x) = \sum_{n \leq x} a_n$ we integrate by parts, as we did in treating the zeta function. This works especially well if $w(x)$ is monotonic, or at least has bounded variation. Unfortunately, this avenue is not immediately available, because $w_0(u)$ has large variation (see Exercise 1). This is illustrated in Figure 25.1 where $w_0(u)$ is first depicted as a curve in the complex plane, and then its real part is graphed, along with its asymptotic shape.

Fortunately, we can avoid this disaster by introducing a simple averaging. Since we are working in the multiplicative group of positive real numbers, instead of the usual arithmetic mean, $\frac{1}{b-a} \int_a^b f(x) dx$, we use its multiplicative analogue, $\frac{1}{\log b/a} \int_a^b f(x) \frac{dx}{x}$. We put

$$w_1(s, a) = \frac{1}{\Gamma(s) \log 4} \int_{1/2}^2 \Gamma(s, axe^{i\phi}) \frac{dx}{x} \quad (25.29) \quad \boxed{\text{E:Defw1}}$$

where ϕ is defined by (25.4). In Figure 25.2 we see that the situation appears to be considerably improved. We now establish that this is indeed the case.

L:w1Est **Lemma 25.3.** *Suppose that $0 < \sigma \leq A$, that ϕ is given by (25.4), and*

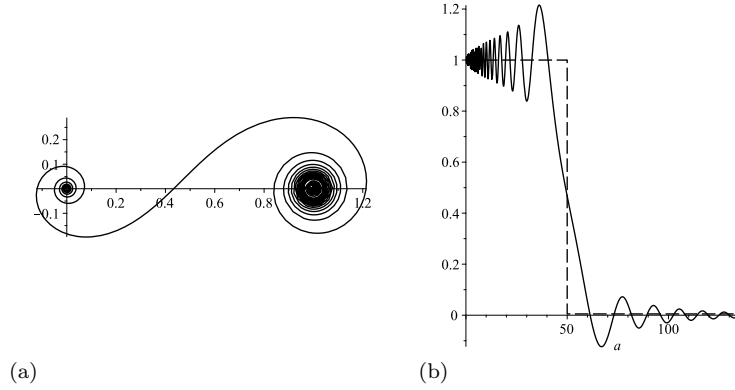


Figure 25.1 Graphs of (a) $w_0(\frac{1}{2} + 50i, a)$ and (b) $\text{Re } w_0(\frac{1}{2} + 50i, a)$, for $0 \leq a \leq 150$.

Fi:w0

that w_1 is defined as above. For $a > 0$,

$$\frac{\partial}{\partial a} w_1(s, a) \ll e^{-a/(2\tau)} a^{\sigma-1} \tau^{-\sigma}$$

where the implicit constant may depend on A .

It follows immediately from the above that

$$\begin{aligned} \text{Var}_{[0, \infty)} w_1(s, a) &= \int_0^\infty \left| \frac{\partial}{\partial a} w_1(s, a) \right| da \ll \int_0^\infty e^{-a/(2\tau)} \left(\frac{a}{\tau} \right)^\sigma \frac{da}{a} \\ &= \int_0^\infty e^{-x} (2x)^\sigma \frac{dx}{x} = \Gamma(\sigma) 2^\sigma, \end{aligned}$$

which is not only finite, but also uniformly bounded for $0 < \delta \leq \sigma \leq A$.

Proof By Leibniz's rule we know that

$$\frac{\partial}{\partial a} w_1(s, a) = \frac{1}{\Gamma(s) \log 4} \int_{1/2}^2 \frac{\partial}{\partial a} \Gamma(s, axe^{i\phi}) \frac{dx}{x}.$$

By the parameterization (25.11), this is

$$\begin{aligned} &= \frac{e^{i\phi s}}{\Gamma(s) \log 4} \int_{1/2}^2 \frac{\partial}{\partial a} \int_{ax}^\infty \exp(-ue^{i\phi}) u^{s-1} du \frac{dx}{x} \\ &= \frac{-e^{i\phi s}}{\Gamma(s) \log 4} \int_{1/2}^2 \exp(-axe^{i\phi}) (ax)^{s-1} dx \\ &= \frac{-e^{i\phi s}}{a\Gamma(s) \log 4} \int_{a/2}^{2a} \exp(-ue^{i\phi}) u^{s-1} du. \end{aligned}$$

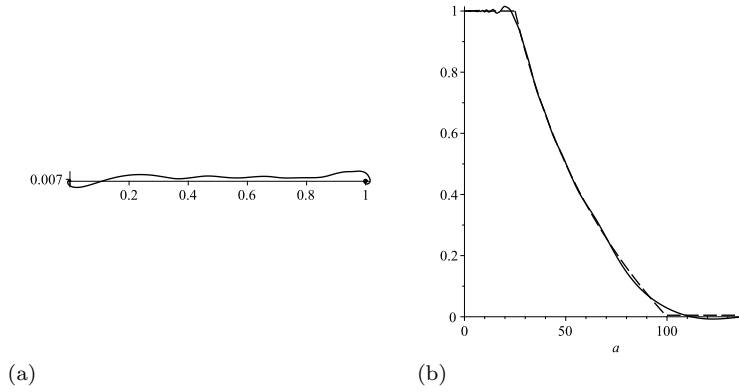


Figure 25.2 Graphs of (a) $w_1(\frac{1}{2} + 50i, a)$ and (b) $\operatorname{Re} w_1(\frac{1}{2} + 50i, a)$ and its asymptotic shape, for $0 \leq a \leq 150$.

Fi:w1

By (25.11) again, this is

$$= \frac{\Gamma(s, 2ae^{i\phi}) - \Gamma(s, \frac{1}{2}ae^{i\phi})}{a\Gamma(s) \log 4}.$$

The right hand side of (25.15) is

$$\ll |\Gamma(s)| e^{-a/\tau} \left(\frac{a}{\tau}\right)^\sigma.$$

We apply this twice to obtain the stated bound. $\square$

T:LafeEst

Theorem 25.4. Put

$$A(s, \chi; x) = \sum_{n \leq x} \chi(n) n^{-s}. \quad (25.30)$$

E:DefA

Suppose that χ is a primitive character modulo q with $q > 1$, and that $0 < \delta \leq 1/2$. Then

$$L(s, \chi) \ll \int_0^\infty (|A(s, \chi; u)| + |A(1-s, \bar{\chi}; u)|) e^{-u^2/(q\tau)} \left(\frac{u^2}{q\tau}\right)^{\delta/2} \frac{du}{u}.$$

uniformly for $\delta \leq \sigma \leq 1 - \delta$.

By the change of variable $v = u^2/(q\tau)$ we find that

$$\int_0^\infty \exp\left(\frac{-u^2}{q\tau}\right) \left(\frac{u^2}{q\tau}\right)^{\delta/2} \frac{du}{u} = 2 \int_0^\infty e^{-v} v^{\delta/2-1} dv = 2\Gamma(\delta).$$

Thus our bound for $|L(s, \chi)|$ is comparable to a weighted average of the

quantity $|A(s, \chi; u)| + |A(1-s, \bar{\chi}; u)|$. The bulk of the weight in this average is placed on sums of length $u \ll \sqrt{q\tau}$.

To prepare for the proof of the above, we make some preliminary remarks. The functional equation of $L(s, \chi)$ in its symmetric form asserts that

$$\begin{aligned} L(s, \chi) \Gamma((s + \kappa)/2) (q/\pi)^{(s+\kappa)/2} \\ = \varepsilon(\chi) L(1-s, \bar{\chi}) \Gamma((1-s + \kappa)/2) (q/\pi)^{(1-s+\kappa)/2}. \end{aligned}$$

(See Corollary 10.8.) Let $\Delta(s, \chi)$ denote the quotient of the cofactors,

$$\begin{aligned} \Delta(s, \chi) &= \frac{\varepsilon(\chi) \Gamma((1-s + \kappa)/2) (q/\pi)^{(1-s+\kappa)/2}}{\Gamma((s + \kappa)/2) (q/\pi)^{(s+\kappa)/2}} \\ &= \varepsilon(\chi) 2^s \pi^{s-1} q^{1/2-s} \Gamma(1-s) \sin \frac{\pi}{2} (s + \kappa). \end{aligned} \quad (25.31) \quad \boxed{\text{E:DefDelta}(s, \chi)}$$

Thus the asymmetric form of the functional equation (as in Corollary 10.9) asserts that $L(s, \chi) = \Delta(s, \chi) L(1-s, \bar{\chi})$. Since $|\varepsilon(\chi)| = 1$ it follows from (25.3) that

$$|\Delta(s, \chi)| \asymp (q\tau)^{1/2-\sigma} \quad (25.32) \quad \boxed{\text{E:Delta}(s, \chi)\text{Est}}$$

uniformly for $-A \leq \sigma \leq A$, $|t| \geq 1$. On dividing both sides of (25.27) by $\Gamma((s + \kappa)/2) (q/\pi)^{(s+\kappa)/2}$, we find that

$$\begin{aligned} L(s, \chi) &= \sum_{n=1}^{\infty} \frac{\chi(n)}{n^s} \cdot \frac{\Gamma((s + \kappa)/2, \pi n^2 z / q)}{\Gamma((s + \kappa)/2)} \\ &+ \Delta(s, \chi) \sum_{n=1}^{\infty} \frac{\bar{\chi}(n)}{n^{1-s}} \cdot \frac{\Gamma((1-s + \kappa)/2, \pi n^2 / (qz))}{\Gamma((1-s + \kappa)/2)}. \end{aligned} \quad (25.33) \quad \boxed{\text{E:LfcnFE2}}$$

If we take $z = re^{i\phi}$ with ϕ given by (25.4), then the above reads

$$\begin{aligned} L(s, \chi) &= \sum_{n=1}^{\infty} \frac{\chi(n)}{n^s} w_0((s + \kappa)/2, \pi n^2 r / q) \\ &+ \Delta(s, \chi) \sum_{n=1}^{\infty} \frac{\bar{\chi}(n)}{n^{1-s}} w_0((1-s + \kappa)/2, \pi n^2 / (rq)). \end{aligned} \quad (25.34) \quad \boxed{\text{E:afew0}}$$

Here the factor $e^{-i\phi}$ that arises in the second sum is appropriate, since the imaginary part of $1-s$ is the negative of that of s . We divide both sides of this by r , integrate from $1/2$ to 2 , and divide by $\log 4$ to find

that

$$\begin{aligned}
 L(s, \chi) &= \sum_{n=1}^{\infty} \frac{\chi(n)}{n^s} w_1((s + \kappa)/2, \pi n^2/q) \\
 &+ \Delta(s, \chi) \sum_{n=1}^{\infty} \frac{\bar{\chi}(n)}{n^{1-s}} w_1((1 - s + \kappa)/2, \pi n^2/q).
 \end{aligned} \tag{25.35}$$

In calculating the second term we have used the fact that

$$\int_{1/2}^2 f(c/r) \frac{dr}{r} = \int_{1/2}^2 f(cr) \frac{dr}{r}.$$

Proof The first sum on the right hand side of (25.35) is

$$\int_0^{\infty} w_1((s + \kappa)/2, \pi u^2/q) dA(s, \chi; u). \tag{25.36}$$

When we integrate this by parts we must consider the size of

$$w_1((s + \kappa)/2, \pi u^2/q) A(s, \chi; u)$$

for u near 0 and as $u \rightarrow \infty$. The lower endpoint is easy to treat, since $A(s, \chi, u) = 0$ when $u < 1$. As $u \rightarrow \infty$ we have $A(s, \chi; u) \ll u^A$ for some A . From (25.15) we see that $w_1((s + \kappa)/2, \pi u^2/q) \ll \exp(-cu^2)$ for some $c > 0$. Here A and c may depend on various parameters, but are independent of u . Thus the product of the two bounds tends to 0 as $u \rightarrow \infty$. Hence by integration by parts the expression (25.36) is

$$\begin{aligned}
 & - \int_0^{\infty} A(s, \chi, u) dw_1((s + \kappa)/2, \pi u^2/q) \\
 &= - \int_0^{\infty} A(s, \chi, u) \frac{\partial}{\partial u} w_1((s + \kappa)/2, \pi u^2/q) du \\
 &= \frac{-2\pi}{q} \int_0^{\infty} A(s, \chi, u) u \frac{\partial}{\partial a} w_1((s + \kappa)/2, a) \Big|_{a=\pi u^2/q} du.
 \end{aligned}$$

By Lemma 25.3 this is

$$\ll (q\tau)^{-(\sigma+\kappa)/2} \int_0^{\infty} |A(s, \chi; u)| \exp\left(\frac{-\pi u^2}{2q\tau}\right) u^{\sigma+\kappa-1} du.$$

By considering separately the ranges $0 < u \leq \sqrt{q\tau}$ and $\sqrt{q\tau} \leq u < \infty$ we find that $\exp(-\pi u^2/(2q\tau))(u^2/(q\tau))^{(\sigma+\kappa)/2} \ll \exp(-u^2/(q\tau))(u^2/(q\tau))^{\delta/2}$ uniformly for $u > 0$. Thus the above is

$$\ll \int_0^{\infty} |A(s, \chi; u)| e^{-u^2/(q\tau)} \left(\frac{u^2}{q\tau}\right)^{\delta/2} \frac{du}{u}.$$

□

Suppose that k is a fixed positive integer. Then by Hölder's Inequality,

$$\begin{aligned}
 & \left| \int_0^\infty |A(s, \chi; u)| e^{-u^2/(q\tau)} \left(\frac{u^2}{q\tau} \right)^{\delta/2} \frac{du}{u} \right|^{2k} \\
 & \leq \left(\int_0^\infty |A(s, \chi; u)|^{2k} e^{-u^2/(q\tau)} \left(\frac{u^2}{q\tau} \right)^{\delta/2} \frac{du}{u} \right) \\
 & \quad \times \left(\int_0^\infty e^{-u^2/(q\tau)} \left(\frac{u^2}{q\tau} \right)^{\delta/2} \frac{du}{u} \right)^{2k-1} \\
 & \ll \int_0^\infty |A(s, \chi; u)|^{2k} e^{-u^2/(q\tau)} \left(\frac{u^2}{q\tau} \right)^{\delta/2} \frac{du}{u}. \quad (25.37) \quad \boxed{\text{E:Bndqtau}}
 \end{aligned}$$

If $Q/2 \leq q \leq Q$ and $T/2 \leq |t| \leq T$, then the above is

$$\ll \int_0^\infty |A(s, \chi; u)|^{2k} e^{-u^2/(2QT)} \left(\frac{u^2}{QT} \right)^{\delta/2} \frac{du}{u}. \quad (25.38) \quad \boxed{\text{E:BndQT}}$$

This latter form will be useful in forming moment estimates.

??.1 Exercises

Exer:VarIncGam

1. Suppose that $\sigma \geq \delta > 0$ and that $t > 0$.

- (a) Suppose that $0 \leq \phi < \pi/2$. Show that the variation of $\Gamma(s, ue^{i\phi})$ as u runs from 0 to ∞ is

$$\frac{e^{-\phi t}}{(\cos \phi)^\sigma} \Gamma(\sigma).$$

- (b) Show that the above is minimized by taking $\phi = \arctan t/\sigma$.
(c) Conclude that

$$\text{Var}_{[0, \infty)} \Gamma(s, ue^{i\phi}) \gg \tau^{1/2} |\Gamma(s)|$$

uniformly in all choices of ϕ .

2. Suppose that $c > 0$. Show that (25.35) is a special case of the more

general identity

$$L(s, \chi) = \sum_{n=1}^{\infty} \frac{\chi(n)}{n^s} w_1((s + \kappa)/2, \pi n^2/(cq)) \\ + \Delta(s, \chi) \sum_{n=1}^{\infty} \frac{\bar{\chi}(n)}{n^{1-s}} w_1((1 - s + \kappa)/2, c\pi n^2/q).$$

3. If $\int_0^\infty |f(x)|x^{\sigma-1} dx < \infty$, then we call the function

$$F(s) = \int_0^\infty f(x)x^{s-1} dx$$

the *Mellin transform* of f , and usually one can recover f from F by means of the *inverse Mellin transform*,

$$f(x) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} F(s)x^{-s} ds$$

for suitable values of c . In Chapter 5 we see several useful pairs f, F of this kind. Suppose now that $\int_0^\infty |f(x)|x^{\sigma-1} dx < \infty$, and that $\int_0^\infty |g(x)|x^{\sigma-1} dx < \infty$ and set

$$F(s) = \int_0^\infty f(x)x^{s-1} dx, \quad G(s) = \int_0^\infty g(x)x^{s-1} dx.$$

Define h by the convolution formula

$$h(x) = (f * g)(x) = \int_0^\infty f(y)g(x/y) \frac{dy}{y}.$$

- (a) Show that

$$\int_0^\infty |h(x)|x^{\sigma-1} dx \leq \int_0^\infty |f(x)|x^{\sigma-1} dx \int_0^\infty |g(x)|x^{\sigma-1} dx.$$

- (b) Put

$$H(s) = \int_0^\infty h(x)x^{s-1} dx.$$

Show that $H(s) = F(s)G(s)$.

4. (a) By integrating by parts, or otherwise, show that if $\operatorname{Re} z > 0$ and $\operatorname{Re} s > 0$, then

$$\int_0^\infty \Gamma(z, a)a^{s-1} da = \frac{\Gamma(s+z)}{s}.$$

(b) Suppose that $\operatorname{Re} z > 0$ and that $\operatorname{Re} s > 0$. For $c > 0$ let

$$f(z, a) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \frac{\Gamma(s+z)}{s} a^{-s} ds.$$

(i) By using Theorem C.4, or otherwise, show that $\frac{\partial}{\partial a} f(z, a) = -e^{-a} a^z$.

(ii) Show that $\frac{\partial}{\partial a} \Gamma(z, a) = -e^{-a} a^z$.

(iii) Show that when z is fixed, $\lim_{a \rightarrow \infty} f(z, a) = 0$.

(iv) Show that when z is fixed, $\lim_{a \rightarrow \infty} \Gamma(z, a) = 0$.

(v) (Mellin) Deduce that

$$\frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \frac{\Gamma(s+z)}{s} a^{-s} ds = \Gamma(z, a)$$

for $\operatorname{Re} z > 0$, $\operatorname{Re} a > 0$.

5. For $\operatorname{Re} z > 0$ and w in the slit complex plane $|\arg w| < \pi$, let

$$\gamma(z, w) = \Gamma(z) - \Gamma(z, w) = \int_0^w e^{-u} u^{z-1} du$$

be the complementary incomplete gamma function, and set $g(z, w) = w^{-z} \gamma(z, w) = \int_0^1 e^{-wv} v^{z-1} dv$.

(a) Show that

$$g(z, w) = \frac{e^{-w}}{z} + \frac{w}{z} g(z+1, w).$$

(b) Show that

$$g(z, w) = e^{-w} \sum_{k=0}^K \frac{w^k}{z(z+1) \cdots (z+k)} + \frac{w^{K+1} g(z+K+1, w)}{z(z+1) \cdots (z+K)}.$$

(c) Deduce that

$$g(z, w) = e^{-w} \sum_{k=0}^{\infty} \frac{w^k}{z(z+1) \cdots (z+k)}.$$

(d) Note that

$$\frac{g(z, w)}{\Gamma(z)} = e^{-w} \sum_{k=0}^{\infty} \frac{w^k}{\Gamma(z+k+1)}$$

is an entire function of z , and of w .

25.1 Notes

S: MeanLargeVals Notes

Section 21.1. Concerning majorant inequalities, antecedents of Theorem 26.13 are found in the work of Wiener (unpublished — see Theorem 12.6.12 of Boas (1954), Erdős and Fuchs (1956), Wiener and Wintner (1956), and Halász (1968). Logan (1988) showed that the constant 3 is best possible. For a general discussion of majorant principles, see Shapiro (1975).

Section 21.2. ^{Tem79}Temme (1979) gave a detailed account of the asymptotics of the incomplete gamma function $\Gamma(s, a)$. ^{Rub05}Rubinstein (2005) developed a variety of tools for the computation of various sorts of L -functions.

Section 21.3. Should mention the mean value theorem of Nigel Watt (NW95), which depends on Kloosterman sums. See Heath-Brown's review. Section 21.4. For an account of the Phragmén–Lindelöf Theorem see, for example, Theorem 5.1.9 in ^{BS2A}Simon (2015).

25.2 References

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26

Mean Values of Dirichlet Polynomials

C:MeanVals

26.1 Mean value estimates

S:MVEst

Suppose that $\lambda_1 < \lambda_2 < \cdots < \lambda_N$. Then

$$\begin{aligned} \int_0^T \left| \sum_{n=1}^N a_n e(\lambda_n t) \right|^2 dt &= T \sum_{n=1}^N |a_n|^2 \\ &\quad + \sum_{\substack{1 \leq m, n \leq N \\ m \neq n}} a_m \overline{a_n} \frac{e((\lambda_m - \lambda_n)T) - 1}{2\pi i(\lambda_m - \lambda_n)}. \end{aligned} \quad (26.1) \quad \text{E: Int | S | 2}$$

Here the last term is a bounded function of T , so that

$$\lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T \left| \sum_{n=1}^N a_n e(\lambda_n t) \right|^2 dt = \sum_{n=1}^N |a_n|^2.$$

This gives us a sort of asymptotic Parseval identity, but we want something more quantitative for a given finite T . We note that if λ_n is very close to λ_{n+1} , then the terms $a_n e(\lambda_n t)$ and $a_{n+1} e(\lambda_{n+1} t)$ will reinforce each other over certain long stretches of t , and then subtract from each other over in other long stretches of t . Thus any bound that is to depend on the numbers $|a_n|^2$ must also involve some information concerning the spacing of the numbers λ_n .

T: MV1

Theorem 26.1. *Suppose that $\lambda_1, \lambda_2, \dots, \lambda_N$ are distinct real numbers, and that $\delta > 0$ has the property that $|\lambda_m - \lambda_n| \geq \delta$ for $m \neq n$. For any real $T \geq 0$, any positive integer N , and any real or complex numbers a_n there is a real number θ , $-1 \leq \theta \leq 1$, such that*

$$\int_0^T \left| \sum_{n=1}^N a_n e(\lambda_n t) \right|^2 dt = (T + \theta/\delta) \sum_{n=1}^N |a_n|^2.$$

By replacing a_n by $a_n e(\lambda_n T_0)$, we find that

$$\int_{T_0}^{T_0+T} \left| \sum_{n=1}^N a_n e(\lambda_n t) \right|^2 dt = (T + \theta/\delta) \sum_{n=1}^N |a_n|^2$$

for some θ . Thus our estimate depends on the length of the interval of integration, but not on its position.

First Proof Let S_+ and S_- be functions with the properties described in Theorem ??, with $I = [0, T]$. Then

$$\int_0^T \left| \sum_{n=1}^N a_n e(\lambda_n t) \right|^2 dt \leq \int_{\mathbb{R}} \left| \sum_{n=1}^N a_n e(\lambda_n t) \right|^2 S_+(t) dt$$

by Theorem ??(b). On expanding the right hand side and integrating term by term, we see that the above is

$$= \sum_{m,n} a_m \overline{a_n} \widehat{S}_+(\lambda_n - \lambda_m) \leq (T + 1/\delta) \sum_{n=1}^N |a_n|^2$$

by Theorem ??(c),(d). Similarly,

$$\begin{aligned} \int_0^T \left| \sum_{n=1}^N a_n e(\lambda_n t) \right|^2 dt &\geq \int_{\mathbb{R}} \left| \sum_{n=1}^N a_n e(\lambda_n t) \right|^2 S_-(t) dt \\ &= \sum_{m,n} a_m \overline{a_n} \widehat{S}_-(\lambda_n - \lambda_m) \geq (T - 1/\delta) \sum_{n=1}^N |a_n|^2. \end{aligned}$$

□

Second Proof We estimate the contribution of the non-diagonal terms on the right hand side of (26.1) via two applications of Hilbert's inequality in the form of Theorem ??. Specifically, by taking $y_m = a_m e(\lambda_m T)$ and $x_n = \overline{a_n} e(-\lambda_n T)$ we deduce that

$$\left| \sum_{\substack{1 \leq m, n \leq N \\ m \neq n}} a_m \overline{a_n} \frac{e((\lambda_m - \lambda_n)T)}{2\pi i(\lambda_m - \lambda_n)} \right| \leq \frac{1}{2\delta} \sum_{n=1}^N |a_n|^2.$$

Similarly,

$$\left| \sum_{\substack{1 \leq m, n \leq N \\ m \neq n}} \frac{a_m \overline{a_n}}{2\pi i(\lambda_m - \lambda_n)} \right| \leq \frac{1}{2\delta} \sum_{n=1}^N |a_n|^2,$$

so the desired estimate now follows by the triangle inequality. □

Cor: MVDirl **Corollary 26.2.** For any positive real T , any positive integer N , and any real or complex numbers $a_1, a_2, \dots, a_N$,

$$\int_0^T \left| \sum_{n=1}^N a_n n^{-it} \right|^2 dt = (T + O(N)) \sum_{n=1}^N |a_n|^2. \quad (26.2) \quad \text{E: MVEst1}$$

Proof We note that

$$\log(n+1) - \log n = \int_n^{n+1} \frac{1}{u} du \geq \frac{1}{n+1}.$$

Thus we can appeal to Theorem 26.1 with $\delta = 1/(2\pi N)$. $\square$

Exam: T+O(N) **Example 26.1.** Put $D(s) = \sum_{n=1}^N n^{-s}$. By integrating by parts we find that

$$D(s) = \frac{N^{1-s} - 1}{1-s} + 1 - s \int_1^N \frac{\{u\}}{u^{s+1}} du.$$

Hence in particular,

$$D(it) = \frac{N^{1-it}}{1-it} + O(\tau \log 2N)$$

where $\tau = |t| + 4$. Thus $|D(it)| \asymp N$ for $0 \leq t \leq 1$, and so $\int_0^1 |D(it)|^2 dt \asymp N^2 = N \sum_{n=1}^N |a_n|^2$ since $a_n = 1$ for all n in the present case. Thus the error term $O(N) \sum_{n=1}^N |a_n|^2$ in (26.2) is best-possible.

The proofs of Theorem 26.1 are short and elegant, due to the substantial material found in Appendices ?? and ??. In some cases of interest, the λ_n are irregularly spaced, so that one λ_n might be closer to its nearest neighbor than another one. In such a situation, we have a more delicate weighted estimate, although with a slightly inferior constant.

T: MV2 **Theorem 26.3.** Let $\lambda_1, \lambda_2, \dots$ be distinct real numbers, and put

$$\delta_n = \min_{\substack{m \\ m \neq n}} |\lambda_m - \lambda_n|.$$

For any real $T \geq 0$ and any real or complex numbers a_n with $\sum_{n=1}^\infty |a_n| < \infty$ there is a real number θ , $-1 \leq \theta \leq 1$, such that

$$\int_0^T \left| \sum_{n=1}^\infty a_n e(\lambda_n t) \right|^2 dt = T \sum_{n=1}^\infty |a_n|^2 + \frac{3}{2} \theta \sum_{n=1}^\infty \frac{|a_n|^2}{\delta_n}.$$

Proof We proceed as in the second proof of Theorem 26.1, but now

appeal to Theorem ?? in place of Theorem ?? to see that for any positive N there is a θ_N such that

$$\int_0^T \left| \sum_{n=1}^N a_n e(\lambda_n t) \right|^2 dt = T \sum_{n=1}^N |a_n|^2 + \frac{3}{2} \theta_N \sum_{n=1}^N \frac{|a_n|^2}{\delta_n}.$$

The stated result then follows by allowing N to tend to infinity. $\square$

Cor:MVDir2

Corollary 26.4. *For any positive real T and any real or complex numbers $a_1, a_2, \dots$ with $\sum_{n=1}^{\infty} |a_n| < \infty$,*

$$\int_0^T \left| \sum_{n=1}^{\infty} a_n n^{-it} \right|^2 dt = T \sum_{n=1}^{\infty} |a_n|^2 + O\left(\sum_{n=1}^{\infty} n |a_n|^2\right).$$

Proof This follows from Theorem 26.3 with $\delta_n = 1/(2\pi(n+1))$. $\square$

The following extension of the above is occasionally useful.

Cor:MVDir3

Corollary 26.5. *Suppose that $T > 0$, and that $a_1, a_2, \dots$ and $b_1, b_2, \dots$ are sequences with $\sum_{n=1}^{\infty} |a_n| < \infty$ and $\sum_{n=1}^{\infty} |b_n| < \infty$, and $a_1 b_1 = 0$. Then*

$$\int_0^T \left| \sum_{n=1}^{\infty} (a_n n^{-it} + b_n n^{it}) \right|^2 dt = \sum_{n=1}^{\infty} (|a_n|^2 + |b_n|^2) (T + O(n)).$$

Proof This follows from Theorem 26.3 in the same way as Corollary 26.4, but now some of the frequencies are of the form $\frac{1}{2\pi} \log n$ while others are of the form $-\frac{1}{2\pi} \log n$. $\square$

In Corollary 26.4 it would typically be the case that the a_n are nonzero for most n . However, when the a_n are nonzero only for n in a sparse set, we can obtain a slightly better estimate, as follows.

Cor:MVDir4

Corollary 26.6. *Suppose that $a_1, a_2, \dots$ are real or complex numbers such that $\sum_{n=1}^{\infty} |a_n| < \infty$. and let d_n be an integer such that $d_n \leq n/2$ and with the property that $a_m = 0$ whenever $0 < |m - n| < d_n$. Then for any $T > 0$,*

$$\int_0^T \left| \sum_{n=1}^{\infty} a_n n^{-it} \right|^2 dt = T \sum_{n=1}^{\infty} |a_n|^2 + O\left(\sum_{n=1}^{\infty} \frac{n |a_n|^2}{d_n}\right).$$

This extension of Corollary 26.4 can be applied to either or both of the sums in Corollary 26.5.

Proof It suffices to apply Theorem 26.3 with $\delta_n = d_n/(4\pi n)$. $\square$

Primes and primepowers are somewhat sparse, but only irregularly so. When considering a sum over primepowers whose coefficients (on the primepowers) are slowly varying, the following estimate is useful.

T:primepowergaps

Theorem 26.7. *Let Q denote the set of primepowers. For $n \in Q$, let d_n be the minimum of $|m - n|$ for $m \in Q$, $m \neq n$. Then for $U \geq 4$,*

$$\sum_{\substack{n \in Q \\ U \leq n \leq 2U}} \frac{1}{d_n} \ll \frac{U \log \log U}{(\log U)^2}.$$

Since $d_n \geq 1$ for all n , it is trivial that the above sum is $\ll U/\log U$. If it were the case that $d_n \gg \log n$ for all $n \in Q$, then the sum would be $\asymp U/(\log U)^2$. If twin primes occur with the conjectured frequency, then the above estimate is best possible.

Proof For the purposes of this proof, let $\mathcal{P}$ denote the set of primes, and Q^* the set of proper primepowers, which is to say the set of numbers of the form p^k with $k > 1$. Thus Q is a disjoint union of $\mathcal{P}$ and Q^* . For primes p set $d_0(p)$ be the minimum of $|p - p'|$ for $p' \in \mathcal{P}$, $p' \neq p$. We show first that

$$\sum_{U \leq p \leq 2U} \frac{1}{d_0(p)} \ll \frac{U \log \log U}{(\log U)^2}. \quad (26.3) \quad \text{E:d0(p) sum}$$

The number of summands in the above sum is $\ll U/\log U$, so the contribution made by those primes for which $d_0(p) > \log U$ is $\ll U/(\log U)^2$. The contributions to the above sum made by primes for which $d_0(p) \leq \log U$ is

$$\ll \sum_{r \leq \log U} \frac{1}{r} \#\{p \in [U, 2U] : p + r \in \mathcal{P}\}.$$

By the sieve estimate of Corollary 3.14 the above is

$$\ll \sum_{r \leq \log U} \frac{c(r)U}{(\log U)^2}$$

where $c(r) = \prod_{p|r, p>2} \left(\frac{p-1}{p-2}\right)$. From (2.32) it follows that $\sum_{r \leq R} c(r) \ll R$, so (26.3) follows by partial summation.

For all primepowers $n \in Q$ we define $d_1(n)$ as follows. If $n \in Q^*$, then set $d_1(n) = 1$; let p_1 denote the largest prime $< n$, and p_2 denote the least prime $> n$, and set $d_1(p_1) = d_1(p_2) = 1$. The number of $n \in Q^*$ such that $U \leq n \leq 2U$ is $\ll U^{1/2}/\log U$. It p is a prime such that the

greatest primepower $< p$ is a prime, and the least primepower $> p$ is also a prime, then we set $d_1(p) = d_0(p)$. Thus

$$\sum_{\substack{n \in Q \\ U \leq n \leq 2U}} \frac{1}{d_1(n)} = \sum_{U \leq p \leq 2U} \frac{1}{d_0(p)} + O\left(\frac{U^{1/2}}{\log U}\right) \ll \frac{U \log \log U}{(\log U)^2}.$$

Since $d_1(n) \leq d_n$ for all primepowers n , the desired bound follows. $\square$

In Chapter ?? we found that it is fruitful to estimate the mean square of a trigonometric polynomial at well-spaced points in terms of the continuous mean square. Similarly, it is useful to estimate the mean square of a Dirichlet polynomial at well-spaced points.

T:DiscreteDP1

Theorem 26.8. *Let*

$$D(s) = \sum_{n=1}^N a_n n^{-s}. \quad (26.4) \quad \text{E:DefD}$$

Suppose that $0 < \delta \leq 1$, $T \geq 1$, and that $t_1, t_2, \dots, t_R$ are real numbers such that $A \leq t_1 < t_2 < \dots < t_R \leq A + T$ and $t_{r+1} - t_r \geq \delta$ for $r = 1, 2, \dots, R-1$. If $D(s)$ is defined as in (26.4), then

$$\sum_{r=1}^R |D(it_r)|^2 \ll (\log N + 1/\delta)(T + N) \sum_{n=1}^N |a_n|^2$$

uniformly for any complex numbers a_n and any real number A .

Proof Let $\mathfrak{M}_r = [t_r - \delta/2, t_r + \delta/2]$ for $r = 1, 2, \dots, R$. By Lemma ?? it follows that

$$|D(it_r)|^2 \leq \frac{1}{\delta} \int_{\mathfrak{M}_r} |D(it)|^2 dt + \int_{\mathfrak{M}_r} |D(it)D'(it)| dt.$$

The intervals $\mathfrak{M}_r$ are disjoint and all lie in the interval $[A - \delta, A + T + \delta]$, so

$$\sum_{r=1}^R |D(it_r)|^2 \leq \frac{1}{\delta} \int_{A-\delta}^{A+T+\delta} |D(it)|^2 dt + \int_{A-\delta}^{A+T+\delta} |D(it)D'(it)| dt.$$

By the Cauchy–Schwarz inequality the above is

$$\begin{aligned} &\leq \frac{1}{\delta} \int_{A-\delta}^{A+T+\delta} |D(it)|^2 dt \\ &\quad + \left(\int_{A-\delta}^{A+T+\delta} |D(it)|^2 dt \right)^{1/2} \left(\int_{A-\delta}^{A+T+\delta} |D'(it)|^2 dt \right)^{1/2}. \end{aligned}$$

By Corollary 26.2 it follows that this is

$$\ll \frac{T+N}{\delta} \sum_{n=1}^N |a_n|^2 + (T+N) \left(\sum_{n=1}^N |a_n|^2 \right)^{1/2} \left(\sum_{n=1}^N |a_n \log n|^2 \right)^{1/2}.$$

This gives the stated result. $\square$

We now extend the above to allow points $s_r = \sigma_r + it_r$ whose abscissæ are not all equal.

T:DiscreteDP2

Theorem 26.9. *Let $D(s)$ be defined as in (26.4), let A be a real number, and suppose that $T \geq 2$. Let $s_1, s_2, \dots, s_R$ be distinct complex numbers in the half-strip $\sigma_r \geq 0$, $A \leq t_r \leq A+T$, and suppose that $\delta \leq 1$ has been chosen so that $|t_{r_1} - t_{r_2}| \geq \delta$ whenever $r_1 \neq r_2$. Then*

$$\sum_{r=1}^R |D(s_r)|^2 \ll (\log N + 1/\delta)(T+N) \sum_{n=1}^N |a_n|^2 \left(1 + \log \frac{\log 2N}{\log 2n} \right).$$

Proof Let

$$S(s; u) = \sum_{2 \leq n \leq N} a_n n^{-s},$$

so that if $\sigma \geq 0$, then

$$D(s) = a_1(1 - N^{-\sigma}) + D(it)N^{-\sigma} + \sigma \int_2^N S(it; u) u^{-\sigma-1} du.$$

Hence

$$|D(s)|^2 \ll |a_1|^2 + |D(it)|^2 + \left(\sigma \int_2^N |S(it; u)| u^{-\sigma-1} du \right)^2.$$

By the Cauchy–Schwarz inequality the last term above is

$$\leq \left(\sigma^2 \int_2^N \frac{\log u}{u^{2\sigma+1}} du \right) \left(\int_2^N \frac{|S(it, u)|^2}{u \log u} du \right).$$

By integrating by parts twice we find that

$$\sigma^2 \int_1^N \frac{\log u}{u^{2\sigma+1}} du = \frac{1}{4} - \frac{1}{4N^{2\sigma}} - \frac{\sigma}{2} \log N \leq \frac{1}{4}.$$

Thus

$$|D(s)|^2 \ll |a_1|^2 + |D(it)|^2 + \int_2^N \frac{|S(it; u)|^2}{u \log u} du \quad (26.5) \quad \text{E:D(sigma)est}$$

uniformly for $\sigma \geq 0$. When we take $s = s_r$ and sum over r , the contribution of $|a_1|^2$ is $\ll |a_1|^2 R \ll |a_1|^2 T/\delta$. The contribution of $|D(it_r)|^2$ is exactly as it was in Theorem 26.8, and

$$\sum_{r=1}^R \int_2^N \frac{|S(it; u)|^2}{u \log u} du \ll (\log N + 1/\delta)(T + N) \int_2^N \sum_{2 \leq n \leq u} |a_n|^2 (u \log u)^{-1} du.$$

Here the integral on the right is

$$= \sum_{n=2}^N |a_n|^2 \int_n^N \frac{du}{u \log u} = \sum_{n=2}^N |a_n|^2 \log \frac{\log N}{\log n}.$$

This gives the result. $\square$

When $N > T$ we do not obtain good bounds for the mean square in terms of $\sum_{n=1}^N |a_n|^2$, but we now show that quite reasonable upper bounds can be obtained in terms of short sums of the coefficients.

T:gallagher1

Theorem 26.10 (Gallagher). *Let $\mathcal{L}$ be a countable set of real numbers and for $\lambda \in \mathcal{L}$ let $a(\lambda)$ be complex numbers such that $\sum_{\lambda \in \mathcal{L}} |a(\lambda)| < \infty$. For real t and $\delta > 0$ let*

$$S(t) = \sum_{\lambda \in \mathcal{L}} a(\lambda) e(\lambda t), \quad A_\delta(x) = \delta^{-1} \sum_{\substack{\lambda \in \mathcal{L} \\ |\lambda - x| < \delta/2}} a(\lambda).$$

Then

$$\int_{-\infty}^{\infty} |S(t)|^2 \left(\frac{\sin \pi \delta t}{\pi \delta t} \right)^2 dt = \int_{-\infty}^{\infty} |A_\delta(x)|^2 dx.$$

Proof Let

$$F_\delta(x) = \begin{cases} \delta^{-1} & \text{when } |x| < \delta/2, \\ 0 & \text{when } |x| \geq \delta/2. \end{cases}$$

Then

$$A_\delta(x) = \sum_{\lambda} a(\lambda) F_\delta(x - \lambda).$$

We note that $A_\delta \in L^1(\mathbb{R})$, since by the triangle inequality

$$\|A_\delta\|_{L^1(\mathbb{R})} \leq \sum_{\lambda} \|a(\lambda) F_\delta(x - \lambda)\|_{L^1(\mathbb{R})} = \sum_{\lambda} |a(\lambda)| < \infty.$$

Let $\widehat{A}_\delta(t)$ denote the Fourier transform of $A_\delta(x)$, i.e.,

$$\widehat{A}_\delta(t) = \int_{-\infty}^{\infty} A_\delta(x) e(-xt) dx.$$

Then

$$\begin{aligned}\widehat{A}_\delta(t) &= \sum_{\lambda \in \mathcal{L}} a(\lambda) \int_{-\infty}^{\infty} F_\delta(x - \lambda) e(-xt) dx \\ &= \sum_{\lambda \in \mathcal{L}} a(\lambda) e(-\lambda t) \widehat{F}_\delta(t) = S(-t) \widehat{F}_\delta(t).\end{aligned}$$

Next we show that $A_\delta \in L^2(\mathbb{R})$. To see why this is so, we first note that

$$|A_\delta(x)|^2 = \sum_{\substack{\lambda_i \in \mathcal{L} \\ |x - \lambda_i| \leq \delta/2}} a(\lambda_1) \overline{a(\lambda_2)} \leq \sum_{\substack{\lambda_i \in \mathcal{L} \\ |x - \lambda_i| \leq \delta/2}} |a(\lambda_1)| |a(\lambda_2)|.$$

If $|\lambda_1 - \lambda_2| > \delta$, then there is no x for which both inequalities $|x - \lambda_1| \leq \delta/2$, $|x - \lambda_2| \leq \delta/2$ are satisfied. If $|\lambda_1 - \lambda_2| \leq \delta$, then the set of x for which $|x - \lambda_1| \leq \delta/2$ and $|x - \lambda_2| \leq \delta/2$ is an interval of length $\delta - |\lambda_1 - \lambda_2|$. Thus

$$\begin{aligned}\int_{-\infty}^{\infty} |A_\delta(x)|^2 dx &\leq \sum_{\lambda_1, \lambda_2 \in \mathcal{L}} \max(0, \delta - |\lambda_1 - \lambda_2|) |a(\lambda_1)| |a(\lambda_2)| \\ &\leq \delta \left(\sum_{\lambda \in \mathcal{L}} |a(\lambda)| \right)^2 < \infty.\end{aligned}$$

By Plancherel's identity for functions in $L^1(\mathbb{R}) \cap L^2(\mathbb{R})$, it follows that

$$\int_{-\infty}^{\infty} |A_\delta(x)|^2 dx = \int_{-\infty}^{\infty} |\widehat{A}_\delta(t)|^2 dt = \int_{-\infty}^{\infty} |S(-t) \widehat{F}_\delta(t)|^2 dt.$$

Since

$$\widehat{F}_\delta(t) = \frac{\sin \pi \delta t}{\pi \delta t}$$

we have the stated result. $\square$

When the above is applied to ordinary Dirichlet series, this gives

Cor:gallagher2

Corollary 26.11 (Gallagher). *Suppose that $\sum_{n=1}^{\infty} |a_n| < \infty$, that $\delta > 0$, and that $\kappa = e^{2\pi\delta}$. Then*

$$\int_{-\infty}^{\infty} \left| \sum_{n=1}^{\infty} \frac{a_n}{n^{it}} \right|^2 \left(\frac{\sin \pi \delta t}{\pi \delta t} \right)^2 dt = \frac{1}{2\pi\delta^2} \int_0^{\infty} \left| \sum_{\substack{n=1 \\ y < n < \kappa y}}^{\infty} a_n \right|^2 \frac{dy}{y}. \quad (26.6) \quad \text{E:GallEst2}$$

Proof Take $\lambda_n = \frac{\log n}{2\pi}$ in Theorem 26.10, and set $y = e^{2\pi x - \pi\delta}$. Thus $dx = \frac{dy}{2\pi dy}$. $\square$

Cor:gallagher3

Corollary 26.12 (Gallagher). Suppose that $\sum_{n=1}^{\infty} |a_n| < \infty$, and let $\kappa = \exp(T^{-1})$ where $T \geq 1$. Then

$$\int_{-T}^T \left| \sum_{n=1}^{\infty} \frac{a_n}{n^{it}} \right|^2 dt \ll T^2 \int_0^{\infty} \left| \sum_{y < n \leq \kappa y} a_n \right|^2 \frac{dy}{y}. \quad (26.7) \quad \text{E:GallEst3}$$

The sum on the right hand side above is empty when $y < 1/\kappa$, so in effect the range of integration is $[1/\kappa, \infty)$.

Proof We take $\delta = 1/(2\pi T)$ in Corollary 26.11. Put $g(u) = \frac{\sin u}{u}$ for $u \neq 0$, and $g(0) = 1$. Then $|g(u)| \gg 1$ uniformly for $0 \leq u \leq 1/2$, so the left hand side of (26.7) is majorized by the left hand side of (26.6). $\square$

In addition to the mean value estimates we have already considered, the following *majorant principle* is sometimes useful.

T:majorant

Theorem 26.13. Suppose that $\lambda_1, \lambda_2, \dots$ are real numbers, and that real or complex numbers a_j and A_j have the property that $|a_j| \leq A_j$ for all j . Suppose also that $\sum_{j=1}^{\infty} A_j < \infty$. Then for any $T > 0$,

$$\int_{-T}^T \left| \sum_{j=1}^{\infty} a_j e(\lambda_j t) \right|^2 dt \leq 3 \int_{-T}^T \left| \sum_{j=1}^{\infty} A_j e(\lambda_j t) \right|^2 dt. \quad (26.8) \quad \text{E:majorant0}$$

Proof For $T > 0$ let

$$K_T(t) = \max(0, 1 - |t|/T). \quad (26.9) \quad \text{E:DefK_T}$$

We show first that

$$\int_{-T}^T K_T(t) \left| \sum_{j=1}^{\infty} a_j e(\lambda_j t) \right|^2 dt \leq \int_{-T}^T K_T(t) \left| \sum_{j=1}^{\infty} A_j e(\lambda_j t) \right|^2 dt. \quad (26.10) \quad \text{E:wtdmajorant1}$$

To see why this is so, and observe that by integration by parts it is immediate that

$$\widehat{K}_T(\alpha) = \int_{-\infty}^{\infty} K_T(t) e(-t\alpha) dt = \frac{1}{T} \left(\frac{\sin \pi T \alpha}{\pi \alpha} \right)^2 \geq 0.$$

By multiplying out the modulus-squared we see that the left hand side of (26.10) is

$$= \sum_{j,k} a_j \overline{a_k} \int_{-\infty}^{\infty} K_T(t) e((\lambda_j - \lambda_k)t) dt = \sum_{j,k} a_j \overline{a_k} K_T(\lambda_k - \lambda_j).$$

Since $\widehat{K}_T(\alpha) \geq 0$ for all α , the above is

$$\leq \sum_{j,k} A_j A_k \widehat{K}_T(\lambda_k - \lambda_j) = \int_{-\infty}^{\infty} K_T(t) \left| \sum_{j=1}^{\infty} A_j e(\lambda_j t) \right|^2 dt.$$

Thus we have (26.10). We note that if we apply (26.10) with a_j replaced by $a_j e(\lambda_j U)$, then we have

$$\begin{aligned} \int_{-\infty}^{\infty} K_T(t - U) \left| \sum_{j=1}^{\infty} a_j e(\lambda_j t) \right|^2 dt \\ \leq \int_{-\infty}^{\infty} K_T(t) \left| \sum_{j=1}^{\infty} A_j e(\lambda_j t) \right|^2 dt \end{aligned} \quad (26.11) \quad \boxed{\text{E:wtdmajorant2}}$$

for any real number U . We note that the function

$$w(t) = K_T(t + T) + K_T(t) + K_T(t - T)$$

majorizes the characteristic function of the interval $[-T, T]$. Thus by three applications of (26.11), with $U = -T$, $U = 0$, and $U = T$, we find that

$$\int_{-T}^T \left| \sum_{j=1}^{\infty} a_j e(\lambda_j t) \right|^2 dt \leq 3 \int_{-T}^T K_T(t) \left| \sum_{j=1}^{\infty} A_j e(\lambda_j t) \right|^2 dt,$$

and (26.8) follows from this. $\square$

To see how we might use Theorem 26.13, suppose that we have two Dirichlet series,

$$\alpha(s) = \sum_{n=1}^{\infty} a_n n^{-s}, \quad A(s) = \sum_{n=1}^{\infty} A_n n^{-s} \quad (26.12) \quad \boxed{\text{E:DefDSs}}$$

with $|a_n| \leq A_n$ for all n and suppose that σ is chosen so that $\sum_{n=1}^{\infty} A_n n^{-\sigma} < \infty$. Then

$$\int_{-T}^T |\alpha(\sigma + it)|^2 dt \leq 3 \int_{-T}^T |A(\sigma + it)|^2 dt.$$

Specializing this still further, we see in particular that if $|a_n| \leq 1$ for all n , and $1 < \sigma \leq 2$, then

$$\int_{-1}^1 |\alpha(\sigma + it)|^2 dt \leq 3 \int_{-1}^1 |\zeta(\sigma + it)|^2 dt \asymp \frac{1}{\sigma - 1}. \quad (26.13) \quad \boxed{\text{E:SpecMaj}}$$

26.1.1 Exercises

1. (Goldston 1981) Let $S(t) = \sum_{\mu \in \mathcal{M}} c(\mu) e(\mu t)$ where $\mathcal{M}$ is a countable set of real numbers and $\sum_{\mu \in \mathcal{M}} |c(\mu)| < \infty$. Suppose that $T > 0$, $\delta > 0$, and take $I = [0, T]$ in Theorem E.2.

(a) Explain why

$$\int_0^T |S(t)|^2 dt \leq \int_{-\infty}^{\infty} S_+(t) |S(t)|^2 dt.$$

(b) Deduce that

$$\begin{aligned} \int_0^T |S(t)|^2 dt &\leq \left(T + \frac{1}{\delta}\right) \sum_{\mu \in \mathcal{M}} |c(\mu)|^2 \\ &\quad + \left(\max_t |\widehat{S}_+(t)|\right) \sum_{\substack{\mu, \nu \in \mathcal{M} \\ |\mu - \nu| < \delta}} |c(\mu)c(\nu)|. \end{aligned}$$

(c) Show similarly that

$$\begin{aligned} \int_0^T |S(t)|^2 dt &\geq \left(T + \frac{1}{\delta}\right) \sum_{\mu \in \mathcal{M}} |c(\mu)|^2 \\ &\quad - \left(\max_t |\widehat{S}_-(t)|\right) \sum_{\substack{\mu, \nu \in \mathcal{M} \\ |\mu - \nu| < \delta}} |c(\mu)c(\nu)|. \end{aligned}$$

(d) Write $S_{\pm}(x) = \chi_I(x) + (S_{\pm}(x) - \chi_I(x))$, and deduce that

$$|\widehat{S}_{\pm}(t)| \leq \|S_{\pm}\|_{L^1(\mathbb{R})} \leq \|\chi_I\|_{L^1(\mathbb{R})} + \|S_{\pm} - \chi_I\|_{L^1(\mathbb{R})} = T + \frac{1}{\delta}.$$

(e) Conclude that there is a number θ , $|\theta| \leq 1$, such that

$$\int_0^T |S(t)|^2 dt = \left(T + \frac{\theta}{\delta}\right) \left(\sum_{\mu \in \mathcal{M}} |c(\mu)|^2 + \sum_{\substack{\mu, \nu \in \mathcal{M} \\ |\mu - \nu| < \delta}} |c(\mu)c(\nu)| \right).$$

2. Let Q and d_n be defined as in Theorem 26.7.

(a) Show that

$$\sum_{\substack{n \in Q \\ U \leq n \leq 2U}} d_n \ll U.$$

(b) By Cauchy's inequality, or otherwise, deduce that

$$\sum_{\substack{U \leq n \leq 2U \\ n \in Q}} \frac{1}{d_n} \gg \frac{U}{(\log U)^2}.$$

3. Suppose that f is analytic in a domain that contains the closed disc $|z| \leq R$.

(a) Show that if $0 < r \leq R$, then

$$f(0) = \frac{1}{2\pi} \int_0^{2\pi} f(re^{i\theta}) d\theta.$$

(b) Deduce that

$$\frac{1}{2}R^2 f(0) = \frac{1}{2\pi} \int_0^R \int_0^{2\pi} f(re^{i\theta}) r dr d\theta.$$

(c) Deduce that

$$f(0) = \frac{1}{\pi R^2} \iint_{\mathcal{D}} f(\sigma + it) d\sigma dt$$

where $\mathcal{D} = \{s := \sigma^2 + t^2 \leq R^2\}$.

(d) Deduce that

$$|f(0)| \leq \frac{1}{\pi R^2} \iint_{\mathcal{D}} |f(s)| d\sigma dt.$$

- (e) Suppose that $D(s)$ is defined as in (26.4), that the complex numbers $s_1, s_2, \dots, s_R$ lie in the rectangle $0 \leq \sigma_r \leq 1/\log N$, $A \leq t_r \leq A + T$, and that $|s_{r_1} - s_{r_2}| \geq 1/\log N$ whenever $r_1 \neq r_2$. Show that

$$\sum_{r=1}^R |D(s_r)|^2 \ll (T + N)(\log N) \sum_{n=1}^N |a_n|^2.$$

- (f) Suppose that $D(s)$ and the s_r are defined as above, except that now the s_r lie in the rectangle $0 \leq \sigma_r \leq 1$, $A \leq t_r \leq A + T$. Show that

$$\sum_{r=1}^R |D(s_r)|^2 \ll (T + N)(\log N)^2 \sum_{n=1}^N \frac{|a_n|^2}{\log(n+1)}.$$

4. ^{JBC83}
(Conrey, 1983)

- (a) Suppose that f is a bounded function of bounded variation on the interval $[a, b]$. Show that

$$i(\log y) \int_a^b f(t) y^{it} dt = \left[f(t) y^{it} \right]_a^b - \int_a^b y^{it} df(t)$$

for any real number y . (Material in Appendix A may be useful here.)

- (b) With f as above, show that

$$(\log y) \int_a^b f(t) y^{it} dt \ll \sup_{a \leq t \leq b} |f(t)| + \text{var}_{[a,b]} f.$$

- (c) Suppose that functions f and g are bounded and of bounded variation on an interval $[a, b]$. Show that

$$\text{var}_{[a,b]} fg \leq \sup_{a \leq t \leq b} |f(t)| \text{var}_{[a,b]} g + \text{var}_{[a,b]} f \sup_{a \leq t \leq b} |g(t)|.$$

- (d) Show that if m and n are positive integers with $m < n \leq 2m$, then $\log \frac{n}{m} \asymp \frac{n-m}{m}$.
- (e) Show that if m and n are positive integers with $n > 2m$, then $\log \frac{n}{m} \gg 1$.
- (f) Deduce that if m and n are positive integers with $m < n$, then $\log \frac{n}{m} \gg \min(1, \frac{n-m}{m})$.
- (g) Show that if N is a positive integer and $1 \leq m \leq N$, then

$$\sum_{\substack{1 \leq n \leq N \\ n \neq m}} \frac{1}{|\log n/m|} \ll N \log N.$$

- (h) By means of Theorem ??, or otherwise, show that

$$\sum_{\substack{1 \leq m, n \leq N \\ m \neq n}} \frac{a_m a_n}{|\log n/m|} \ll N(\log N) \sum_{n=1}^N |a_n|^2$$

uniformly for arbitrary real or complex numbers a_n .

- (i) Let $f_1, f_2, \dots, f_N$ be functions defined on an interval $I = [A, A+T]$, and suppose that C is a number such that $\sup_I |f_n| \leq C$ and $\text{var}_I f_n \leq C$ for all n . Note that

$$\begin{aligned} \int_I \left| \sum_{n=1}^N a_n f_n(t) n^{-it} \right|^2 dt &= \sum_{n=1}^N |a_n|^2 \int_I |f_n(t)|^2 dt \\ &\quad + \sum_{\substack{1 \leq m, n \leq N \\ m \neq n}} a_m \overline{a_n} \int_I f_m(t) \overline{f_n(t)} \left(\frac{n}{m} \right)^{it} dt. \end{aligned}$$

For given m and n with $m \neq n$, put $f(t) = f_m(t) \overline{f_n(t)}$. Show that $\sup_I |f(t)| \leq C^2$ and that $\text{var}_I f \leq C^2$.

(j) Show that in the above situation,

$$\begin{aligned} \int_I \left| \sum_{n=1}^N a_n f_n(t) n^{-it} \right|^2 dt &= \sum_{n=1}^N |a_n|^2 \int_I |f_n(t)|^2 dt \\ &\quad + O(N \log N) \sum_{n=1}^N |a_n|^2. \end{aligned}$$

This has many uses. For example, if $N(t)$ is an increasing function with $N(T) = N$, then the integral

$$\int_A^{A+T} \left| \sum_{n \leq N(t)} a_n n^{-it} \right|^2 dt$$

can be estimated, with an error term that is only one logarithm larger than in Corollary 26.2.

5. Suppose that $\lambda_1, \lambda_2, \dots, \lambda_N$ are distinct real numbers and that $\delta > 0$ has the property that $|\lambda_m - \lambda_n| \geq \delta$ whenever $m \neq n$. Put

$$A(t) = \sum_{n=1}^N a_n e(\lambda_n t), \quad B(t) = \sum_{n=1}^N b_n e(\lambda_n t).$$

(a) Show that for any $T \geq 0$ there is a number θ with $|\theta| \leq 1$ such that

$$\int_0^T A(t) \overline{B(t)} dt = T \sum_{n=1}^N a_n \overline{b_n} + \frac{\theta}{\delta} \left(\sum_{n=1}^N |a_n|^2 \right)^{1/2} \left(\sum_{n=1}^N |b_n|^2 \right)^{1/2}.$$

(b) Suppose that δ_n is defined as in Theorem 26.3. Show that for any $T \geq 0$ there is a θ with $|\theta| \leq 1$ such that

$$\int_0^T A(t) \overline{B(t)} dt = T \sum_{n=1}^N a_n \overline{b_n} + \frac{3}{2} \theta \left(\sum_{n=1}^N \frac{|a_n|^2}{\delta_n} \right)^{1/2} \left(\sum_{n=1}^N \frac{|b_n|^2}{\delta_n} \right)^{1/2}.$$

6. We consider the situation of Theorem 26.1.

(a) Show that if $\lambda_{n+1} - \lambda_n = \delta$ for all n , and $\delta T = 1$, then

$$\frac{1}{T} \int_0^T \left| \sum_{n=1}^N a_n e(\lambda_n t) \right|^2 dt = \sum_{n=1}^N |a_n|^2.$$

(b) Show that if $\lambda_{n+1} - \lambda_n \geq \delta$ for all n , $\varepsilon > 0$, and $\delta T \geq 1 + \varepsilon$, then

$$\frac{1}{T} \int_0^T \left| \sum_{n=1}^N a_n e(\lambda_n t) \right|^2 dt \geq C(\varepsilon) \sum_{n=1}^N |a_n|^2 \quad (26.14) \quad \boxed{\text{E:MSLB}}$$

where $C(\varepsilon) = \varepsilon/(1 + \varepsilon)$.

7. ^{Ing36} (Ingham, 1936) The object of this exercise is to show that if $\lambda_{n+1} - \lambda_n \geq \delta$ for all n and $\delta T = 1$, then the inequality (26.14) is false when $C(\varepsilon)$ is replaced by any constant $C > 0$. For $|z| < 1$ and $0 < \alpha < 1/2$, write

$$(1 + z)^{-\alpha} = \sum_{m=0}^{\infty} b_m z^m.$$

For $0 < r < 1$ and $\lambda > 0$, put

$$f_r(t) = e(\lambda t/2)(1 + re(t))^{-\alpha} + e(-\lambda t/2)(1 + re(-t))^{-\alpha}.$$

(a) Show that

$$f_r(t) = \sum_{m=0}^{\infty} b_m r^m (e((m + \lambda/2)t) + e(-(m + \lambda/2)t)).$$

(Note that the frequencies in the above are spaced by at least 1 if $\lambda \geq 1$.)

(b) Show that

$$\int_{-1/2}^{1/2} |(1 + re(t))|^{2\alpha} dt = \sum_{m=0}^{\infty} |b_m|^2 r^{2m}.$$

(c) Note that $(1 + z)^{-\alpha} = z^{-\alpha/2}(z^{1/2} + z^{-1/2})^{-\alpha}$, and deduce that

$$\lim_{r \rightarrow 1^-} (1 + re(t))^{-\alpha} = e(-\alpha t/2)(2 \cos \pi t)^{-\alpha}$$

for $-1/2 < t < 1/2$.

(d) Show that $|g(re(t))|^2 < (\cos \pi t)^{-2\alpha}$ when $-1/2 < t < 1/2$ and $0 \leq r < 1$.

(e) Explain how you know that

$$\begin{aligned} \lim_{r \rightarrow 1^-} \int_{-1/2}^{1/2} |g(re(t))|^2 dt &= \int_{-1/2}^{1/2} \lim_{r \rightarrow 1^-} |g(re(t))|^2 dt \\ &= \int_{-1/2}^{1/2} (2 \cos \pi t)^{-2\alpha} dt. \end{aligned}$$

- (f) Show that the right hand side above is

$$\sim \frac{2(2\pi)^{-2\alpha}}{1-2\alpha}$$

as $\alpha \rightarrow (1/2)^-$.

- (g) Show that

$$\begin{aligned} \lim_{r \rightarrow 1^-} \int_{-1/2}^{1/2} |f_r(t)|^2 dt \\ &= \int_{-1/2}^{1/2} \left| e\left(\frac{1}{2}(\lambda - \alpha)t\right) + e\left(\frac{1}{2}(\alpha - \lambda)t\right) (2 \cos \pi t)^{-\alpha} \right|^2 dt \\ &= \int_{-1/2}^{1/2} (2 \cos \pi(\lambda - \alpha)t)^2 (2 \cos \pi t)^{-2\alpha} dt. \end{aligned}$$

- (h) Take
- $\lambda = \alpha + 1$
- . Then the above integral is

$$\int_{-1/2}^{1/2} (2 \cos \pi t)^{2-2\alpha} dt.$$

Explain why this is bounded, and show that it tends to $4/\pi$ as $\alpha \rightarrow (1/2)^-$. Note that the frequencies are the numbers $\pm(m + \frac{\alpha+1}{2})$ for $m = 0, 1, 2, \dots$. Thus the gaps between the frequencies are all 1, except for the gap between $-(\alpha+1)/2$ and $(\alpha+1)/2$, which is nearly $3/2$.

- (i) Explain why the asymptotic mean square of
- f
- is

$$2 \sum_{m=0}^{\infty} b_m^2 = 2^{1-2\alpha} \int_{-1/2}^{1/2} (\cos \pi t)^{-2\alpha} dt.$$

- (j) Theorem 26.1 relates to finite sums, but our construction thus far has involved infinite sums. Explain why

$$\sum_{m=0}^M \left(1 - \frac{m}{M}\right) b_m (e((m + \lambda/2)t) + e(-(m + \lambda/2)t))$$

behaves similarly to the infinite sum, when M is large.

8. In the situation of Theorem 26.8, apart from the ordering of the t_r , the hypothesis as to spacing is equivalent to asserting that no interval $[t, t + \delta)$ contains more than 1 of the t_r . Suppose we weaken this hypothesis by asserting only that no interval $[k\delta, (k+1)\delta)$ contains more than 1 of the t_r . Show that the same conclusion still follows (with a larger implicit constant).

Exer:Sobolev2d

9. Suppose that $f(x_1, x_2)$ has continuous partial derivatives through the second order, and let $f_1 = \frac{\partial f}{\partial x_1}$, $f_2 = \frac{\partial f}{\partial x_2}$, and $f_{12} = \frac{\partial^2 f}{\partial x_1 \partial x_2}$.

(a) Let $U = [0, 1]$. Show that if $(a_1, a_2) \in U^2$, then

$$|f(a_1, a_2)| \leq \int_{U^2} |f(x_1, x_2)| + |f_1(x_1, x_2)| + |f_2(x_1, x_2)| + |f_{12}(x_1, x_2)| dx_1 dx_2.$$

(b) Show that

$$|f(1/2, 1/2)| \leq \int_{U^2} |f(x_1, x_2)| + \frac{1}{2}|f_1(x_1, x_2)| + \frac{1}{2}|f_2(x_1, x_2)| + \frac{1}{4}|f_{12}(x_1, x_2)| dx_1 dx_2.$$

(c) Suppose that $\delta_1 > 0$ and that $\delta_2 > 0$. Show that

$$|f(a_1, a_2)| \leq \int_{a_1 - \frac{1}{2}\delta_1}^{a_1 + \frac{1}{2}\delta_1} \int_{a_2 - \frac{1}{2}\delta_2}^{a_2 + \frac{1}{2}\delta_2} \frac{|f(x_1, x_2)|}{\delta_1 \delta_2} + \frac{|f_1(x_1, x_2)|}{2\delta_2} + \frac{|f_2(x_1, x_2)|}{2\delta_1} + \frac{|f_{12}(x_1, x_2)|}{4} dx_2 dx_1.$$

10. Show that

$$\int_0^T \left| \operatorname{Re} \sum_{n=1}^N a_n n^{-it} \right|^2 = \sum_{n=1}^N |a_n|^2 \left(\frac{1}{2}T + O(n) \right).$$

11. ^{Ing36}(Ingham, 1936) Suppose that $\{\lambda_n\}$ is a sequence of real numbers with $\lambda_{n+1} > \lambda_n$ for all n , and put

$$S(t) = \sum_{-\infty}^{\infty} a_n e(\lambda_n t)$$

where $\sum_n |a_n| < \infty$.

(a) Show that if $\lambda_n = n$ for all n , then

$$\max_n |a_n| \leq \int_{-1/2}^{1/2} |S(t)| dt.$$

Our object, in what follows, is to explore what might be said in this direction for more general well-spaced λ_n .

- (b) Set $K(t) = \cos \pi t$ for $-1/2 \leq t \leq 1/2$ and $K(t) = 0$ for $|t| > 1/2$. Show that

$$\hat{K}(\alpha) = \frac{2 \cos \pi \alpha}{\pi(1 - 4\alpha^2)}.$$

(c) Show that

$$\int_{-\infty}^{\infty} K(t)e(ct)S(t) dt = \sum_{n=-\infty}^{\infty} a_n \hat{K}(c + \lambda_n)$$

for any real number c .

(d) Explain why

$$\left| \int_{-\infty}^{\infty} K(t)e(ct)S(t) dt \right| \leq \int_{-1/2}^{1/2} |S(t)| dt.$$

(e) Let k be an integer such that $|a_k| = \max_n |a_n|$. Show that

$$\left| \sum_n a_n \hat{K}(\lambda_n - \lambda_k) \right| \geq |a_k| \left(\hat{K}(0) - \sum_{\substack{n \\ n \neq k}} |\hat{K}(\lambda_n - \lambda_k)| \right).$$

(f) Suppose that $\varepsilon > 0$ and that $\lambda_{n+1} - \lambda_n \geq 1 + \varepsilon$ for all n . Show that the expression in large parentheses above is

$$\geq \frac{2}{\pi} \left(1 - \frac{1}{2} \sum_{r=1}^{\infty} \frac{1}{((1+\varepsilon)r)^2 - 1/4} \right).$$

(g) Show that the expression in large parentheses above is

$$\geq 1 - \frac{1}{2(1+\varepsilon)^2} \sum_{r=1}^{\infty} \frac{1}{r^2 - 1/4}.$$

(h) Show that the sum over r above is $= 2$.

(i) By choosing a suitable value of c , show that

$$\int_{-1/2}^{1/2} |S(t)| dt \geq C(\varepsilon) \max_n |a_n|$$

where

$$C(\varepsilon) = \frac{2\varepsilon(2+\varepsilon)}{(1+\varepsilon)^2\pi}.$$

(j) Show that $C(\varepsilon) > \varepsilon$ if $0 < \varepsilon \leq 1/6$.

(k) Suppose that $T > 0$, that $\lambda_{n+1} - \lambda_n \geq \delta > 0$ for all n , and that $\delta T \geq 1 + \varepsilon$. Show that

$$\frac{1}{T} \int_{-T/2}^{T/2} |S(t)| dt \geq C(\varepsilon) \max_n |a_n|.$$

- (1) Let δ and T be as above, and let A be an arbitrary real number. Show that

$$\int_A^{A+T} |S(t)| dt \geq C(\varepsilon) \max_n |a_n|.$$

The argument above fails to give a positive lower bound when $\delta T = 1$. One might wonder whether a better result could be obtained by the above method, if the kernel K were replaced by a different kernel. Note that by the Poisson summation formula,

$$\sum_n K(n + 1/2) = \sum_k (-1)^k \widehat{K}(k).$$

We need $\widehat{K}$ to be absolutely integrable, so it is necessary that K be continuous. Since $K(t) = 0$ for $|t| > 1/2$, it follows that $K(\pm 1/2) = 0$. Thus all terms in the sum on the left above are 0. But the right hand side is

$$\geq \widehat{K}(0) - \sum_{n \neq 0} |\widehat{K}(n)|.$$

However, we need this last expression to be positive in order to obtain a positive lower bound, so it seems that this method cannot succeed when $\delta T = 1$. However, see Exercise 13 below.

12. For $0 < \varepsilon \leq 1/2$ let $K_\varepsilon(x) = \max(0, 1/\varepsilon - \|x\|)$.
- Show that $\widehat{K}_\varepsilon(0) = 1$.
 - By integrating by parts, show that

$$\widehat{K}_\varepsilon(n) = \left(\frac{\sin \pi n \varepsilon}{\pi n \varepsilon} \right)^2$$

for $n \neq 0$.

- In the notation of the preceding exercise, take $a_n = (-1)^n \widehat{K}_\varepsilon(n)$. Deduce that $S(t) = K_\varepsilon(t + 1/2)$.
- Set $T = 1 - 2\varepsilon$. Note that $S(t) = 0$ for $-T/2 \leq t \leq T/2$, and that $\max_n |a_n| = a_0 = 1$. Here $\delta = 1$, so $\delta T = 1 - 2\varepsilon$, but $\int_{-T/2}^{T/2} |S(t)| dt$ is not bounded below by $c a_0$ with $c > 0$.

Exer: Ingham 13. ^{Ing50}(Ingham, 1950) The object of this exercise is to show that if

$$f(t) = \sum_{n=1}^N a_n e(\lambda_n t)$$

where $\lambda_{n+1} - \lambda_n \geq 1$ for $1 \leq n < N$, then

$$\max_n |a_n| \leq 2 \int_{-1/2}^{1/2} |f(t)| dt.$$

- (a) Explain why the following reformulation is equivalent to the above. Let m and n be nonnegative integers, and put

$$f(t) = \sum_{r=-m}^n a_r e(\lambda_r t)$$

where $\lambda_{r+1} - \lambda_r \geq 1$ for $-m \leq r < n$. Suppose also that $\lambda_0 = 0$. Then $|a_0| \leq 2 \int_{-1/2}^{1/2} |f(t)| dt$.

- (b) The sum $f(t)$ has $m+n+1$ terms. Let $\mathcal{M}$ be any set of $m+n+1$ integers with $0 \in \mathcal{M}$, and put

$$g(t) = \sum_{\mu \in \mathcal{M}} c(\mu) e(\mu t)$$

where the coefficients $c(\mu)$ are to be determined so $e(\lambda_r t)$ is orthogonal on the interval $[-1/2, 1/2]$ to $g(t)$ for all $r \neq 0$. This is a system of $m+n$ homogeneous equations in $m+n+1$ variables. To solve this system, put

$$G(u) = \int_{-1/2}^{1/2} g(t) e(-ut) dt$$

for real u .

- (c) Show that

$$G(u) = \frac{\sin \pi u}{\pi} \sum_{\mu} \frac{(-1)^\mu c(\mu)}{u - \mu} = \frac{\sin \pi u}{\pi} F(u),$$

say.

- (d) Show that $G(u)$ is an entire function.
 (e) We want the rational function $F(u)$ to have poles at the μ and to be zero at the λ_r , so we set $F(u) = P(u)/Q(u)$ where

$$P(u) = \prod_{r \neq 0} (u - \lambda_r), \quad Q(u) = \prod_{\mu \in \mathcal{M}} (u - \mu).$$

- (f) It is clear that $F(\lambda_r) = 0$. Explain why it follows that $G(\lambda_r) = 0$.
 (g) Show that the partial fraction expansion of F is

$$\sum_{\mu} \frac{P(\mu)}{Q'(\mu)(u - \mu)}.$$

(h) Show that

$$Q'(\mu) = \prod_{\substack{\mu' \\ \mu' \neq \mu}} (\mu - \mu').$$

(i) Show that

$$c(\mu) = (-1)^\mu \frac{\prod_{r \neq 0} (u - \lambda_r)}{\prod_{\substack{\mu' \\ \mu' \neq \mu}} (\mu - \mu')}.$$

(j) Show that

$$\int_{-1/2}^{1/2} f(t)g(-t) dt = \sum_r G(\lambda_r) = a_0 c(0).$$

(k) Show that

$$\left| \int_{-1/2}^{1/2} f(t)g(-t) dt \right| \leq \int_{-1/2}^{1/2} |f(t)| dt \sum_{\mu} |c(\mu)|.$$

(l) Assign indices to the μ , so that $\mu_0 = 0$, $\mu_r = \lceil \lambda_r \rceil$ for $r < 0$, and $\mu_r = \lfloor \lambda_r \rfloor$ for $r > 0$. Show that

$$c(0) = \prod_r r \neq 0 \frac{\lambda_r}{\mu_r} \geq 1.$$

(m) Show that $(-1)^{\mu_r} c(\mu_r) < 0$ for all $r \neq 0$.

(n) Show that

$$1 = \lim_{u \rightarrow \infty} \frac{uP(u)}{Q(u)} = \sum_{\mu} (-1)^\mu c(\mu).$$

(o) Conclude that $|a_0| \leq 2 - 1/c(0)$.

(p) Let $\Delta_K(t)$ be the Fejér kernel, and set $f(t) = \Delta_K(\lambda t)$ where λ is slightly less than 2, and K is large. Here the largest coefficient is 1. Show that $\int_{-1/2}^{1/2} |f(t)| dt$ is approximately $1/\lambda$. Deduce that the constant 2 in (a) is best possible.

14. Let $K_T(t)$ be defined as in (26.9). By some form of the inversion theorem for Fourier transforms, it follows that

$$\int_{-\infty}^{\infty} \frac{1}{T} \left(\frac{\sin \pi T \alpha}{\pi \alpha} \right)^2 d\alpha = \max(0, 1 - |t|/T).$$

Give a direct computational proof of the above identity.

15. Suppose that $A_n \geq 0$ for all n , and that $\sum_n A_n < \infty$. Also, let λ_n be arbitrary real numbers, and let $K_T(t)$ be defined as in (26.9).

(a) Show that

$$\int_{-\infty}^{\infty} K_T(t) \left| \sum_{j=1}^{\infty} A_j e(\lambda_j t) \right|^2 dt \geq T \sum_{j=1}^{\infty} A_j^2.$$

(b) Deduce that

$$\int_{-\infty}^{\infty} \left| \sum_{j=1}^{\infty} A_j e(\lambda_j t) \right|^2 dt \geq T \sum_{j=1}^{\infty} A_j^2.$$

- Exer:Jutila** 16. (a) Let numbers λ_j , a_j , and A_j be as described in Theorem 26.13. Show that

$$\sum_{j,k} \left| \sum_{n=1}^{\infty} a_n e(\lambda_j - \lambda_k) \right|^2 \leq \sum_{j,k} \left| \sum_{n=1}^{\infty} A_n e(\lambda_j - \lambda_k) \right|^2.$$

- (b) Let the Dirichlet series $\alpha(s)$, $A(s)$ be defined as in (26.12), with $|a_n| \leq A_n$ for all n , and suppose that σ is chosen so that $\sum A_n n^{-\sigma} < \infty$. Let $t_1, t_2, \dots, t_R$ be real numbers. Show that

$$\sum_{1 \leq j, k \leq R} |\alpha(\sigma + i(t_j - t_k))|^2 \leq \sum_{1 \leq j, k \leq R} |A(\sigma + i(t_j - t_k))|^2.$$

- (c) Suppose that C is a positive real number such that $|a_n| \leq C A_n$ for all n . Show that if Dirichlet characters χ_j and complex numbers s_j are chosen, then

$$\begin{aligned} \sum_{1 \leq j, k \leq R} \left| \sum_{n=1}^N a_n \chi_j(n) \overline{\chi_k(n)} n^{-s_j - \overline{s_k}} \right|^2 \\ \leq C^2 \sum_{1 \leq j, k \leq R} \left| \sum_{n=1}^N A_n \chi_j(n) \overline{\chi_k(n)} n^{-s_j - \overline{s_k}} \right|^2. \end{aligned}$$

17. (a) Let $C = [c_{nj}]$ be an arbitrary $N \times J$ matrix, and suppose that $|a_n| \leq A_n$ for all n . Show that

$$\sum_{j=1}^J \sum_{k=1}^J \left| \sum_{n=1}^N a_n c_{nj} \overline{c_{nk}} \right|^2 \leq \sum_{j=1}^J \sum_{k=1}^J \left| \sum_{n=1}^N A_n c_{nj} \overline{c_{nk}} \right|^2.$$

- (b) Let $\mathbf{u}_1, \dots, \mathbf{u}_M$ and $\mathbf{v}_1, \dots, \mathbf{v}_N$ be arbitrary members of an inner product space. Show that

$$\sum_{m=1}^M \sum_{n=1}^N |\langle \mathbf{u}_m, \mathbf{v}_n \rangle|^2 \leq \left(\sum_{m=1}^M \sum_{\mu=1}^M |\langle \mathbf{u}_m, \mathbf{u}_\mu \rangle|^2 \right)^{1/2} \times \left(\sum_{n=1}^N \sum_{\nu=1}^N |\langle \mathbf{v}_n, \mathbf{v}_\nu \rangle|^2 \right)^{1/2}$$

18. Let numbers a_j , A_j and λ_j be as described in Theorem 26.13. Show that

$$\int_{-T}^T \left| \sum_{j=1}^{\infty} a_j e(\lambda_j t) \right|^{2k} dt \leq 3 \int_{-T}^T \left| \sum_{j=1}^{\infty} A_j e(\lambda_j t) \right|^{2k} dt$$

for any positive integer k .

19. Let a_j , A_j , and the λ_j be as in Theorem 26.13, and consider the possibility of a more general inequality of the form

$$\int_{-cT}^{cT} \left| \sum_{j=1}^{\infty} a_j e(\lambda_j t) \right|^2 dt \leq K(c) \int_{-T}^T \left| \sum_{j=1}^{\infty} A_j e(\lambda_j t) \right|^2 dt$$

where $c > 0$ and $K(c)$ depends on c .

- (a) Show that the above is valid when $K(c) = 1 + \lceil 2c \rceil$.
 (b) Show that if the above is valid, then it must be the case that $K(c) \geq \lceil 2c \rceil$.

26.2 Character sums and hybrids

S:hybrids

From the basic orthogonality properties of Dirichlet characters (cf Corollary 4.5) we know that

$$\sum_{\chi \bmod q} \left| \sum_{n=1}^q a_n \chi(n) \right|^2 = \varphi(q) \sum_{\substack{n=1 \\ (n,q)=1}}^q |a_n|^2 \quad (26.15) \quad \text{E:chiorthog}$$

for arbitrary complex numbers a_n . Sometimes we want to sum over all χ modulo q , but evaluate our character sum not at χ but rather at the primitive character χ^* that induces χ . For this situation we do not have an identity, but at least we have an upper bound.

L:S(chi*) **Lemma 26.14.** *Let χ^* denote the primitive character that induces χ . Then*

$$\sum_{\chi \bmod q} \left| \sum_{n=1}^q a_n \chi^*(n) \right|^2 \leq q \sum_{n=1}^q |a_n|^2 \quad (26.16) \quad \text{E:S(chi*)}$$

for arbitrary complex numbers a_n .

The bound here is sharp, since equality is attained when all the a_n are equal. (Consider the contribution on the left hand side made by the character χ_0^* .)

Proof Each character χ modulo q is induced by a unique primitive character χ^* . Let d denote the conductor of χ^* . Then $d|q$. This is a one-to-one correspondence, so

$$\sum_{\chi \bmod q} \left| \sum_{n=1}^q a_n \chi^*(n) \right|^2 = \sum_{d|q} \sum_{\chi \bmod d}^* \left| \sum_{n=1}^q a_n \chi(n) \right|^2.$$

By Lemma ?? with q replaced by d , we know that

$$\sum_{\chi \bmod d}^* \left| \sum_{n=1}^q a_n \chi(n) \right|^2 \leq \sum_{\substack{b=1 \\ (b,d)=1}}^d \left| \sum_{n=1}^q a_n e(bn/d) \right|^2.$$

On summing this over $d|q$ we deduce that

$$\sum_{\chi \bmod q} \left| \sum_{n=1}^q a_n \chi^*(n) \right|^2 \leq \sum_{a=1}^q \left| \sum_{n=1}^q a_n e(an/q) \right|^2.$$

By the Parseval identity for the Discrete Fourier Transform (which is equation (4.4) in Chapter 4), the right hand side above is

$$= q \sum_{n=1}^q |a_n|^2,$$

which is the desired estimate. $\square$

For similar sums over intervals of arbitrary length we argue less precisely, as follows.

T:charmodq **Theorem 26.15.** *Let*

$$S(\chi) = \sum_{M+1}^{M+N} a_n \chi(n)$$

where the a_n are complex numbers. Then

$$\sum_{\chi \bmod q} |S(\chi)|^2 \leq \left(\left\lfloor \frac{N-1}{q} \right\rfloor + 1 \right) \varphi(q) \sum_{\substack{n=M+1 \\ (n,q)=1}}^{M+N} |a_n|^2 \quad (26.17) \quad \boxed{\text{E:chimodq}}$$

and

$$\sum_{\chi \bmod q} |S(\chi^*)|^2 \leq \left(\left\lfloor \frac{N-1}{q} \right\rfloor + 1 \right) q \sum_{n=M+1}^{M+N} |a_n|^2 \quad (26.18) \quad \boxed{\text{E:primchimodq}}$$

where χ^* is the primitive character that induces χ .

Proof Let

$$Z(q, h) = \sum_{\substack{n=M+1 \\ n \equiv h \bmod q}}^{M+N} a_n,$$

as in (??). Then

$$\sum_{\chi \bmod q} |S(\chi)|^2 = \sum_{\chi \bmod q} \left| \sum_{h=1}^q Z(q, h) \chi(h) \right|^2 = \varphi(q) \sum_{\substack{h=1 \\ (h,q)=1}}^q |Z(q, h)|^2$$

by (26.15). The sum that defines $Z(q, h)$ has at most $\left\lfloor \frac{N-1}{q} \right\rfloor + 1$ terms, so by Cauchy's inequality it follows that

$$|Z(q, h)|^2 \leq \left(\left\lfloor \frac{N-1}{q} \right\rfloor + 1 \right) \sum_{\substack{n=M+1 \\ n \equiv h \bmod q}}^{M+N} |a_n|^2,$$

so we have (26.17). The estimate (26.18) is proved in the same way, using Lemma 26.14. $\square$

We now establish a series of fundamental estimates concerning

$$S(s; \chi) = \sum_{n=1}^N a_n \chi(n) n^{-s} \quad (26.19) \quad \boxed{\text{E:DefPol1}}$$

where the a_n are arbitrary complex numbers, χ is a Dirichlet character.

$\boxed{\text{T:smoothhybrid}}$

Theorem 26.16. *Let $S(s; \chi)$ be given by (26.19), and let A be a real number and T be a positive real number. Then*

$$\sum_{\chi \bmod q} \int_A^{A+T} |S(it; \chi)|^2 dt \ll \frac{\varphi(q)}{q} \sum_{\substack{n=1 \\ (n,q)=1}}^N |a_n|^2 (qT + n), \quad (26.20) \quad \boxed{\text{E:chit}}$$

$$\sum_{\chi \bmod q} \int_A^{A+T} |S(it; \chi^*)|^2 dt \ll \sum_{n=1}^N |a_n|^2 (qT + n) \quad (26.21) \quad \boxed{\text{E:qchi}}$$

where χ^* denotes the primitive character that induces χ , and

$$\sum_{q \leq Q} \frac{q}{\varphi(q)} \sum_{\chi \bmod q}^* \int_A^{A+T} |S(it; \chi)|^2 dt \ll \sum_{n=1}^N |a_n|^2 (Q^2 T + n) \quad (26.22) \quad \boxed{\text{E:Qchi}}$$

where $\sum^*$ indicates that the sum is restricted to primitive characters.

Proof It suffices to prove the theorem with $A = 0$ for then the more general conclusion follows on replacing a_n by $a_n n^{-iA}$. When $T \leq 1$ the conclusions follow at once from Theorem 26.15 for the first two inequalities and the large sieve inequality (Theorem ??) for the third. Thus we may suppose that $T \geq 1$. By Corollary 26.12 we know that

$$\int_0^T |S(it; \chi)|^2 dt \ll T^2 \int_{-\infty}^{\infty} \left| \sum_{\substack{n=1 \\ y < n \leq \tau y}}^N a_n \chi(n) \right|^2 \frac{dy}{y}$$

where $\tau = \exp(\frac{1}{T})$. Thus $\tau - 1 \asymp 1/T$. We sum the above over $\chi \pmod{q}$. By (26.17) we see that

$$\sum_{\chi} \left| \sum_{y < n \leq \tau y} a_n \chi(n) \right|^2 \ll \frac{\varphi(q)}{q} (q + (\tau - 1)y) \sum_{y < n \leq \tau y} |a_n|^2.$$

Thus

$$\begin{aligned} \sum_{\chi \bmod q} \int_0^T |S(it; \chi)|^2 dt &\ll \frac{\varphi(q)}{q} \sum_n |a_n|^2 \int_{n/\tau}^n \left(\frac{qT^2}{y} + T \right) dy \\ &= \frac{\varphi(q)}{q} \sum_n |a_n|^2 (qT + Tn(1 - 1/\tau)), \end{aligned}$$

which gives (26.20).

To obtain (26.21) we apply (26.18), from which we see that

$$\sum_{\chi} \left| \sum_{y < n \leq \tau y} a_n \chi^*(n) \right|^2 \ll (q + (\tau - 1)y) \sum_{y < n \leq \tau y} |a_n|^2.$$

It then suffices to argue as in the first case.

Finally, to obtain (26.22) we apply the large sieve (Theorem ??), from which we see that

$$\sum_{q \leq Q} \frac{q}{\varphi(q)} \sum_{\chi}^* \left| \sum_{y < n \leq \tau y} a_n \chi(n) \right|^2 \ll (Q^2 + (\tau - 1)y) \sum_{y < n \leq \tau y} |a_n|^2,$$

and then the rest of the argument is the same as before. $\square$

The method of proof of Theorem 26.8 combines well with the estimates of Theorem 26.16 to yield a variant with discrete values of t .

T:discretehybrid1

Theorem 26.17. *Let $S(s; \chi)$ be given by (26.19), let $A, T \geq 2$ and $0 < \delta \leq 1$ be a real numbers. Further, for $r = 1, 2, \dots, R$ let (t_r, χ_r) be a pair in which χ_r is a character (mod q), $A \leq t_r \leq A + T$, and $|t_{r_1} - t_{r_2}| \geq \delta$ if $\chi_{r_1} = \chi_{r_2}$. Then*

$$\sum_{r=1}^R |S(it_r; \chi_r)|^2 \ll \frac{\varphi(q)}{q} (\log N + 1/\delta) \sum_{\substack{n=1 \\ (n,q)=1}}^N |a_n|^2 (qT + n), \quad (26.23) \quad \text{E:chirtr}$$

and

$$\sum_{r=1}^R |S(it_r; \chi_r^*)|^2 \ll (\log N + 1/\delta) \sum_{n=1}^N |a_n|^2 (qT + n) \quad (26.24) \quad \text{E:qrchir}$$

where χ^* denotes the primitive character that induces χ . Finally, let (t_r, χ_r, q_r) be a triple with χ_r a primitive character (mod q_r) for some $q_r \leq Q$, $A \leq t_r \leq A + T$, and $|t_{r_1} - t_{r_2}| \geq \delta$ if $\chi_{r_1} = \chi_{r_2}$. Then

$$\sum_{r=1}^R \frac{q_r}{\varphi(q_r)} |S(it_r; \chi_r)|^2 \ll (\log N + 1/\delta) \sum_{n=1}^N |a_n|^2 (Q^2 T + n). \quad (26.25) \quad \text{E:Qchirtr}$$

Proof As in the proof of Theorem 26.16, we may assume that $A = 0$. We consider first (26.26). Let $\mathcal{R}(\chi) = \{r : \chi_r = \chi\}$. Let $\mathfrak{M}_r = [t_r - \delta/2, t_r + \delta/2]$ for $r = 1, 2, \dots, R$. By Lemma ?? it follows that

$$|S(it_r; \chi_r)|^2 \leq \frac{1}{\delta} \int_{\mathfrak{M}_r} |S(it; \chi_r)|^2 dt + \int_{\mathfrak{M}_r} |S(it; \chi_r) S'(it; \chi_r)|^2 dt.$$

For any χ (mod q), the intervals $\mathfrak{M}_r$ with $r \in \mathcal{R}(\chi)$ are disjoint and lie in the interval $[-\delta, T + \delta]$, so

$$\sum_{r \in \mathcal{R}(\chi)} |S(it_r; \chi)|^2 \leq \frac{1}{\delta} \int_{-\delta}^{T+\delta} |S(it; \chi)|^2 dt + \int_{-\delta}^{T+\delta} |S(it; \chi) S'(it; \chi)| dt.$$

We sum this over χ to see that

$$\begin{aligned} \sum_{r=1}^R |S(it_r; \chi_r)|^2 &\leq \frac{1}{\delta} \sum_{\chi} \int_{-\delta}^{T+\delta} |S(it; \chi)|^2 dt \\ &\quad + \sum_{\chi} \int_{-\delta}^{T+\delta} |S(it; \chi) S'(it; \chi)| dt. \end{aligned}$$

By the Cauchy–Schwarz inequality, the last term above is

$$\leq \left(\sum_{\chi} \int_{-\delta}^{T+\delta} |S(it; \chi)|^2 dt \right)^{1/2} \left(\sum_{\chi} \int_{-\delta}^{T+\delta} |S'(it; \chi)|^2 dt \right)^{1/2}.$$

From (26.20) it follows that

$$\begin{aligned} \sum_{r=1}^R |S(it_r; \chi_r)|^2 &\ll \frac{\varphi(q)}{q} \left(\frac{1}{\delta} \sum_{n=1}^N |a_n|^2 (qT + n) \right. \\ &\quad \left. + \left(\sum_{n=1}^N |a_n|^2 (qT + n) \right)^{1/2} \left(\sum_{n=1}^N |a_n|^2 (qT + n) (\log n)^2 \right)^{1/2} \right), \end{aligned}$$

which gives the stated estimate. The estimates (26.27) and (26.28) are derived similarly from (26.21) and (26.22), respectively. $\square$

For points s_r with nonnegative real part we have the following further estimates.

T:discretehybrid2

Theorem 26.18. *Let $S(s; \chi)$ be given by (26.19), let $A, T \geq 2$ and $0 < \delta \leq 1$ be a real numbers. Further, for $r = 1, 2, \dots, R$ let (s_r, χ_r) be a pair in which χ_r is a character (mod q), $\sigma_r \geq 0$, $A \leq t_r \leq A + T$, and $|t_{r_1} - t_{r_2}| \geq \delta$ if $\chi_{r_1} \neq \chi_{r_2}$. Then*

$$\begin{aligned} \sum_{r=1}^R |S(s_r; \chi_r)|^2 &\ll \frac{\varphi(q)}{q} (\log N + 1/\delta) \\ &\quad \times \sum_{\substack{n=1 \\ (n,q)=1}}^N |a_n|^2 (qT + n) \left(1 + \log \frac{\log 2N}{\log 2n} \right), \end{aligned} \tag{26.26} \quad \text{E:chirtr}$$

and

$$\begin{aligned} \sum_{r=1}^R |S(s_r; \chi_r^*)|^2 &\ll (\log N + 1/\delta) \\ &\quad \times \sum_{n=1}^N |a_n|^2 (qT + n) \left(1 + \log \frac{\log 2N}{\log 2n} \right) \end{aligned} \tag{26.27} \quad \text{E:qrchir}$$

where χ^* denotes the primitive character that induces χ . Finally, let (s_r, χ_r, q_r) be a triple with χ_r a primitive character (mod q_r) for some

$q_r \leq Q$, $\sigma_r \geq 0$, $A \leq t_r \leq A + T$, and $|t_{r_1} - t_{r_2}| \geq \delta$ if $\chi_{r_1} \neq \chi_{r_2}$. Then

$$\sum_{r=1}^R \frac{q_r}{\varphi(q_r)} |S(s_r; \chi_r)|^2 \ll (\log N + 1/\delta) \sum_{n=1}^N |a_n|^2 (Q^2 T + n) \left(1 + \log \frac{\log 2N}{\log 2n}\right). \quad (26.28) \quad \boxed{\text{E:Qchirtr}}$$

Proof We argue in the same way that we did in deriving Theorem 26.9 from Theorem 26.8. In particular, we again use the inequality (26.5). We have $R \ll \varphi(q)T/\delta$ in the case of (26.26) and (26.27), and $R \ll Q^2 T/\delta$ for (26.28). Thus the proof is entirely parallel to the former one. $\square$

26.1.1 Exercises

1. As in Lemma 26.14, let χ^* denote the primitive character that induces χ .

(a) Let χ denote a Dirichlet character modulo q . Explain why the assertion that

$$\sum_{\chi} \left| \sum_{n=1}^q a_n \chi^*(n) \right|^2 \leq q \sum_{n=1}^q |a_n|^2$$

for all choices of the a_n is equivalent to the assertion that

$$\sum_{n=1}^q \left| \sum_{\chi} b(\chi) \chi^*(n) \right|^2 \leq q \sum_{\chi} |b(\chi)|^2$$

for all choices of the $b(\chi)$.

(b) Note that the left hand side above is

$$= \sum_{\chi_1, \chi_2} b(\chi_1) \overline{b(\chi_2)} \sum_{n=1}^q \chi_1^*(n) \overline{\chi_2^*(n)}.$$

(c) Show that the sum over n above is $= q\varphi(d)/d$ if $\chi_1 = \chi_2$ and their conductor is d , and is $= 0$ otherwise.

(d) Deduce that

$$\sum_{n=1}^q \left| \sum_{\chi} b(\chi) \chi^*(n) \right|^2 = q \sum_{d|q} \frac{\varphi(d)}{d} \sum_{\chi \bmod d}^* |b(\chi)|^2.$$

(e) Explain why this gives a second proof of Lemma 26.14.

26.3 Notes

S: MeanVals Notes

Section 21.1. Concerning majorant inequalities, antecedents of Theorem 26.13 are found in the work of Wiener (unpublished — see Theorem 12.6.12 of Boas (1954), Erdős and Fuchs (1956), Wiener and Wintner (1956), and Halász (1968). Logan (1988) showed that the constant 3 is best possible. For a general discussion of majorant principles, see Shapiro (1975).

26.4 References

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26.5 Mean values of L -functions

S:MVLFcns

Suppose that $\sigma > 1$. Then the Dirichlet series for $\zeta(s)$ is absolutely convergent, and hence the Dirichlet series for $\zeta(s)^k$ is also absolutely convergent:

$$\zeta(s)^k = \sum_{n=1}^{\infty} \frac{d_k(n)}{n^s} = \prod_p \left(1 - \frac{1}{p^s}\right)^{-k} = \prod_p \left(\sum_{r=0}^{\infty} \binom{k+r-1}{r} p^{-rs}\right).$$

Here $d_k(n)$ is known as the k^{th} divisor function. It is the unique multiplicative function with the property that $d_k(p^r) = \binom{k+r-1}{r}$. Thus (ignoring for the moment the sharp estimates we derived in §21.1),

$$\begin{aligned} \int_0^T |\zeta(\sigma + it)|^{2k} dt &= \int_0^T \left| \sum_{n=1}^{\infty} \frac{d_k(n)}{n^{\sigma}} n^{-it} \right|^2 dt \\ &= \sum_{m,n} \frac{d_k(m)d_k(n)}{(mn)^{\sigma}} \int_0^T \left(\frac{m}{n}\right)^{it} dt. \end{aligned}$$

The functions m^{-it} and n^{-it} are asymptotically orthogonal to the extent that

$$\frac{1}{T} \int_0^T \left(\frac{m}{n}\right)^{it} dt = \begin{cases} 1 & (m = n), \\ O_{m,n}(1/T) & (m \neq n). \end{cases}$$

The double sum of the coefficients,

$$\sum_{m,n} \frac{d_k(m)d_k(n)}{(mn)^{\sigma}},$$

is absolutely convergent (with value $\zeta(\sigma)^{2k}$), so by the principle of dominated convergence it follows that

$$\lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T |\zeta(\sigma + it)|^{2k} dt = \sum_{n=1}^{\infty} \frac{d_k(n)^2}{n^{2\sigma}} \quad (26.29) \quad \text{E:zeta2kthmv}$$

for any fixed $\sigma > 1$. The question before us is to try to determine what combination of values of k and $\sigma \leq 1$ (if any) the above relation continues to be valid. In this connection, the following principle is often useful.

T:alpha2sigma

Theorem 26.19. *Let k be a positive integer, and suppose that $\alpha \geq 1/2$ is a number such that*

$$\int_0^T |\zeta(\alpha + it)|^{2k} dt \ll_{\varepsilon} T^{1+\varepsilon} \quad (26.30) \quad \text{E:zetamvEst}$$

for every $\varepsilon > 0$. Then the relation (26.29) holds for all $\sigma > \alpha$.

Proof We may suppose that $\sigma \leq 1$. We suppose also that $2 \leq x \leq T^A$ for some constant A . Then by the formula (5.25) for the inverse Mellin transform with abelian weights we know that

$$\sum_{n=1}^{\infty} \frac{d_k(n)}{n^s} e^{-n/x} = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \zeta(s+w)^k \Gamma(w) x^w dw$$

for $c > 1 - \sigma$. Write $w = u + iv$. When we move the contour to the abscissa $u = \alpha - \sigma$, we pass two poles: One at $w = 0$ with residue $\zeta(s)^k$, and the other at $w = 1 - s$. To estimate the residue at this second pole, let C be a circle of radius $1/\log x$ centered at $1 - s$. For w on this circle, $|\zeta(s+w)^k| \asymp (\log x)^k$, $|x^w| \asymp x^{1-\sigma}$, and $|\Gamma(w)| \asymp e^{-\pi\tau/2} \tau^{1/2-\sigma}$ by (25.3). Hence the residue at $1 - s$ is $\ll e^{-\pi\tau/2} \tau^{1/2-\sigma} x^{1-\sigma} (\log x)^{k-1}$. Thus the above is

$$\begin{aligned} &= \zeta(s)^k + \frac{1}{2\pi} \int_{-\infty}^{\infty} \zeta(\alpha + it + iv)^k \Gamma(\alpha - \sigma + iv) x^{\alpha-\sigma+iv} dv \\ &\quad + O(e^{-\pi\tau/2} \tau^{1/2-\sigma} x^{1-\sigma} (\log x)^{k-1}), \end{aligned}$$

and consequently

$$\begin{aligned} &\int_T^{2T} \left| \zeta(s)^k - \sum_{n=1}^{\infty} \frac{d_k(n)}{n^s} e^{-n/x} \right|^2 dt \\ &\ll x^{2(\alpha-\sigma)} \int_T^{2T} \left| \int_{-\infty}^{\infty} |\zeta(\alpha + it + iv)^k \Gamma(\alpha - \sigma + iv)| dv \right|^2 dt + e^{-T}. \end{aligned}$$

By the Cauchy-Schwarz inequality we see that the square of the modulus of the above integral over v is

$$\leq \int_{-\infty}^{\infty} |\Gamma(\alpha - \sigma + iv)| dv \int_{-\infty}^{\infty} |\zeta(\alpha + it + iv)|^{2k} |\Gamma(\alpha - \sigma + iv)| dv.$$

The first integral above is bounded, so

$$\begin{aligned} &\int_T^{2T} \left| \zeta(s)^k - \sum_{n=1}^{\infty} \frac{d_k(n)}{n^s} e^{-n/x} \right|^2 dt \\ &\ll x^{2(\alpha-\sigma)} \int_{-\infty}^{\infty} \int_T^{2T} |\zeta(\alpha + it + iv)|^{2k} dt |\Gamma(\alpha - \sigma + iv)| dv + e^{-T}. \end{aligned}$$

To estimate the double integral we consider two ranges of v . First, if $-4T \leq v \leq 4T$, then the integral over t is $\ll T^{1+\varepsilon}$, so the contribution of such v is also $\ll T^{1+\varepsilon}$. Secondly, if $|v| > 4T$, then by the trivial bound $\zeta(\alpha + it) \ll \tau^{1/4}$ we see that the integral over t is $\ll Tv^{k/2}$, but by (25.3) the resulting contribution is $\ll e^{-T}$. We conclude that the above is

$$\ll x^{2(\alpha-\sigma)} T^{1+\varepsilon}. \quad (26.31) \quad \boxed{\text{E:ApproxEst}}$$

By Corollary 26.4 we see that

$$\int_T^{2T} \left| \sum_{n=1}^{\infty} \frac{d_k(n)}{n^s} e^{-n/x} \right|^2 dt = \sum_{n=1}^{\infty} \frac{d_k(n)^2}{n^{2\sigma}} e^{-2n/x} (T + O(n)).$$

Since $e^{-2n/x} = 1 + O(n/x)$ for $n \leq x$, we find that the right hand side above is

$$\begin{aligned} &= T \sum_{n=1}^{\infty} \frac{d_k(n)^2}{n^{2\sigma}} + O\left(T \sum_{n>x} \frac{d_k(n)^2}{n^{2\sigma}}\right) \\ &\quad + O\left(\frac{T}{x} \sum_{n \leq x} \frac{d_k(n)^2}{n^{2\sigma-1}}\right) + O\left(\sum_{n=1}^{\infty} \frac{d_k(n)^2}{n^{2\sigma-1}} e^{-2n/x}\right). \end{aligned}$$

From Corollary 2.15 we deduce that

$$\sum_{U < n \leq 2U} d_k(n)^2 \ll U(\log 2U)^{k^2-1},$$

and from this it follows that

$$\sum_{n \leq x} \frac{d_k(n)^2}{n^{2\sigma-1}} \ll x^{2-2\sigma} (\log x)^{k^2}$$

uniformly for $1/2 \leq \sigma \leq 1$. Consequently,

$$\int_T^{2T} \left| \sum_{n=1}^{\infty} \frac{d_k(n)}{n^s} e^{-n/x} \right|^2 dt = T \sum_{n=1}^{\infty} \frac{d_k(n)^2}{n^{2\sigma}} + O((T+x)x^{1-2\sigma}(\log x)^{k^2}).$$

We take $x = T$ and combine this with (26.31) to obtain the stated result. $\square$

We argued qualitatively above. If we wanted a quantitative estimate, we would need to employ the triangle inequality for L^2 norms; see Exercise 26.5.1.2.

We recall that the Lindelöf Hypothesis (LH) asserts that

$$\zeta(1/2 + it) \ll_{\varepsilon} \tau^{\varepsilon} \quad (26.32) \quad \boxed{\text{E:DefLH}}$$

for every $\varepsilon > 0$. As we discussed already in §§10.1,13.2, RH implies LH.

Cor:LH2mvt

Corollary 26.20. *Assume the Lindelöf Hypothesis. Then the relation (26.29) holds for all positive integers k , and all fixed $\sigma > 1/2$.*

We have some nontrivial bounds for the zeta function, but they fall short of LH, and so it is to be expected that our results concerning mean values are a bit fragmentary and unsatisfactory. We now establish a first mean value theorem for the zeta function on the $1/2$ -line.

T:zeta(1/2+it)^2

Theorem 26.21. *For $T \geq 2$,*

$$\int_0^T |\zeta(1/2 + it)|^2 dt = T \log T + O(T).$$

Proof It suffices to show that

$$\int_T^{2T} |\zeta(1/2 + it)|^2 dt = T \log T + O(T),$$

since we can replace T by $T/2^k$ in the above, and then sum over k to obtain the stated result. We write the formula (??) as

$$\zeta(s) = \sum_{n \leq x} n^{-s} + R(s),$$

and take $x = 4T$. Thus for $s = 1/2 + it$ with $T \leq t \leq 2T$ we have $R(s) \ll T^{-1/2}$. By Corollary 26.4 we find that

$$\int_T^{2T} \left| \sum_{n \leq 4T} n^{-1/2 - it} \right|^2 dt = T \sum_{n \leq 4T} \frac{1}{n} + O\left(\sum_{n \leq 4T} 1 \right) = T \log T + O(T).$$

Hence, as in Exercise 26.5.1.2(b), we have

$$\begin{aligned} \int_T^{2T} |\zeta(1/2 + it)|^2 dt &= (T \log T + O(T)) \left(1 + O((T \log T)^{-1/2}) \right) \\ &= T \log T + O(T). \end{aligned}$$

□

By combining the above with Theorem 26.19 we obtain the following further result.

Cor:zetamv2

Corollary 26.22. *Let $\sigma > 1/2$ be fixed. Then*

$$\int_0^T |\zeta(\sigma + it)|^2 dt \sim \zeta(2\sigma)T$$

as $T \rightarrow \infty$.

By using our averaged form of the approximate functional equation for the zeta function, we obtain the following useful estimate.

T:zeta4mean **Theorem 26.23.** *Suppose that $T \geq 60$. Then*

$$\int_0^T |\zeta(\sigma + it)|^4 dt \ll T(\log T)^4$$

uniformly for $|\sigma - 1/2| \leq 2/\log T$.

Proof From (25.22) we see that

$$|\zeta(s)|^4 \ll \left| \sum_n w(n/\sqrt{\tau}) n^{-s} \right|^4 + \left| \sum_n w(n/\sqrt{\tau}) n^{s-1} \right|^4 + \tau^{-1} (\log \tau)^4$$

for $|\sigma - 1/2| \leq 2/\log \tau$. Here $w(u)$ is defined as in (25.23). By taking $k = 2$ in (25.26) we find that

$$\left| \sum_n w(n/\sqrt{\tau}) n^{-s} \right|^4 \leq \int_{T^{1/2}}^{8T^{1/2}} |A(s, x)|^4 \frac{dx}{x} \quad (26.33) \quad \text{E:MainStep}$$

uniformly for $T \leq t \leq 2T$. Here $A(s, x)$ is defined as in (25.24). Note that

$$A(s, x)^2 = \sum_{n \leq x^2} c_n n^{-s}$$

where $c_n = c_{n,x}$ counts only some of the divisors of n , so that $0 \leq c_n \leq d(n)$ for all x . Hence when we integrate both sides of (26.33) with respect to t and apply Corollary 26.2, we find that

$$\int_T^{2T} \left| \sum_n w(n/\sqrt{\tau}) n^{-s} \right|^4 dt \ll \int_{T^{1/2}}^{8T^{1/2}} \sum_{n \leq x^2} \frac{d(n)^2}{n^{2\sigma}} (T + x^2) \frac{dx}{x}.$$

From Corollary 2.15 we know that $\sum_{n \leq U} d(n)^2 \ll U(\log U)^3$, and hence that $\sum_{n \leq U} d(n)^2/n \ll (\log U)^4$. Thus the above is $\ll T(\log T)^4$. The same bound applies with s replaced by $1 - s$, so we have the desired result. $\square$

Cor:4thmom **Corollary 26.24.** *Suppose that $\sigma > 1/2$. Then*

$$\lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T |\zeta(\sigma + it)|^4 dt = \frac{\zeta(2\sigma)^4}{\zeta(4\sigma)}.$$

Here we have used the identity

$$\sum_{n=1}^{\infty} d(n)^2/n^s = \zeta(s)^4/\zeta(2s), \quad (26.34) \quad \text{E:RamId}$$

which is a special case of the identity discussed in Exercise 4.3.1.4.

Of the many consequences of Theorem 26.23, we note one particularly useful example.

Cor:2mzeta*zeta'

Corollary 26.25. *Suppose that $T \geq 60$. Then*

$$\int_0^T |\zeta(\sigma + it)\zeta'(\sigma + it)|^2 dt \ll T(\log T)^6$$

uniformly for $|\sigma - 1/2| \leq 1/\log T$.

Proof If $f(z)$ is analytic in a neighborhood of $z = 0$, then by Cauchy's formula we know that

$$f'(0) = \frac{1}{2\pi i} \oint \frac{f(z)}{z^2} dz.$$

We take $f(z) = \zeta(s + z)^2$ and set $z = re^{i\theta}$ where $r = 1/\log T$. By the Cauchy–Schwarz inequality we find that

$$|\zeta(s)\zeta'(s)|^2 \ll (\log T)^2 \int_0^{2\pi} |\zeta(s + re^{i\theta})|^4 d\theta.$$

We integrate this over $0 \leq t \leq T$ to obtain the stated result. $\square$

Cor:disczeta4mv

Corollary 26.26. *Suppose that $T \geq 60$, that $\delta > 0$, and that $\delta/2 \leq t_1 < t_2 < \dots < t_R < T - \delta/2$ where $t_{r+1} - t_r \geq \delta$ for $r = 1, 2, \dots, R-1$. Then*

$$\sum_{r=1}^R |\zeta(1/2 + it_r)|^4 \ll (1/\delta + \log T)T(\log T)^4.$$

Proof By the Sobolev inequality (??) of Lemma ?? we see that

$$\begin{aligned} |\zeta(1/2 + it_r)|^4 &\leq \delta^{-1} \int_{t_r - \frac{1}{2}\delta}^{t_r + \frac{1}{2}\delta} |\zeta(1/2 + it)|^4 dt \\ &\quad + 2 \int_{t_r - \frac{1}{2}\delta}^{t_r + \frac{1}{2}\delta} |\zeta(1/2 + it)^3 \zeta'(1/2 + it)| dt. \end{aligned}$$

The intervals of integration are disjoint, so on summing over r we find that

$$\sum_{r=1}^R |\zeta(1/2 + it_r)|^4 \ll \delta^{-1} \int_0^T |\zeta(1/2 + it)|^4 dt + \int_0^T |\zeta(1/2 + it)\zeta'(1/2 + it)| dt.$$

By the Cauchy–Schwarz inequality the second integral above is

$$\leq \left(\int_0^T |\zeta(1/2 + it)|^4 dt \right)^{1/2} \left(\int_0^T |\zeta(1/2 + it)\zeta'(1/2 + it)|^2 dt \right)^{1/2}.$$

The desired result now follows from the Theorem and the preceding Corollary. $\square$

When we form a weighted average a Dirichlet series $\alpha(s) = \sum a_n n^{-s}$ over a vertical line, its coefficients are diminished. For example, in Exercise 5.1.1.5 we saw that if $\sigma_c < 0$, then

$$\frac{1}{2\pi}(\log N) \int_{-\infty}^{\infty} \alpha(it) \left(\frac{\sin \frac{1}{2}t \log N}{\frac{1}{2}t \log N} \right)^2 dt = \sum_{n \leq N} a_n \left(1 - \frac{\log n}{\log N} \right). \quad (26.35) \quad \boxed{\text{E:Riesz1}}$$

We also know that averaging a function causes its norm to decrease. For example, suppose that $w(x)$ is a weight function such that $w(x) \geq 0$ for all x and $\int_{-\infty}^{\infty} w(x) dx = 1$. If $f \in L^p(\mathbb{R})$ for some real number $p \geq 1$, and we define F to be the convolution

$$F(x) = (w * f)(x) = \int_{-\infty}^{\infty} w(u) f(x - u) du,$$

then not only is $F \in L^p(\mathbb{R})$, but also $\|F\|_p \leq \|f\|_p$. (See Exercise 5.) We use these ideas to derive the following useful complement to our upper bounds.

T:zetakmvlb

Theorem 26.27. *Suppose that the real number σ and a positive integer k are fixed, with $1/2 < \sigma < 1$. Then*

$$\liminf_{T \rightarrow \infty} \frac{1}{T} \int_0^T |\zeta(\sigma + it)|^{2k} dt \geq \sum_{n=1}^{\infty} \frac{d_k(n)^2}{n^{2\sigma}}. \quad (26.36) \quad \boxed{\text{E:zetakmvlb}}$$

Proof Since σ and k are fixed, implicit constants in this proof may depend on these quantities. For our present purposes we need a formula similar to (26.35), but in which the kernel decay more quickly. To this end we start by putting

$$K(w) = c_k \left(\frac{\sinh \frac{w}{4k}}{w} \right)^{4k} = c_k \left(\frac{e^{w/(4k)} - e^{-w/(4k)}}{2w} \right)^{4k} \quad (26.37) \quad \boxed{\text{E:DefK}}$$

where $w = u + iv$ is a complex variable, and c_k is chosen so that

$$\frac{1}{2\pi} \int_{-\infty}^{\infty} K(iv) dv = 1. \quad (26.38) \quad \boxed{\text{E:IntKiv}}$$

We note that $K(w)$ is an entire function. Since $|\sinh w| \leq e^{|u|}$, it follows that

$$K(w) \ll \frac{e^{|u|}}{|w|^{4k}}. \quad (26.39) \quad \boxed{\text{E:KEst}}$$

The implicit constant in the above may depend on k , but we suppress

this, here and elsewhere, since k is considered to be fixed. We note that $K(-w) = K(w)$ for all complex w , and that

$$K(iv) = c_k \left(\frac{\sin \frac{v}{4k}}{v} \right)^{4k} \geq 0$$

for all real v . To create a kernel that yields a weighted partial sum of a Dirichlet series, we rescale $K(w)$ and put

$$\begin{aligned} K_N(w) &= K(w \log N) \log N \\ &= c_k (\log N) \left(\frac{N^{w/(4k)} - N^{-w/(4k)}}{2w \log N} \right)^{4k}. \end{aligned} \quad (26.40) \quad \boxed{\text{E:DefKN}}$$

For real α let

$$W(\alpha) = \frac{1}{2\pi i} \int_{a-i\infty}^{a+i\infty} K(w) e^{\alpha w} dw = \frac{1}{2\pi} \int_{-\infty}^{\infty} K(iv) e^{i\alpha v} dv. \quad (26.41) \quad \boxed{\text{E:DefW}}$$

Thus W is the inverse Laplace transform of K . We note that the value of the above integrals are independent of the value of the real number a . Suppose that $a > 0$. From (26.39) it follows that

$$|W(\alpha)| \leq \frac{e^{a(1+\alpha)}}{2\pi} \int_{-\infty}^{\infty} \frac{dv}{|a+iv|^{4k}} \ll \frac{e^{a(1+\alpha)}}{a^{4k-1}}.$$

If $\alpha \leq -1$, then this upper bound tends to 0 as $a \rightarrow \infty$. Hence $W(\alpha) = 0$ if $\alpha \leq -1$. Since K is even it follows that W is also even, and hence that $W(\alpha) = 0$ if $\alpha \geq 1$. We now take $a = 0$ to see that

$$W(\alpha) = \frac{1}{2\pi} \int_{-\infty}^{\infty} K(iv) e^{i\alpha v} dv.$$

From this it follows that $W(0) = 1$, that $|W(\alpha)| \leq 1$ for all α , and that W is continuous. We now set

$$W_N(x) = W\left(-\frac{\log x}{\log N}\right) = \frac{1}{2\pi i} \int_{a-i\infty}^{a+i\infty} K_N(s) x^{-s} ds. \quad (26.42) \quad \boxed{\text{E:DefWN}}$$

Thus W_N is the inverse Mellin transform of K_N . On expanding the binomial on the right hand side of (27.36) we see that W_N is a linear combination of Riesz typical means of order $4k-1$ with truncations at $N^{j/(2k)}$ for $j = 1, 2, \dots, 2k$.

If $\alpha(s) = \sum a_n n^{-s}$ is a Dirichlet series with abscissa of convergence σ_c , then

$$\sum_{n \leq N} \frac{a_n}{n^s} W_N(n) = \frac{1}{2\pi i} \int_{a-i\infty}^{a+i\infty} \alpha(s+w) K_N(w) dw$$

for $a > \sigma_c$. In particular, if $\alpha(s) = \zeta(s)^k$ and $1/2 < \sigma \leq 1$, then

$$\sum_{n \leq N} \frac{d_k(n)}{n^s} W_N(n) = \frac{1}{2\pi i} \int_{1-i\infty}^{1+i\infty} \zeta(s+w)^k K_N(w) dw.$$

On moving the contour to $a = 0$ we see that this is

$$\begin{aligned} &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \zeta(s+iv)^k K_N(iv) dv \\ &\quad + \text{Res} [\zeta(s+w)^k K_N(w)]_{w=1-s}. \end{aligned} \tag{26.43} \quad \boxed{\text{E:MainId}}$$

Here the first term above is a weighted average of the numbers $\zeta(s+iv)^k$, with the bulk of the weight attached to points for which $v \ll 1/\log N$.

Let $A \geq 1$ and $N \geq 1$ be parameters that will eventually be chosen to be functions of T , with $A = o(T)$ and $N = o(T)$. Then by Corollary 26.2 we see that

$$\int_A^{T-A} \left| \sum_{n \leq N} \frac{d_k(n)}{n^{\sigma+it}} W_N(n) \right|^2 dt = (T - 2A + O(N)) \sum_{n \leq N} \frac{d_k(n)^2}{n^{2\sigma}} W_N(n)^2.$$

Since W is continuous at 0 and $W(0) = 1$, it follows that for any $\varepsilon > 0$ there is a $\delta > 0$ such that $|W_N(n) - 1| < \varepsilon$ for all $n < N^\delta$. Thus if $N = T^{1/2}$, then

$$\int_A^{T-A} \left| \sum_{n \leq N} \frac{d_k(n)}{n^{\sigma+it}} W_N(n) \right|^2 dt \sim T \sum_{n=1}^{\infty} \frac{d_k(n)^2}{n^{2\sigma}}.$$

To assess the residue in (26.43) let $\mathcal{C}$ be a circle of radius $1/\log N$ centered at $1-s$. Thus the residue is

$$\begin{aligned} \frac{1}{2\pi i} \int_{\mathcal{C}} \zeta(s+w)^k K_N(w) dw &\ll (\log N)^{k-1} \max_{w \in \mathcal{C}} |K_N(w)| \\ &= (\log N)^k \max_{w \in \mathcal{C}} |W(w \log N)| \\ &\ll \frac{N^{1-\sigma}}{\tau^{4k} (\log N)^{3k}}. \end{aligned}$$

The mean square of this for $A \leq t \leq T-A$ is $\ll N^{2-2\sigma} A^{-8k+1} \ll N = o(T)$. Thus

$$\int_A^{T-A} \left| \frac{1}{2\pi} \int_{-\infty}^{\infty} \zeta(s+iv)^k K_N(iv) dv \right|^2 dt \sim T \sum_{n=1}^{\infty} \frac{d_k(n)^2}{n^{2\sigma}}.$$

By the Cauchy–Schwarz inequality,

$$\left| \frac{1}{2\pi} \int_{-\infty}^{\infty} \zeta(s+iv)^k K_N(iv) dv \right|^2 \leq \frac{1}{2\pi} \int_{-\infty}^{\infty} K_N(iv) dv \\ \times \frac{1}{2\pi} \int_{-\infty}^{\infty} |\zeta(\sigma+i(t+v))|^{2k} K_N(iv) dv.$$

Here the first integral on the right hand side is equal to 1. We integrate the above with respect to t to see that

$$T \sum_{n=1}^{\infty} \frac{d_k(n)^2}{n^{2\sigma}} \lesssim \frac{1}{2\pi} \int_A^{T-A} \int_{-\infty}^{\infty} |\zeta(\sigma+i(t+v))|^{2k} K_N(iv) dv dt.$$

With the change of variable $y = t + v$ we see that the right hand side above is

$$= \int_{-\infty}^{\infty} |\zeta(\sigma+iy)|^{2k} \left(\frac{1}{2\pi} \int_{y-T+A}^{y-A} K_N(iv) dv \right) dy \\ = \int_{-\infty}^0 dy + \int_0^T dy + \int_T^{\infty} dy = I_1 + I_2 + I_3,$$

say. Since $K_N(iv) \geq 0$ for all v , it follows from (26.38) that

$$\frac{1}{2\pi} \int_{y-T+A}^{y-A} K_N(iv) dv \leq \frac{1}{2\pi} \int_{-\infty}^{\infty} K_N(iv) dv = \frac{1}{2\pi} \int_{-\infty}^{\infty} K(iv) dv = 1$$

for all y . Thus

$$I_2 \leq \int_0^T |\zeta(\sigma+it)|^{2k} dt.$$

From Corollary 1.17 we know that $\zeta(\sigma+it) \ll \tau^{1/2}$. Hence

$$I_1 \ll \int_{-\infty}^0 (|y|+4)^k \int_{-\infty}^{y-A} |v|^{-4k} dv dy (\log N)^{-4k+1} \\ \ll \int_{-\infty}^0 (|y|+4)^k (|y|+A)^{-4k+1} dy \ll A^{-3k+2} \ll A^{-1} = o(T).$$

Similarly,

$$I_3 \ll \int_T^{\infty} y^k \int_{y-T+A}^{\infty} v^{-4k} dv dy (\log N)^{-4k+1} \\ \ll \int_T^{\infty} y^k (y-T+A)^{1-4k} dy \ll T^k A^{2-4k}.$$

We take $A = T^{1/2}$, with the result that the above is $\ll T^{1-k} \ll 1 = o(T)$. Thus we have the result. $\square$

For many purposes the upper bound of Theorem 26.23 for the fourth moment of the zeta function suffices. By taking more care, we now show that on the $1/2$ -line we can derive a more precise estimate.

T:zeta4mean2

Theorem 26.28. For $T \geq 2$,

$$\int_0^T |\zeta(1/2 + it)|^4 dt = \left(\frac{1}{2\pi^2} + O((\log x)^{-1/2}) \right) T(\log T)^4.$$

Proof We first derive a useful formula for $\zeta(1/2 + it)^2$. Put

$$M_1(t) = \sum_{n \leq x} d(n)n^{-1/2-it} \quad (26.44) \quad \text{E:DefM1}$$

where x is a parameter whose value will be chosen later. Weighted partial sums are more easily manipulated than unweighted ones, so we set

$$S_1(t) = \sum_{n \leq x} d(n)n^{-1/2-it} \left(1 - \frac{n}{x}\right)^2$$

with the result that $M_1(t) = S_1(t) + R_1(t)$ where

$$R_1(t) = \sum_{n \leq x} d(n)n^{-1/2-it} \left(1 - \left(1 - \frac{n}{x}\right)^2\right). \quad (26.45) \quad \text{E:DefR1}$$

Let $K(x, w) = 2x^w / (w(w+1)(w+2))$. By the formulæ (5.17), (5.19) concerning the Mellin and inverse Mellin transforms relating to Cesàro partial sums, we see that

$$S_1(t) = \frac{1}{2\pi i} \int_{1-i\infty}^{1+i\infty} \zeta(1/2 + it + w)^2 K(x, w) dw.$$

From Corollary 10.5 we know that if α is fixed, $\alpha < 0$, then $\zeta(s) \ll \tau^{1/2-\alpha}$ uniformly for $\sigma \geq \alpha$, $|t| \geq 1$. Let ϕ be fixed, $-1 < \phi < -1/2$, and write $w = u + iv$. If $u \geq \phi$, then

$$\frac{\zeta(1/2 + it + u + iv)^2}{w(w+1)(w+2)} \ll \frac{(|t+v|+4)^{-2\phi}}{(|v|+4)^3}.$$

Thus in the integral above we may move the path of integration from the 1 -line to the abscissa ϕ . In doing so, we pass poles at $w = 0$ and at $w = 1/2 - it$. Hence $S_1(t) = \zeta(1/2 + it)^2 + S_2(t) + R_2(t)$ where

$$S_2(t) = \frac{1}{2\pi i} \int_{\phi-i\infty}^{\phi+i\infty} \zeta(1/2 + it + w)^2 K(x, w) dw + \zeta(1/2 + it)^2$$

and

$$R_2(t) = \text{Res} \left[\zeta(1/2 + it + w)^2 K(x, w) \right]_{w=1/2-it}. \quad (26.46) \quad \text{E:DefR2}$$

We write the functional equation of the zeta function in the asymmetric form as $\zeta(s) = \Delta(s)\zeta(1-s)$ where $\Delta(s) = 2^s \pi^{s-1} \Gamma(1-s) \sin \pi s/2$ by Corollary 10.4. Hence

$$\zeta(1/2 + \phi + it + iv)^2 = \Delta(1/2 + \phi + it + iv)^2 \zeta(1/2 - \phi - it - iv)^2.$$

We write

$$\zeta(1/2 - \phi - it - iv)^2 = \sum_{n \leq y} d(n) n^{-1/2 + \phi + it + iv} + \sum_{n > y} d(n) n^{-1/2 + \phi + i(t+v)}.$$

Thus $S_2(t) = S_3(t) + R_3(t)$ where

$$S_3(t) = \frac{1}{2\pi i} \int_{\phi - i\infty}^{\phi + i\infty} \Delta(1/2 + it + w)^2 \left(\sum_{n \leq y} d(n) n^{-1/2 + it + w} \right) K(x, w) dw \quad (26.47) \quad \boxed{\text{E:DefS3}}$$

and

$$R_3(t) = \frac{1}{2\pi i} \int_{\phi - i\infty}^{\phi + i\infty} \Delta(1/2 + it + w)^2 \left(\sum_{n > y} d(n) n^{-1/2 + it + w} \right) K(x, w) dw. \quad (26.48) \quad \boxed{\text{E:DefR3}}$$

Let θ be fixed with $0 < \theta < 1/2$. We move the contour in (26.47) from the abscissa ϕ to the abscissa θ . In doing so we pass a pole at $w = 0$. Thus $S_3(t) = R_4(t) - M_2(t)$ where

$$R_4(t) = \frac{1}{2\pi i} \int_{\theta - i\infty}^{\theta + i\infty} \Delta(1/2 + it + w)^2 \left(\sum_{n \leq y} d(n) n^{-1/2 + it + w} \right) K(x, w) dw \quad (26.49) \quad \boxed{\text{E:DefR4}}$$

and

$$M_2(t) = \Delta(1/2 + it)^2 \sum_{n \leq y} d(n) n^{-1/2 + it}. \quad (26.50) \quad \boxed{\text{E:DefM2}}$$

Thus we have shown that

$$\zeta(1/2 + it)^2 = M_1(t) + M_2(t) - R_1(t) - R_2(t) - R_3(t) - R_4(t).$$

It suffices to show that

$$\int_T^{2T} |\zeta(1/2 + it)|^4 dt = \left(\frac{1}{2\pi^2} + O((\log T)^{-1/2}) \right) T(\log T)^4, \quad (26.51) \quad \boxed{\text{E:zeta4meanEst2}}$$

so we consider the mean square size of the M_i and R_j over the interval $[T, 2T]$.

Since $1 - (1 - n/x)^2 \ll n/x$ for $1 \leq n \leq x$, it follows by Corollary 26.2

that

$$\int_T^{2T} |R_1(t)|^2 \ll \sum_{n \leq x} d(n)^2 n x^{-2} (T+x) \left(\frac{T}{x} + 1\right) \sum_{n \leq x} d(n)^2 \ll (T+x)(\log x)^3 \quad (26.52) \quad \boxed{\text{E:R1Est}}$$

since $\sum_{n \leq x} d(n)^2 \ll x(\log x)^3$ by (2.31).

Let $\mathcal{C}$ be a circle centered at $1/2 - it$ and radius $1/\log x$. Then

$$R_2(t) = \frac{1}{2\pi i} \int_{\mathcal{C}} \frac{2\zeta(1/2 + it + w)^2 x^w}{w(w+1)(w+2)} dw.$$

For $w \in \mathcal{C}$, we see that $|\zeta(1/2 + it + w)| \asymp (\log x)^2$, $|x^w| \asymp x^{1/2}$, and $|w(w+1)(w+2)| \asymp \tau^3$. Hence $R_2(t) \ll x^{1/2} \tau^{-3} (\log x)^3$, and so

$$\int_T^{2T} |R_2(t)|^2 dt \ll \frac{x(\log x)^6}{T^5}. \quad (26.53) \quad \boxed{\text{E:R2Est}}$$

Let $W_\phi(x; t, v) = |\Delta(1/2 + \phi + i(t+v))^2 K(x, \phi + iv)|$. By the Cauchy–Schwarz inequality,

$$\begin{aligned} |R_3(t)|^2 &\leq \int_{-\infty}^{\infty} W_\phi(x; t, v) dv \\ &\quad \times \int_{-\infty}^{\infty} \left| \sum_{n > y} d(n) n^{-1/2 + \phi + i(t+v)} \right|^2 W_\phi(x; t, v) dv. \end{aligned}$$

From Corollary 10.5 we deduce that $\Delta(1/2 + \phi + i(t+v))^2 \ll (|t+v| + 4)^{-2\phi}$. It is also clear that $K(x; \phi + iv) \ll x^\phi (|v| + 4)^{-3}$. Suppose that $T \leq t \leq 2T$. Then the first integral above is $\ll x^\phi T^{-2\phi}$. By Corollary 26.4 we see that

$$\begin{aligned} \int_A^{A+T} \left| \sum_{n > y} d(n) n^{-1/2 + \phi + it} \right|^2 dt &= \sum_{n > y} d(n)^2 n^{-1+2\phi} (T + O(n)) \\ &\ll T y^{2\phi} (\log y)^3 + y^{1+2\phi} (\log y)^3. \end{aligned}$$

uniformly for any real number A . Hence

$$\begin{aligned} \int_T^{2T} |R_3(t)|^2 dt &\ll x^\phi T^{-2\phi} \int_{-\infty}^{\infty} \int_T^{2T} \left| \sum_{n > y} d(n) n^{-1/2 + \phi + i(t+v)} \right|^2 dt W_\phi(x; t, v) dv \\ &\ll (T+y) \left(\frac{T^2}{xy}\right)^{-2\phi} (\log y)^3. \end{aligned} \quad (26.54) \quad \boxed{\text{E:R3Est}}$$

Similarly, by the Cauchy–Schwarz inequality,

$$\begin{aligned} |R_4(t)|^2 &\leq \int_{-\infty}^{\infty} W_{\theta}(x; t, v) dv \\ &\quad \times \int_{-\infty}^{\infty} \left| \sum_{n \leq y} d(n) n^{-1/2+\theta+i(t+v)} \right|^2 W_{\theta}(x; t, v) dv. \end{aligned}$$

The estimates for $\Delta(1/2 + \phi + i(t+v))$ and for $K(x; \phi + iv)$ derived above are still valid when ϕ is replaced by θ , so $W_{\theta}(x; t, v) \ll x^{\theta}(|t+v|+4)^{-2\theta}(|v|+4)^{-3}$, and hence the first integral above is $\ll x^{\theta}T^{-2\theta}$ uniformly for $T \leq t \leq 2T$. By Corollary 26.2 we see that

$$\begin{aligned} \int_A^{A+T} \left| \sum_{n \leq y} d(n) n^{-1/2+\theta+it} \right|^2 &= (T + O(y)) \sum_{n \leq y} d(n)^2 n^{-1+2\theta} \\ &\ll (T+y)y^{2\theta}(\log y)^3 \end{aligned}$$

uniformly in A . Hence

$$\begin{aligned} \int_T^{2T} |R_4(t)|^2 dt &\ll x^{\theta}T^{-2\theta} \int_{-\infty}^{\infty} \int_T^{2T} \left| \sum_{n \leq y} d(n) n^{-1/2+\theta+i(t+v)} \right|^2 dt W_{\theta}(x; t, v) dv \\ &\ll (T+y) \left(\frac{xy}{T^2} \right)^{2\theta} (\log y)^3. \end{aligned} \tag{26.55} \quad \boxed{\text{E:R4Est}}$$

To treat the main terms we first observe that

$$|M_1(t) + M_2(t)|^2 = |M_1(t)|^2 = 2 \operatorname{Re} M_1(t) \overline{M_2(t)} + |M_2(t)|^2.$$

By Corollary 26.4 we see that

$$\int_T^{2T} |M_1(t)|^2 dt = \sum_{n \leq x} \frac{d(n)^2}{n} (T + O(n)) = T \sum_{n \leq x} \frac{d(n)^2}{n} + O(x(\log x)^3).$$

We now show that

$$\sum_{n \leq x} \frac{d(n)^2}{n} = \frac{1}{4\pi^2} (\log x)^4 + O((\log x)^3). \tag{26.56} \quad \boxed{\text{E:MainSumEst}}$$

To see why this is so, we first observe that by computing Euler products it is clear that

$$\sum_{n=1}^{\infty} d(n)^2 n^{-s} = \frac{\zeta(s)^4}{\zeta(2s)}$$

for $\sigma > 1$. Hence by Perron's formula (Theorem 5.1) we see that

$$\sum_{n \leq x} \frac{d(n)^2}{n} = \frac{1}{2\pi i} \int_{\varepsilon - i\infty}^{\varepsilon + i\infty} \frac{\zeta(s+1)^4 x^s}{\zeta(2s+2)s} ds.$$

From the trivial estimate of Corollary 1.17 we know that $\zeta(s) \ll \tau^{1/4}$ for $\sigma \geq 3/4$, $|t| \geq 1$. Hence by moving the contour to the rectilinear path with vertices $\varepsilon - i\infty$, $\varepsilon - iT$, $3/4 - iT$, $3/4 + it$, $\varepsilon + iT$, $\varepsilon + i\infty$ with $T = x^{1/8}$ we see that

$$\sum_{n \leq x} \frac{d(n)^2}{n} = \text{Res} \left[\frac{\zeta(s+1)^4 x^s}{2\pi i \zeta(2s+2)s} \Big|_{s=0} \right] + O(x^{-1/8+2\varepsilon}).$$

This gives (26.56), and so we see that

$$\int_T^{2T} |M_1(t)|^2 dt = \frac{1}{4\pi^2} T(\log x)^4 + O((T+x)(\log x)^3). \quad (26.57) \quad \boxed{\text{E:M1Est}}$$

Since $|\Delta(1/2 + it)| = 1$, it follows similarly that

$$\int_T^{2T} |M_2(t)|^2 dt = \frac{1}{4\pi^2} T(\log y)^4 + O((T+y)(\log y)^3). \quad (26.58) \quad \boxed{\text{E:M2Est}}$$

In (26.54) we see that we want $xy \ll T^2$, while in (26.55) we see that we want $xy \gg T^2$. Also, in our estimates we see that it would be useful to have $x \ll T$ and $y \ll T$. Hence we now take $x = y = cT$ where c is a positive constant. To complete the final estimate, below, it is convenient to take c to be rather small, say $c = 1/8$. With our parameters chosen in this way, we now show that

$$\int_T^{2T} M_1(t) \overline{M_2(t)} dt \ll T(\log T)^2. \quad (26.59) \quad \boxed{\text{E:M1M2Est}}$$

To this end we first note that $\overline{M_2(t)} = M_1(t) \overline{\Delta(1/2 + it)^2}$. As for the second factor, we recall that the functional equation for the zeta function in the symmetric form asserts that

$$\zeta(1/2 + it) \Gamma(1/4 + it/2) \pi^{-it/2} = \zeta(1/2 - it) \Gamma(1/4 - it/2) \pi^{it/2}.$$

We divide both sides by $|\Gamma(1/4 + it/2)| = |\Gamma(1/4 - it/2)|$, and let $Z(t)$ denote the result:

$$\begin{aligned} Z(t) &= \zeta(1/2 + it) \frac{\Gamma(1/4 + it/2) \pi^{-it/2}}{|\Gamma(1/4 + it/2)|} \\ &= \zeta(1/2 - it) \frac{\Gamma(1/4 - it/2) \pi^{it/2}}{|\Gamma(1/4 - it/2)|}. \end{aligned} \quad (26.60) \quad \boxed{\text{E:DefZ}}$$

This is *Hardy's Z-function*, as we defined it in §14.2. Since the second formula above is the complex conjugate of the first, it is clear that $Z(t)$ is a real-valued function of the real variable t . Define $\theta(t)$ so that the cofactor of $\zeta(1/2 + it)$ above is $e^{i\theta(t)}$. Thus $\theta(t)$ is a smooth function of t and $\zeta(1/2 + it) = e^{-2i\theta(t)}\zeta(1/2 - it)$. Hence $\Delta(1/2 + it) = e^{-2i\theta(t)}$, and so $\overline{\Delta(1/2 + it)^2} = e^{4i\theta(t)}$. From our discussion of Stirling's formula in Appendix C we see that

$$\log \Gamma(s) = s \log s - s - \frac{1}{2} \log s + \frac{1}{2} \log(2\pi) + O(1/|s|)$$

uniformly for $|s| \geq 1$ and $|\arg s| \leq \pi - \delta$. With a little calculation, we deduce from this that

$$\theta(t) = \frac{t}{2} \log \frac{t}{2\pi} - \frac{t}{2} - \frac{\pi}{8} + O(1/\tau). \quad (26.61) \quad \boxed{\text{E:thetaEst}}$$

Let $F(t)$ denote the main term above. Then the left hand side of (26.59) is

$$\begin{aligned} &= \int_T^{2T} M_1(t)^2 (e^{4iF(t)} + O(1/t)) dt \\ &= \int_T^{2T} M_1(t)^2 e^{4iF(t)} dt + O\left(\left(\sum_{n \leq T/8} d(n)n^{-1/2}\right)^2\right). \end{aligned} \quad (26.62) \quad \boxed{\text{E:M1M2Est2}}$$

Here the error term is $\ll T(\log T)^2$. The integral above is

$$\sum_{m \leq T/8} \sum_{n \leq T/8} \frac{d(m)d(n)}{\sqrt{mn}} \int_T^{2T} e^{i(4F(t) - t \log mn)} dt.$$

For a given pair m, n let $f(t) = 4F(t) - t \log mn$. Then

$$f'(t) = \log \frac{t^2}{4\pi^2 mn}.$$

This is an increasing function, and on the interval $[T, 2T]$ it is bounded below by

$$f'(T) = \log \frac{T^2}{4\pi^2 mn} \geq \log \frac{16}{\pi^2} \gg 1.$$

By Corollary ?? it follows that the integrals above are uniformly bounded, and hence that the integral in (26.62) is

$$\ll \left(\sum_{n \leq T/8} \frac{d(n)}{n^{1/2}}\right)^2 \ll T(\log T)^2.$$

This completes the proof of (26.59). To obtain the Theorem, it remains

only to combine this with the estimates (26.52), (26.53), (26.54), (26.55), (30.14), and (30.15). $\square$

We now establish q -analogues of Theorem 26.23.

T:LqT4mean

Theorem 26.29. *Let χ denote a character modulo q , $q \geq 2$, and χ^* the primitive character that induces χ . If $T \geq 2$ and $|\sigma - 1/2| \leq 1/(4 \log qT)$, then*

$$\sum_{\chi} \int_0^T |L(\sigma + it, \chi^*)|^4 dt \ll qT(\log qT)^4.$$

Proof We apply Theorem 25.4 with $\delta = 1/4$. Let

$$W(u, r) = \int_0^\infty \exp(-u^2/r)(u^2/r)^{1/8} u^{-1} du.$$

From Theorem 25.4 via Hölder's inequality, we see that

$$\begin{aligned} |L(s, \chi^*)|^4 &\ll \int_0^\infty (|A(s, \chi^*; u)|^4 + |A(1-s, \overline{\chi^*}; u)|^4) W(u, q\tau) du \\ &\quad \times \left(\int_0^\infty W(u, q\tau) du \right)^3. \end{aligned}$$

We have already remarked that $\int_0^\infty W(u, r) du \ll 1$ uniformly in r . We note further that if $0 < r_1 \leq r_2 \leq 4r_1$, then $W(r_1, u) \ll W(r_2, u)$ uniformly for $u > 0$. Since $\tau = |t| + 4$, it follows that if $2 \leq T \leq t \leq 2T$, then $\tau \leq 4T$, so the above is

$$\ll \int_0^\infty (|A(s, \chi^*; u)|^4 + |A(1-s, \overline{\chi^*}; u)|^4) W(u, 4qT) du.$$

Hence

$$\begin{aligned} &\sum_{\chi} \int_T^{2T} |L(\sigma + it, \chi^*)|^4 dt \\ &\ll \int_0^\infty \sum_{\chi} \int_T^{2T} (|A(s, \chi^*)|^4 + |A(1-s, \overline{\chi^*})|^4) dt W(u, 4qT) du. \end{aligned}$$

(26.63) **E:L4momEst1**

We note that

$$A(s, \chi^*; u)^2 = \sum_{k \leq u^2} \frac{\chi^*(k)}{k^s} \sum_{\substack{m, n \leq u \\ mn=k}} 1 = \sum_{k \leq u^2} \frac{\chi^*(k) d(k, u)}{k^s},$$

say. Thus by Theorem 26.16 we see that the expression (26.63) is

$$\ll \int_0^\infty \sum_{k \leq u^2} d(k, u)^2 \left(\frac{1}{k^{2\sigma}} + \frac{1}{k^{2-2\sigma}} \right) (qT + u^2) W(u, 4qT) du.$$

Put $\kappa = 1/\log qT$. Since $0 \leq d(k, u) \leq d(k)$ for all k , and $\sum_{k \leq y} d(k)^2/k \ll (\log 2y)^4$ for $y \geq 1$, the above is

$$\ll \int_1^\infty (\log 2u)^4 u^{4\kappa} (qT + u^2) W(u, 4qT) du.$$

By means of the change of variable $v = u^2/r$ we see that if $r \geq 1$ and $\alpha \geq 0$, then

$$\begin{aligned} \int_0^\infty (\log 2u)^4 u^\alpha W(u, r) du &\ll \int_0^\infty (\log 2rv)^4 (rv)^{\alpha/2} v^{1/8} e^{-v} \frac{dv}{v} \\ &\ll (\log 2r)^4 r^{\alpha/2} \int_0^\infty v^{\alpha/2+1/8} e^{-v} \frac{dv}{v} + r^{\alpha/2} \int_0^\infty (\log v)^4 v^{\alpha/2+1/8} e^{-v} \frac{dv}{v} \\ &= (\log 2r)^4 r^{\alpha/2} \Gamma(\alpha/2 + 1/8) + r^{\alpha/2} \Gamma^{(4)}(\alpha/2 + 1/8) \ll r^{\alpha/2} (\log 2r)^4 \end{aligned}$$

for bounded values of α . We apply this with $\alpha = 4\kappa$ and with $\alpha = 2 + 4\kappa$, and conclude that

$$\sum_\chi \int_T^{2T} |L(\sigma + it, \chi^*)|^4 dt \ll qT (\log 2qT)^4$$

for $T \geq 2$. We take $T = 2, 4, 8, \dots$ and sum to deduce that

$$\int_2^T \sum_\chi |L(\sigma + it, \chi^*)|^4 dt \ll qT (\log 2qT)^4.$$

It remains to treat the interval $0 \leq t \leq 2$. Our method applies to any interval in which τ changes by at most a bounded factor. When $t = 0$ we have $\tau = 4$, and when $t = 2$ we have $\tau = 6$. Thus our method, when applied to the interval $[0, 2]$ yields the estimate

$$\int_0^2 \sum_\chi |L(\sigma + it, \chi^*)|^4 dt \ll q (\log 2q)^4.$$

This completes the proof. $\square$

Cor: 2mqTL*L'

Corollary 26.30. *Let χ denote a character modulo q , $q \geq 2$, and χ^* the primitive character that induces χ . If $T \geq 2$, then*

$$\int_0^T \sum_\chi |L(1/2 + it, \chi^*) L'(1/2 + it, \chi^*)|^2 dt \ll qT (\log qT)^6.$$

Proof This follows from Theorem 26.29 by the same method that we used to derive Corollary 26.25 from Theorem 26.23. $\square$

Cor:discqT4mv

Corollary 26.31. *Suppose that $\delta > 0$, that $T \geq 2$, and that for each character χ modulo q we have numbers $t_{j,\chi}$, $j = 1, 2, \dots, J_\chi$ in the interval $[\delta/2, T - \delta/2]$ such that $|t_{j,\chi} - t_{k,\chi}| \geq \delta$ when $j \neq k$. Then*

$$\sum_{\chi} \sum_{j=1}^{J_\chi} |L(1/2 + it_{j,\chi}, \chi^*)|^4 \ll (1/\delta + \log qT)qT(\log qT)^4.$$

Proof This follows from Theorem 26.29 and Corollary 26.31 by the same method that we used to derive Corollary 26.26 from Theorem 26.23 and Corollary 26.25. $\square$

T:LQ2T4mean

Theorem 26.32. *Suppose that $Q \geq 2$, $T \geq 2$, and that $|\sigma - 1/2| \leq 1/(4 \log QT)$. Then*

$$\sum_{q \leq Q} \sum_{\chi}^* \int_0^T |L(\sigma + it, \chi)|^4 \ll Q^2 T (\log QT)^4.$$

Proof We show first that

$$\sum_{Q/2 < q \leq Q} \sum_{\chi}^* \int_{T/2}^T |L(\sigma + it, \chi)|^4 \ll Q^2 T (\log QT)^4,$$

and then sum over dyadic blocks. The above is proved by the same method used to prove Theorem 26.29, but with an appeal to (26.22) instead of (26.21). $\square$

Cor:2mQ2TL*L'

Corollary 26.33. *Suppose that $Q \geq 2$ and that $T \geq 2$. Then*

$$\sum_{q \leq Q} \sum_{\chi}^* \int_0^T |L(1/2 + it, \chi) L'(1/2 + it, \chi)|^2 \ll Q^2 T (\log QT)^6.$$

discQ2T4mv

Corollary 26.34. *Suppose that $\delta > 0$, that $Q \geq 2$, $T \geq 2$, and that for each primitive character χ modulo q with $q \leq Q$ we have numbers $t_{j,\chi}$, $j = 1, 2, \dots, J_\chi$ in the interval $[\delta/2, T - \delta/2]$ such that $|t_{j,\chi} - t_{k,\chi}| \geq \delta$ when $j \neq k$. Then*

$$\sum_{q \leq Q} \sum_{\chi}^* \sum_{j=1}^{J_\chi} |L(1/2 + it_{j,\chi}, \chi)|^4 \ll (1/\delta + \log QT)Q^2 T (\log QT)^4.$$

26.5.1 Exercises

1. By arguing in the same style that we used to derive (26.29), show that if $\alpha > 1$, $\beta > 1$, and μ and ν are nonnegative integers, then

$$\lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T \zeta^{(\mu)}(\alpha + it) \zeta^{(\nu)}(\beta - it) dt = \zeta^{(\mu+\nu)}(\alpha + \beta).$$

Exer:Lptriineq

2. Let p be a real number, $1 \leq p < \infty$, and let

$$\|f\|_p = \left(\int_a^b |f(x)|^p dx \right)^{1/p}$$

denote the L^p norm of a function f on the interval $[a, b]$. We recall that such norms have a triangle inequality: $\|f + g\|_p \leq \|f\|_p + \|g\|_p$.

- (a) Suppose that $f = f_1 + f_2$, and that $\|f_2\|_p < \|f_1\|_p$. Show that

$$\|f_1\|_p \left(1 - \frac{\|f_2\|_p}{\|f_1\|_p} \right) \leq \|f\|_p \leq \|f_1\|_p \left(1 + \frac{\|f_2\|_p}{\|f_1\|_p} \right).$$

- (b) Deduce in particular that

$$\int_a^b |f(x)|^2 dx = \int_a^b |f_1(x)|^2 dx \left(1 + O\left(\frac{\|f_2\|_2}{\|f_1\|_2} \right) \right).$$

- (c) In the context of Theorem 26.19, by choosing the parameter x appropriately, show that if $1/2 \leq \alpha < \sigma \leq 1$, then

$$\int_0^T |\zeta(\sigma + it)|^{2k} dt = T \left(\sum_{n=1}^{\infty} \frac{d_k(n)^2}{n^{2\sigma}} \right) \left(1 + O(T^{-\frac{\sigma-\alpha}{2-\alpha-\sigma} + \epsilon}) \right).$$

3. Suppose that $-1/\log T \leq \alpha \leq 1/\log T$, that $-1/\log T \leq \beta \leq 1/\log T$, and that $-1 \leq \delta \leq 1$.

- (a) Show that

$$\begin{aligned} \int_0^T \zeta(1/2 + \alpha + it) \overline{\zeta(1/2 + \beta + i\delta + it)} dt \\ = T \int_1^T u^{-1-\alpha-\beta+i\delta} du + O(T). \end{aligned}$$

- (b) Show that

$$\begin{aligned} \int_0^T \zeta'(1/2 + \alpha + it) \overline{\zeta(1/2 + \beta + i\delta + it)} dt \\ = -T \int_1^T u^{-1-\alpha-\beta+i\delta} \log u du + O(T \log T). \end{aligned}$$

4. (a) By mimicking the proof of Theorem 26.23, show that if $k \geq 2$ is an integer and $1/2 \leq \sigma \leq 1$, then

$$\int_0^T |\zeta(\sigma + it)|^{2k} dt \ll T + T^{k(1-\sigma)+\varepsilon}.$$

- (b) Deduce that the relation (26.29) holds for $\sigma > 1 - 1/k$.
 (c) Show that

$$\lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T |\zeta(1 + it)|^{2k} dt = \sum_{n=1}^{\infty} \frac{d_k(n)^2}{n^2}$$

for all positive integers k .

Exer:pNormIneq

5. Let p be a real number, $1 \leq p < \infty$.
 (a) Suppose that $w \in L^1(\mathbb{R})$, and that $f \in L^p(\mathbb{R})$. Put $F(x) = (w * f)(x) = \int_{-\infty}^{\infty} w(u)f(x-u) du$. Show that $\|F\|_p \leq \|w\|_1 \|f\|_p$.
 (b) Let $\mathbb{T} = \mathbb{R}/\mathbb{Z}$ denote the circle group. Suppose that $w \in L^1(\mathbb{T})$, and that $f \in L^p(\mathbb{T})$. Put $F(x) = (w * f)(x) = \int_0^1 w(u)f(x-u) du$. Show that $\|F\|_p \leq \|w\|_1 \|f\|_p$.
 6. (Balasubramanian & Ramachandra 1990) Show that if k is a positive integer, then

$$\int_0^T |\zeta(1/2 + it)|^{2k} dt \gg_k T(\log T)^{k^2}$$

for $T \geq 2$.

7. Suppose that k is a positive integer, that $1/2 < \sigma < 1$, and that $\delta > 0$. Show that if $H \geq T^\delta$, then

$$\int_T^{T+H} |\zeta(\sigma + it)|^{2k} dt \geq (1 + o(1))H \sum_{n=1}^{\infty} \frac{d_k(n)^2}{n^{2\sigma}}.$$

8. Show that

$$\int_0^T |\zeta(1/2 + it)\zeta'(1/2 + it)|^2 dt \gg T(\log T)^6$$

for $T \geq 2$.

27

Large Values of Dirichlet Polynomials

C:LargeVals

Let $m(V) = \text{meas}\{x \in [a, b] : |f(x)| \geq V\}$ where f is a measurable function defined on $[a, b]$ and $V > 0$. Then

$$V^p m(V) = \int_{\substack{a \leq x \leq b \\ |f(x)| \geq V}} V^p dx \leq \int_a^b |f(x)|^p dx = \|f\|_p^p.$$

Hence

$$m(V) \leq \frac{\|f\|_p^p}{V^p}.$$

For a general function this is all that we can say about the measure of the set on which it is large, based on its L^p norm. If we have bounds for $\|f\|_p$ and for $\|f\|_q$, then

$$m(V) \leq \min \left(\frac{\|f\|_p^p}{V^p}, \frac{\|f\|_q^q}{V^q} \right),$$

and there is still not much more that we can say, even if f is analytic. However, if f is a Dirichlet polynomial, then there is much more that we can say about its large values beyond what follows from mean value estimates.

Suppose that

$$D(s) = \sum_{n=1}^N a_n n^{-s}, \tag{27.1} \quad \text{E:DefD}(s)$$

that $t_r \in [0, T]$ for $r = 1, 2, \dots, R$, that $|t_{r_1} - t_{r_2}| \geq 1$ for $r_1 \neq r_2$. Let Δ have the property that

$$\sum_{r=1}^R \left| \sum_{n=1}^N a_n n^{-it_r} \right|^2 \leq \Delta^2 \sum_{n=1}^N |a_n|^2 \tag{27.2} \quad \text{E:HM-1}$$

for all choices of the a_n . This is a bilinear form inequality, so by the duality theorem (Theorem ??), Δ has the equivalent property that

$$\sum_{n=1}^N \left| \sum_{r=1}^R y_r n^{-it_r} \right|^2 \leq \Delta^2 \sum_{r=1}^R |y_r|^2 \quad (27.3) \quad \boxed{\text{E:HMO}}$$

for all choices of the y_r . We expand the square on the left hand side above and take the sum over n inside, to see that it is

$$= \sum_{1 \leq r_1, r_2 \leq R} y_{r_1} \overline{y_{r_2}} \sum_{n=1}^N n^{i(t_{r_2} - t_{r_1})}. \quad (27.4) \quad \boxed{\text{E:HMEst0}}$$

Now $|y_{r_1} \overline{y_{r_2}}| \leq \frac{1}{2} |y_{r_1}|^2 + \frac{1}{2} |y_{r_2}|^2$ by the geometric–arithmetic mean inequality, so the above is

$$\leq \sum_{r_1=1}^R |y_{r_1}|^2 \sum_{r_2=1}^R \left| \sum_{n=1}^N n^{i(t_{r_2} - t_{r_1})} \right|. \quad (27.5) \quad \boxed{\text{E:HMEst1}}$$

Thus

$$\Delta^2 \leq \max_{1 \leq r_1 \leq R} \sum_{r_2=1}^R \left| \sum_{n=1}^N n^{i(t_{r_2} - t_{r_1})} \right|.$$

Our argument here is reminiscent of one of our approaches to the large sieve, where we found that it was fruitful to introduce weights. We can introduce weights here, also. Suppose that $w_n \geq 0$ for all n , that $w_n \geq 1$ for $1 \leq n \leq N$, and that $\sum_{n=1}^{\infty} w_n < \infty$. Then the left hand side of (27.3) is

$$\leq \sum_{n=1}^{\infty} w_n \left| \sum_{r=1}^R y_r n^{-it_r} \right|^2.$$

On continuing as above, we find that

$$\Delta^2 \leq \max_{1 \leq r_1 \leq R} \sum_{r_2=1}^R |W(i(t_{r_1} - t_{r_2}))| \quad (27.6) \quad \boxed{\text{E:HM1}}$$

where $W(s) = \sum_{n=1}^{\infty} w_n n^{-s}$ for $\sigma \geq 0$. Suppose that we take $w_n = 2 - n/N$ for $1 \leq n \leq 2N$, and $w_n = 0$ for $n > 2N$. Then by the formula (5.19) for the inverse Mellin transform with Cesàro weights we see that

$$W(it) = \frac{1}{\pi i} \int_{2-i\infty}^{2+i\infty} \zeta(z+it) \frac{(2N)^z}{z(z+1)} dz.$$

On moving the path of integration to the abscissa $\operatorname{Re} z = 1/2$, we see that the above is

$$= \frac{1}{\pi} \int_{-\infty}^{\infty} \zeta(1/2 + i(t+y)) \frac{(2N)^{1/2+iy}}{(1/2+iy)(3/2+iy)} dy \\ + \frac{2(2N)^{1-it}}{(1-it)(2-it)}.$$

If the Lindelöf Hypothesis (LH) is true, then the above is

$$\ll C(\varepsilon) N^{1/2} \tau^\varepsilon + \frac{N}{\tau^2}.$$

Hence

$$\sum_{r_2=1}^R |W(i(t_{r_1} - t_{r_2}))| \ll N + C(\varepsilon) N^{1/2} T^\varepsilon R,$$

for all r_1 , so

$$\sum_{r=1}^R |D(it_r)|^2 \ll (N + C_1(\varepsilon) N^{1/2} T^\varepsilon R) \sum_{n=1}^N |a_n|^2. \quad (27.7) \quad \boxed{\text{E:HalaszLH}}$$

Suppose that V is a number such that $|D(it_r)| \geq V$ for all r . Then

$$V^2 R \ll N \sum_{n=1}^N |a_n|^2 + C_1(\varepsilon) N^{1/2} T^\varepsilon R \sum_{n=1}^N |a_n|^2. \quad (27.8) \quad \boxed{\text{E:V}^2\text{JEst}}$$

Hence there is a constant $C_2(\varepsilon)$ with the property that if

$$V^2 \geq C_2(\varepsilon) N^{1/2} T^\varepsilon \sum_{n=1}^N |a_n|^2, \quad (27.9) \quad \boxed{\text{E:V1b}}$$

then the second term on the right hand side of (27.8) does not majorize the left hand side, and hence $V^2 R \ll N \sum_{n=1}^N |a_n|^2$, which is to say that we have proved

T:HalaszLH **Theorem 27.1.** *Assume the Lindelöf Hypothesis. Let $D(s)$ be a Dirichlet polynomial as in (27.1), and $\mathcal{T}$ be a set of R real numbers in the interval $[0, T]$ such that $|t - t'| \geq 1$ whenever $t, t' \in \mathcal{T}$ and $t \neq t'$. There is a constant $C_2(\varepsilon)$ such that if V satisfies (27.9), then*

$$R \ll \frac{NG}{V^2} \quad (27.10) \quad \boxed{\text{E:HEst}}$$

where

$$G = \sum_{n=1}^N |a_n|^2. \quad (27.11) \quad \boxed{\text{E:DefG}}$$

Suppose that $|a_n| \asymp 1$ for all n . Then $\sup |D(it)| \ll N$, the asymptotic root-mean-value of $|D(it)|$ is $\asymp N^{1/2}$, and from the above we see that if V is a little larger than $T^\varepsilon N^{3/4}$, then

$$R \ll \frac{N^2}{V^2}. \quad (27.12) \quad \boxed{\text{E:HMEst}}$$

From Theorem 26.9 we also know that

$$R \ll \frac{(T+N)N \log N}{V^2}.$$

If N is small compared with T , say of the size of a small fractional power of T , then the first upper bound above is enormously better than the second. Of course the first depends on LH and applies only to large values of V , while the second is unconditional and is valid for all $V > 0$.

In passing from (27.4) to (27.5) we used the triangle inequality, and we would expect that in doing so a great deal of cancellation has been lost. This would suggest that something substantially stronger than (27.7) should hold. But this is false! The surprise here is that (27.7) is within a factor T^ε of being best possible. We see this from the following

Exam:Bourgain

Example 27.1. (Bourgain) Suppose that $\delta > 0$ is a small positive absolute constant, that $H \leq \delta N^{1/2}$, that

$$D(s) = \sum_{N-H < n \leq N} n^{-s},$$

and that $\mathcal{T} = \mathcal{T}_\delta \{a + 2\pi Nb : a, b \in \mathbb{Z}, -A \leq a \leq A, -B \leq b \leq B\}$ where $A = \delta N/H$, $B = \delta N/H^2$. If $N-H < n \leq N$, then

$$n^{-it} = N^{-it} \exp\left(-it \log \frac{n}{N}\right) = N^{-it} \exp\left(-it \log\left(1 - \frac{N-n}{N}\right)\right).$$

For $0 \leq u \leq 1/2$ let r be defined by the equation $-\log(1-u) = u + r$. Then r is real and $r \ll u^2$, so that if $u = (N-n)/N$ we see that the above is

$$= N^{-it} e^{itu} e^{itr} = N^{-it} \exp\left(it \frac{N-n}{N}\right) \left(1 + O(|t|(N-n)^2/N^2)\right).$$

If $t = a + 2\pi Nb \in \mathcal{T}_\delta$, then the above is

$$\begin{aligned} &= N^{-it} \exp(ia(N-n)/N) \exp(2\pi ib(N-n)) \left(1 + O((A+NB)H^2/N^2)\right) \\ &= N^{-it} (1 + O(AH/N)) (1 + O((A+NB)H^2/N^2)). \end{aligned}$$

In view of the definitions of A and of B , the above is

$$= N^{-it} (1 + O(\delta)).$$

We sum over n to see that if $t \in \mathcal{T}$, then $D(it) = HN^{-it}(1 + O(\delta))$. If we apply (27.7) to the points $\mathcal{T}$, then $R = \text{card } \mathcal{T}$, the left hand side is of the order RH^2 , while the right hand side is comparable to $N^{1/2}T^\varepsilon RH$. If $H \sim \delta N^{1/2}$, then the right hand side is larger than the left by T^ε . To prove Theorem 27.1 we introduced weights, but the two sides of the resulting relation are essentially unchanged.

Before Bourgain invented his Example, it was conjectured that $\Delta^2 \ll N + RT^\varepsilon$, but this is false because it is much stronger than (27.7), which we now recognize is essentially best possible.

The Dirichlet polynomial in Bourgain's Example is just a short sum, and is therefore not typical of the Dirichlet polynomials that we most often encounter. We avoid Bourgain's Example by proposing a bound in terms of the maximum of the coefficients instead of their mean square size.

Conj:Halaszlinfty

Conjecture 27.2. *Let $D(s)$ be a Dirichlet polynomial as in (27.1), and suppose that $|a_n| \leq 1$ for all n . Let $\mathcal{T}$ be a set of R real numbers in the interval $[0, T]$ such that $|t - t'| \geq 1$ whenever $t, t' \in \mathcal{T}$ and $t \neq t'$. Then*

$$\sum_{t \in \mathcal{T}} |D(it)|^2 \ll_\varepsilon (N + R)N^{1+\varepsilon}.$$

If $|D(it)| \geq V$ for all $t \in \mathcal{T}$, and $V > N^{1/2+\varepsilon}$, then it follows from the above that

$$R \ll_\varepsilon \frac{N^{2+\varepsilon}}{V^2}. \quad (27.13) \quad \text{E:HMConj}$$

To obtain unconditional results using our new ideas, we could simply replace the appeal to LH by one to a known bound for $|\zeta(1/2 + it)|$, such as the one found in Theorem ???. One could also proceed along these lines using a different abscissa instead of $1/2$, as discussed in Exercise 2. For larger values of N , say $T^{1/2} \leq N \leq T$, a further approach works well: From Theorem ??? we see that

$$\sum_{n \leq x} n^{-it} = \frac{x^{1-it}}{1-it} + O(\tau^{1/2} \log \tau).$$

We integrate this over $0 \leq x \leq N$, and then divide by N to see that

$$\begin{aligned} \sum_{n=1}^N (1 - n/N)n^{-it} &= \frac{N^{1-it}}{(1-it)(2-it)} + O(\tau^{1/2} \log \tau) \\ &\ll \tau^{1/2} \log \tau + \frac{N}{\tau^2}. \end{aligned} \quad (27.14) \quad \text{E:whitedn-it}$$

By replacing N by $2N$ we obtain the same weights that we already used, and we see that if the numbers t_r are well-spaced in the interval $[0, T]$, then

$$\sum_{r_2=1}^R |W(i(t_{r_1} - t_{r_2}))| \ll N + RT^{1/2} \log T.$$

In view of (27.6), this gives

T:HM2 **Theorem 27.3.** *Let $D(s)$ be a Dirichlet polynomial as in (27.1), let $\mathcal{T}$ be a set of R real numbers in the interval $[0, T]$ such that $|t - t'| \geq 1$ whenever $t, t' \in \mathcal{T}$ and $t \neq t'$, and let G be defined as in (27.11). Then*

$$\sum_{t \in \mathcal{T}} |D(it)|^2 \ll (N + RT^{1/2} \log T)G. \quad (27.15) \quad \text{E:HM2}$$

From the above we see that there is an absolute constant C such that if $|D(it)| \geq V$ for all $t \in \mathcal{T}$ and

$$V^2 \geq CT^{1/2}(\log T)G, \quad (27.16) \quad \text{E:V1b2}$$

then we have (27.10). At first sight it would seem that for smaller V we have no bound, but by exercising a little care we obtain the following comprehensive result.

T:HM3 **Theorem 27.4.** *Let $D(s)$ be a Dirichlet polynomial as in (27.1), let $t_1, t_2, \dots, t_R$ be real numbers in the interval $[0, T]$ such that $|t_{r_1} - t_{r_2}| \geq 1$ whenever $r_1 \neq r_2$, and let G be defined by (27.11). If $|D(it_r)| \geq V > 0$ for all r , then*

$$R \ll \frac{NG}{V^2} + \frac{NTG^3(\log T)^2}{V^6}. \quad (27.17) \quad \text{E:HM3}$$

Proof If the condition (27.16) holds, then there is nothing to prove. Thus we may assume that (27.16) fails. Let T_1 be the number for which $V^2 = CT_1^{1/2}(\log T_1)G$ where C is the constant in (27.16). Thus $T_1 < T$. We divide the interval $[0, T]$ into $\lceil T/T_1 \rceil$ intervals, each one of length $T/\lceil T/T_1 \rceil \leq T_1$. Thus Theorem ^{T:HM2} applies to each subinterval. Since $\lceil T/T_1 \rceil \asymp T/T_1$, it follows that the total number of points t , summed over all subintervals, is

$$R \ll \left(1 + \frac{T}{T_1}\right) \frac{NG}{V^2} = \frac{NG}{V^2} + \frac{TNG}{T_1 V^2}.$$

Since $T_1(\log T_1)^2 \asymp V^4/G^2$, it follows that the second term on the right above is

$$\frac{TNG(\log T_1)^2}{T_1(\log T_1)^2 V^2} \asymp \frac{TNG^3(\log T_1)^2}{V^6} \leq \frac{TNG^3(\log T)^2}{V^6}$$

since $T_1 \leq T$. This gives the result. $\square$

The bound $R \leq T$ is trivial, and we expect that $|D(it)| \ll G^{1/2}$ a positive proportion of t . Thus we expect to have a nontrivial bound for R only when V^2 is somewhat larger than G . On combining the estimates derived from Theorems 26.8 and 27.4, we find the following:

$$R \ll \begin{cases} T & (V^2 \leq G \log N), \\ \frac{TG \log N}{V^2} & (G \log N \leq V^2 \leq N^{1/2} G (\log T)^{1/2}), \\ \frac{TNG^3 (\log T)^2}{V^6} & (N^{1/2} G (\log T)^{1/2} \leq V^2 \leq T^{1/2} \log T), \\ \frac{NG}{V^2} & (V^2 \geq T^{1/2} G \log T). \end{cases}$$

We now extend our discussion in two directions. First, we replace n^{-it} by $\chi(n)$, or by $\chi(n)n^{-it}$. Secondly, in applications it frequently happens that we want to sample not just at points it on the imaginary axis, but at points $s = \sigma + it$ with well-spaced t and $\sigma \geq 0$. These extensions are easily obtained from the following useful

L:weightedhybridPV

Lemma 27.5. *Suppose that $s = \sigma + it$ with $\sigma \geq 0$, and that χ is a character modulo q . Then*

$$\sum_{n=1}^N \left(1 - \frac{n}{N}\right) \chi(n) n^{-s} \ll (q\tau)^{1/2} \log q\tau + E_0(\chi) \frac{N}{\tau^2} \quad (27.18) \quad \text{E:weightedhybridPV}$$

where $E_0(\chi) = 1$ if $\chi = \chi_0$, and $E_0(\chi) = 0$ otherwise.

Proof We recall that Theorem ?? asserts that

$$\sum_{n \leq u} \chi(n) n^{-it} = E_0(\chi) \frac{\varphi(q)}{q} \cdot \frac{u^{1-it}}{1-it} + O((q\tau)^{1/2} \log q\tau).$$

We integrate both sides of this over the interval $0 \leq u \leq x$, and then divide both sides by x to see that

$$\begin{aligned} \sum_{n \leq x} \left(1 - \frac{n}{x}\right) \chi(n) n^{-it} &= E_0(\chi) \frac{\varphi(q) x^{1-it}}{q(1-it)(2-it)} + O((q\tau)^{1/2} \log q\tau) \\ &\ll E_0(\chi) \frac{x}{\tau^2} + (q\tau)^{1/2} \log q\tau. \end{aligned} \quad (27.19) \quad \text{E:qthybridEst}$$

By restricting the real parameter x to the integer value N , we obtain the desired result when $\sigma = 0$. To extend the above to allow $\sigma > 0$, we express the weight $\max(0, 1 - n/N)n^{-\sigma}$ as a nonnegative linear combination of the weights $\max(0, 1 - n/x)$ with $0 \leq x \leq N$. Specifically, we

show that if $a_1, a_2, \dots$ is a sequence of real or complex numbers and

$$A(x) = \sum_{1 \leq n \leq x} a_n, \quad B(x) = \sum_{1 \leq n \leq x} (x - n)a_n,$$

then

$$\begin{aligned} \sum_{1 \leq n \leq x} (x - n)a_n n^{-\sigma} &= \frac{B(x)}{x^\sigma} + 2\sigma \int_0^x B(v)v^{-\sigma-1} dv \\ &\quad + \sigma(\sigma + 1) \int_0^x B(v)(x - v)v^{-\sigma-2} dv \end{aligned} \quad (27.20) \quad \boxed{\text{E:lcwhts}}$$

for $\sigma \geq 0$. To this end we first note that

$$\sum_{1 \leq n \leq x} (x - n)a_n n^{-\sigma} = \int_0^x \sum_{1 \leq n \leq u} a_n n^{-\sigma} du. \quad (27.21) \quad \boxed{\text{E:IntA=B}}$$

By Riemann–Stieltjes integration by parts (as in the proof of Theorem 1.3) we see that the integrand above is

$$= \frac{A(u)}{u^\sigma} + \sigma \int_1^u A(v)v^{-\sigma-1} dv.$$

Hence the right hand side of (27.21) is

$$= \int_1^x A(u)u^{-\sigma} du + \sigma \int_1^x A(v)(x - v)v^{-\sigma-1} dv.$$

From (27.21) with $\sigma = 0$ we see that $B(x) = \int_0^x A(u) du$. We integrate the two integrals above by parts (integrating A and differentiating the rest) to obtain (27.20). We now take $a_n = \chi(n)n^{-it}$ in (27.20). From (27.19) we deduce that

$$\sum_{n \leq x} (x - n)\chi(n)n^{-s} = W_1(x)(q\tau)^{1/2} \log q\tau + W_2(x)E_0(\chi)\tau^{-2}$$

where

$$W_j(x) = x^{j-\sigma} + 2\sigma \int_1^x v^{j-\sigma-1} dv + \sigma(\sigma + 1) \int_1^x (x - v)v^{j-\sigma-2} dv.$$

We need to show that $W_j(x) \ll x^j$ for $j = 1, 2$. The bound (27.18) is trivial if $\sigma \geq 2$, so we may suppose that $0 \leq \sigma \leq 2$. For $j = 1$ we see that

$$W_1(x) \ll x + \int_1^x 1 dv + \sigma \int_1^x (x - v)v^{-\sigma-1} dv.$$

It is necessary to estimate the last term above only when $\sigma > 0$, and for such σ we have that it is

$$\leq \sigma x \int_1^x v^{-\sigma-1} dv = x(1 - x^{-\sigma}) \ll x.$$

Finally,

$$W_2(x) \ll x^2 + \int_1^x v dv + x \int_1^x 1 dv \ll x^2.$$

□

In the same way that we derived theorem 27.3 from (27.14), the following more general results are immediate.

T:HMqT **Theorem 27.6.** *Suppose that for $r = 1, 2, \dots, R$ we have a character χ_r modulo q and a point $s_r = \sigma_r + it_r$ with $\sigma_r \geq 0$, $0 \leq t_r \leq T$, and with the further property that $|t_{r_1} - t_{r_2}| \geq 1$ if $\chi_{r_1} = \chi_{r_2}$. Let*

$$D(s, \chi) = \sum_{n=1}^N a_n \chi(n) n^{-s}, \quad (27.22) \quad \text{E:DefD}(s, \text{chi})$$

and let G be defined as in (27.11). Then

$$\sum_{r=1}^R |D(s_r, \chi_r)|^2 \ll (N + R(qT)^{1/2} \log qT) G.$$

If $t_r = 0$ for all r , then we find that

$$\sum_{r=1}^R \left| \sum_{n=1}^N a_n \chi_r(n) \right|^2 \ll (N + Rq^{1/2} \log q) G,$$

which is an exact q -analogue of (27.15).

T:HMqT **Theorem 27.7.** *Suppose that for $r = 1, 2, \dots, R$ we have a primitive character χ_r modulo q_r with $1 \leq q_r \leq Q$ and a point $s_r = \sigma_r + it_r$ with $\sigma_r \geq 0$, $0 \leq t_r \leq T$, and with the further property that $|t_{r_1} - t_{r_2}| \geq 1$ if $\chi_{r_1} = \chi_{r_2}$. Let $D(s, \chi)$ be defined as in (27.22), and let G be defined as in (27.11). Then*

$$\sum_{r=1}^R |D(s_r, \chi_r)|^2 \ll (N + RQT^{1/2} \log QT) G.$$

The device by which we derived Theorem 27.4 from Theorem 27.3 was invented by M. N. Huxley, and is known as ‘*Huxley’s Trick*’. It seems that there is no way to similarly partition characters into useful subsets,

so we lack a q analogue of Theorem 27.4. However, to address this issue, Huxley introduced further ideas, now known as the *Huxley Reflection Method*, which we now pursue. Roughly speaking, the idea is to start a sum of length N , and then use the functional equations of L -functions to create a corresponding sum of length Q^2T/N . The product of these two sums will be of length Q^2T , and hence the mean square can be efficiently estimated by means of Theorem 26.17.

Suppose that χ is a primitive character modulo q . The functional equation of $L(s, \chi)$ (cf Corollary 10.8) asserts that

$$L(s, \chi) \Gamma\left(\frac{s+\kappa}{2}\right) \left(\frac{q}{\pi}\right)^{\frac{s}{2}} = \varepsilon(\chi) L(1-s, \bar{\chi}) \Gamma\left(\frac{1-s+\kappa}{2}\right) \left(\frac{q}{\pi}\right)^{\frac{1-s}{2}}$$

where $\kappa = (1 - \chi(-1))/2$ and $\varepsilon(\chi) = \tau(\chi)/(i^\kappa \sqrt{q})$. Hence

$$L(s, \chi) = \varepsilon(\chi) \left(\frac{q}{\pi}\right)^{\frac{1}{2}-s} \frac{\Gamma\left(\frac{1-s+\kappa}{2}\right)}{\Gamma\left(\frac{s+\kappa}{2}\right)} L(1-s, \bar{\chi}).$$

Now suppose that χ is a character modulo q , but not necessarily primitive. Then χ is induced by a primitive character χ^* where χ^* is a character modulo d for some $d|q$, and

$$L(s, \chi) = L(s, \chi^*) \prod_{p|q} \left(1 - \frac{\chi^*(p)}{p^s}\right).$$

Thus

$$L(s, \chi) = \varepsilon(\chi) L(1-s, \bar{\chi}) \gamma(s, \chi) P(s, \chi) \quad (27.23) \quad \boxed{\text{E:asymFE}}$$

where

$$\gamma(s, \chi) = \pi^{s-1/2} \frac{\Gamma\left(\frac{1-s+\kappa}{2}\right)}{\Gamma\left(\frac{s+\kappa}{2}\right)} = (2\pi)^s \Gamma(1-s) \sin \frac{\pi}{2}(s+\kappa), \quad (27.24) \quad \boxed{\text{E:Defgam(s)}}$$

$$P(s, \chi) = d^{\frac{1}{2}-s} \prod_{p|q} \frac{1 - \frac{\chi^*(p)}{p^s}}{1 - \frac{\chi^*(p)}{p^{1-s}}}. \quad (27.25) \quad \boxed{\text{E:DefP(s)}}$$

If $\kappa = 0$, then $\gamma(s, \chi)$ has simple zeros at $0, -2, -4, \dots$ and simple poles at $1, 3, 5, \dots$. If $\kappa = 1$, then $\gamma(s, \chi)$ has simple zeros at $-1, -3, -5, \dots$ and simple poles at $2, 4, 6, \dots$. Thus in either case $\gamma(s, \chi)$ is analytic for $\sigma < 1$. In the product that defines $P(s, \chi)$ we may restrict p to those $p|q$ such that $p \nmid d$, for if $p|d$ then $\chi^*(p) = 0$. Suppose that $p|q$ and $p \nmid d$.

Then $\chi^*(p)$ is a root of unity. Choose θ so that $\chi^*(p) = e^{i\theta}$. Then

$$\frac{1 - \frac{\chi^*(p)}{p^s}}{1 - \frac{\chi^*(p)}{p^{1-s}}} \quad (27.26) \quad \boxed{\text{E:pfactor}}$$

has simple zeros on the imaginary axis and simple poles on the 1-line at the points

$$\frac{i(\theta + 2k\pi)}{\log p}, \quad 1 + \frac{i(\theta + 2k\pi)}{\log p}, \quad (k \in \mathbb{Z})$$

respectively. Thus $P(s, \chi)$ is analytic for $\sigma < 1$. We are especially interested in the size of $\gamma(s, \chi)$ and of $P(s, \chi)$ when $\sigma \leq 1/2$. In Chapter 10 we observed that $|\gamma(s, \chi_0)| \asymp \tau^{1/2-\sigma}$ uniformly for $|t| \geq 1$, $-A \leq \sigma \leq A$. We now refine this.

L:gam(s)est

Lemma 27.8. *Let $\gamma(s, \chi)$ be defined as in (27.24). Then $|\gamma(1/2+it, \chi)| = 1$ for all real t ,*

$$\gamma(s, \chi) \ll \left(\frac{|1-s|}{2\pi} \right)^{1/2-\sigma} \quad (27.27) \quad \boxed{\text{E:gam(s)est1}}$$

uniformly for all s in the halfplane $\sigma \leq 1/2$, and

$$\gamma(s, \chi) \ll \tau^{1/2-\sigma} \quad (27.28) \quad \boxed{\text{E:gam(s)est2}}$$

when $-\tau^{2/3} \leq \sigma \leq 1/2$.

Proof The first assertion follows from the first formula for $\gamma(s, \chi)$ in (27.24), since the denominator of the fraction is the complex conjugate of the numerator, when $\sigma = 1/2$. For the two estimates we may assume that $t \geq 0$, since $\gamma(\bar{s}, \chi) = \overline{\gamma(s, \chi)}$. We now argue from the second formula for $\gamma(s, \chi)$ in (27.24). We note that

$$\left| \sin \frac{\pi}{2}(s + \kappa) \right| = \left| \frac{e^{i\frac{\pi}{2}(s+\kappa)} - e^{-i\frac{\pi}{2}(s+\kappa)}}{2i} \right| = \frac{1}{2} e^{\pi t/2} + O(e^{-\pi t/2}).$$

Stirling's formula, as stated in Theorem C.1, applies to $\Gamma(1-s)$ uniformly for $\sigma \leq 1/2$. In the logarithmic form it asserts that

$$\log \Gamma(1-s) = \left(\frac{1}{2} - s \right) \log(1-s) + s - 1 + \frac{1}{2} \log 2\pi + O\left(\frac{1}{|1-s|} \right).$$

Hence

$$\log |\gamma(s, \chi)| = \sigma \log 2\pi + \left(\frac{1}{2} - \sigma \right) \log |1-s| + \sigma + t \arg(1-s) + \frac{\pi}{2} t + O(1).$$

Now

$$\arg(1-s) = -\arctan \frac{t}{1-\sigma} = -\frac{\pi}{2} + \arctan \frac{1-\sigma}{t} \leq -\frac{\pi}{2} + \frac{1-\sigma}{t}.$$

Thus

$$\log |\gamma(s, \chi)| \leq \left(\frac{1}{2} - \sigma\right) \log \frac{|1-s|}{2\pi} + O(1),$$

which gives (27.27).

We note that

$$\begin{aligned} \log |1-s| &= \frac{1}{2} \log ((1-\sigma)^2 + t^2) \\ &= \log t + \frac{1}{2} \log \left(1 + \left(\frac{1-\sigma}{t}\right)^2\right) \leq \log t + \frac{1}{2} \left(\frac{1-\sigma}{t}\right)^2. \end{aligned}$$

Hence

$$\log |\gamma(s, \chi)| \leq \left(\frac{1}{2} - \sigma\right) \log \frac{t}{2\pi} + \frac{(1-\sigma)^3}{2t^2} + O(1).$$

This gives (27.28) when $\sigma \geq -t^{2/3}$ and $t \geq 1$. Finally, $\gamma(s, \chi) \ll 1$ when $-4 \leq \sigma \leq 1/2$ and $0 \leq t \leq 1$.

With further analysis, outlined in Exercise 10, one finds that (27.28) fails when $(1-\sigma)/t^{2/3}$ is unbounded as $t \rightarrow \infty$. $\square$

L:P(s)est

Lemma 27.9. For $P(s, \chi)$ defined by (27.25) we have $|P(1/2 + it, \chi)| = 1$ for all real t , and $|P(s, \chi)| \leq q^{\frac{1}{2}-\sigma}$ for $\sigma \leq 1/2$.

Proof The first assertion is clear, since the denominator in (27.26) is the complex conjugate of the numerator, when $\sigma = 1/2$. Set

$$f(s) = f_p(s) = p^{s-1/2} \frac{1+p^{-s}}{1+p^{s-1}}.$$

Since the fraction (27.26) is periodic with period $2\pi i / \log p$, the fraction above is the same apart from a translation. We note that

$$f(\sigma) = \frac{p^{\sigma/2} + p^{-\sigma/2}}{p^{(1-\sigma)/2} + p^{(\sigma-1)/2}} = \frac{\cosh \frac{\sigma}{2} \log p}{\cosh \frac{1-\sigma}{2} \log p}.$$

Now $\cosh x$ is an even function and is strictly increasing for $x \geq 0$. Since $|\sigma/2| < (1-\sigma)/2$ for $\sigma < 1/2$, it follows that $0 < f(\sigma) < 1$ for $\sigma < 1/2$. We also note that

$$\frac{1-p^\sigma}{1+p^{\sigma-1}} p^{-1/2} \leq |f(s)| \leq \frac{1+p^\sigma}{1-p^{\sigma-1}} p^{-1/2}$$

when $\sigma \leq 0$. The two fractions tend to 1 as $\sigma \rightarrow -\infty$ so $|f(s)|$ is uniformly close to $p^{-1/2}$ when σ is large and negative. By applying the maximum modulus principle to f on a rectangle with vertices $-C, 1/2, 1/2 + 2\pi i/\log p, -C + 2\pi i/\log p$ we deduce that $|f(s)| \leq 1$ throughout this rectangle, and by periodicity for all s with $\sigma \leq 1/2$. Let r denote the product of those primes that divide q but do not divide d . We have shown that $|P(s, \chi)| \leq (dr)^{1/2-\sigma}$ when $\sigma \leq 1/2$. Since $dr|q$, we have the stated bound. $\square$

L:Lfcnbnds **Lemma 27.10.** *Let χ be a character mod q . Then*

$$L(s, \chi) - E(\chi) \frac{\varphi(q)s}{q(s-1)} \ll \min\left(\frac{1}{\sigma-1}, \log q\tau\right) \quad (27.29) \quad \text{E:Lbnd1}$$

uniformly for $\sigma \geq 1$,

$$L(s, \chi) - E(\chi) \frac{\varphi(q)s}{q(s-1)} \ll (q\tau)^{\frac{1-\sigma}{2}} \log q\tau \quad (27.30) \quad \text{E:Lbnd2}$$

uniformly for $0 \leq \sigma \leq 1$, and

$$L(s, \chi) \ll (q|1-s|)^{\frac{1}{2}-\sigma} \min\left(\frac{-1}{\sigma}, \log q\tau\right) \quad (27.31) \quad \text{E:Lbnd3}$$

for $\sigma \leq 0$.

Proof Let

$$S(x; \chi) = \sum_{1 \leq n \leq x} \chi(n), \quad R(x; \chi) = S(x; \chi) - E(\chi) \frac{\varphi(q)}{q} x.$$

Thus $R(x; \chi) \ll q$ uniformly for $x > 0$. By integrating by parts (as in the proof of Theorem 1.12) we see that if $x > 0$ and $\sigma > 1$, then

$$\begin{aligned} L(s, \chi) &= \sum_{n \leq x} \frac{\chi(n)}{n^s} + \int_x^\infty u^{-s} dS(u; \chi) \\ &= \sum_{n \leq x} \frac{\chi(n)}{n^s} + E(\chi) \frac{\varphi(q)}{q} \cdot \frac{x^{1-s}}{s-1} \\ &\quad - \frac{R(x; \chi)}{x^s} + s \int_x^\infty \frac{R(u; \chi)}{u^{s+1}} du. \end{aligned} \quad (27.32) \quad \text{E:Lform0}$$

This formula provides an analytic continuation to the halfplane $\sigma > 0$. For the present we assume that $\sigma \geq 1$. In this case,

$$L(s, \chi) - E(\chi) \frac{\varphi(q)}{q} \cdot \frac{1}{s-1} \ll \int_1^x \frac{du}{u^\sigma} + \frac{q\tau}{x^\sigma}.$$

We take $x = q\tau$. For the integral we have two bounds:

$$\int_1^x \frac{du}{u^\sigma} = \frac{1 - x^{1-\sigma}}{\sigma - 1} \leq \frac{1}{\sigma - 1}, \quad \int_1^x \frac{du}{u^\sigma} \leq \int_1^x \frac{du}{u} = \log x.$$

This gives the bound (27.29).

To obtain (27.31) we appeal to (27.23), from which we see that if $\sigma \leq 0$, then

$$\begin{aligned} L(s, \chi) &= \varepsilon(\chi) \left(L(1-s, \bar{\chi}) + E(\chi) \frac{\varphi(q)}{qs} \right) \gamma(s, \chi) P(s, \chi) \\ &\quad - E(\chi) \frac{\varphi(q)}{sq} \gamma(s, \chi) P(s, \chi). \end{aligned}$$

From (27.29), (27.27), and Lemma 27.9 we see that the first term on the right above is $\ll (q|1-s|)^{1/2-\sigma} \min(\frac{1}{-\sigma}, \log q\tau)$. The second term on the right above occurs only when $\chi = \chi_0$. Since χ_0 is an even character, it follows that $\gamma(0, \chi_0) = 0$. Thus $\gamma(s, \chi_0)/s \ll |1-s|^{1/2-\sigma}/(1+|s|)$ for $\sigma \leq 0$. Thus we have (27.31).

To obtain the bound (27.30) we appeal to the bounds (27.29), (27.31), and argue ‘by convexity’, which is to say by using the Phragmén–Lindelöf Theorem, which states that if $f(s)$ is analytic in the strip $S = \{s : 0 < \sigma < 1\}$, if f is continuous on $\bar{S}$, if $|f(s)| \leq 1$ for $s \in \partial S$, and if there are constants $A > 0$ and α , $0 < \alpha < \pi$ such that

$$|f(s)| \leq \exp(A \exp(\alpha|t|)), \quad (27.33) \quad \boxed{\text{E:P-LCritBnd}}$$

then $|f(s)|$ is bounded and $\sup_{s \in S} |f(s)| = \sup_{s \in \partial S} |f(s)|$. We take

$$f(s) = \left(L(s, \chi) - E(\chi) \frac{\varphi(q)}{q} \cdot \frac{s}{s-1} \right) \frac{q^{\frac{s-1}{2}} \Gamma(\frac{s}{2} + 1) \cos \frac{\pi s}{4}}{(2-s) \log q(3-s)}.$$

From (27.29) and (27.31) it follows that $f(1+it) \ll 1$ and $f(it) \ll 1$, in view of the orders of magnitude indicated of the following functions, each of which is analytic in the strip $-1/2 \leq \sigma \leq 3/2$:

$$\begin{aligned} |q^{(s-1)/2}| &\asymp q^{(\sigma-1)/2}, & |\Gamma(\frac{s}{2} + 1)| &\asymp \tau^{(\sigma+1)/2} \exp(-\frac{\pi\tau}{4}), \\ |\cos \frac{\pi s}{4}| &\asymp \exp(\frac{\pi\tau}{4}), & |2-s| &\asymp \tau, & |\log q(3-s)| &\asymp \log q\tau. \end{aligned}$$

Thus (27.30) follows from the Phragmén–Lindelöf Theorem, provided that we can show a weak upper bound of the form

$$L(s, \chi) - E(\chi) \frac{\varphi(q)}{q} \cdot \frac{s}{s-1} \ll (q\tau)^A. \quad (27.34) \quad \boxed{\text{E:Lweakbnd}}$$

To this end, we take $x = 1^-$ in (27.32) and note that $R(1^-; \chi) = -E(\chi)\varphi(q)/q$. Thus

$$L(s, \chi) - E(\chi) \frac{\varphi(q)}{q} \cdot \frac{s}{s-1} \ll q\tau \int_1^\infty u^{-\sigma-1} du \ll q\tau$$

uniformly for $1/2 \leq \sigma \leq 1$. For $0 \leq \sigma \leq 1/2$ we again appeal to (27.23), from which with Lemmas 27.8 and 27.9 we see that

$$\begin{aligned} L(s, \chi) - E(\chi) \frac{\varphi(q)}{q} \cdot \frac{s}{s-1} &\ll |L(1-s, \overline{\chi i}) \gamma(s, \chi) P(s, \chi)| + 1 \\ &\ll (q\tau)^{\frac{3}{2}-\sigma} + E(\chi) \left| \frac{1-s}{-s} \right| |\gamma(s, \chi) P(s, \chi)|. \end{aligned}$$

As we already noted, $\gamma(s, \chi_0)/s$ is bounded for s near 0. Thus we have (27.34) with $A = 3/2$, so the proof is complete. $\square$

We also prepare a handy kernel:

L:kernel **Lemma 27.11.** For $u \geq 0$ let

$$\kappa(u) = \sum_{\substack{0 \leq j \leq 4 \\ 2u-1 \leq j}} (-1)^j \binom{4}{j} (j+1-2u)^3, \quad (27.35) \quad \text{Defk(u)}$$

and set

$$K(s) = \frac{48 \sum_{j=0}^4 (-1)^j \binom{4}{j} \left(\frac{j+1}{2}\right)^{s+3}}{s(s+1)(s+2)(s+3)}. \quad (27.36) \quad \text{E:DefK}$$

Then $K(s)$ is an entire function, and if $\alpha(s) = \sum_{n=1}^\infty a_n n^{-s}$ has abscissa of convergence σ_c and $x > 0$, then

$$\sum_{n=1}^\infty a_n \kappa\left(\frac{n}{x}\right) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \alpha(s) K(s) x^s ds \quad (27.37) \quad \text{E:k(u) <-> K(s)}$$

for $c > \max(0, \sigma_c)$.

To prepare for the proof of this lemma, we construct notation to express forward differences. If $f(x)$ is defined on the real line, then $\Delta f(x) = f(x+1) - f(x)$. If a forward difference with a step size $h > 0$ is desired, we write $\Delta_h f(x) = f(x+h) - f(x)$. These operations can be iterated; for example, $(\Delta^2 f)(x) = \Delta(\Delta f(x)) = (f(x+2) - f(x+1)) - (f(x+1) - f(x)) = f(x) - 2f(x+1) + f(x+2)$. In general, $\Delta^k f = \Delta(\Delta^{k-1} f)$.

We refer to this as a k^{th} order forward difference. By an easy induction we see that

$$(\Delta_h^k f)(x) = \sum_{j=0}^k (-1)^{k-j} \binom{k}{j} f(x + jh).$$

We note that if f is a polynomial of degree d , then Δf is a polynomial of degree $d - 1$, and hence that $(\Delta^k f)(x) = 0$ identically if $k > d$.

Proof In §5.1 we remarked that the Cesàro partial sums of order k of a Dirichlet series $\alpha(s)$ are given by the formulæ

$$C_k(x) = \frac{1}{k!} \sum_{n \leq x} a_n (x - n)^k = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \alpha(s) \frac{x^{s+k}}{s(s+1)\cdots(s+k)} ds$$

for $c > \max(0, \sigma_c)$. We take $k = 3$ and form the 4th-order forward difference of both sides of the above, starting at $\frac{1}{2}x$ and taking steps of size $\frac{1}{2}x$. The left hand side then becomes

$$\begin{aligned} & \sum_{j=0}^4 (-1)^j \binom{4}{j} C_3\left(\frac{1}{2}(j+1)x\right) \\ &= \frac{1}{6} \sum_n a_n \sum_{j=0}^4 (-1)^j \binom{4}{j} \left(\max\left(0, \frac{1}{2}(j+1)x - n\right)\right)^3 \\ &= \frac{1}{6} \sum_n a_n \sum_{\substack{0 \leq j \leq 4 \\ (j+1)x > 2n}} (-1)^j \binom{4}{j} \left(\frac{1}{2}(j+1)x - n\right)^3 = \frac{x^3}{48} \sum_n a_n \kappa\left(\frac{n}{x}\right). \end{aligned}$$

The sum that defines $\kappa(u)$ is empty when $u > 5/2$. When $u < 1/2$, j runs over the full range $0 \leq j \leq 4$, and then the expression represents a fourth order forward difference of u^3 . Thus $\kappa(u) = 0$ for $u < 1/2$. The weight $\kappa(u)$ is piecewise polynomial, and is equal to a single polynomial in each of the intervals $(-\infty, \frac{1}{2}]$, $[\frac{1}{2}, 1]$, $[1, \frac{3}{2}]$, $[\frac{3}{2}, 2]$, $[2, \frac{5}{2}]$, $[\frac{5}{2}, \infty)$. At each of the transition points, the summand that is introduced or removed vanishes to the third order at that point. Hence $\kappa''(u)$ is continuous and piecewise linear.

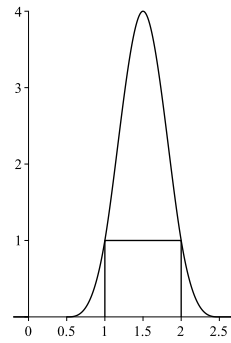


Figure 22.3 Graphs of $\kappa(u)$ and $\chi_{[1,2]}(u)$.

Thomas ^{Simp}Simpson (1757) showed that if $X_1, X_2, \dots, X_n$ are inde-

pendent random variables, each one uniformly distributed on $[0, 1]$, and $X = X_1 + X_2 + \cdots + X_n$, then X is distributed with the density

$$f_n(x) = \frac{1}{(n-1)!} \sum_{x \leq j \leq n} (-1)^{n-j} \binom{n}{j} (j-x)^{n-1}$$

for $n > 1$. Thus $\kappa(u) = 6f_4(2u-1)$. With this interpretation, we see that $\kappa(u) = \kappa(3-u)$, that $\int_{1/2}^{5/2} \kappa(u) du = 3$, and that $\kappa(u)$ is increasing for $1/2 \leq u \leq 3/2$. These same observations can also be derived by means of elementary calculations, without recourse to the results of Simpson.

When we apply the same fourth order forward difference to the integral representation of $C_3(x)$, we find that the integrand becomes

$$\frac{\alpha(s)}{s(s+1)(s+2)(s+3)} \sum_{j=0}^4 (-1)^j \binom{4}{j} \left(\frac{j+1}{2}x\right)^{s+3} = \frac{x^3}{48} \alpha(s) K(s) x^s.$$

Thus we have (27.37).

Let $S(s)$ denote the sum over j in the definition (27.36) of $K(s)$. To show that $K(s)$ is entire, it suffices to show that $S(-r) = 0$ for $r = 0, 1, 2, 3$. But this is obvious, since $S(-r)$ is a forward difference of order 4 of x^{3-r} , which is a polynomial of degree $3-r < 4$. $\square$

L:G2est **Lemma 27.12.** *Suppose that*

$$A(s) = \sum_{m=1}^M a_m m^{-s}, \quad B(s) = \sum_{n=1}^N b_n n^{-s}, \quad C(s) = A(s)B(s) = \sum_{k=1}^{MN} c_k k^{-s}.$$

Then

$$\sum_{k=1}^{MN} |c_k|^2 \leq \left(\sum_{m=1}^M d(m) |a_m|^2 \right) \left(\sum_{n=1}^N d(n) |b_n|^2 \right).$$

Proof The sequence $\{c_k\}$ is the Dirichlet convolution $a * b$ of the sequences $\{a_m\}$ and $\{b_n\}$:

$$c_k = \sum_{\substack{m,n \\ mn=k}} a_m b_n.$$

By Cauchy's inequality,

$$|c_k|^2 \leq \left(\sum_{\substack{m,n \\ mn=k}} 1 \right) \left(\sum_{\substack{m,n \\ mn=k}} |a_m|^2 |b_n|^2 \right) = \sum_{\substack{m,n \\ mn=k}} d(mn) |a_m|^2 |b_n|^2.$$

Write $m = \prod_p p^\mu$ and $n = \prod_p p^\nu$. Since $\mu + \nu + 1 \leq (\mu + 1)(\nu + 1)$ for nonnegative μ, ν , it follows that $d(mn) \leq d(m)d(n)$. We substitute this inequality in the above, and sum over k to obtain the stated result. $\square$

T:HMqT2

Theorem 27.13. *Let q be a positive integer, and $T \geq 2$ be real. Suppose that for $r = 1, 2, \dots, R$ we have a character χ_r modulo q and a point $s_r = \sigma_r + it_r$ with $\sigma_r \geq 0$, $0 \leq t_r \leq T$, and with the further property that $|t_{r_1} - t_{r_2}| \geq 1$ if $\chi_{r_1} = \chi_{r_2}$. Let $D(s, \chi)$ be defined as in (27.22), and set*

$$G = \sum_{n=1}^N |a_n|^2, \quad G_2 = \sum_{n=1}^N d(n) |a_n|^2. \quad (27.38) \quad \text{E:DefG,G2}$$

If $N \leq qT$, then

$$\begin{aligned} & \sum_{r=1}^R |D(s_r, \chi_r)|^2 \\ & \ll NG + R^{2/3} N^{1/3} q^{1/3} T^{1/3} G^{2/3} G_2^{1/3} \log qT. \end{aligned} \quad (27.39) \quad \text{E:HMqT2}$$

When $N \geq qT$ we already know that the sum is $\ll NG \log N$ by the estimate (26.26) found in Theorem 26.17.

Proof We first establish the desired bound under the assumptions that $\sigma_r = 0$ for all r and that the sum runs not from 1 to N but rather from $N+1$ to $2N$. Thus our first object is to show that

$$\begin{aligned} & \sum_{r=1}^R \left| \sum_{n=N+1}^{2N} a_n \chi_r(n) n^{-it_r} \right|^2 \\ & \ll NG + R^{2/3} N^{1/3} q^{1/3} T^{1/3} G^{2/3} G_2^{1/3} \log qT. \end{aligned} \quad (27.40) \quad \text{E:Step1}$$

In this context,

$$G = \sum_{n=N+1}^{2N} |a_n|^2, \quad G_2 = \sum_{n=N+1}^{2N} d(n) |a_n|^2. \quad (27.41) \quad \text{E:tempDefG,G2}$$

Let b_r be determined by the equation

$$\overline{b_r} = \sum_{n=N+1}^{2N} a_n \chi_r(n) n^{-it_r}. \quad (27.42) \quad \text{E:Defbr}$$

Thus the left hand side of (27.40) is

$$= \sum_{r=1}^R \sum_{n=N+1}^{2N} a_n \chi_r(n) n^{-it_r} b_r = \sum_{n=N+1}^{2N} a_n \sum_{r=1}^R b_r \chi_r(n) n^{-it_r}.$$

Since the left hand side of (27.40) is equal to the right hand side of the

above, their squares are also equal. That is,

$$\left(\sum_{r=1}^R \left| \sum_{n=N+1}^{2N} a_n \chi_r(n) n^{-it_r} \right|^2 \right)^2 = \left(\sum_{n=N+1}^{2N} a_n \sum_{r=1}^R b_r \chi_r(n) n^{-it_r} \right)^2.$$

We apply Cauchy's inequality to the right hand side above, to see that it is

$$\leq G \sum_{n=N+1}^{2N} \left| \sum_{r=1}^R b_r \chi_r(n) n^{-it_r} \right|^2. \quad (27.43) \quad \boxed{\text{E: CritIneq}}$$

Let $\kappa(u)$ be defined as in Lemma 27.11. Then $\kappa(u) \geq 0$ for all $u \geq 0$, and $\kappa(u) \geq 1$ for $1 \leq u \leq 2$, so the sum over n in (27.43) is

$$\leq \sum_{n=1}^{\infty} \left(\frac{2N}{n} \right)^{1/2} \kappa\left(\frac{n}{N}\right) \left| \sum_{r=1}^R b_r \chi_r(n) n^{-it_r} \right|^2.$$

We expand the modulus-squared and take the sum over n inside to see that the above is

$$\begin{aligned} &= (2N)^{1/2} \sum_{r_1=1}^R \sum_{r_2=1}^R b_{r_1} \overline{b_{r_2}} \sum_{n=1}^{\infty} \kappa\left(\frac{n}{N}\right) \chi_{r_1} \overline{\chi_{r_2}}(n) n^{-\frac{1}{2}-i(t_{r_1}-t_{r_2})} \\ &\ll N^{1/2} \sum_{r_2=1}^R |b_{r_2}| \sum_{r_1=1}^R |b_{r_1}| \left| \sum_{n=1}^{\infty} \kappa\left(\frac{n}{N}\right) \chi_{r_1} \overline{\chi_{r_2}}(n) n^{-\frac{1}{2}-i(t_{r_1}-t_{r_2})} \right|. \end{aligned} \quad (27.44) \quad \boxed{\text{E: Exp2}}$$

From (27.37) we deduce that

$$\begin{aligned} &\sum_{n=1}^{\infty} \kappa\left(\frac{n}{N}\right) \chi_{r_1} \overline{\chi_{r_2}}(n) n^{-\frac{1}{2}-i(t_{r_1}-t_{r_2})} \\ &= \frac{1}{2\pi i} \int_{1-i\infty}^{1+i\infty} L\left(\frac{1}{2} + i(t_{r_1} - t_{r_2}) + w, \chi_{r_1} \overline{\chi_{r_2}}\right) K(w) N^w dw. \end{aligned} \quad (27.45) \quad \boxed{\text{E: Exp3}}$$

Write $w = u + iv$. The above integrand has a pole at $w = \frac{1}{2} - i(t_{r_1} - t_{r_2})$ if $\chi_{r_1} = \chi_{r_2}$. By moving the contour to the abscissa $u = -1$ we find that the above is

$$\begin{aligned} &= \frac{1}{2\pi i} \int_{-1-i\infty}^{-1+i\infty} L\left(\frac{1}{2} + i(t_{r_1} - t_{r_2}) + w, \chi_{r_1} \overline{\chi_{r_2}}\right) K(w) N^w dw \\ &\quad + E(\chi_{r_1} \overline{\chi_{r_2}}) \frac{\varphi(q)}{q} K\left(\frac{1}{2} - i(t_{r_1} - t_{r_2})\right) N^{\frac{1}{2}-i(t_{r_1}-t_{r_2})} \end{aligned} \quad (27.46) \quad \boxed{\text{E: Residue}}$$

where $E(\chi) = 1$ if $\chi = \chi_0$, and $E(\chi) = 0$ otherwise. Concerning the convergence of this integral as $|t| \rightarrow \infty$, we note that

$$L\left(\frac{1}{2} + i(t_{r_1} - t_{r_2}) + w, \chi_{r_1} \bar{\chi}_{r_2}\right) \ll q(T + |v|)$$

for $u \geq -1$, by Lemma 27.10. This suffices, since $K(w) \ll v^{-4}$. By the functional equation in the form (27.23), we see that the integral above is

$$\begin{aligned} &= \frac{\varepsilon(\chi_{r_1} \bar{\chi}_{r_2})}{2\pi i} \int_{-1-i\infty}^{-1+i\infty} L\left(\frac{1}{2} - i(t_{r_1} - t_{r_2}) - w, \bar{\chi}_{r_1} \chi_{r_2}\right) \\ &\quad \times \gamma\left(\frac{1}{2} + i(t_{r_1} - t_{r_2}) + w, \chi_{r_1} \bar{\chi}_{r_2}\right) \\ &\quad \times P\left(\frac{1}{2} + i(t_{r_1} - t_{r_2}) + w, \chi_{r_1} \bar{\chi}_{r_2}\right) K(w) N^w dw. \end{aligned} \quad (27.47) \quad \boxed{\text{E:Exp4}}$$

For $\sigma > 1$ we write

$$\begin{aligned} L(s, \bar{\chi}_{r_1} \chi_{r_2}) &= \sum_{m=1}^{\infty} \bar{\chi}_{r_1} \chi_{r_2}(m) m^{-s} = \sum_{m \leq M} + \sum_{m > M} \\ &= S_1(s, \bar{\chi}_{r_1} \chi_{r_2}) + S_2(s, \bar{\chi}_{r_1} \chi_{r_2}), \end{aligned} \quad (27.48) \quad \boxed{\text{E:DefS1S2}}$$

say, where M is an integer such that $MN \asymp qT$. When we replace L by S_i in (27.47) we obtain an integral $I_i(r_1, r_2)$ for $i = 1, 2$. To complete the estimation of (27.44) we estimate the contributions made by the I_i and the residue in (27.46).

To estimate I_1 we move the contour from the abscissa $u = -1$ to $u = 0$. This is justified by our bounds for $\gamma(s)$, $P(s)$, and $K(s)$. Since $|\gamma(s)| = |P(s)| = 1$ on the new contour, we deduce that

$$I_1(r_1, r_2) \ll \int_{-\infty}^{\infty} |S_1\left(\frac{1}{2} - i(t_{r_1} - t_{r_2}) - iv, \bar{\chi}_{r_1} \chi_{r_2}\right)| |K(iv)| dv.$$

After replacing b_{r_1} and S_1 by their complex conjugates, we see that

$$\begin{aligned} &|b_{r_1}| |S_1\left(\frac{1}{2} - i(t_{r_1} - t_{r_2}) - iv, \bar{\chi}_{r_1} \chi_{r_2}\right)| \\ &= \left| \left(\sum_{n=N+1}^{2N} a_n \chi_{r_1}(n) n^{-it_{r_1}} \right) \left(\sum_{m=1}^M m^{-\frac{1}{2} - i(t_{r_1} - t_{r_2}) - iv} \chi_{r_1}(m) \bar{\chi}_{r_2}(m) \right) \right| \\ &= \left| \sum_{k=N}^{2MN} c_k \chi_{r_1}(k) k^{-it_{r_1}} \right| = |C(it_{r_1}, \chi_{r_1})| \end{aligned}$$

say, where

$$c_k = c_{k, r_2, v} = \sum_{\substack{1 \leq m \leq M \\ N < n \leq 2N \\ mn=k}} a_n m^{-\frac{1}{2} + it_{r_2} - iv} \bar{\chi}_{r_2}(m).$$

Hence

$$|c_k| \leq \sum_{\substack{1 \leq m \leq M \\ N < n \leq 2N \\ mn=k}} |a_n| m^{-1/2}$$

for all r_2 and v . By Cauchy's inequality,

$$\begin{aligned} \sum_{r_1=1}^R |b_{r_1}| \left| S_1 \left(\frac{1}{2} - i(t_{r_1} - t_{r_2}) - iv, \bar{\chi}_{r_1} \chi_{r_1} \right) \right| \\ = \sum_{r_1=1}^R |C(it_{r_1}, \chi_{r_1})| \leq R^{1/2} \left(\sum_{r_1=1}^R |C(it_{r_1}, \chi_{r_1})|^2 \right)^{1/2}. \end{aligned}$$

From the estimate (26.26) of Theorem 26.17 we see that the above is

$$\ll R^{1/2} \left(qT(\log qT) \sum_{k=1}^{2MN} |c_k|^2 \right)^{1/2}.$$

By Lemma 27.12 we deduce that

$$\sum_{k=N}^{2MN} |c_k|^2 \leq \left(\sum_{n=N+1}^{2N} d(n) |a_n|^2 \right) \left(\sum_{m=1}^M \frac{d(m)}{m} \right) \ll G_2(\log qT)^2.$$

Thus

$$\sum_{r_1=1}^R |b_{r_1}| \left| S_1 \left(\frac{1}{2} - i(t_{r_1} - t_{r_2}) - iv, \bar{\chi}_{r_1} \chi_{r_2} \right) \right| \ll (RqTG_2)^{1/2} (\log qT)^{3/2}$$

uniformly for all r_2 and v , so the contribution of I_1 to (27.44) is

$$\ll N^{1/2} \left(\sum_{r_2=1}^R |b_{r_2}| \right) (RqTG_2)^{1/2} (\log qT)^{3/2} \int_{-\infty}^{\infty} |K(iv)| dv.$$

From the definition (27.36) of $K(s)$ we see that $K(iv) \ll (|v| + 1)^{-4}$, and hence that the integral above is bounded. By a further application of Cauchy's inequality we see that the above is

$$\ll (NqT)^{1/2} R G_2^{1/2} (\log qT)^{3/2} \left(\sum_{r=1}^R |b_r|^2 \right)^{1/2}. \quad (27.49) \quad \boxed{\text{E:11contrib}}$$

Apart from the need to partition $S_2(s, \chi)$ into subsums, we treat I_2

in the same way that we treated I_1 . By our bounds from Lemmas 27.8, 27.9, 27.11 we see that

$$I_2(r_1, r_2) \ll \frac{1}{N} \int_{-\infty}^{\infty} q(T+|v|)(|v|+1)^{-4} |S_2(\tfrac{3}{2}-i(t_{r_1}-t_{r_2})-iv, \bar{\chi}_{r_1} \chi_{r_2})| dv.$$

Our main task is to show that

$$\begin{aligned} \sum_{r_1=1}^R |b_{r_1}| |S_2(\tfrac{3}{2}-i(t_{r_1}-t_{r_2})-iv, \bar{\chi}_{r_1} \chi_{r_2})| \\ \ll R^{\frac{1}{2}} M^{-\frac{1}{2}} N^{\frac{1}{2}} G_2^{\frac{1}{2}} \log 2MN \end{aligned} \quad (27.50) \quad \boxed{\text{E: sum } |b_r| |S_2| \text{ est}}$$

uniformly for all r_2 and v . Since $MN \asymp qT$, it then follows that

$$\sum_{r_1=1}^R |b_{r_1}| |I_2(r_1, r_2)| \ll (RqTG_2)^{\frac{1}{2}} \log qT,$$

which implies that

$$N^{\frac{1}{2}} \sum_{r_2=1}^R |b_{r_2}| \sum_{r_1=1}^R |b_{r_1}| |I_2(r_1, r_2)| \ll \left(\sum_{r_2=1}^R |b_{r_2}| \right) (RNqTG_2)^{\frac{1}{2}} \log qT,$$

and by a further application of Cauchy's inequality we see that this is

$$\ll R(NqTG_2)^{\frac{1}{2}} \left(\sum_{r_2=1}^R |b_{r_2}| \right)^{1/2} \log qT.$$

This quantity is slightly smaller than our estimate (27.49) for the contribution of I_1 in (27.44).

To prove (27.50) we first put $M_j = 2^j M$ for $j = 0, 1, 2, \dots$, and set

$$S_{2,j} = S_{2,j}(r_1, r_2, v) = \sum_{M_{j-1} < m \leq M_j} \bar{\chi}_{r_1} \chi_{r_2}(m) m^{-\frac{3}{2}+i(t_{r_1}-t_{r_2})+iv}$$

for $j = 1, 2, \dots$. By the triangle inequality,

$$|b_{r_1}| |S_2(\tfrac{3}{2}-i(t_{r_1}-t_{r_2})-iv, \bar{\chi}_{r_1} \chi_{r_2})| \leq \sum_{j=1}^{\infty} |b_{r_1}| |S_{2,j}|.$$

On replacing b_{r_1} and $S_{2,j}$ by their complex conjugates, we see that

$$\begin{aligned} |b_{r_1}| |S_{2,j}| &= \left| \sum_{n=N+1}^{2N} a_n \chi_{r_1}(n) n^{-it_{r_1}} \right| \\ &\quad \times \left| \sum_{M_{j-1} < m \leq M_j} m^{-\frac{3}{2}-it_{r_1}} \chi_{r_1}(m) \left(m^{i(v+it_{r_2})} \overline{\chi_{r_2}} \right) \right| \\ &= |C_j(r_1)| = |C_j(r_1, r_2, v)| \end{aligned}$$

where

$$C_j(r_1) = C_j(r_1, r_2, v) = \sum_{M_{j-1}N < m \leq 2M_jN} c_{j,k} \chi_{r_1}(k) k^{-it_{r_1}}$$

and

$$c_{j,k} = c_{j,k,r_2,v} = \sum_{\substack{M_{j-1} < m \leq M_j \\ N < n \leq 2N \\ mn=k}} a_n m^{-\frac{3}{2}} \left(m^{i(v+it_{r_2})} \overline{\chi_{r_2}}(m) \right).$$

Hence

$$|c_{j,k}| \leq \sum_{\substack{M_{j-1} < m \leq M_j \\ N < n \leq 2N \\ mn=k}} |a_n| m^{-\frac{3}{2}}$$

for all r_2 and v . Thus by Cauchy's inequality,

$$\sum_{r_1=1}^R \sum_{j=1}^{\infty} |b_{r_1}| |S_{2,j}| \leq \left(\sum_{r_1,j} j^{-2} \right)^{1/2} \left(\sum_{j=1}^{\infty} j^2 \sum_{r_1=1}^R |C_j(r_1)|^2 \right)^{1/2}.$$

Since $M_j N \gg qT$, we see by the estimate (26.26) of Theorem 26.17 that the above is

$$\ll R^{1/2} \left(\sum_{j=1}^{\infty} j^2 (\log M_j N) M_j N \sum_{M_{j-1}N < k \leq 2M_jN} |c_{j,k}|^2 \right)^{1/2}. \quad (27.51) \quad \boxed{\text{E:Exp5}}$$

By Lemma 27.12 we see that

$$\begin{aligned} \sum_{M_{j-1}N < k \leq 2M_jN} |c_{j,k}|^2 &\leq \left(\sum_{M_{j-1} < m \leq M_j} d(m) m^{-3} \right) \left(\sum_{n=N+1}^{2N} d(n) |a_n|^2 \right) \\ &\ll M_j^{-2} (\log M_j) G_2. \end{aligned}$$

Thus the expression (27.51) is

$$\begin{aligned} &\ll R^{1/2} \left(\sum_{j=1}^{\infty} j^2 (j + \log MN) 2^{-j} M^{-1} N (j + \log M) G_2 \right)^{1/2} \\ &\ll R^{\frac{1}{2}} M^{-\frac{1}{2}} N^{\frac{1}{2}} G_2^{\frac{1}{2}} \log qT, \end{aligned}$$

and hence we have (27.50).

From the definition (27.36) of $K(s)$ it is clear that $K(\frac{1}{2} + it) \ll \tau^{-4}$. Hence the contribution to (27.44) made by the residue in (??) is

$$\ll N \sum_{\substack{1 \leq r_1, r_2 \leq R \\ \chi_{r_1} = \chi_{r_2}}} \frac{|b_{r_1} b_{r_2}|}{(1 + |t_{r_1} - t_{r_2}|)^4}.$$

By the arithmetic-geometric mean inequality we know that $|b_{r_1} b_{r_2}| \leq \frac{1}{2} |b_{r_1}|^2 + \frac{1}{2} |b_{r_2}|^2$. Hence the above is

$$\leq N \sum_{r_1=1}^R |b_{r_1}|^2 \sum_{\substack{1 \leq r_2 \leq R \\ \chi_{r_1} = \chi_{r_2}}} (1 + |t_{r_1} - t_{r_2}|)^{-4}.$$

For a given r_1 , the numbers t_{r_2} for which $\chi_{r_2} = \chi_{r_1}$ are spaced apart from each other by a distance of at least 1. Hence the sum over r_2 above is $\ll 1$, and so the quantity above is

$$\ll N \sum_{r=1}^R |b_r|^2.$$

Thus the quantity (27.44) is bounded by the above plus the estimate (27.49) for the contribution of I_1 , since our estimate for the contribution of I_2 is smaller. From the definition (27.42) of b_r we know that

$$\sum_{r=1}^R |b_r|^2 = \sum_{r=1}^R \left| \sum_{n=N+1}^{2N} a_n \chi_r(n) n^{-it_r} \right|^2.$$

Hence when we insert our bound for the expression (27.44) into the inequality (27.43) we find that

$$\begin{aligned} \left(\sum_{r=1}^R |b_r|^2 \right)^2 &\ll R(NqT)^{1/2} G G_2^{1/2} (\log qT)^{3/2} \left(\sum_{r=1}^R |b_r|^2 \right)^{1/2} \\ &\quad + NG \sum_{r=1}^R |b_r|^2. \end{aligned} \tag{27.52} \quad \boxed{\text{E:Exp6}}$$

If the second term on the right hand side is larger than the first, then it majorizes the left hand side, so we have

$$\sum_{r=1}^R |b_r|^2 \ll NG.$$

If the first term on the right hand side of (27.52) is at least as large as the second one, then it majorizes the left hand side, with the result that

$$\left(\sum_{r=1}^R |b_r|^2 \right)^{3/2} \ll R(NqT)^{1/2} G G_2^{1/2} (\log qT)^{3/2}$$

and so

$$\sum_{r=1}^R |b_r|^2 \ll R^{2/3} N^{1/3} q^{1/3} T^{1/3} G^{2/3} G_2^{1/3} \log qT$$

in this case. Hence in either case we have (27.40)

We now drop the condition that $\sigma_r = 0$ for all r , and instead allow $\sigma_r \geq 0$, while still restricting n to lie between $N+1$ and $2N$. Thus our new goal is to show that

$$\begin{aligned} \sum_{r=1}^R \left| \sum_{n=N+1}^{2N} a_n \chi_r(n) n^{-\sigma_r - it_r} \right|^2 \\ \ll NG + R^{2/3} N^{1/3} q^{1/3} T^{1/3} G^{2/3} G_2^{1/3} (\log 2qT)^A. \end{aligned} \quad (27.53) \quad \boxed{\text{E:Step2}}$$

Here G and G_2 are still defined as in (27.41). We employ the same integration by parts method that we used to prove Theorems 26.9 and 26.18. This approach is more successful in the present setting because the sum over n is shorter. For a given character χ and real number t , let

$$S(u) = S(u; t, \chi) = \sum_{N+1 \leq n \leq u} a_n \chi(n) n^{-it}.$$

Then by integrating by parts we see that if $\sigma \geq 0$, then

$$\sum_{n=N+1}^{2N} a_n \chi(n) n^{-\sigma - it} = \frac{S(2N)}{(2N)^\sigma} + \sigma \int_N^{2N} \frac{S(u)}{u^{\sigma+1}} du.$$

Hence

$$\left| \sum_{n=N+1}^{2N} a_n \chi(n) n^{-\sigma - it} \right| \leq |S(2N)| + \frac{\sigma}{N^{\sigma+1}} \int_N^{2N} |S(u)| du.$$

For $\sigma \geq 0$, the quantity σ/N^{sigma} achieves its maximum when $\sigma = 1/\log N$. Thus the above is

$$\ll |S(2N)| + \frac{1}{N \log N} \int_N^{2N} |S(u)| du.$$

By squaring both sides, and then applying the Cauchy–Schwarz inequality we deduce that

$$\left| \sum_{n=N+1}^{2N} a_n \chi(n) n^{-\sigma-it} \right|^2 \ll |S(2N)|^2 + \frac{1}{N(\log N)^2} \int_N^{2N} |S(u)|^2 du.$$

We now take $\chi = \chi_r$, $\sigma = \sigma_r$, $t = t_r$, and sum over r . The contribution of the $|S(2N; s_r, \chi_r)|^2$ is the same as in (27.40). From (27.40) we see that if $N \leq u \leq 2N$, then

$$\sum_{r=1}^R |S(u; t_r, \chi_r)|^2 \ll NG(u) + R^{2/3} q^{1/3} T^{1/3} N^{1/3} G(u)^{2/3} G_2(u)^{1/3} \log 2qT$$

where

$$G(u) = \sum_{N+1 \leq n \leq u} |a_n|^2 \leq G, \quad G_2(u) = \sum_{N+1 \leq n \leq u} d(n) |a_n|^2 \leq G_2,$$

Thus we have the same bound as in (27.40) for all u , and hence we have (27.53).

Finally, let $D(s, \chi)$ be defined as in (27.22), and write

$$D(s, \chi) = \sum_{j=1}^J D_j(s, \chi)$$

where $J = \lfloor (\log N)/\log 2 \rfloor$

$$D_j(s, \chi) = \sum_{N/2^{j+1} < n \leq N/2^j} a_n \chi(n) n^{-s},$$

$$G(j) = \sum_{N/2^{j+1} < n \leq N/2^j} |a_n|^2, \quad G_2(j) = \sum_{N/2^{j+1} < n \leq N/2^j} d(n) |a_n|^2.$$

Let δ be fixed, $0 < \delta < 1$. Then by Cauchy's inequality,

$$|D(s_r, \chi_r)|^2 = \left| \sum_{j=0}^J D_j(s_r, \chi_r) \right|^2 \leq \left(\sum_{j=0}^J \delta^j \right) \left(\sum_{j=0}^J \delta^{-j} |D_j(s_r, \chi_r)|^2 \right)$$

$$\ll \sum_{j=0}^J \delta^{-j} |D_j(s_r, \chi_r)|^2.$$

We sum over r and apply (27.53) to see that

$$\sum_{r=1}^R |D(s_r, \chi_r)|^2 \ll C_1 N G(j) + C_2 R^{2/3} N^{1/3} q^{1/3} T^{1/3} G(j)^{2/3} G_2(j)^{1/3} \log qT$$

where

$$C_1 = \sum_{j=0}^J (2\delta)^{-j}, \quad C_2 = \sum_{j=0}^J (2^{1/3}\delta)^{-j}.$$

We observe that $G(j) \leq G$ and $G_2(j) \leq G_2$ for all j where G and G_2 are defined as in (27.38). We set $\delta = 9/10$, and note that $2\delta > 2^{1/3}\delta = 1.1339\dots > 1$, so that $C_1 \ll 1$ and $C_2 \ll 1$. Thus we have (27.39), and the proof is complete. $\square$

Cor:HMLVqT2

Corollary 27.14. *Suppose that q is a positive integer, and that $T \geq 2$ is real. Suppose that $D(s, \chi)$ is defined as in (27.22), and that for $r = 1, 2, \dots, R$ we have pairs (s_r, χ_r) with the property that $s_r = \sigma_r + it_r$, $\sigma_r \geq 0$, $0 \leq t_r \leq T$, that χ_r is a character mod q , and that $|t_{r_1} - t_{r_2}| \geq 1$ if $\chi_{r_1} = \chi_{r_2}$. Let G and G_2 be defined as in (27.38). Then*

$$R \ll \frac{NG}{V^2} + \frac{NqTG^2G_2(\log qT)^3}{V^6}. \quad (27.54) \quad \text{E:HMLVqT2}$$

The bound here is strikingly similar to that obtained by a totally different method for $q = 1$ in Theorem 27.4.

Proof In (27.39), the left hand side is $\geq RV^2$. If NG is the larger of the two terms on the right hand side, then it follows that $R \ll NG/V^2$. If NG is the smaller of the two terms on the right hand side, then $RV^2 \ll R^{2/3}N^{1/3}q^{1/3}T^{1/3}G^{2/3}G_2^{1/3}\log qT$, which gives

$$R \ll \frac{NqTG^2G_2(\log qT)^3}{V^6}.$$

Thus (27.54) holds in either case. $\square$

T:HMQT2

Theorem 27.15. *Suppose that for $r = 1, 2, \dots, R$ we have a primitive character χ_r modulo q_r with $1 \leq q_r \leq Q$ and a point $s_r = \sigma_r + it_r$ with $\sigma_r \geq 0$, $0 \leq t_r \leq T$, and with the further property that $|t_{r_1} - t_{r_2}| \geq 1$ if $\chi_{r_1} = \chi_{r_2}$. Let $D(s, \chi)$ be defined as in (27.22), and let G and G_2 be defined as in (27.38). Then*

$$\sum_{r=1}^R |D(s_r, \chi_r)|^2 \ll NG + R^{2/3}N^{1/3}Q^{2/3}T^{1/3}G^{2/3}G_2^{1/3}\log QT. \quad (27.55) \quad \text{E:HMQT2}$$

Proof This theorem is proved in the same way as the preceding one, with just a few obvious changes. For example, since χ_{r_s} and χ_{r_2} are primitive, if $\chi_{r_1}\bar{\chi}_{r_2} = \chi_0$, then $\chi_{r_1} = \chi_{r_2}$. The character $\chi_{r_1}\bar{\chi}_{r_2}$ is a character modulo $[q_{r_1}, q_{r_2}] \leq q_{r_1}q_{r_2} \leq Q^2$. In places where we appealed to (26.26) in Theorem 26.17, we instead appeal to (26.28), which is found in the same theorem. $\square$

??.1 Exercises

1. Suppose that f is a measurable function on $[0, 1]$, that $V > 0$, and that $m(V)$ denotes the measure of the set of those $x \in [0, 1]$ for which $|f(x)| \geq V$. Suppose further that we know that

$$\int_0^1 |f(x)| dx \leq 1, \quad \text{and that} \quad \int_0^1 |f(x)|^3 dx \leq 4.$$

- (a) Show that

$$\int_0^1 |f(x)|^2 dx \leq 2.$$

- (b) Show that $m(V) \leq 1$, that $m(V) \leq 1/V$, that $m(V) \leq 4/V^3$, and that $m(V) \leq 2/V^2$ for all $V > 0$.
- (c) Show that $m(V) \leq \min(1, 1/V, 4/V^3) \leq 2/V^2$ for all $V > 0$.

Exer:M(alpha,4T)

2. ^{HL71}(Montgomery, 1971) Let α be fixed, $0 < \alpha < 1$, and set

$$M(\alpha, T) = \max_{\substack{\sigma \geq \alpha \\ 0 \leq t \leq T \\ |s-1| \geq 1}} |\zeta(s)|. \quad (27.56) \quad \text{E:DefM(alpha,T)}$$

- (a) Let $w_n = \max(0, 2 - n/N)$. With $W(s)$ defined as in the proof of Theorem 27.1, show that if $|t| \leq T$, then $W(it) \ll M(\alpha, 4T)N^\alpha + N/\tau^2$.
- (b) Let $D(s)$ be a Dirichlet polynomial as in (27.1), and let $t_1, t_2, \dots, t_R$ be real numbers in the interval $[0, T]$ such that $|t_{r_1} - t_{r_2}| \geq 1$ whenever $r_1 \neq r_2$. Show that if G is defined as in (27.11), then

$$\sum_{r=1}^R |D(it_r)|^2 \ll (N + M(\alpha, 4T)N^\alpha R)G.$$

- (c) Show that there is a constant $C = C(\alpha) > 0$ such that if $V^2 \geq CM(\alpha, 4T)N^\alpha$ and $|D(it_r)| \geq V$ for all r , then (27.10) holds.

3. (a) Show that when Bourgain's Example is inserted into the bilinear form inequality (27.2), the conclusion is that $\Delta^2 \gg RH$. By duality, this must hold also in (27.3). Our object now is to give a direct proof of this latter inequality.
- (b) Let $\mathcal{T}_\delta$ be defined as in Bourgain's Example. Show that $\mathcal{T}_\delta - \mathcal{T}_\delta = \mathcal{T}_\delta + \mathcal{T}_\delta \subseteq \mathcal{T}_{2\delta}$. (By definition, $\mathcal{A} + \mathcal{B} = \{a + b : a \in \mathcal{A}, b \in \mathcal{B}\}$.)
- (c) Note that the left hand side of (27.3) is

$$\begin{aligned} &\geq \sum_{N-H < n \leq N} \left| \sum_{r=1}^R y_r n^{-it_r} \right|^2 \\ &= \sum_{1 \leq r_1, r_2 \leq R} y_{r_1} \overline{y_{r_2}} \sum_{N-H < n \leq N} n^{-i(t_{r_1} - t_{r_2})}. \end{aligned}$$

- (d) Explain why this last sum over n is $= HN^{-i(t_{r_1} - t_{r_2})}(1 + O(\delta))$.
- (e) Take $y_r = N^{it_r}$ for all r . Explain why the lower expression displayed above is $\gg HR^2$, while the right hand side of (27.3) is $= \Delta^2 R$. Deduce that $\Delta^2 \gg HR$.
4. We now develop a variant of Halász's Method. Let $D(s)$, $\mathcal{T}$, and G be defined as in Theorem 27.1. Let $t_1, t_2, \dots, t_R$ denote the members of $\mathcal{T}$, and let y_r be a unimodular number with the property that $|D(it_r)| = y_r D(it_r)$.

- (a) Show that

$$\sum_{r=1}^R |D(it_r)| = \sum_{n=1}^N a_n \sum_{r=1}^R y_r n^{-it_r}.$$

- (b) Deduce that

$$\left(\sum_{r=1}^R |D(it_r)| \right)^2 \leq G \left(\sum_{n=1}^N \left| \sum_{r=1}^R y_r n^{-it_r} \right|^2 \right).$$

- (c) Suppose that $w_1, w_2, \dots$ are nonnegative numbers such that $w_n \geq 1$ for $1 \leq n \leq N$, and $\sum_{n=1}^\infty w_n < \infty$. Put $W(s) = \sum_{n=1}^\infty w_n n^{-s}$ for $\sigma \geq 0$. Explain why the second factor on the right hand side above is

$$\leq \sum_{1 \leq r_1, r_2 \leq R} |W(i(t_{r_1} - t_{r_2}))|.$$

- (d) Conclude that

$$\left(\sum_{r=1}^R |D(it_r)| \right)^2 \leq G \sum_{1 \leq r_1, r_2 \leq R} |W(i(t_{r_1} - t_{r_2}))|.$$

(e) Suppose that $|D(it_r)| \geq V$ for all r . Deduce that

$$R \leq \frac{W(0)G}{V^2} + \frac{G}{V^2 R} \sum_{\substack{1 \leq r_1 \leq R \\ r_1 \neq r_2}} |W(i(t_{r_1} - t_{r_2}))|.$$

(f) Note that

$$\frac{1}{R} \sum_{\substack{1 \leq r_1 \leq R \\ r_1 \neq r_2}} |W(i(t_{r_1} - t_{r_2}))| \leq \max_{1 \leq r_1 \leq R} \sum_{\substack{1 \leq r_2 \leq R \\ r_2 \neq r_1}} |W(i(t_{r_1} - t_{r_2}))|.$$

When combined with the bound from the preceding part, we obtain the same bound that we achieved from our original method. In the case that the numbers $t_{r_1} - t_{r_2}$ are distinct and well-spaced, we might be able to derive an upper bound for the left hand side above than the one on the right hand side, and thus obtain a better overall bound for, such t_r .

5. Suppose that $D(s)$ is defined as in (27.1). Suppose that $N^{1/2} \leq N \leq T$. From Corollary 26.2 and Lemma 27.12 we know that

$$\begin{aligned} \int_0^T |D(it)|^2 dt &\ll T \sum_{n=1}^N |a_n|^2, \\ \int_0^T |D(it)|^4 dt &\ll N^2 \left(\sum_{n=1}^N d(n) |a_n|^2 \right)^2. \end{aligned}$$

It would be helpful if we could derive similar estimates for fractional exponents between 2 and 4. In particular, let ν be determined by the equation $N^\nu = T$. Thus $1 \leq \nu \leq 2$. It would be useful if it were the case that

$$\int_0^T |D(it)|^{2\nu} dt \ll T \left(\sum_{n=1}^N d(n) |a_n|^2 \right)^\nu.$$

Suppose that $D(s)$ is taken as in Bourgain's Example, with $H \leq \delta N$. Take $T = N^2/H^2$. Show that in the interval $[0, T]$ there are $\asymp N/H^2$ disjoint intervals, each of length $\asymp N/H$ on which $|D(it)| \gg H$. Deduce that the left hand side above is $\gg N^2 H^{2\nu-3}$ and that the right hand side is $\ll N^2 H^{\nu-2} (\log N)^\nu$. Note the resulting contradiction.

One of course might conjecture that if $|a_n| \leq 1$ for all n , then

$$\int_0^T |D(it)|^{2\nu} dt \ll_\varepsilon T^{2+\varepsilon}. \quad (27.57) \quad \boxed{\text{E: Conj |D| 2nu}}$$

6. Let $\mathcal{T}$ be a finite set of real numbers. Show that the following two statements about a number $\Delta(N, \mathcal{T})$ are equivalent:

(i) If $|a_n| \leq 1$ for all $n \leq N$, then

$$\sum_{t \in \mathcal{T}} \left| \sum_{n=1}^N a_n n^{-it} \right| \leq \Delta(N, \mathcal{T}).$$

(ii) If $|y_t| \leq 1$ for all $t \in \mathcal{T}$, then

$$\sum_{n=1}^N \left| \sum_{t \in \mathcal{T}} y_t n^{-it} \right| \leq \Delta(N, \mathcal{T}).$$

7. To prove the identity (27.20), it suffices to note that both sides are linear forms in the a_n , and then to note that for each n the coefficient of a_n is the same on both sides.

- (a) Show that if $n > x$, then a_n makes no contribution to either side of (27.20).
 (b) Show that if $n \leq x$, then the contribution of a_n to the right hand side of (27.20) is a_n multiplied by

$$\frac{x-n}{x^\sigma} + 2\sigma \int_n^x (v-n)v^{-\sigma-1} + \sigma(\sigma+1) \int_n^x (v-n)(x-v)v^{-\sigma-2} dv.$$

- (c) Show that the above expression is equal to $(x-n)n^{-\sigma}$.

Exer:HMGLH

8. The *Generalized Lindelöf Hypothesis* (GLI) asserts that if χ is a character modulo q , then $L(1/2 + it, \chi) \ll_\varepsilon (q\tau)^\varepsilon$. If $L(s, \chi)$ is an L -function whose nontrivial zeros all lie on the $1/2$ -line, then it also satisfies this estimate.

- (a) Adopt the notation and hypotheses of Theorem 27.6. Suppose, in addition, that GLI is valid for all L -functions mod q . Show that

$$\sum_{r=1}^R |D(s_r, \chi_r)|^2 \ll (N + C(\varepsilon)RN^{1/2}(qT)^\varepsilon)G.$$

- (b) Adopt the notation and hypotheses of Theorem 27.7. Suppose, in addition, that GLI is valid for all L -functions. Show that

$$\sum_{r=1}^R |D(s_r, \chi_r)|^2 \ll (N + C(\varepsilon)RN^{1/2}(QT)^\varepsilon)G.$$

9. ^{DRHB79a}(?) The q -analogue of Halász's bound is a bilinear form inequality of the form

$$\sum_{r=1}^R \left| \sum_{n=1}^N a_n \chi_r(n) \right|^2 \leq \Delta^2 \sum_{n=1}^N |a_n|^2.$$

Our object is to establish an analogue of Bourgain's Example for the bilinear form above. Let $p_1, p_2, \dots, p_R$ denote the first R odd primes. Let χ_r denote the quadratic character modulo p_r . Take $a_n = 1$ if n is a square, and $a_n = 0$ otherwise.

- Show that the left hand side above is $\asymp NR$.
- Show that the right hand side above is $\asymp \Delta^2 N^{1/2}$.
- Deduce that $\Delta^2 \gg RN^{1/2}$. Compare this with the result of Exercise 8(a).

Exer:gam(s)est3

- Show that $\arctan \delta = \delta - \frac{1}{3}\delta^3 + O(\delta^5)$ for $0 \leq \delta \leq 1$.
 - Show that $\log(1 + \delta) = \delta + O(\delta^2)$ for $0 \leq \delta \leq 1$.
 - Deduce that there is a small positive constant c such that

$$|\gamma(s, \chi)| \asymp \left(\frac{t}{2\pi}\right)^{\frac{1}{2}-\sigma} \exp\left(\frac{(1-\sigma)^3}{8t^2}\right)$$

uniformly for $t^{2/3} \leq -\sigma \leq ct$.

- Show that if $s = \sigma + it$, then

$$\sin \pi s = (\sin \pi \sigma) \cosh \pi t + i(\cos \pi \sigma) \sinh \pi t.$$

- Let $f(s) = \exp(\sin \pi s)$. Show that $|f(it)| = |f(1 + it)| = 1$ for all t .
 - Show that $f(\frac{1}{2} + it) = \exp(\cosh \pi t) > \exp(\frac{1}{2} \exp(\pi|t|))$.
 - One of the hypotheses in the Phragmén–Lindelöf Theorem is that $0 < \alpha < \pi$. Show that this constraint on α cannot be relaxed.
- For a positive integer k , let $f(x) = \frac{1}{k!} \max(0, (1-x)^k)$. Let $F(s) = \int_0^\infty f(x)x^{s-1} dx$ be the Mellin transform of f . By induction on k , show that the integral that defines F converges for $\sigma > 0$, and that in that halfplane

$$F(s) = \frac{1}{s(s+1)\cdots(s+k)}.$$

- (b) For a positive integer k let $F(s)$ be defined as above, and let $f(x)$ be the inverse Mellin transform of F ,

$$f(x) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} F(s)x^{-s} ds$$

where $c > 0$. Show that $f(x) = 0$ if $x \geq 1$, and that $f(x) = \frac{1}{k!}(1-x)^k$ if $0 \leq x \leq 1$.

13. Let $\kappa(u)$ be defined as in (??).

- (a) Show that

$$\kappa(u) = \sum_{\substack{0 \leq j \leq 4 \\ 2u-1 \geq j}} (-1)^j \binom{4}{j} (2u-j-1)^3.$$

- (b) By swapping j with $4-j$, show that

$$\kappa(u) = \sum_{\substack{0 \leq j \leq 4 \\ 5-2u \leq j}} (-1)^j \binom{4}{j} (2u+j-5)^3.$$

- (c) Deduce that $\kappa(3-k) = \kappa(u)$.
 (d) Show that $\kappa(u) = (2u-1)^3$ for $1/2 \leq u \leq 1$.
 (e) Show that $\kappa(u) = 31 - 90u + 84u^2 - 24u^3$ for $1 \leq u \leq 3/2$.
 (f) Show that $\kappa(u) = -131 + 234u - 132u^2 + 24u^3$ for $3/2 \leq u \leq 2$.
 (g) Show that $\kappa(u) = (5-2u)^3$ for $2 \leq u \leq 5/2$.
 (h) Show that $\kappa(1) = \kappa(2) = 1$.
 (i) Show that $\kappa'(u)6(3-2u)(6u-5) > 0$ for $1 \leq u < 3/2$.

14. Suppose that $D_i(s) = \sum_{k_i=1}^{K_i} a_i(k_i)k_i^{-s}$ for $i = 1, 2, 3$, and that $D(s) = D_1(s)D_2(s)D_3(s) = \sum_{n=1}^{K_1K_2K_3} c(n)n^{-s}$.

- (a) Show that

$$c(n) = \sum_{\substack{k_1, k_2, k_3 \\ k_1 k_2 k_3 = n}} a_1(k_1)a_2(k_2)a_3(k_3).$$

- (b) Show that

$$|c(n)|^2 \leq d_3(n) \sum_{\substack{k_1, k_2, k_3 \\ k_1 k_2 k_3 = n}} |a_1(k_1)|^2 |a_2(k_2)|^2 |a_3(k_3)|^2.$$

- (c) Show that if $k_1 k_2 k_3 = n$, then $d_3(n) \leq d_3(k_1)d_3(k_2)d_3(k_3)$.

(d) Deduce that

$$\sum_{n=1}^{K_1 K_2 K_3} |c(n)|^2 \leq \prod_{i=1}^3 \left(\sum_{k_i=1}^{K_i} d_3(k_i) |a_i(k_i)|^2 \right).$$

27.1 Notes

S:LargeVals Notes

For an account of the Phragmén–Lindelöf Theorem see, for example, Theorem 5.1.9 in ^{BS2A}Simon (2015).

27.2 References

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28

Zero Density Theorems

C:ZDT

28.1 Zero counting functions

S:ZDT ZCF

There are many situations where, in lieu of the Riemann Hypothesis, an unconditional conclusion can be achieved *via* a bound for one of the functions

$$N(\sigma, T) = \text{card}\{\rho = \beta + i\gamma : \zeta(\rho) = 0, \beta \geq \sigma, |\gamma| \leq T\}, \quad (28.1) \quad \boxed{\text{E:NT}}$$

$$N_1(\sigma, q, T) = \sum_{\chi \bmod q} N(\sigma, \chi, T), \quad (28.2) \quad \boxed{\text{E:NqT}}$$

or

$$N_2(\sigma, Q, T) = \sum_{q \leq Q} \sum_{\chi \bmod q}^* N(\sigma, \chi, T), \quad (28.3) \quad \boxed{\text{E:NQT}}$$

where

$$N(\sigma, \chi, T) = \text{card}\{\rho = \beta + i\gamma : L(\rho; \chi^*) = 0, \beta \geq \sigma, |\gamma| \leq T\}, \quad (28.4) \quad \boxed{\text{E:NchiT}}$$

χ^* denotes the primitive character inducing χ and $\sum^*$ denotes a sum restricted to primitive characters. The underlying methods for dealing with each are closely related, and we will mostly work out the details for (28.3). Later we will see that in some aspects the theory for (28.1) can be pushed a bit further. It is also possible to cover all possible bases by considering

$$N(\sigma, q, Q, T) = \sum_{\substack{k \leq Q \\ (k, q) = 1}} \sum_{\chi \bmod qk}^\dagger N(\sigma, \chi, T), \quad (28.5) \quad \boxed{\text{E:NqQT}}$$

where $\sum^\dagger$ indicates a sum over primitive characters with conductor dk and $d|q$.

In earlier chapters estimates for the number of zeros of an analytic function of interest in a given region have depended upon standard theorems from classical complex analysis. To make progress here more sophisticated methods are required which depend on the special properties of L -functions.

When $\sigma > 1$ the lack of zeros is an immediate consequence of the Euler product. It can also be seen through the observation that

$$L(s; \chi)^{-1} = \sum_{n=1}^{\infty} \frac{\mu(n)\chi(n)}{n^s} \quad (28.6) \quad \boxed{\text{E:L-1}}$$

converges when $\sigma > 1$ and so is analytic in that half-plane. In particular

$$L(s; \chi)L(s; \chi)^{-1} = 1.$$

When $\sigma \leq 1$ we do not have the luxury of immediately knowing that (28.6) converges. However if we write

$$M(s; \chi) = \sum_{n \leq K} \mu(n)\chi(n)n^{-s}, \quad (28.7) \quad \boxed{\text{E:MSchi}}$$

then we can have some expectation that for a suitable parameter K the expression

$$L(s; \chi)M(s; \chi) - 1 \quad (28.8) \quad \boxed{\text{E:LM-1}}$$

is small most of the time. Such a function M is often termed a *mollifier*. On the other hand if $L(s; \chi) = 0$, then the expression is -1 and so the expectation is that this happens at most infrequently.

In order to quantify this observation we will need to approximate $L(s; \chi)$ by a series of the kind

$$\sum_n \chi(n)n^{-s}w(n)$$

with a weight function $w(n)$ which is close to 1 for smaller n . One such example which we have used in the past is given by

$$e^{-1/x} = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} x^w \Gamma(w) dw$$

where $c > 0$, and its inverse transform

$$\Gamma(w) = \int_0^\infty x^{w-1} e^{-x} dx = \int_0^\infty x^{-w-1} e^{-1/x} dx.$$

Although this is simple and useful, it would be handier if we could move part of our contour well to the left. However the gamma function has

the disadvantage of possessing singularities at the negative integers. A small modification removes this deficiency.

28.2 A Mellin transform

E:Melling

THIS SECTION SHOULD BE IN AN APPENDIX.

Let

$$g(w) = \frac{1}{\varpi w} \int_0^\infty x^{-w-1} \exp(-x - x^{-1}) dx \quad (28.9) \quad \text{E:gfn}$$

where

$$\varpi = \int_0^\infty x^{-1} \exp(-x - x^{-1}) dx. \quad (28.10) \quad \text{E:resg}$$

Then integral in (28.9) is an entire function, so that the function g is analytic for $w \neq 0$ and has a simple pole at $w = 0$ with residue 1. For $x \geq 0$, let

$$f(x) = \frac{1}{\varpi} \int_x^\infty y^{-1} \exp(-y - y^{-1}) dy. \quad (28.11) \quad \text{E:ffn}$$

Then, by integration by parts and a change of variable,

$$g(w) = \int_0^\infty x^{-w-1} f(x) dx = \int_0^\infty y^{w-1} f(1/y) dy. \quad (28.12) \quad \text{E:gintf}$$

A change of variable also shows that

$$f(1/x) = \frac{1}{\varpi} \int_0^x y^{-1} \exp(-y - y^{-1}) dy. \quad (28.13) \quad \text{E:f1overx}$$

Thus

$$f(x) + f(1/x) = 1 \quad (28.14) \quad \text{E:f+f}$$

and so

$$f(x) \ll x^{-1} e^{-x}, \quad f(1/x) = 1 + O(x^{-1} e^{-x}) \quad (28.15) \quad \text{E:f=1+o}$$

The function $g(w)$ behaves in a similar way to the gamma function, but with the added advantage that its only singularity is at 1. Thus we have the following estimate, which is similar to bounds following from Stirling's formula. It is perhaps not the most precise bound that can be established, but it will suffice for our purposes.

T:omegag

Lemma 28.1. *Suppose that $w \in \mathbb{C}$, $\operatorname{Re} w = u$ and $\operatorname{Im} w = v$. Then there is a constant $C \geq 1$ such that*

$$wg(w) \ll (C + C|w|)^{\max(1, |u|)} \exp(-\pi|v|/2).$$

Proof If necessary by considering the complex conjugate we may suppose that $v \geq 0$. Let $\delta = (1 + |u|)/(2 + |w|)$, $R > 0$ and C_R denote the contour consisting of the line segments joining

$$\{-R, R, R + i(\pi/2 - \delta), -R + i(\pi/2 - \delta), -R\}.$$

Then, by Cauchy's theorem, we have

$$\int_{C_R} \exp(wz - e^z - e^{-z}) dz = 0.$$

Moreover

$$\begin{aligned} \int_{\pm R}^{\pm R + i(\pi/2 - \delta)} \exp(wz - e^z - e^{-z}) dz &\ll \exp(\pm Ru - e^R \sin \delta) \\ &\rightarrow 0 \text{ as } R \rightarrow \infty. \end{aligned}$$

Thus

$$\begin{aligned} \varpi w g(w) &= \int_0^\infty y^{w-1} \exp(-y - 1/y) dy \\ &= \int_{-\infty}^\infty \exp(wx - e^x - e^{-x}) dx \\ &= \int_{-\infty}^\infty \exp\left(wx + w\left(i\frac{\pi}{2} - i\delta\right) - 2 \cosh\left(x + i\frac{\pi}{2} - i\delta\right)\right) dx. \end{aligned}$$

Hence

$$|\varpi w g(w)| \leq \int_{-\infty}^\infty \exp\left(ux - v\pi/2 + v\delta - e^x \sin \delta - e^{-x} \sin \delta\right) dx$$

Therefore

$$|\varpi w g(w)| \leq 2 \exp(-v\pi/2 + v\delta) \int_0^\infty \exp(|u|x - e^x \sin \delta) dx.$$

The integral here is

$$\int_1^\infty t^{|u|-1} e^{-t \sin \delta} dt. \quad (28.16) \quad \boxed{\text{E:sindelta}}$$

When $|u| \geq 1$ it is

$$\leq (\sin \delta)^{-|u|} \Gamma(|u|) \ll \left(\frac{|u|}{e \sin \delta}\right)^{|u|} |u|^{-1/2}$$

and

$$\sin \delta = \delta - \int_0^\delta \int_0^\theta \sin \beta d\beta d\theta \geq \delta - \frac{1}{2} \delta^2 \sin \delta$$

so that

$$\sin \delta \geq \frac{2\delta}{2 + \delta^2}.$$

Also $v\delta \leq 1 + |u|$. Hence

$$e^{v\delta} \left(\frac{|u|}{e \sin \delta} \right)^{|u|} \ll \left(\frac{|u|(2 + \delta^2)}{\delta} \right)^{|u|} \ll (C + C|w|)^{|u|}$$

as desired.

When $|u| \leq 1$ the integral (28.16) is

$$\leq \frac{1}{\sin \delta}$$

and we have

$$e^{v\delta} (\sin \delta)^{-1} \ll 1 + |w|$$

which once more suffices. $\square$

We also need to know more about the inverse Mellin transform of g .

T:f1X **Theorem 28.2.** *Suppose that $X > 0$ and $\nu > 0$. Then*

$$\frac{1}{2\pi i} \int_{\nu-i\infty}^{\nu+i\infty} X^w g(w) dw = f(1/X)$$

Proof By (28.9) and the lemma, the integral above is

$$\frac{1}{\varpi} \int_0^\infty x^{-1} \exp(-x - x^{-1}) \frac{1}{2\pi i} \int_{\nu-i\infty}^{\nu+i\infty} \frac{(X/x)^w}{w} dw dx.$$

The inner integral is $2\pi i$ when $x < X$ and 0 when $x > X$. Thus the above is

$$\frac{1}{\varpi} \int_0^X x^{-1} \exp(-x - x^{-1}) dx = f(1/X)$$

by (28.13) $\square$

28.3 A bound for $L(s; \chi)$

THIS SECTION SHOULD BE IN CHAPTER 22.

Let χ be a primitive character modulo q and suppose that $\frac{1}{2} \leq \sigma \leq 1$ and $Y > 0$. We need an effective upper bound for $L(s; \chi)$ in terms of Dirichlet polynomials on the σ -line. We start by imitating the process used to establish approximate functional equations.

When $c = \operatorname{Re} w > \sigma$, by Theorem 28.2 we have

$$\sum_{m=1}^{\infty} \frac{\chi(m)}{m^s} f(m/Y) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} L(s+w; \chi) Y^w g(w) dw$$

We now move the path to the line $\operatorname{Re} w = -1$, picking up the residue at 0, and another one at $w = 1-s$ when $q = 1$ and apply the functional equation for $L(s; \chi)$ (Corollary 10.9). Thus the above is

$$\begin{aligned} & L(s; \chi) + E(\chi) x^{1-s} g(1-s) \\ & + \frac{1}{2\pi i} \int_{-1-i\infty}^{-1+i\infty} \psi(s+w; \chi) L(1-s-w; \bar{\chi}) Y^w g(w) dw \end{aligned} \quad (28.17) \quad \boxed{\text{eq:E:Lapprox1}}$$

where

$$\psi(z; \chi) = \epsilon(\chi) 2^z \pi^{z-1} q^{1/2-z} \Gamma(1-z) \sin(\pi(z+\kappa)/2). \quad (28.18) \quad \boxed{\text{E:fepsi}}$$

and $E(\chi) = 0$ unless $q = 1$ in which case $E(\chi) = 1$.

We apply Stirling's formula, Theorem C.1, (C.18), to bound ψ . When $\operatorname{Re} w \leq -1$ and $\frac{1}{2} \leq \sigma \leq 1$ we have $|1-s-w| \geq 1$ and $|\arg(1-s-w)| \leq \pi/2$ so that uniformly for such $s+w$ we have

$$\psi(s+w; \chi) \ll (3q + q|s+w|)^{\frac{1}{2}-\sigma-u}. \quad (28.19) \quad \boxed{\text{eq:E:psibound}}$$

where $w = u + iv$.

The third term in (28.17) is

$$\sum_{m=1}^{\infty} \bar{\chi}(m) m^{s-1} h(mY; s, \chi) \quad (28.20) \quad \boxed{\text{E:sumhnY}}$$

where, for $V > 0$,

$$h(V; s, \chi) = \frac{1}{2\pi i} \int_{-1-i\infty}^{-1+i\infty} \psi(s+w; \chi) V^w g(w) dw. \quad (28.21) \quad \boxed{\text{E:hVschi}}$$

Let $\lambda \geq 3$ be a parameter at our disposal. We move the contour to the vertical line $u = -\lambda$. Then, by Lemma 28.1, for a suitable constant $c_1 \geq 1$ the integrand is

$$\ll (4q + q|t| + q\lambda + q|v|)^{\frac{1}{2}-\sigma+\lambda} V^{-\lambda} (c_1 + c_1\lambda + c_1|v|)^{\lambda} e^{-\frac{\pi|v|}{2}}.$$

We suppose henceforward that

$$|t| \leq T$$

where $T \geq 3$. Then the integrand is

$$\ll P^{\lambda} V^{-\lambda} (\lambda + |v|)^{2\lambda} e^{-\pi|v|/2}$$

where

$$P \geq CqT \quad (28.22) \quad \boxed{\text{eq:E:psibound1}}$$

and C is a suitably large absolute constant. Hence

$$h(V; s, \chi) \ll P^\lambda V^{-\lambda} \lambda^{5\lambda}.$$

Suppose that $V \geq (3e)^5 P$. Then the function of a positive real variable ν ,

$$F(\nu) = P^\nu V^{-\nu} \nu^{5\nu} = \exp(5\nu \log \nu - \nu \log(V/P))$$

has a minimum when $\nu = \nu_0$ where

$$\nu_0 = e^{-1}(V/P)^{1/5} \geq 3$$

and

$$F(\nu_0) = \exp(5\nu_0) = \exp(-5e^{-1}(V/P)^{1/5}).$$

Let $\lambda = \nu_0$. Then

$$h(V; s, \chi) \ll \exp(-5e^{-1}(V/P)^{1/5}). \quad (28.23) \quad \boxed{\text{E:hVschi1}}$$

Suppose that $Y \leq P/\log P$ and $Z \geq 2PY^{-1}(3e \log P)^5$. The function

$$\nu(\theta) = 5e^{-1}(\theta Y/P)^{1/5} - 2 \log \theta$$

is increasing for $\theta \geq PY^{-1}(2e)^5$ and so if $m > Z$, then we have

$$\begin{aligned} \nu(m) &> \nu(Z) \\ &\geq 15 \cdot 2^{1/5} \log P - 2 \log \left(\frac{2P}{Y} (3e \log P)^5 \right) \\ &> 3 \log P. \end{aligned}$$

Recall that $P > C$ is suitably large. Thus when $m > Z$ we have

$$5e^{-1}(mY/P)^{1/5} > 3 \log P + 2 \log m.$$

Hence, by (28.23),

$$\begin{aligned} \sum_{m>Z} \bar{\chi}(m) m^{s-1} h(mY; s, \chi) &\ll \sum_{m>Z} \exp(-5e^{-1}(mY/P)^{1/5}) \\ &\quad \sum_{m>Z} m^{-2} \exp(-\nu(m)) \end{aligned}$$

so that

$$\sum_{m>Z} \bar{\chi}(m) m^{s-1} h(mY; s, \chi) \ll P^{-3}. \quad (28.24) \quad \boxed{\text{E:sumhV}}$$

We now rewrite the contribution from $m \leq Z$ as

$$\sum_{m \leq Z} \bar{\chi}(m) n^{s-1} h(mY; s, \chi) = \frac{1}{2\pi i} \int_{-1-i\infty}^{-1+i\infty} \sum_{m \leq Z} \bar{\chi}(m) m^{s+w-1} \psi(s+w; \chi) Y^w g(w) dw.$$

When $\sigma \geq \theta > \frac{1}{2}$ we then move the path to the line $\operatorname{Re} w = \theta + \frac{1}{2} - 2\sigma$. Thus

$$-1 < \theta - \frac{3}{2} \leq \operatorname{Re} w \leq \frac{1}{2} - \theta < 0,$$

and to summarize so far, we have

$$\begin{aligned} \sum_{m=1}^{\infty} \frac{\chi(m)}{m^s} f(m/Y) &= L(s; \chi) + E(\chi) Y^{1-s} g(1-s) + \\ &\frac{1}{2\pi i} \int_{\theta-\frac{1}{2}-i\infty}^{\theta-\frac{1}{2}+i\infty} \sum_{m \leq Z} \frac{\bar{\chi}(m)}{m^{\bar{s}-w}} \psi(1-\bar{s}+w; \chi) Y^{1-2\sigma+w} g(1-2\sigma+w) dw \\ &\quad + O(P^{-3}). \end{aligned} \quad (28.25) \quad \boxed{\text{E: sumfm/Y}}$$

Suppose that $X \geq 3Y \log(YP)$. Then, by (28.15)

$$\sum_{m > X} \chi(m) m^{-1/2-it} f(m/Y) \ll Y(YP)^{-3} \ll P^{-3}. \quad (28.26) \quad \boxed{\text{E: sumfm>X}}$$

Thus we obtain

$$\begin{aligned} L(s; \chi) &= \sum_{m \leq X} \frac{\chi(m)}{m^s} f(m/Y) - E(\chi) Y^{1-s} g(1-s) - \\ &\frac{1}{2\pi i} \int_{\theta-\frac{1}{2}-i\infty}^{\theta-\frac{1}{2}+i\infty} \sum_{m \leq Z} \frac{\bar{\chi}(m)}{m^{\bar{s}-w}} \psi(1-\bar{s}+w; \chi) Y^{1-2\sigma+w} g(1-2\sigma+w) dw \\ &\quad + O(P^{-3}). \end{aligned}$$

If $E(\chi) \neq 0$, then $q = 1$. Thus, by Lemma 28.1

$$E(\chi) Y^{1-s} g(1-s) \ll Y^{1-\sigma} |1-s|^{-1} (1+|s|) \exp(-\pi|t|/2).$$

Thus we have

$$\begin{aligned}
 L(s; \chi) &= \sum_{m \leq X} \frac{\chi(m)}{m^s} f(m/Y) - \\
 &\frac{1}{2\pi i} \int_{\theta - \frac{1}{2} - i\infty}^{\theta - \frac{1}{2} + i\infty} \sum_{m \leq Z} \frac{\bar{\chi}(m)}{m^{\bar{s}-w}} \psi(1 - \bar{s} + w; \chi) Y^{1-2\sigma+w} g(1 - 2\sigma + w) dw \\
 &\quad + O\left(P^{-3} + E(\chi) Y^{1-\sigma} |1-s|^{-1} e^{-|t|}\right). \quad (28.27) \quad \boxed{\text{E:sumfm}<\text{X}}
 \end{aligned}$$

When $\text{Re } w = \theta - \frac{1}{2}$ we also have

$$\begin{aligned}
 &\psi(1 - \bar{s} + w; \chi) Y^{1-2\sigma+w} g(1 - 2\sigma + w) \\
 &\ll \left| \theta - \frac{1}{2} \right|^{-1} Y^{\theta + \frac{1}{2} - 2\sigma} P^{\theta - \sigma} e^{-|v|}.
 \end{aligned}$$

Putting it all together we have the following.

T:Lapprox **Theorem 28.3.** *There is a constant $C \geq 3$ such that whenever $T \geq 2$, χ is a primitive character modulo q , $P \geq CqT$,*

$$Y \leq P/\log P, \quad X \geq 3Y \log(YP) \text{ and } Z \geq 2PY^{-1}(3e \log P)^5$$

we have

$$\begin{aligned}
 &L(s; \chi) - \sum_{m \leq X} \frac{\chi(m)}{m^s} f(m/Y) \ll \\
 &Y^{\frac{1}{2}-\theta} (Y^2 P)^{\theta-\sigma} \left| \theta - \frac{1}{2} \right|^{-1} \int_{-\infty}^{\infty} \left| \sum_{m \leq Z} \chi(m) m^{-s+\theta-\frac{1}{2}+iv} \right| e^{-|v|} dv \\
 &\quad + E(\chi) Y^{\theta+\frac{1}{2}-2\sigma} |1-s|^{-1} \exp(-|t|) + P^{-3}
 \end{aligned}$$

uniformly for $|t| \leq T$, $\frac{1}{2} < \theta \leq \sigma \leq 1$, $E(\chi)s \neq 1$.

28.4 Zero density estimates

S:ZDE1

We can now apply the results in the previous sections to give non-trivial bounds for the functions N , N_1 , N_2 . The basic bounds we give here have many applications.

T:ZDE1a **Theorem 28.4.** *Suppose $q \geq 1$ and $T \geq 1$. Then*

$$N_1(\theta, q, T) \ll (qT)^{\frac{3(1-\theta)}{2-\theta}} (\log(2qT))^6 \quad (28.28) \quad \boxed{\text{E:ZDE1a}}$$

uniformly when $\frac{1}{2} \leq \theta \leq 1$.

Suppose further that $\eta > 0$. Then there is a positive number $C(\eta)$ such that

$$N_1(\theta, q, T) \ll_{\eta} (\log(2qT))^{C(\eta)} ((qT)^{(2+\eta)(1-\theta)} + (qT)^{\frac{3(1-\theta)}{3\theta-1}}) \quad (28.29) \quad \boxed{\text{E:ZDE2a}}$$

uniformly when $\frac{2}{3} \leq \theta \leq 1$,

T:ZDE1b **Corollary 28.5.** *There is a positive constant c such that, whenever $\frac{1}{2} \leq \theta \leq 1$,*

$$N_1(\theta, q, T) \ll (qT)^{\frac{12}{5}(1-\theta)} (\log 2qT)^c.$$

T:ZDE1 **Theorem 28.6.** *Suppose $q \geq 1$ and $T \geq 1$. Then*

$$N_2(\theta, Q, T) \ll (Q^2T)^{\frac{3(1-\theta)}{2-\theta}} (\log(2QT))^6 \quad (28.30) \quad \boxed{\text{E:ZDE1}}$$

uniformly when $\frac{1}{2} \leq \theta \leq 1$.

Suppose further that $\eta > 0$. Then there is a positive number $C(\eta)$ such that

$$N_2(\theta, Q, T) \ll_{\eta} (\log(2QT))^{C(\eta)} ((QT)^{(2+\eta)(1-\theta)} + (Q^2T)^{\frac{3(1-\theta)}{3\theta-1}}) \quad (28.31) \quad \boxed{\text{E:ZDE2}}$$

uniformly when $\frac{2}{3} \leq \theta \leq 1$.

T:ZDE1c **Corollary 28.7.** *There is a positive constant c such that whenever $\frac{1}{2} \leq \theta \leq 1$*

$$N_2(\theta, Q, T) \ll (Q^2T)^{\frac{12}{5}(1-\theta)} (\log 2QT)^c.$$

The conjecture that the above corollaries hold with $12/5$ replaced by ξ for any $\xi > 2$ is known at the density hypothesis. On inspection of the Theorems one can see that this does indeed hold in the restricted range $\theta \geq 5/6$.

We concentrate on the proof of Theorem 28.6. Theorem 28.4 follows in the same way using the concomitant mean value theorems.

By Corollary 14.7,

$$N_2(1/2, Q, T) \ll Q^2T \log(2QT).$$

Also, by Corollary 11.10 we know that there is a positive constant c such that there is at most one zero $\rho = \beta + i\gamma$ to be counted with $\beta \geq 1 - c/\log 2QT$. Thus in the proof of Theorem 28.6 we may suppose that

$$\frac{1}{2} - \frac{\log \log(2QT)}{\log(2QT)} \leq \theta \leq 1 - \frac{1}{c \log 2QT}. \quad (28.32) \quad \boxed{\text{E:ZDEalpha}}$$

Similar considerations to the above pertain with regard to Theorem 28.4.

We consider first (28.30) as this has the simplest proof and yet illustrates the main ideas.

28.4.1 The Ingham Bound (28.30)

For brevity write

$$H = CQ^2T \quad (28.33) \quad \boxed{\text{E: IngD}}$$

where C is a large constant, and for

$$C \leq K \leq H \quad (28.34) \quad \boxed{\text{E: IngK}}$$

let $M(s; \chi)$ be as in (28.7), and define

$$a(n) = \sum_{\substack{l|n \\ l \leq K}} \mu(l) \quad (28.35) \quad \boxed{\text{E: Ingan}}$$

so that

$$|a(n)| \leq d(n). \quad (28.36) \quad \boxed{\text{E: Ingand}}$$

Let

$$Y \geq 1 \quad (28.37) \quad \boxed{\text{E: IngY}}$$

and suppose that $\frac{1}{2} \leq \phi < \sigma \leq 1$. Then, by (5.25)

$$\sum_{n=1}^{\infty} a(n) \chi(n) n^{-s} e^{-n/Y} = \frac{1}{2\pi i} \int_{2-i\infty}^{2+i\infty} L(s+w; \chi) M(s+w; \chi) Y^w \Gamma(w) dw.$$

The integrand has singularities at 0, and, when $L(s; \chi) = \zeta(s)$, at $w = 1 - s$. In view of the bounds given by Corollaries 10.5 and 10.10, Lemma 10.15 and (C.19) of Theorem C.1 we are able to move the path of integration to the line $\text{Re } w = \phi - \sigma$ and pick up the residues at 0 and, when $L(s; \chi) = \zeta(s)$, at $w = 1 - s$. Thus

$$\begin{aligned} \sum_{n=1}^{\infty} a(n) \chi(n) n^{-s} e^{-n/Y} = & E(\chi) M(1; \chi) Y^{1-s} \Gamma(1-s) + L(s; \chi) M(s; \chi) + \\ & \frac{1}{2\pi} \int_{-\infty}^{\infty} L(\tfrac{1}{2} + it + iv; \chi) M(\tfrac{1}{2} + it + iv; \chi) Y^{\frac{1}{2}-\sigma+iv} \Gamma(\tfrac{1}{2} - \sigma + iv) dv. \end{aligned}$$

Let

$$Z = Y(\log H)^2. \quad (28.38) \quad \boxed{\text{E: IngZ}}$$

Then

$$\sum_{n>Z} a(n)\chi(n)n^{-s}e^{-n/Y} \ll H^{-1}.$$

When $E(\chi) = 1$ and $|t| \geq (\log H)^2$, by Stirling's formula (C.19),

$$E(\chi)M(1; \chi)Y^{1-s}\Gamma(1-s) \ll H^{-1}.$$

We also have $a(n) = 0$ when $2 \leq n \leq K$, and $a(1) = 1$ and $e^{-1/Y} = 1 + O(1/Y)$. Hence

$$\begin{aligned} 1 + \sum_{K < n \leq Z} a(n)\chi(n)n^{-s}e^{-n/Y} &= L(s; \chi)M(s; \chi) + O(1/H) + \\ \frac{1}{2\pi} \int_{-\infty}^{\infty} L(\tfrac{1}{2} + it + iv; \chi)M(\tfrac{1}{2} + it + iv; \chi)Y^{\frac{1}{2}-\sigma+iv}\Gamma(\tfrac{1}{2} - \sigma + iv)dv. \end{aligned} \quad (28.39) \quad \boxed{\text{E: ZDELM1}}$$

By Corollary 14.3 the number of zeros $\rho = \beta + i\gamma$ of the Riemann zeta function with $|\gamma| \leq (\log H)^2$ is $\ll (\log H)^3$ which is an acceptable bound in our theorem. Thus we can exclude such zeros from our subsequent analysis and (28.39) holds for all remaining zeros s . When χ is a primitive character modulo q , let $\mathcal{R}_0(\theta, \chi, T)$ denote the set of zeros ρ of $L(s; \chi)$ with $\beta \geq \theta$, $|\gamma| \leq T$, except that when $q = 1$ we exclude the zeros with $|\gamma| \leq (\log H)^2$. Then

$$N_2(\theta, Q, T) \leq \sum_{q \leq Q} \sum_{\chi \bmod q}^* \text{card } \mathcal{R}_0(\theta, \chi, T) + O((\log H)^3)$$

and for $\rho \in \mathcal{R}_0(\theta, \chi, T)$ we have

$$\begin{aligned} 1 \ll & \left| \sum_{K < n \leq Z} a(n)\chi(n)n^{-\rho}e^{-n/Y} \right| + \\ & Y^{\frac{1}{2}-\beta} \int_{-\infty}^{\infty} |L(\tfrac{1}{2} + i\gamma + iv; \chi)M(\tfrac{1}{2} + i\gamma + iv; \chi)|e^{-|v|}dv. \end{aligned}$$

By Corollary 14.7, given $q \leq Q$, a primitive character χ modulo q and $|t| \leq T$ there are at most $\ll \log 2QT$ zeros ρ of $L(s; \chi)$ with $|t - \text{Im } \rho| \leq 1$. Thus we can partition $\mathcal{R}_0(\theta, \chi, T)$ into $\ll \log H$ subsets in each of which any distinct pair ρ and ρ' of zeros of a given $L(s; \chi)$ satisfy $|\text{Im}(\rho -$

$\rho')| \geq 1$. Let $\mathcal{R}(\theta, \chi, T)$ denote such a subset with the largest number of elements. Thus

$$N_2(\theta, Q, T) \ll (\log H) \sum_{q \leq Q} \sum_{\chi \bmod q}^* \text{card } \mathcal{R}(\theta, \chi, T) + O((\log H)^3). \quad (28.40) \quad \boxed{\text{E:NQTlog}}$$

For each $\rho \in \mathcal{R}(\theta, Q, T)$ at least one of the following holds

$$1 \ll \left| \sum_{K < n \leq Z} a(n) \chi(n) n^{-\rho} e^{-n/Y} \right|^2,$$

$$1 \ll Y^{\frac{2}{3} - \frac{4}{3}\theta} \int_{-\infty}^{\infty} |L(\frac{1}{2} + i\gamma + iv; \chi) M(\frac{1}{2} + i\gamma + iv; \chi)|^{4/3} e^{-|v|} dv.$$

The number of zeros in the first case is

$$\ll \sum_{q \leq Q} \sum_{\chi \bmod q}^* \sum_{\rho \in \mathcal{R}(\theta, \chi, T)} \left| \sum_{K < n \leq Z} a(n) \chi(n) n^{-\rho} e^{-n/Y} \right|^2$$

and in the second is $\ll$

$$Y^{\frac{2}{3} - \frac{4}{3}\theta} \int_{-\infty}^{\infty} \sum_{q \leq Q} \sum_{\chi \bmod q}^* \sum_{\rho \in \mathcal{R}(\theta, \chi, T)} |L(\frac{1}{2} + i\gamma + iv; \chi) M(\frac{1}{2} + i\gamma + iv; \chi)|^{\frac{4}{3}} \frac{dv}{e^{|v|}}.$$

By (26.28) the first expression is

$$\begin{aligned} &\ll (\log H) \sum_{K < n \leq Z} a(n)^2 n^{-2\theta} e^{-2n/Y} (n + H) \\ &\ll (\log H)^4 ((1 - \theta)^{-1} Y^{2-2\theta} + (2\theta - 1)^{-1} H K^{1-2\theta}) \\ &\ll (\log H)^5 (Y^{2-2\theta} + H K^{1-2\theta}) \end{aligned}$$

and, by Hölder's inequality and Corollary 26.31 and (26.28), the second is

$$\begin{aligned} &\ll Y^{\frac{2}{3} - \frac{4}{3}\theta} \int_{-\infty}^{\infty} \left(\sum_{q \leq Q} \sum_{\chi \bmod q}^* \sum_{\rho \in \mathcal{R}(\theta, \chi, T)} |L(\frac{1}{2} + i\gamma + iv; \chi)|^4 \right)^{1/3} \times \\ &\quad \left(\sum_{q \leq Q} \sum_{\chi \bmod q}^* \sum_{\rho \in \mathcal{R}(\theta, \chi, T)} |M(\frac{1}{2} + i\gamma + iv; \chi)|^2 \right)^{2/3} e^{-|v|} dv \\ &\ll Y^{\frac{2}{3} - \frac{4}{3}\theta} (\log H)^{5/3} H^{1/3} \left((\log H) \sum_{n \leq K} n^{-1} (n + H) \right)^{2/3} \\ &\ll Y^{\frac{2}{3} - \frac{4}{3}\theta} (\log H)^3 H^{1/3} (K + H)^{2/3}. \end{aligned}$$

The choices

$$K = H, \quad Y = H^{\frac{3}{4-2\theta}}$$

with (28.40) secure the required bound (28.28).

28.4.2 The Huxley Bound (28.31)

SS:THB

We now concentrate on (28.31). Since the bound in (28.30) is smaller when $\theta \leq 3/4$ we may suppose that

$$\theta \geq 3/4. \quad (28.41) \quad \text{E:ZDEalphab}$$

In (28.31) the first term on the right only comes into play when, essentially, $\theta > \frac{5}{6}$.

Define H as before and

$$P = CQT \quad (28.42) \quad \text{E:ZDE31}$$

where C is as in Theorem 28.3, and in that theorem take

$$X = Z = C_1 P^{1/2} (\log P)^3, Y = P^{1/2} (\log P)^2 \quad (28.43) \quad \text{E:ZDEXYU}$$

where C_1 is a sufficiently large constant.

We may certainly suppose that

$$0 < \eta < 1/4, \quad K = P^{\eta/5} \quad (28.44) \quad \text{E:HuxX}$$

and we use the same mollifier (28.7) as before.

When $\sigma \geq \theta$ and $|t| \leq T$, by Theorem 28.3 and (28.41),

$$\begin{aligned} L(s; \chi)M(s; \chi) - \sum_{n \leq KX} a(n)\chi(n)n^{-s} &\ll \\ E(\chi)Y^{\frac{1}{2}-\theta}K^{1-\theta}(\log K)|1-s|^{-1}\exp(-|t|) + \\ Y^{\frac{1}{2}-\theta} \int_{-\infty}^{\infty} \left| \sum_{n \leq KX} b(n; v)\chi(n)n^{-s} \right| e^{-|v|} dv + P^{-1} \end{aligned}$$

where

$$a(n) = \sum_{\substack{l \leq K, m \leq X \\ lm=n}} \mu(l)f(m/Y) \quad (28.45) \quad \text{E:ZDEan}$$

and

$$b(n; v) = \sum_{\substack{l \leq K, m \leq X \\ lm=n}} \mu(l)m^{\theta-\frac{1}{2}+iv}. \quad (28.46) \quad \text{E:ZDEbn}$$

By (28.15) we have $f(m/Y) = 1 + O(mY^{-1} \exp(-Y/m))$. By (28.43) and (28.44) we have $K \leq Y(\log P)^{-2} < X$ and, when $n \leq K$,

$$\begin{aligned} a(n) &= \sum_{lm=n} \mu(l) f(m/Y) \\ &= \sum_{lm=n} \mu(l) + O(d(n) \exp(-(\log P)^2)). \end{aligned}$$

The main term here is 0 unless $n = 1$ in which case it is 1. Thus

$$\begin{aligned} L(s; \chi) M(s; \chi) - 1 &\ll \left| \sum_{K < n \leq KX} a(n) \chi(n) n^{-s} \right| + H^{-1} + \\ &E(\chi) Y^{\frac{1}{2}-\theta} K^{1-\theta} (\log K) |1-s|^{-1} \exp(-|t|) + \\ &Y^{\frac{1}{2}-\theta} \int_{-\infty}^{\infty} \left| \sum_{n \leq KX} \chi(n) n^{-s} b(n; v) \right| e^{-|v|} dv \end{aligned}$$

When $n \leq K$, since $\sigma \geq \theta$, we have, by (28.46),

$$\begin{aligned} \sum_{n \leq K} \chi(n) n^{-s} b(n; v) &\ll \sum_{l \leq K} l^{-\sigma} \sum_{m \leq K/l} m^{\theta-\sigma-\frac{1}{2}} \\ &\ll \sum_{l \leq K} l^{-\theta} \sum_{m \leq K/l} m^{-\frac{1}{2}}. \end{aligned}$$

Thus

$$\sum_{n \leq K} \chi(n) n^{-s} b(n; v) \ll \sum_{l \leq K} l^{-\frac{1}{2}-\theta} K^{\frac{1}{2}} \ll K^{\frac{1}{2}} \log 2K.$$

By (28.41) and (28.43) we have

$$(X/Y)^{\theta-1/2} (\log H)^{-1/2} \ll 1.$$

Hence

$$\begin{aligned} L(s; \chi) M(s; \chi) - 1 &\ll \left| \sum_{K < n \leq KX} a(n) \chi(n) n^{-s} \right| + P^{-1} \\ &+ Y^{\frac{1}{2}-\theta} K^{\frac{1}{2}} (\log K) (1 + E(\chi) |1-s|^{-1}) \\ &+ \int_{-\infty}^{\infty} \left| \sum_{K < n \leq KX} b^*(n; v) \chi(n) n^{-s} \right| e^{-|v|} dv \end{aligned}$$

where

$$b^*(n; v) = (\log H)^{1/2} \sum_{\substack{l \leq K, m \leq X \\ lm = n}} \mu(l) (m/Z)^{\theta - \frac{1}{2} + iv} \quad (28.47) \quad \boxed{\text{E:ZDEb*n}}$$

and we have used (28.43).

We record for future use that

$$a(n) \ll d(n), \quad b^*(n; v) \ll (\log P)^{1/2} d(n). \quad (28.48) \quad \boxed{\text{E:ZDEab}}$$

Now we suppose that if $q = 1$, then $|1 - s| \gg 1$, which we are certainly entitled to do when s is a zero since in this case $L(s; \chi) = \zeta(s)$. Hence

$$\begin{aligned} L(s; \chi)M(s; \chi) - 1 &\ll \left| \sum_{K < n \leq KX} a(n) \chi(n) n^{-s} \right| + H^{-1} \\ &\quad + Y^{\frac{1}{2} - \theta} K^{\frac{1}{2}} \log P \\ &\quad + \int_{-\infty}^{\infty} \left| \sum_{K < n \leq KX} b^*(n; v) \chi(n) n^{-s} \right| e^{-|v|} dv \end{aligned}$$

By (28.41) and (28.43) we have

$$Y^{\frac{1}{2} - \theta} K^{\frac{1}{2}} \log P \ll (\log P)^{-1}.$$

Therefore, if $\rho = \beta + i\gamma$ is a zero of $L(s; \chi)$ with $\beta = \operatorname{Re} \rho \geq \theta$, then

$$\begin{aligned} 1 &\ll \left| \sum_{K < n \leq KX} a(n) \chi(n) n^{-\rho} \right| \\ &\quad + \int_{-\infty}^{\infty} \left| \sum_{K < n \leq KX} b^*(n; v) \chi(n) n^{-\rho} \right| e^{-|v|} dv \end{aligned}$$

We now partition the interval $(K, KX]$ into

$$\ll \log H$$

dyadic intervals $I_j = (K_j, K'_j]$ where $K_j = K2^{j-1}$ and

$$K'_j = \min(K2^j, KX).$$

Thus

$$1 \ll \sum_{j \ll \log H} \left(\left| \sum_{n \in I_j} a(n) \chi(n) n^{-\rho} \right| + \int_{-\infty}^{\infty} \left| \sum_{n \in I_j} b^*(n; v) \chi(n) n^{-\rho} \right| e^{-|v|} dv \right)$$

Let $\mathcal{P}(\chi)$ be the set of zeros ρ of $L(s; \chi)$ being counted. Then for each such ρ there is a $j \ll \log H$ such that

$$1 \ll (\log H) \left(\left| \sum_{n \in I_j} a(n) \chi(n) n^{-\rho} \right| + \int_{-\infty}^{\infty} \left| \sum_{n \in I_j} b^*(n; v) \chi(n) n^{-\rho} \right| e^{-|v|} dv \right).$$

Let $\mathcal{P}_j(\chi)$ be the set of $\rho \in \mathcal{P}(\chi)$ for which this holds. Then, for any fixed k and $\rho \in \mathcal{P}_j(\chi)$,

$$1 \ll (\log H)^{2k} \left(\left| \sum_{n \in I_j} a(n) \chi(n) n^{-\rho} \right|^{2k} + \int_{-\infty}^{\infty} \left| \sum_{n \in I_j} b^*(n; v) \chi(n) n^{-\rho} \right|^{2k} e^{-|v|} dv \right).$$

Hence

$$1 \ll (\log H)^{3k} \left(\left| \sum_{n \in J_j} c_1(n) \chi(n) n^{-\rho} \right|^2 + \int_{-\infty}^{\infty} \left| \sum_{n \in J_j} c_2(n; v) \chi(n) n^{-\rho} \right|^2 e^{-|v|} dv \right).$$

where

$$J_j \subset (K_j^k, K_j'^k], \quad |c_1(n)| \ll d_{2k}(n), \quad |c_2(n; v)| \ll d_{2k}(n).$$

Let

$$N_j = \sum_{q \leq Q} \sum_{\chi \bmod q}^* \text{card } \mathcal{P}_j(\chi).$$

Then, by Theorem 27.15 we have

$$N_j \ll (\log H)^{C_1(k)} \left(K_j^{k(2-2\theta)} + K_j^{k(\frac{4}{3}-2\theta)} H^{\frac{1}{3}} N_j^{\frac{2}{3}} \right).$$

Thus

$$N_j \ll (\log 2QT)^{C(k)} \left(K_j^{k(2-2\theta)} + K_j^{k(4-6\theta)} H \right).$$

If $K_j \leq H^{\frac{1/2}{3\theta-1}}$, then we choose k so that

$$H^{\frac{1}{3\theta-1}} \leq K_j^k \leq H^{\frac{3/2}{3\theta-1}}.$$

Note that by (28.44), $K_j \geq K = P^{\eta/5}$. Thus $k \ll 1$ as required and

$$N_j \ll (\log H)^{C_2(k)} H^{\frac{3(1-\theta)}{3\theta-1}}.$$

If $K_j > H^{\frac{1/2}{3\theta-1}}$, then we take $k = 2$. By 28.43 and 28.44, $KX \leq P^{\frac{1}{2}+\eta/4}$. Thus

$$\begin{aligned} N_j &\ll (\log H)^{C(k)} \left(P^{(2+\eta)(1-\theta)} + H^{\frac{4-6\theta}{3\theta-1}} H \right) \\ &\ll (\log H)^{C(k)} \left(P^{(2+\eta)(1-\theta)} + H^{\frac{3(1-\theta)}{3\theta-1}} \right). \end{aligned}$$

Since there are $\ll \log H$ possibilities for j , (28.31) follows.

Theorem 28.3 follows in the same way. We then define $P = H = CqT$ and use Theorems 26.18 and 27.13.

28.4.3 Exercises

1. Suppose that there is no exceptional zero of any L -function formed from a character with conductor dividing q . Prove that there are positive constants c_1 and c_2 such that if

$$q \leq \exp \left(\frac{\log x}{c_2 \log \log x} \right),$$

then

$$\psi(x; q, a) = \frac{x}{\phi(q)} \left(1 + O \left(\exp \left(- \frac{(\log x)}{c_1 (\log q + (\log x)^{\frac{2}{5}} (\log \log x)^{\frac{1}{5}})} \right) \right) \right).$$

28.5 Primes in Short Intervals

S:PinShorts

One of the most important applications of zero density estimates, and a driving force for their development, concerns the distribution of primes

in short intervals. It is a routine consequence of the Riemann Hypothesis that when $2 \leq h \leq x$ we have

$$\pi(x+h) - \pi(x) = \int_x^{x+h} \frac{du}{\log u} + O(x^{1/2} \log x).$$

It is, perhaps, surprising that without any unproven hypothesis the exponent $\frac{7}{12}$ below is so close to $\frac{1}{2}$.

T:HuxleyinShorts

Theorem 28.8. *There is a positive number c such that if $G(x) \geq 1$ and $x^{7/12}(\log x)^c G(x)^{7/2} \leq h \leq x$, then*

$$\psi(x+h) - \psi(x) = h + O\left(\frac{h}{G(x)} + h \exp\left(-\frac{(\log x)^{\frac{1}{3}}}{(c \log \log x)^{\frac{1}{3}}}\right)\right) \quad (28.49) \quad \text{eq:Huxpsi}$$

and

$$\pi(x+h) - \pi(x) - \int_x^{x+h} \frac{du}{\log u} \ll \frac{h}{G(x)} + h \exp\left(-\frac{(\log x)^{\frac{1}{3}}}{(c \log \log x)^{\frac{1}{3}}}\right). \quad (28.50) \quad \text{eq:Huxpi}$$

If Corollary 28.5 were to hold with $12/5$ replaced by ξ for some $\xi > 2$, then the above holds with $7/12$ replaced by $1 - 1/\xi$. Thus one sees that the density hypothesis is practically as good as the Riemann Hypothesis in this context.

Proof By Theorem 12.5, when $2 \leq T \leq x^{1/2}$ and $x \geq c$ we have

$$\psi(x+h) - \psi(x) = h - \sum_{\substack{\rho \\ |\gamma| \leq T}} \int_x^{x+h} u^{\rho-1} du + O(xT^{-1}(\log x)^2).$$

Here the sum is over zeros $\rho = \beta + i\gamma$ of $\zeta(s)$ with $0 < \beta < 1$. By Corollary 14.3, Theorem 6.6 and Corollary 10.3 we may restrict the sum to zeros with $\beta \geq 1/2$. Hence

$$\psi(x+h) - \psi(x) - h \ll hx^{-1} \sum_{\substack{|\gamma| \leq T \\ \beta \geq 1/2}} x^\beta + xT^{-1}(\log x)^2. \quad (28.51) \quad \text{E:HuxEst}$$

We recall that, by (28.1) and (28.2), $N(\theta, T) = N_1(\theta, 1, T)$. The sum above is

$$x^{1/2}N(1/2, T) + \int_{1/2}^1 x^u N(u, T)(\log x) du.$$

By Theorem 24.18, $N(u, T) = 0$ when $u \geq 1 - \delta$ where

$$\delta = \frac{1}{c_1(\log T)^{2/3}(\log \log T)^{1/3}} \quad (28.52) \quad \text{E:Huxdelta}$$

and c_1 is a positive constant. Thus the above is

$$x^{1/2}N(1/2, T) + \int_{1/2}^{1-\delta} x^u N(u, T)(\log x) du.$$

Then, by Theorem 28.4 and Corollary 28.5 with $q = 1$, this is

$$\begin{aligned} &\ll \int_{1/2}^{5/6} x^u T^{12(1-u)/5} (\log x)^{c_2} du + \int_{5/6}^{1-\delta} x^u T^{11(1-u)/5} (\log x)^{c_2} du \\ &\ll (\log x)^{c_2} (x^{1/2} T^{6/5} + (\log x)^{c_2} x^{5/6} T^{2/5} + x(T^{11/5}/x)^\delta). \end{aligned}$$

We now make the choice

$$T = xh^{-1}(\log x)^{-2}G(x)$$

so that

$$T \leq x^{5/12}(\log x)^{-c}G(x)^{-7/2}.$$

Then, by (28.51)

$$\psi(x+h) - \psi(x) - h \ll hG(x)^{-1} + h(\log x)^{c_2}(T^{11/5}/x)^\delta.$$

By (28.52),

$$(T^{11/5}/x)^\delta \leq \exp\left(-\frac{\log x}{c_3(\log x)^{2/3}(\log \log x)^{1/3}}\right),$$

which gives (28.49).

It is immediate from Chebyshev's inequality that

$$\psi(x+h) - \psi(x) - \vartheta(x+h) + \vartheta(x) \ll x^{1/2},$$

and so (28.49) holds with ψ replaced by ϑ . Moreover

$$\pi(x+h) - \pi(x) = \frac{\vartheta(x+h) - \vartheta(x)}{\log(x+h)} + \int_x^{x+h} \frac{\vartheta(t) - \vartheta(x)}{t \log^2 t} dt.$$

Then substituting (28.49) with ψ replaced by ϑ when

$$t \geq x + x^{7/12}(\log x)^c G(x)^{7/2}$$

gives (28.50) as required. $\square$

Whilst it is speculated that results of the above kind persist for h significantly smaller than $\frac{7}{12}$, or $\frac{1}{2}$, we can only establish that such results hold for most, but not necessarily all, pairs x and h with h smaller.

T:2momshorts

Theorem 28.9. *There is a positive number c such that if $x \geq 2$ and $h \geq 1$, then*

$$\begin{aligned} & \int_0^x (\psi(y+h) - \psi(y) - h)^2 dy \\ & \ll xh^2(x^{1/6}h^{-1})^{2/5}(\log x)^c + xh^2 \exp\left(-\frac{(\log x)^{\frac{1}{3}}}{(c \log \log x)^{\frac{1}{3}}}\right). \end{aligned} \quad (28.53) \quad \text{eq:2momshorts}$$

Proof We will show that

$$\begin{aligned} & \int_{x/2}^x (\psi(y+h) - \psi(y) - h)^2 dy \\ & \ll xh^2(x^{1/6}h^{-1})^{2/5}(\log x)^c + xh^2 \exp\left(-\frac{(\log x)^{\frac{1}{3}}}{(c' \log \log x)^{\frac{1}{3}}}\right). \end{aligned} \quad (28.54) \quad \text{E:shortdyadic}$$

Then it follows that

$$\begin{aligned} & \int_{x^{\frac{2}{3}}}^x (\psi(y+h) - \psi(y) - h)^2 dy \\ & \leq \sum_{k \leq \frac{\log x}{3 \log 2}} \int_{x2^{-k}}^{x2^{1-k}} (\psi(u+h) - \psi(u) - h)^2 du \\ & \ll \sum_{k \leq \frac{\log x}{3 \log 2}} \frac{x}{2^k} h^2 (x^{1/6}h^{-1})^{2/5} (\log x)^c \\ & \quad + \frac{x}{2^k} h^2 \exp\left(-\frac{(\log x)^{\frac{1}{3}}}{(c \log \log x)^{\frac{1}{3}}}\right) \end{aligned}$$

which is acceptable, and we also have trivially

$$\int_0^{x^{2/3}} (\psi(y+h) - \psi(y) - h)^2 dy \ll x^{2/3} h^2 (\log x)^2.$$

Thus the theorem would follow from (28.54).

To prove (28.54) we can suppose that $h \leq x/4$, since otherwise the conclusion follows from Theorem 6.9. Suppose that $h/2 \leq g \leq h$. Then the left hand side of (28.54) is

$$\int_{\frac{x}{2}-g}^{x-g} (\psi(y+g+h) - \psi(y+g) - h)^2 dy$$

and on integrating over g we see that it is

$$2h^{-1} \int_{h/2}^h \int_{\frac{x}{2}-g}^{x-g} (\psi(y+g+h) - \psi(y+g) - h)^2 dy dg.$$

We have

$$\begin{aligned} & (\psi(y+g+h) - \psi(y+g) - h)^2 \\ & \ll (\psi(y+g+h) - \psi(y) - g - h)^2 + (\psi(y+g) - \psi(y) - g)^2. \end{aligned}$$

Hence the left hand side of (28.54) is

$$\ll h^{-1} \int_{h/2}^{2h} \int_{x/4}^x (\psi(y+g) - \psi(y) - g)^2 dy dg.$$

On inverting the order of integration and making the substitution $g = hu$ we obtain

$$\begin{aligned} & h^{-1} \int_{x/4}^x \int_{h/2}^{2h} (\psi(y+g) - \psi(y) - g)^2 dg dy \\ & = h^{-1} \int_{x/4}^x \int_{h/(2y)}^{2h/y} (\psi(y+yu) - \psi(y) - yu)^2 y du dy \\ & \ll x h^{-1} \int_{x/4}^x \int_{h/(2x)}^{8h/x} (\psi(y+yu) - \psi(y) - yu)^2 du dy \\ & = h^{-1} \int_{h/(2x)}^{8h/x} \int_{x/4}^x (2x-y)(\psi_0(y+yu) - \psi_0(y) - yu)^2 dy du \end{aligned}$$

where $\psi_0(v) = \psi(v)$ unless v is a positive integer, in which case it is $\psi(v) - \frac{1}{2}\Lambda(v)$.

By Theorem 12.5 and the discussion following that theorem we see that when $v \geq 2$ we have

$$\psi_0(v) = v - \lim_{T \rightarrow \infty} \sum_{|\gamma| \leq T} \frac{v^\rho}{\rho} - \log 2\pi - \frac{1}{2} \log(1 - v^{-2})$$

and the series is boundedly convergent. We also have

$$\frac{1}{2} \log \left(\frac{1 - (y+yu)^{-2}}{1 - y^{-2}} \right) = \frac{1}{2} \log \left(1 - \frac{(1+u)^2 - 1}{y^2(1+u)^2 - 1} \right) \ll uy^{-2}.$$

Hence the above is

$$\ll x + h^2 x^{-5} + h^{-1} \int_{h/(2x)}^{8h/x} \int_{x/4}^x (2x-y) \left| \sum_{\rho} y^\rho \frac{(1+u)^\rho - 1}{\rho} \right|^2 dy du$$

Therefore the left hand side of (28.54) is

$$\ll x + h^{-1} \int_{h/(2x)}^{8h/x} \int_0^{2x} (2x-y) \left| \sum_{\rho} y^\rho \frac{(1+u)^\rho - 1}{\rho} \right|^2 dy du$$

and expanding the integrand and integrating by parts we obtain

$$x + \frac{1}{h} \sum_{\rho_1} \sum_{\rho_2} \int_{\frac{h}{2x}}^{\frac{8h}{x}} \frac{(2x)^{2+\rho_1+\bar{\rho}_2} ((1+u)^{\rho_1} - 1) ((1+u)^{\bar{\rho}_2} - 1)}{(1+\rho_1+\bar{\rho}_2)(2+\rho_1+\bar{\rho}_2)\rho_1\bar{\rho}_2} du.$$

We have $\rho = \beta + i\gamma$ with $0 < \beta < 1$ and $|\gamma| > 1$. Hence

$$\left| \frac{(1+u)^\rho - 1}{\rho} = \int_1^{1+u} v^{\rho-1} dv \right| \ll \min(u, |\gamma|^{-1}).$$

Then the inequality $|z_1 z_2| \leq |z_1|^2 + |z_2|^2$ and the symmetry of ρ_1 and ρ_2 establishes that the above is

$$\begin{aligned} &\ll x + \frac{1}{h} \sum_{\rho_1} \int_{h/(2x)}^{8h/x} \sum_{\rho_2} \frac{x^{2+2\beta_1} \min(u^2, |\gamma_1|^{-2})}{(1+|\gamma_1 - \gamma_2|)^2} du \\ &\ll x + x \sum_{\rho_1} x^{2\beta_1} \min(h^2 x^{-2}, \gamma_1^{-2}) \sum_{\rho_2} \frac{1}{(1+|\gamma_1 - \gamma_2|)^2}. \end{aligned}$$

By Theorem 10.13 the innermost sum is $\ll \log |\gamma_1|$. Thus the left hand side of (28.54) is

$$\ll x + x \sum_{\rho} x^{2\beta} (\log |\gamma|) \min(h^2 x^{-2}, \gamma^{-2}).$$

The sum here is

$$\ll h^2 x^{-1} (\log x) \sum_{\substack{\rho \\ |\gamma| \leq x/h}} x^{2\beta} + x \sum_{\substack{\rho \\ |\gamma| > x/h}} \frac{x^{2\beta} \log |\gamma|}{\gamma^2}.$$

By the symmetry of the zeros we can suppose that $\beta \geq 1/2$. The first term here is

$$\begin{aligned} &\ll h^2 (\log x) N(1/2, x/h) + h^2 x^{-1} (\log x) \int_{1/2}^1 x^{2\theta} (2 \log x) N(\theta, x/h) d\theta \\ &\ll x h (\log x)^2 + \int_{\frac{1}{2}}^{1-\delta} x^{2\theta} \frac{h^2}{x} (\log x)^c \left(\frac{x}{h}\right)^{\lambda(\theta)} d\theta \end{aligned}$$

where

$$\lambda(\theta) = \begin{cases} \frac{12}{5}(1-\theta) & (1/2 \leq \theta \leq 5/6), \\ (2+\eta)(1-\theta) & (5/6 \leq \theta \leq 1-\delta), \end{cases}$$

and the second is

$$\begin{aligned}
&\ll x \sum_{\substack{\rho \\ \beta \geq 1/2 \\ |\gamma| > x/h}} \left(x + \int_{1/2}^{\beta} x^{2\theta} (2 \log x) d\theta \right) \left(\int_{|\gamma|}^{\infty} \frac{2 \log t - 1}{t^3} dt \right) \\
&\ll x^2 \int_{x/h}^{\infty} N(1/2; t) \frac{2 \log t - 1}{t^3} dt \\
&\quad + x(\log x) \int_{1/2}^1 x^{2\theta} \int_{x/h}^{\infty} N(\theta; t) \frac{2 \log t - 1}{t^3} dt d\theta \\
&\ll xh(\log x)^2 + \int_{\frac{1}{2}}^{1-\delta} x^{2\theta+1} \int_{\frac{x}{h}}^{\infty} (\log t)^c t^{\lambda(\theta)-3} dt d\theta
\end{aligned}$$

By (28.31) with $\eta = 1/5$ this is

$$\begin{aligned}
&\ll xh^2(\log x)^c \left(\frac{1}{h} \int_{\frac{1}{2}}^{\frac{5}{6}} \left(\frac{x^{\frac{1}{6}}}{h} \right)^{\frac{12(1-\theta)}{5}} d\theta + \int_{\frac{5}{6}}^{1-\delta} \left(\frac{x^{\eta}}{h^{2+\eta}} \right)^{1-\theta} d\theta \right) \\
&\ll xh^2(\log x)^c \left(\left(\frac{x^{\frac{1}{6}}}{h} \right)^{\frac{6}{5}} + \left(\frac{x^{\frac{1}{6}}}{h} \right)^{\frac{2}{5}} + xh^2 \left(\frac{x^{\frac{1}{11}}}{h} \right)^{\frac{11}{30}} + xh^2 \left(\frac{x^{\eta}}{h^{2+\eta}} \right)^{\delta} \right).
\end{aligned}$$

If $h \leq x^{1/6}$, then (28.54) is trivial. Thus we may suppose $h > x^{1/6}$. Then the above is

$$\ll xh^2(x^{1/6}/h)^{2/5}(\log x)^c + xh^2 \exp \left(- \frac{(\log x)^{\frac{1}{3}}}{(c' \log \log x)^{\frac{1}{3}}} \right)$$

as required, and this completes the proof of the theorem. $\square$

28.5.1 Exercises

1. (a) Prove that if $x \geq 2$ and

$$x^{-5/6} \exp \left(\frac{(\log x)^{1/3}}{(c \log \log x)^{1/3}} \right) < u \leq 1$$

and $x \geq 2$, then

$$\begin{aligned}
&\int_{x/4}^x (\psi_0(y + yu) - \psi_0(y) - yu)^2 dy \\
&\ll x^3 u^2 \exp \left(- \frac{(\log x)^{1/3}}{(c' \log \log x)^{1/3}} \right).
\end{aligned}$$

- (b) Assume the Riemann Hypothesis. Prove that if $x \geq 2$ and $0 < u \leq 1$, then

$$\int_{x/4}^x (\psi_0(y + yu) - \psi_0(y) - yu)^2 dy \ll x^2 u \left(\log \frac{2}{u} \right)^2.$$

2. (a) Prove that if the Riemann Hypothesis is true and $2 \leq h \leq x$, then

$$\int_0^x (\vartheta(y + h) - \vartheta(y) - h)^2 \ll xh (\log(2x/h))^2.$$

- (b) Prove that if p_n is the n -th prime in order of magnitude, then

$$\sum_{\substack{p_n \leq x \\ p_{n+1} - p_n > h}} (p_{n+1} - p_n) \ll xh^{-1}(\log x)^2.$$

and

$$\sum_{p_n \leq x} (p_{n+1} - p_n)^2 \ll x(\log x)^3.$$

3. Prove that for all large x the interval $(x, x + x^{\frac{7}{12} + \varepsilon})$ contains a sum of two primes. If the Riemann Hypothesis is true show that the interval $(x, x + c(\log x)^2)$ contains a sum of two primes.
4. (a) Prove that there is a positive constant c such that if $q \geq 1$ and $x \geq 2$, then

$$\sum_{\chi \bmod q} \sup_{y \leq x} |\psi(y; \chi)| \ll (\log x)^c (x^{1/2} q + x^{7/11} q^{9/11} + x^{3/4} q^{3/5} + x).$$

- (b) Prove that if $(a, q) = 1$, then

$$\sum_{p \leq x} (\log p) e(ap/q) \ll (\log x)^c (xq^{-1/2} + x^{7/8} + x^{1/2} q^{1/2}).$$

- (c) Prove that if $f(\theta)$ has the property that there is an $x \geq 1$ and $\alpha > 0$ such that whenever $(a, q) = 1$ and $|\theta - a/q| \leq q^{-2}$ we have

$$|f(\theta)| \leq xq^{-\alpha} + x^{1-\alpha} q^{\alpha},$$

then for any pair a, q we have

$$f(\theta) \ll x(q + x|\theta q - a|)^{-\alpha} + x^{1-\alpha}(q + x|\theta q - a|)^{\alpha}$$

(d) Prove that, if $(q, a) = 1$, then

$$\sum_{p \leq x} (\log p) e(\theta p) \ll (\log x)^c (x \Delta^{-1/2} + x^{7/8} + x^{1/2} \Delta^{1/2})$$

where $\Delta = q + x|q\theta - a|$. Compare with Lemma ?? and with Exercise 19.1.7.

5. Prove that there is a positive constant c such that if $Q \geq 1$ and $x \geq 2$, then

$$\sum_{q \leq Q} \sum_{\chi \bmod q}^* \sup_{y \leq x} |\psi(y; \chi)| \ll (\log x)^c (x^{1/2} Q^2 + x^{7/11} Q^{18/11} + x^{3/4} Q^{6/5} + x).$$

Compare this with Theorem ?? and give an alternative proof of the Bombieri-Vinogradov theorem, Theorem ??.

28.6 Zeros near the 1-line

S:Near1

There are several ways in which one can obtain significantly smaller bounds near the 1-line. One way in the special case of the zeta function is to make use of the Korobov-Vinogradov-Richert bound Theorem 24.15 for $\zeta(s)$ near 1.

T:ZDEVin

Theorem 28.10. *Let*

$$\Theta(\sigma, U) = 1 + \sup_{0 < t \leq U} \left| \zeta(\sigma + it) - \frac{1}{\sigma + it} \right|.$$

Then, for $\frac{3}{4} \leq \theta \leq 1$

$$N(\theta, T) \ll \Theta(3\theta - 2, 2T)^3 (\log T)^{15}$$

T:ZDEVincor

Corollary 28.11. *There is a positive constant c such that if $\frac{3}{4} \leq \theta \leq 1$, then*

$$N(\theta, T) \ll T^{c(1-\theta)^{3/2}} (\log T)^c$$

Corollary 28.11 follows by combining the theorem with Theorem 24.15.

Proof By Theorem 24.18 we may certainly suppose that

$$\frac{3}{4} \leq \theta \leq 1 - \frac{1}{C(\log T)^{2/3} (\log \log T)^{1/3}}$$

for some positive constant C . We then follow the proof of (28.30), with

χ identically 1, as far as (28.39) except that we instead move the path of integration to the line $\operatorname{Re} w = \alpha - \beta$ where $\alpha \leq 2\theta - 1$, and we suppose that $s = \rho = \beta + i\gamma$ is a zero of ζ with $\beta \geq \theta$ and $|\gamma| \geq (\log T)^2$. Note that then

$$|\alpha - \beta| = \beta - \alpha \geq \theta - (2\theta - 1) = 1 - \theta \gg 1/\log T.$$

Thus we obtain

$$\begin{aligned} 1 + \sum_{K < n \leq Z} a(n)n^{-\rho}e^{-n/Y} &= O(1/T) + \\ \frac{1}{2\pi} \int_{-\infty}^{\infty} \zeta(\alpha + i\gamma + iv; \chi) M(\alpha + i\gamma + iv) Y^{\alpha - \beta + iv} \Gamma(\alpha - \beta + iv) dv & \\ (28.55) \quad \boxed{\text{E:ZDELM1a}} \end{aligned}$$

and we can observe that

$$\Gamma(\alpha - \beta + iv) \ll \left(1 + \frac{1}{|\alpha - \beta + iv|}\right) e^{-|v|} \ll (\log T) e^{-|v|}.$$

At this stage the only constraints on K, Y, Z are $C \leq K \leq T$, $Y \geq 1$ and $Z = Y(\log T)^2$, and in addition we have $Q = 1$, $H = CT$. Hence

$$\begin{aligned} 1 \ll \left| \sum_{K < n \leq Z} a(n)n^{-\rho}e^{-n/Y} \right| + \\ (\log T)\Theta(\alpha, 2T)Y^{\alpha - \theta} \int_{-(\log T)^2}^{(\log T)^2} |M(\alpha + i\gamma + iv)| e^{-|v|} dv. \end{aligned}$$

Let $\mathcal{N} = \{Z2^{-j} : 0 \leq j \leq (\log(Z/K))/\log 2\}$. Then

$$\left| \sum_{K < n \leq Z} a(n)n^{-\rho}e^{-n/Y} \right| \leq \sum_{N \in \mathcal{N}} \left| \sum_{N' < n \leq N} a(n)\chi(n)n^{-\rho}e^{-n/Y} \right|$$

where $N' = \max(N/2, K)$. Hence

$$\begin{aligned} 1 \ll \sum_{N \in \mathcal{N}} \left| \sum_{N' < n \leq N} a(n)n^{-\rho}e^{-n/Y} \right| + \\ (\log T)\Theta(\alpha, 2T)Y^{\alpha - \theta} \int_{-(\log T)^2}^{(\log T)^2} |M(\alpha + i\gamma + iv)| e^{-|v|} dv. \quad (28.56) \quad \boxed{\text{E:Ubound}} \end{aligned}$$

LEMMA 28.12 AND 28.13 SHOULD GO IN CHAPTER 22.

T:HalLem **Lemma 28.12.** *Suppose that $\xi = (\xi_1, \dots, \xi_N)$ and ϕ_{rn} ($1 \leq r \leq R$), ($r \in \mathbb{N}$) are complex numbers and $\phi_r = (\phi_{r1}, \dots, \phi_{rN})$. Suppose further that $b_n > 0$ ($1 \leq n \leq N$) and $b_n \geq 0$ ($n > N$). Let (ξ, ϕ_r) denote the inner product*

$$\sum_{n=1}^N \xi_n \bar{\phi}_{rn}.$$

Then

$$\sum_{r=1}^R |(\xi, \phi_r)| \leq \left(\sum_{1 \leq n \leq N} |\xi_n|^2 b_n^{-1} \right)^{1/2} \left(\sum_{q=1}^R \sum_{r=1}^R \left| \sum_{n=1}^{\infty} b_n \phi_{qn} \bar{\phi}_{rn} \right| \right)^{1/2}.$$

Proof Choose θ_r so that $e^{i\theta_r}(\xi, \phi_r) = |(\xi, \phi_r)|$. Then the sum in question is

$$\sum_{n=1}^N \xi_n b_n^{-1/2} b_n^{1/2} \sum_{r=1}^R e^{i\theta_r} \bar{\phi}_{rn}.$$

Hence, by the Cauchy-Schwarz inequality, it is

$$\leq \left(\sum_{n=1}^N |\xi_n|^2 b_n^{-1} \right)^{1/2} \left(\sum_{n=1}^{\infty} b_n \left| \sum_{r=1}^R e^{i\theta_r} \bar{\phi}_{rn} \right|^2 \right)^{1/2}.$$

We square out the expression on the right to obtain

$$\sum_{n=1}^{\infty} b_n \left| \sum_{r=1}^R e^{i\theta_r} \bar{\phi}_{rn} \right|^2 = \sum_{q=1}^R \sum_{r=1}^R e^{i(\theta_r - \theta_q)} \sum_{n=1}^{\infty} b_n \phi_{qn} \bar{\phi}_{rn}$$

and the lemma follows. $\square$

T:HalLemD **Lemma 28.13.** *Let*

$$D(s; u, v) = \sum_{u < n \leq v} c(n) n^{-s}$$

where the $c(n)$ are complex numbers, and let s_r ($r = 1, \dots, R$) denote a set of complex numbers $s_r = \sigma_r + it_r$ with the property that $0 \leq \nu \leq \sigma_r \leq 1$, $|t| \leq T$ and $|t_q - t_r| \geq 1$ when $q \neq r$. Suppose also that

$N/2 \leq N' \leq N \leq T^2$ and $\frac{1}{4} \leq \alpha < 1$. Then

$$\sum_{r=1}^R |D(s_r; N', N)| \ll N^{-\nu} ((RN)^{1/2} + R\Theta(\alpha; 2T)^{1/2} N^{\alpha/2}) \left(\sum_{N' < n \leq N} |c(n)|^2 \right)^{1/2}$$

Proof To establish the lemma we note that

$$D(s; N', N) = N^{-\sigma} D(it; N', N) + \int_{N'}^N \sigma u^{-\sigma-1} D(it; N', u) du$$

and so when $0 \leq \sigma \leq 1$ we have

$$D(s; N', N) \ll N^{-\nu} |D(it; N', N)| + N^{-\nu-1} \int_{N'}^N |D(it; N', u)| du.$$

Therefore

$$\sum_{r=1}^R |D(s_r; N', N)| \ll N^{-\nu} \sum_{r=1}^R |D(it_r; N', N)| + N^{-\nu-1} \int_{N'}^N \sum_{r=1}^R |D(it_r; N', u)| du. \quad (28.57) \quad \boxed{\text{E:Dbound}}$$

We now apply the previous lemma with $\xi_n = c(n)$, $b_n = e^{-n/N}$, $\phi_{rn} = n^{it_r}$. Then for $N' \leq u \leq N$ we have

$$\sum_{r=1}^R |D(it_r; N', u)| \leq \left(\sum_{N' < n \leq N} |c(n)|^2 e^{n/N} \right)^{1/2} \left(\sum_{q=1}^R \sum_{r=1}^R \left| \sum_{n=1}^{\infty} e^{-n/N} n^{it_q - it_r} \right| \right)^{1/2}. \quad (28.58) \quad \boxed{\text{E:2Dbound}}$$

For brevity, put $t = t_r - t_q$. Then the sum over n here is

$$\frac{1}{2\pi i} \int_{2-i\infty}^{2+i\infty} \zeta(w+it) N^w \Gamma(w) dw$$

We now move the line to the path $\text{Re } w = \alpha$. In doing so we pick up a residue

$$N^{1-it} \Gamma(1-it) \ll N e^{-|t|}$$

at $w = 1 - it$.

On the α -line we write

$$\zeta(w + it) = \zeta(w + it) - \frac{1}{w + it - 1} + \frac{1}{w + it - 1}.$$

For the third term here we move the path to $\operatorname{Re} w = -\frac{1}{2}$ picking up the residue

$$\frac{1}{it - 1} \ll 1$$

at 0 and the bound

$$\ll \int_{-\infty}^{\infty} \frac{N^{-1/2} e^{-|v|}}{1 + |v + t|} dv \ll \frac{N^{-1/2}}{1 + |t|}$$

from the $-\frac{1}{2}$ -line.

We also have, by Corollary 1.17,

$$\zeta(\alpha + iv + it) - \frac{1}{\alpha + iv + it - 1} \ll 2 + |v + t|.$$

Thus

$$\begin{aligned} \int_{|v| \geq (\log T)^2} \left| \zeta(\alpha + iv + it) - \frac{1}{\alpha + iv + it - 1} \right| N^\alpha |\Gamma(\alpha + iv)| dv \\ \ll N^\alpha \int_{|v| \geq (\log T)^2} (2 + |v + t|) e^{-|v|} dv \ll N^\alpha T^{-1}. \end{aligned}$$

Since $|t| \leq T$ we have

$$\begin{aligned} \int_{|v| \leq (\log T)^2} \left| \zeta(\alpha + iv + it) - \frac{1}{\alpha + iv + it - 1} \right| N^\alpha |\Gamma(\alpha + iv)| dv \\ \ll \Theta(\alpha, 2T) N^\alpha. \end{aligned}$$

Putting these estimates together we find that

$$\sum_{n=1}^{\infty} e^{-n/N} n^{it_q - it_r} \ll \Theta(\alpha, 2T) N^\alpha + N e^{-|t_q - t_r|}.$$

Hence, by (28.58),

$$\begin{aligned} \sum_{r=1}^R |D(it_r; N', u)| \ll \left(\sum_{N' < n \leq N} |c(n)|^2 e^{n/N} \right)^{1/2} \times \\ \left(RN + R^2 \Theta(\alpha, 2T) N^\alpha + \sum_{q=1}^R \sum_{\substack{r=1 \\ r \neq q}}^R N e^{-|t_q - t_r|} \right)^{1/2}. \end{aligned}$$

Inserting this in (28.57) establishes the lemma. $\square$

Let $\mathcal{R}$ be a maximal subset of the zeros $\rho = \beta + i\gamma$ of $\zeta(s)$ with $\beta \geq \theta$, $(\log T)^2 \leq |\gamma| \leq T$ and the γ spaced at least 1 apart, and let $R = \text{card } \mathcal{R}$. Then

$$N(\theta, T) \ll (\log T)^3 + (\log T)R$$

and, by (28.56),

$$\begin{aligned} R \ll \sum_{N \in \mathcal{N}} \sum_{\rho \in \mathcal{R}} \left| \sum_{N' < n \leq N} a(n) n^{-\rho} e^{-n/Y} \right| \\ + (\log T) \Theta(\alpha, 2T) Y^{\alpha-\theta} \int_{-(\log T)^2}^{(\log T)^2} e^{-|v|} \sum_{\rho \in \mathcal{R}} |M(\alpha + i\gamma + iv)| dv. \end{aligned}$$

By Lemma 28.13 with $\nu = \theta$ and the s_r the elements of $\mathcal{R}$, the first sum over $\mathcal{R}$ is bounded by

$$\begin{aligned} N^{-\theta} ((RN)^{1/2} + R\Theta(\alpha, 2T)^{1/2} N^{\alpha/2}) e^{-N/(2Y)} \left(\sum_{N' < n \leq N} d(n)^2 \right)^{1/2} \\ \ll (\log N)^{3/2} R^{1/2} N^{1-\theta} e^{-N/(2Y)} + \\ (\log N)^{3/2} R\Theta(\alpha, 2T)^{1/2} N^{\frac{1+\alpha}{2}-\theta} e^{-N/(2Y)} \end{aligned}$$

and summing this over $N \in \mathcal{N}$ gives the bound

$$\ll (\log T)^{3/2} R^{1/2} Y^{1-\theta} + (\log T)^{5/2} R\Theta(\alpha, 2T)^{1/2} K^{\frac{1+\alpha}{2}-\theta}$$

provided that $1 + \alpha \leq 2\theta$.

The second sum over $\mathcal{R}$ above is

$$\begin{aligned} \sum_{\rho \in \mathcal{R}} |M(\alpha + i\gamma + iv)| \leq \\ \sum_{0 \leq j \leq \frac{\log K}{\log 2}} \sum_{\rho \in \mathcal{R}} \left| \sum_{K2^{-j-1} \leq k \leq K2^{-j}} \mu(k) k^{-\alpha-i\gamma-iv} \right|. \end{aligned}$$

By Lemma 28.13 with $\nu = \alpha$ and the $s_r = \alpha + i \text{Im } \rho$ where $\rho \in \mathcal{R}$, this is

$$\ll (\log T) R^{1/2} K^{1-\alpha} + (\log T) R\Theta(\alpha; 2T)^{1/2} K^{(1-\alpha)/2}.$$

Hence

$$\begin{aligned} R \ll (\log T)^{3/2} R^{1/2} Y^{1-\theta} &+ (\log T)^{5/2} R \Theta(\alpha, 2T)^{1/2} K^{\frac{1+\alpha}{2}-\theta} \\ &+ (\log T)^2 R^{1/2} Y^{\alpha-\theta} \Theta(\alpha; 2T) K^{1-\alpha} \\ &+ (\log T)^2 R Y^{\alpha-\theta} \Theta(\alpha; 2T)^{3/2} K^{\frac{1-\alpha}{2}}. \end{aligned}$$

Let

$$\alpha = \theta - \lambda(1 - \theta)$$

where $\lambda > 1$ is a parameter at our disposal. Then $(1 + \alpha)/2 - \theta = -(\lambda - 1)(1 - \theta)/2$, $\alpha - \theta = -\lambda(1 - \theta)$ and $1 - \alpha = (1 + \lambda)(1 - \theta)$. Let δ be a sufficiently small constant and choose

$$K = (\delta^{-1} (\log T)^{5/2} \Theta(\alpha, 2T)^{1/2})^{\frac{2}{(\lambda-1)(1-\theta)}}. \quad (28.59) \quad \boxed{\text{E: def K}}$$

Then the second term above is $\ll \delta R$. Likewise if we choose

$$Y = (\delta^{-1} (\log T)^2 \Theta(\alpha, 2T)^{3/2} K^{\frac{1-\alpha}{2}})^{\frac{1}{\lambda(1-\theta)}}, \quad (28.60) \quad \boxed{\text{E: def Y}}$$

then the fourth term above is

$$\ll \delta R.$$

Hence

$$R \ll (\log T)^{3/2} R^{1/2} Y^{1-\theta} + (\log T)^2 R^{1/2} Y^{\alpha-\theta} \Theta(\alpha; 2T) K^{1-\alpha}$$

and so

$$R \ll (\log T)^3 Y^{2-2\theta} + (\log T)^4 Y^{2\alpha-2\theta} \Theta(\alpha; 2T)^2 K^{2-2\alpha}.$$

To tidy things up, by the choices (28.59) and (28.60)

$$Y^{2-2\theta} \ll (\log T)^{\frac{9\lambda+1}{\lambda(\lambda-1)}} \Theta(\alpha; 2T)^{\frac{4\lambda-2}{\lambda(\lambda-1)}}$$

and

$$Y^{2\alpha-2\theta} K^{2-2\alpha} \ll (\log T)^{\frac{\lambda+9}{(\lambda-1)}} \Theta(\alpha; 2T)^{\frac{4-2\lambda}{(\lambda-1)}}.$$

Hence

$$R \ll (\log T)^{3+\frac{9\lambda+1}{\lambda(\lambda-1)}} \Theta(\alpha; 2T)$$

The optimal choice of λ depends on whatever bound for $\Theta(\alpha; 2T)$ that we may insert. If θ is restricted to being close to 1, then large values of λ are possible and exponents above would be small, but in the current state of knowledge we have no better bound for Θ than

$$\Theta(\alpha; 2T) = \Theta(\theta - \lambda(1 - \theta)) \ll (\log T)^{c'} T^{c(\lambda+1)^{3/2}(1-\theta)^{3/2}}.$$

Thus for simplicity we suppose that $\lambda = 2$. Note that then the requirement $\frac{1}{4} \leq \alpha = \theta - \lambda(1 - \theta) = 3\theta - 2$ does permit $\frac{3}{4} \leq \theta$ as required. Thus

$$R \ll (\log T)^{13} \Theta(3\theta - 2, 2T)^3 + (\log T)^{15} \Theta(3\theta - 2; 2T)^2.$$

□

28.6.1 Exercises

1 Assume the Lindelöf Hypothesis in the form

$$\Theta(\alpha; T) \ll_{\varepsilon} T^{\varepsilon}$$

uniformly for $\alpha \geq \frac{1}{2}$. Prove that whenever $\theta > \frac{3}{4}$

$$N(\theta, T) \ll_{\varepsilon} T^{\varepsilon}.$$

28.7 A logarithm free bound and the Deuring-Heibronn phenomenon

S:logfree

Another way in which significant improvements can be made is to use Turán's power sum method to remove the logarithmic power. This also comes into play if there were to be an "exceptional zero" of some L -function close to 1, as it can be used to show that the non-exceptional zeros are repelled to the left.

We need to remind ourselves of some basic results concerning the concept of an "exceptional" zero.

Exceptional Zero Statement. *By Corollary 11.10 of Volume 1 there is a positive constant c_1 such that*

$$F(s, T) = \prod_{q \leq T} \prod_{\chi \bmod q}^* L(s, \chi) \quad (28.61) \quad \text{E:LprodT}$$

has at most one zero s with $\operatorname{Re} s > 1 - \frac{1}{c_1 \log T}$, of necessity real and if this "exceptional zero" β_1 exists, then the corresponding character χ_1 is quadratic and, by Corollary 11.12, there is a positive constant c_2 such that $\delta_1 = 1 - \beta_1$ satisfies

$$\frac{1}{c_2 q_1^{1/2} (\log q_1)^2} \leq \delta_1 < \frac{1}{c_1 \log T} \quad (28.62) \quad \text{E:XZ1b}$$

where q_1 is the conductor of χ_1 .

It is convenient to write $E_1 = 0$ if there is no exceptional zero and

$E_1 = 1$ if there is an exceptional zero and to reserve χ_1, β_1, q_1 to denote the corresponding exceptional character, zero and conductor.

Let

$$N^*(\theta, T)$$

denote the number of zeros $\rho = \beta + i\gamma$ of (28.61), with $\beta \geq \theta$ and $|\gamma| \leq T$, other than any exceptional zero.

One can observe that if as T varies there are only a finite number of exceptional moduli, then in principle one could simply adjust the constant c_1 and eliminate the concept of “exceptional”. On the other hand if the exceptional moduli form an infinite sequence $\{q_j\}$ and $\{\beta_j\}$ are the corresponding exceptional zeros, then by the same token one would have to have

$$\limsup_{j \rightarrow \infty} (1 - \beta_j) \log q_j = 0.$$

Thus in principle one could take c_1 to be as small as one pleases. However, this leads inexorably to the non-computability of c_1 , which here we would prefer to avoid, especially in connection with bounding the least prime in an arithmetic progression.

With this in the background we can establish

Theorem 28.14. *There are positive constants c and c_0 such that when $\frac{1}{2} \leq \theta \leq 1$ and $T \geq 2$ we have*

$$N^*(\theta, T) \leq c_0 T^{c(1-\theta)}, \quad (28.63) \quad \text{E:logfree1}$$

and if there is an exceptional real zero β_1 associated with some exceptional primitive character χ_1 with conductor $q_1 \leq T$, then

$$N^*(\theta, T) \leq c_0 \delta_1 (\log T) T^{c(1-\theta)}. \quad (28.64) \quad \text{E:logfree2}$$

We can immediately conclude from this an effective version of the Deuring-Heilbronn phenomenon, which essentially says that if there is an exceptional zero, then the other zeros are repelled away from 1-line.

Corollary 28.15. *There are positive constants c_0, c such that if β_1 is an exceptional zero as defined above, then any other zero $\rho = \beta + i\gamma$ with $|\gamma| \leq T$ of an L -function formed from a primitive character modulo $q \leq T$ satisfies*

$$\beta \leq 1 - \frac{\log \frac{1}{c_0(1-\beta_1) \log T}}{c \log T}.$$

We begin by eliminating some ranges for θ . By Corollary 28.7 there is a positive constant c_2 such that

$$N^*(\theta, T) \ll T^{8(1-\theta)} (\log T)^{C_2}.$$

Suppose that $C > 8$ and $\theta \leq 1 - \frac{1}{c-8}$. If β_1 exists, then, by (28.62), the above is

$$\ll T^{-1/2} (\log T)^{-1} T^{c(1-\theta)} \ll \delta_1 (\log T) T^{c(1-\theta)}.$$

and when β_1 does not exist it is

$$\ll T^{c(1-\theta)}.$$

Also, from the definition above we have

$$N^*(\theta, T) = 0$$

when $\theta \geq 1 - \frac{1}{c_1 \log T}$. Hence we may assume that

$$1 - \frac{1}{c-8} \leq \theta \leq 1 - \frac{1}{c_1 \log T}.$$

Let

$$r = c_3(1 - \theta) \tag{28.65} \quad \boxed{\text{E:rdef}}$$

where

$$c_3 = \max(c_1, 2).$$

Then

$$\frac{1}{\log T} \leq r \leq \frac{1}{10^5} \tag{28.66} \quad \boxed{\text{E:rbound}}$$

provided we take

$$c \geq (8 + 10^5) \max(c_1, 2).$$

Suppose $N^*(\theta, T) > 0$ and let $\rho_0 = \beta_0 + i\gamma_0$ be a non-exceptional zero counted by $N^*(\theta, T)$ so that

$$1 - \beta_0 \leq 1 - \theta = \frac{r}{c_3} \leq \frac{1}{2}.$$

Then consider v satisfying

$$|\gamma_0 - v| \leq \frac{r}{2}, \tag{28.67} \quad \boxed{\text{E:defv}}$$

so that $|v| \leq T + 1$ and

$$|\rho_0 - w| \leq 1 - \beta_0 + |\gamma_0 - v| \leq r, \text{ where } w = 1 + iv. \tag{28.68} \quad \boxed{\text{E:defw}}$$

In particular, when χ is principal we have $|v| \geq 2$.

The core of the proof is a lower bound for expressions of the kind

$$I(X, Y, r, \gamma_0, \chi) = \int_X^Y \int_{\gamma_0 - \frac{r}{2}}^{\gamma_0 + \frac{r}{2}} \left| \sum_{X < p \leq y} \frac{1 + E_1 p^{-\delta_1} \chi_1(p)}{p^{1+iv}} \chi(p) \right|^2 dv \frac{dy}{y}. \quad (28.69) \quad \boxed{\text{E:doubleint}}$$

Suppose that $s = \sigma + it$ satisfies

$$|s - w| \leq \frac{3}{4}, \quad \frac{5}{6} \leq \sigma \leq 2. \quad (28.70) \quad \boxed{\text{E:esssigma}}$$

Then, by Lemma 11.1

$$\frac{L'}{L}(s, \chi) = \sum_{\substack{\rho \\ |\rho - \frac{3}{2} - it| \leq \frac{5}{6}}} \frac{1}{s - \rho} + O(\log T). \quad (28.71) \quad \boxed{\text{E:L'Lapprox}}$$

Note that if χ is principal, since then $|v| \geq 2$, we have $|s - 1| = |s - w + iv| \geq |v| - \frac{3}{4} \geq 1$.

It is useful to restate here an immediate consequence of Theorem 11.5 and Corollary 14.7 of Volume 1.

T:denslem

Lemma 28.16. *Suppose that λ , q and t satisfy $q \ll T$, $|t| \ll T$ and $(\log T)^{-1} \leq \lambda \leq 2$. Then for, each character χ modulo q , $L(s, \chi)$ has $\ll \lambda \log T$ zeros ρ with $|\rho - 1 - it| \leq \lambda$.*

If $|\rho - \frac{3}{2} - it| \leq 5/6$ and $|\rho - w| \geq 1$, then $|\rho - s| = |\rho - w + w - s| \geq 1 - \frac{3}{4} = \frac{1}{4}$. Thus the contribution to the sum in (28.71) from such zeros is $\ll \log T$. Moreover, if $|\rho - \frac{3}{2} - it| > \frac{5}{6}$ and $\frac{5}{6} \leq \sigma \leq 2$ we have $|\rho - s| = |\rho - \frac{3}{2} - it + \frac{3}{2} - \sigma| > \frac{5}{6} - \frac{2}{3} = \frac{1}{6}$ and again the contribution from such zeros is $\ll \log T$. Therefore

$$\frac{L'}{L}(s, \chi) = \sum_{\substack{\rho \\ |\rho - w| < 1}} \frac{1}{s - \rho} + O(\log T).$$

Suppose that

$$|s - w| \leq \frac{5}{8}, \quad \frac{5}{6} \leq \sigma \leq \frac{15}{8}, \quad (28.72) \quad \boxed{\text{E:newess}}$$

If $|\rho - w| > \frac{3}{4}$, then $|s - \rho| = |\rho - w - s + w| \geq \frac{1}{8}$. Hence the contribution from such zeros to the above sum is $\ll \log T$, and we have

$$\frac{L'}{L}(s, \chi) = \sum_{\substack{\rho \\ |\rho - w| \leq \frac{3}{4}}} \frac{1}{s - \rho} + O(\log T).$$

Now suppose that there is an exceptional zero, and the associated character is χ_1 . Then (28.71) with (28.70) holds with s replaced by $s + \delta_1$ and w replaced by $w + \delta_1$. Hence a parallel argument to the above shows that

$$\frac{L'}{L}(s + \delta_1, \chi\chi_1) = \sum_{\substack{\rho' \\ |\rho' - \delta_1 - w| \leq \frac{3}{4}}} \frac{1}{s + \delta_1 - \rho'} + O(\log T).$$

where now the sum is over zeros ρ' of $L(s; \chi\chi_1)$.

Let

$$\begin{aligned} g(z, \chi) &= \frac{L'}{L}(z, \chi) + E_1 \frac{L'}{L}(z + \delta_1, \chi\chi_1) \\ &\quad - \sum_{\substack{\rho \\ |\rho - w| \leq \frac{3}{4}}} \frac{1}{z - \rho} - E_1 \sum_{\substack{\rho' \\ |\rho' - \delta_1 - w| \leq \frac{3}{4}}} \frac{1}{z + \delta_1 - \rho'}. \end{aligned}$$

Suppose that $|s - w| \leq \frac{1}{2}$. The function $g(z, \chi)$ has only removable singularities when $|z - w| \leq \frac{3}{4}$ and by Cauchy's integral formula

$$\frac{g^{(h)}(s, \chi)}{h!} = \frac{1}{2\pi i} \int_C g(z, \chi) \frac{dz}{(z - s)^{h+1}} \ll 4^h \log T$$

where C is the circle, centre w , of radius $\frac{3}{4}$. Let

$$f(z, \chi) = \frac{L'}{L}(z, \chi) + E_1 \frac{L'}{L}(z + \delta_1, \chi\chi_1). \quad (28.73) \quad \boxed{\text{E: Bombf}}$$

Then

$$\begin{aligned} (-1)^h \frac{f^{(h)}(s, \chi)}{h!} &- \sum_{\substack{\rho \\ |\rho - w| \leq \frac{3}{4}}} \frac{1}{(s - \rho)^{h+1}} \\ &- E_1 \sum_{\substack{\rho' \\ |\rho' - \delta_1 - w| \leq \frac{3}{4}}} \frac{1}{(s + \delta_1 - \rho')^{h+1}} \ll 4^h \log T. \end{aligned}$$

Now suppose

$$s_0 = w + r \quad (28.74) \quad \boxed{\text{E: defs0}}$$

and let

$$\lambda = 200r. \quad (28.75) \quad \boxed{\text{E: deflam}}$$

Then, by (28.66),

$$\frac{200}{\log T} < \lambda \leq \frac{1}{4}$$

and so by Lemma 28.16 we have

$$\begin{aligned}
 (-1)^h \frac{f^{(h)}(s_0, \chi)}{h!} &= \sum_{\substack{\rho \\ |\rho-w| \leq \lambda}} \frac{1}{(s_0 - \rho)^{h+1}} \\
 &\quad - E_1 \sum_{\substack{\rho' \\ |\rho' - \delta_1 - w| \leq \lambda}} \frac{1}{(s_0 + \delta_1 - \rho')^{h+1}} \\
 &\ll (\lambda \log T) \lambda^{-h-1} = \lambda^{-h} \log T.
 \end{aligned}$$

Also, if N is the total number of terms in the above sums, then

$$N \ll \lambda \log T \ll \log T. \quad (28.76) \quad \boxed{\text{E:NlambdaT}}$$

Suppose that

$$L \geq N. \quad (28.77) \quad \boxed{\text{E:LgeN}}$$

Then, by Turán's Second Main Theorem, in the form of Corollary K.5, there is a k with $L + 1 \leq k \leq 2L$ such that

$$\begin{aligned}
 &\left| \sum_{\substack{\rho \\ |\rho-w| \leq \lambda}} \frac{1}{(s_0 - \rho)^k} + E_1 \sum_{\substack{\rho' \\ |\rho' - \delta_1 - w| \leq \lambda}} \frac{1}{(s_0 + \delta_1 - \rho')^k} \right| \\
 &\geq 2(16e)^{-k} \max_{|\rho-w| \leq \lambda} |s_0 - \rho|^{-k} \geq (50)^{-k} |s_0 - \rho_0|^{-k}.
 \end{aligned}$$

By (28.68) we have $|s_0 - \rho_0| \leq |s_0 - w| + |\rho_0 - w| \leq 2r$. Thus on taking $h = k - 1$ we have $L \leq h \leq 2L - 1$ and

$$\begin{aligned}
 \left| \frac{f^{(h)}(s_0, \chi)}{h!} \right| &\geq (50)^{-h-1} |s_0 - \rho_0|^{-h-1} - c_4 \lambda^{-h} \log T \\
 &\geq (100r)^{-h-1} - c_4 (200r)^{-h} \log T
 \end{aligned}$$

for some positive absolute constant c_4 .

We choose $c_5 \geq 1$ so that when

$$L \geq c_5 r \log T \quad (28.78) \quad \boxed{\text{E:BombL}}$$

we have

$$\frac{100c_4 r \log T}{2^h} \leq \frac{1}{2},$$

by (28.76) we have (28.77), and at several points below, including (28.82)

and (28.83), we suppose that it is a sufficiently large absolute constant. Then

$$\begin{aligned} \left| \frac{f^{(h)}(s_0, \chi)}{h!} \right| &\geq (100r)^{-h-1} \left(1 - \frac{100c_4 r \log T}{2^h} \right) \\ &\geq \frac{1}{2} (100r)^{-h-1}. \end{aligned}$$

We now return to the definition of f , (28.73). Thus, by (28.74),

$$\begin{aligned} r^h \frac{f^{(h)}(s_0)}{h!} &= (-1)^{h+1} \sum_{n=1}^{\infty} \frac{(r \log n)^h \Lambda(n) \chi(n)}{h! n^{s_0}} (1 + E_1 \chi_1(n) n^{-\delta_1}) \\ &= (-1)^{h+1} \sum_{n=1}^{\infty} \frac{\Lambda(n) \chi(n)}{n^w} \varpi_h(r \log n) (1 + E_1 \chi_1(n) n^{-\delta_1}) \end{aligned}$$

where, for $y > 0$,

$$\varpi_h(y) = \frac{y^h}{h! e^y}.$$

We have

$$\log h! = h \log h - \int_1^h \frac{\lfloor u \rfloor}{u} du \geq h \log h - h.$$

Hence

$$\varpi_h(y) \leq (ey/h)^h e^{-y}.$$

Thus

$$\varpi_h(y) \leq (300)^{-h} \quad (h \geq 900y).$$

The function of y , $h \log(ey/h) - y/2$ is decreasing for $y \geq 2h$. Hence

$$h \log(ey/h) - y/2 \leq h(1 + \log 20 - 10) \quad (y \geq 20h)$$

and so

$$\varpi_h(y) \leq 300^{-h} e^{-y/2} \quad (y \geq 20h). \quad (28.79) \quad \boxed{\text{E:varpiy}}$$

We now consider any x with

$$x \geq T^{c_5} \quad (28.80) \quad \boxed{\text{E:defxC5}}$$

where c_5 is as in (28.78). Then (28.75), (28.76) and the choice $L = \lceil r \log x \rceil$ ensures that (28.78), and so (28.77) holds, and that h satisfies

$$r \log x \leq h \leq 2r \log x. \quad (28.81) \quad \boxed{\text{E:hrlogx}}$$

Thus, when $n \leq x^{1/900}$ we have $r \log n \leq h/900$ and when $n > x^{40}$ we have $r \log n > 20h$. Hence the contribution from these n to

$$\sum_{n=1}^{\infty} \frac{\Lambda(n)\chi(n)}{n^w} \varpi_h(r \log n)(1 + E_1\chi_1(n)n^{-\delta_1})$$

is

$$\begin{aligned} &\ll \sum_{n \leq x^{1/900}} \frac{\Lambda(n)}{n} (300)^{-h} + \sum_{n > x^{40}} \frac{\Lambda(n)}{n^{1+r/2}} (300)^{-h} \\ &\ll (\log x + 1/r)(300)^{-h} \\ &\ll 100^{-h} r^{-1} 3^{-r \log x} (r \log x + 1). \end{aligned}$$

We also require of c_5 that when $y \geq c_5$ we have

$$3^{-y}(y+1) \leq \frac{1}{400} \quad (y \geq c_5). \quad (28.82) \quad \boxed{\text{E:c5ineq}}$$

Thus the total contribution is

$$< \frac{1}{4r100^{h+1}}.$$

It follows that

$$\left| \sum_{x^{1/900} < n \leq x^{40}} \frac{\Lambda(n)\chi(n)}{n^w} \varpi_h(r \log n)(1 + E_1\chi_1(n)n^{-\delta_1}) \right| \geq \frac{1}{4r100^{h+1}}.$$

We also have $w_h(y) \leq 1$. Hence, by Chebyshev's inequalities and (28.80), the contribution from the $n = p^k$ with $k \geq 2$ to this sum is

$$\leq \sum_{\substack{x^{1/900} < p^m \leq x^{40} \\ m \geq 2}} \frac{2(\log p)}{p^m} < x^{-1/2000} \quad (x \geq T^{c_5}, \quad (28.83) \quad \boxed{\text{E:Cheb}})$$

whereas, by (28.66),

$$4r100^{h+1} \leq 800r100^{2r \log x} < x^{2r \log 100} < x^{2(\log 100)/10^5} < x^{1/2000}.$$

Hence

$$\left| \sum_{x^{1/900} < p \leq x^{40}} \frac{(\log p)\chi(p)}{p^w} \varpi_h(r \log p)(1 + E_1\chi_1(p)p^{-\delta_1}) \right| \geq \frac{1}{800r100^h}. \quad (28.84) \quad \boxed{\text{E:Bsum1b}}$$

For convenience write

$$X = x^{1/900}, \quad Y = x^{40}. \quad (28.85) \quad \boxed{\text{E: defXY}}$$

Then

$$\begin{aligned} \sum_{x^{1/900} < p \leq x^{40}} \frac{(\log p)\chi(p)}{p^w} \varpi_h(r \log p)(1 + E_1\chi_1(p)p^{-\delta_1}) \\ = F(Y, v)\varpi_h(r \log Y) - \int_X^Y F(y, v)\varpi'_h(r \log y) \frac{r dy}{y} \end{aligned}$$

where

$$F(y, v) = \sum_{X < p \leq y} \frac{(\log p)\chi(p)}{p^w} (1 + E_1\chi_1(p)p^{-\delta_1}). \quad (28.86) \quad \boxed{\text{E: defF}}$$

We have, by (28.81), $r \log Y = 40r \log x \geq 20h$ and by Mertens theorem, Theorem 2.7 (b),

$$F(Y, v) \ll \log x.$$

Hence, by (28.79),

$$|F(Y, v)\varpi_h(r \log Y)| \ll (\log x)300^{-h}e^{-\frac{\log Y}{2}}$$

and since c_5 is assumed sufficiently large we have

$$|F(Y, v)\varpi_h(r \log Y)| < \frac{1}{1600r100^h}$$

We also have

$$|\varpi'_h(u)| = |\varpi_{h-1}(u) - \varpi_h(u)| \leq 1.$$

Hence

$$\int_X^Y |F(y, v)| \frac{dy}{y} \gg 100^{-h}r^{-2}$$

and so by Schwarz' inequality

$$\int_X^Y |F(y, v)|^2 \frac{dy}{y} \gg (\log x)^{-1}100^{-2h}r^{-4} \quad (28.87) \quad \boxed{\text{E: F21b}}$$

To summarise what we have established so far. We are given $T \geq 2$ and r and θ satisfying (28.65) and (28.66). Then for each non-exceptional zero $\rho_0 = \beta_0 + i\gamma_0$ with $\beta_0 \geq \theta$ and $|\gamma_0| \leq T$, and for each v satisfying (28.67), namely $|\gamma_0 - v| \leq r/2$, we have (28.87) with (28.86) and (28.68) for any x satisfying (28.80) and some h for which (28.81) holds.

By (28.81)

$$(\log x)^{-1} 100^{-2h} r^{-4} \geq (\log x)^3 100^{-4r \log x} (r \log x)^{-4}$$

By (28.80) and (28.66) and that fact that c_5 is sufficiently large, the above is

$$\gg (\log x)^3 x^{-c_6 r}$$

for some positive constant c_6 . Integrating over the v satisfying (28.67) we obtain

$$\int_X^Y \int_{\gamma_0 - r/2}^{\gamma_0 + r/2} |F(y, v)|^2 dv \frac{dy}{y} \gg r (\log x)^3 x^{-c_6 r}.$$

For concision we now drop the suffix 0 and sum over the non-exceptional zeros ρ of $L(s; \chi)$ with $\beta \geq \theta$ and $|\gamma| \leq T$. Thus

$$r (\log x)^3 x^{-c_6 r} N^*(\theta; \chi, T) \ll \int_X^Y \int_{T-r}^{T+r} \sum_{\substack{\rho \\ |\rho-v| \leq r}} \left| \sum_{X < p \leq y} \frac{(\log p) \chi(p)}{p^w} b(p) \right|^2 dv \frac{dy}{y}$$

where $N^*(\theta; \chi, T)$ is the number of such zeros and

$$b(p) = 1 + E_1 \chi_1(p) p^{-\delta_1}$$

. By (28.65), for a each ρ in the sum we we have $|\rho - 1 - iv| = |\beta - 1 + i(\gamma - v)| \leq 1 - \theta + r \ll r$. Hence, by Lemma 28.16 the number of ρ in the sum is $\ll r \log T$. Thus

$$(\log x)^3 x^{-c_6 r} N^*(\theta; \chi, T) \ll (\log T) \int_X^Y \int_{T-r}^{T+r} \left| \sum_{X < p \leq y} \frac{(\log p) \chi(p)}{p^w} b(p) \right|^2 dv \frac{dy}{y}.$$

Therefore, summing over all primitive characters modulo $q \leq T$ we have

$$(\log x)^3 x^{-c_6 r} N^*(\theta; T) \ll (\log T) \int_X^Y \sum_{q \leq T} \sum_{\chi \bmod q}^* \int_{T-r}^{T+r} \left| \sum_{X < p \leq y} \frac{(\log p) \chi(p)}{p^w} b(p) \right|^2 dv \frac{dy}{y}. \quad (28.88) \quad \boxed{\text{E:Nupper}}$$

We now require a lemma arising from the large sieve.

THIS SHOULD GO IN CHAPTER 18.

T:varls **Lemma 28.17.** *Suppose $Q \geq 1$, $N \geq 1$ and $T \geq 2$ and that a_n ($1 \leq n \leq N$) are complex numbers with the property that $a_n = 0$ when n has a prime factor $p \leq Q$. Let*

$$S(s; \chi) = \sum_{n=1}^N a_n \chi(n) n^{-s}.$$

Then

$$\sum_{q \leq Q} \left(\log \frac{Q}{q} \right) \sum_{\chi \bmod q}^* \int_0^T |S(it; \chi)|^2 \ll \sum_{n=1}^N |a_n|^2 (n + Q^2 T)$$

where the sum over χ is over primitive characters modulo q .

To prove the lemma, we start from the observation that, by (9.6), when $(n, q) = 1$ and χ is an arbitrary character modulo q we have

$$\chi(n) \tau(\bar{\chi}) = \sum_{a=1}^q \bar{\chi}(a) e(an/q)$$

and so

$$\frac{1}{\phi(q)} \sum_{\chi \bmod q} |\tau(\bar{\chi}) S(0, \chi)|^2 = \sum_{\substack{a=1 \\ (a, q)=1}}^q |T(a/q)|^2$$

where

$$T(\alpha) = \sum_{n=1}^N a_n e(\alpha n).$$

By Theorem 9.10, when χ is induced by the primitive character χ^* with conductor $d|q$ we have

$$\tau(\bar{\chi}) = \mu(q/d) \chi^*(q/d) \tau(\bar{\chi}^*)$$

and by Theorem 9.7

$$|\tau(\bar{\chi}^*)| = d^{1/2}.$$

Moreover, since $a_n = 0$ when n has a prime factor $p \leq Q$, we have

$$\frac{1}{\phi(q)} \sum_{\chi \bmod q} |\tau(\bar{\chi}) S(0, \chi)|^2 = \sum_{\substack{d|q \\ (d, q/d)=1}} \frac{d}{\phi(q)} \sum_{\chi^* \bmod d}^* |S(0, \chi^*)|^2.$$

We sum over $q \leq Q$ and replace q by dm . Thus, by Corollary ??, the

above is

$$\sum_{d \leq Q} \frac{d}{\phi(d)} \sum_{\substack{m \leq Q/d \\ (m,d)=1}} \frac{\mu(m)^2}{\phi(m)} \sum_{\chi^* \bmod d}^* |S(0, \chi^*)|^2 \leq (N + Q^2) \sum_{n=1}^N |a_n|^2.$$

By Exercise 3.2.1.9 and (3.18)

$$\sum_{\substack{m \leq R \\ (n,d)=1}} \frac{\mu(m)^2}{\phi(m)} \geq \log R.$$

Therefore

$$\sum_{d \leq Q} \left(\log \frac{Q}{d} \right) \sum_{\chi^* \bmod d}^* |S(0, \chi^*)|^2 \leq (N + Q^2) \sum_{n=1}^N |a_n|^2.$$

Now following the proof of (26.22) gives the lemma.

We now return to (28.88). The right hand side is

$$\ll \int_X^Y \sum_{q \leq T^2} \left(\log \frac{T^2}{q} \right) \sum_{\chi \bmod q}^* \int_{T-r}^{T+r} \left| \sum_{X < p \leq y} \frac{(\log p) \chi(p)}{p^w} b(p) \right|^2 dv \frac{dy}{y}.$$

Applying the lemma gives the bound

$$\begin{aligned} N^*(\theta, T) &\ll \frac{x^{c_6 r}}{(\log x)^3} \int_X^Y \sum_{X < p \leq y} \frac{(\log p)^2}{p^2} |b(p)|^2 (p + T^5) \frac{dy}{y} \\ &\ll \frac{x^{c_6 r}}{(\log x)^2} \sum_{X < p \leq Y} \frac{(\log p)^2}{p^2} |b(p)|^2 (p + T^5). \end{aligned}$$

We now have to deal with the sum over p . If $E_1 = 0$, then by Mertens, Theorem 2.7 (b), the sum over p is

$$\ll (\log X)^2 (1 + T^5/X)$$

and we are done. Thus we can henceforward suppose that $E_1 = 1$.

T:hyperb **Lemma 28.18.** *Let*

$$c(n) = \sum_{m|n} \chi_1(n)$$

where $\chi_1(n)$ is an exceptional character with conductor q_1 . Suppose further that $U/\log U \geq q_1$. Then

$$\sum_{n \leq U} \frac{c(n)}{n} = (\log U + \gamma) L(1, \chi_1) + L'(1, \chi_1) + O(q_1^{1/2} (\log U)^{1/2} U^{-1/2}).$$

The proof of the lemma is a routine application of Dirichlet's method of the hyperbola. Thus, with $V = q_1^{1/2} U^{1/2} (\log U)^{1/2}$ we have

$$\sum_{n \leq U} \frac{c(n)}{n} = \sum_{m \leq V} \frac{\chi_1(m)}{m} \sum_{l \leq U/m} \frac{1}{l} + \sum_{l \leq U/V} \frac{1}{l} \sum_{V < m \leq U/l} \frac{\chi_1(m)}{m}.$$

Then applying Euler's estimate to the first sum over l and then partial summation to both sums over m we obtain the desired conclusion.

By the lemma, when $X > q_1^2$ we have

$$\sum_{n \leq X} \frac{c(n)}{n} = L(1, \chi_1) \left(\log X + \gamma + \frac{L'}{L}(1, \chi_1) \right) + O(q_1^{\frac{1}{2}} (\log X)^{\frac{1}{2}} X^{-\frac{1}{2}}).$$

By (11.8) of Theorem 11.4 of volume 1 we have

$$\frac{L'}{L}(1, \chi_1) = \frac{1}{1 - \beta_1} + O(\log q_1).$$

We recall that, by (28.85), $X = x^{1/900}$ and, by (28.80), $x \geq T^{c_5} \geq q_1^{c_5}$. Also, by (11.10) of volume 1, $L(1, \chi_1) \gg 1 - \beta_1$. Hence, as c_5 is sufficiently large, it follows that

$$\sum_{n \leq X} \frac{c(n)}{n} \geq \frac{L(1, \chi_1)}{2(1 - \beta_1)}.$$

On the other hand, also by the lemma, and Theorem 11.11 of volume 1, we have

$$\sum_{X < n \leq XY} \frac{c(n)}{n} = (\log Y) L(1, \chi_1) + O(X^{-1/3}) \ll (\log Y) L(1, \chi_1)$$

Since $c(n)$ is multiplicative and non-negative we have

$$\sum_{n \leq X} \frac{c(n)}{n} \sum_{X < p \leq Y} \frac{c(p)}{p} \leq \sum_{X < n \leq XY} \frac{c(n)}{n}.$$

Combining this with the above inequalities shows that

$$\sum_{\substack{X < p \leq Y \\ \chi_1(p) \neq -1}} \frac{1}{p} \leq \sum_{X < p \leq Y} \frac{c(p)}{p} \ll (1 - \beta_1) \log Y.$$

We have

$$b(n)^2 = 1 + 2p^{-\delta_1} \chi_1(n) + p^{-2\delta_1} \chi_1(p)^2 \leq 2(1 + p^{-\delta_1} \chi_1(n))$$

Hence

$$\sum_{\substack{X < p \leq Y \\ \chi_1(p) \neq -1}} \frac{b(p)^2}{p} \ll (1 - \beta_1) \log Y.$$

We also have

$$1 - p^{-\delta_1} \ll (1 - \beta_1) \log p.$$

Hence, by Mertens once more,

$$\sum_{\substack{X < p \leq Y \\ \chi_1(p) = -1}} \frac{b(p)^2}{p} \ll (1 - \beta_1) \log Y.$$

The theorem now follows.

28.8 Linnik's Theorem on Primes in A.P.

S:Linnikap

An important application of the previous section is to the distribution of prime numbers in arithmetic progressions with relatively large common difference, and in particular to the least prime in an arithmetic progression.

T:PXGpinap

Theorem 28.19 (Gallagher). *There are constants $c \geq 1$ and $\kappa_0 \geq 3$ such that if κ is a constant with $\kappa \geq \kappa_0$ and q and x are such that $1 < Q^{6c} \leq x$, then we have*

$$\sum_{q \leq Q} \sum_{\chi \bmod q}^* |\vartheta(x; \chi) - E_0(\chi)x| \ll x \exp\left(-\frac{\log x}{\kappa \log Q}\right) + \left(\frac{\log x}{\log Q}\right)^2 x Q^{-1}$$

unless

$$F(s, q) = \prod_{q \leq Q} \prod_{\chi \bmod q}^* L(s, \chi) \quad (28.89) \quad \text{E:newLprodT}$$

has an exceptional real zero β_1 with

$$1 - \beta_1 < \frac{1}{\kappa \log Q}$$

in which case the general term on the left is to be replaced by

$$\left| \vartheta(x; \chi_1) + \frac{x^{\beta_1}}{\beta_1} \right|$$

when $\chi = \chi_1$ is the exceptional character and the right hand side is replaced by

$$(1 - \beta_1)(\log x) \left(x \exp \left(-\frac{\log x}{\log Q} \right) + \frac{x \log x}{Q \log Q} \right).$$

Proof By Theorem 12.12 and Corollary 14.7, when $x \geq 2$ and $T \leq x^{1/2}$ we have

$$\vartheta(x; \chi) = E_0(\chi)x - \sum_{\rho \in \mathcal{R}(\chi)} \frac{x^\rho}{\rho} + O\left(\frac{x}{T}(\log x)^2\right) \quad (28.90) \quad \boxed{\text{E:varthchi}}$$

where $\mathcal{R}(\chi)$ is the set of zeros $\rho = \beta + i\gamma$ of $L(s, \chi)$ with $\beta \geq \frac{1}{2}$ and $|\gamma| \leq T$.

Let the constants c, c_0, c_1, c_2 be as in the Exceptional Zero Statement, Theorem 28.14 and Corollary 28.15 and let

$$\kappa_0 = 3 \max(c, c_1, c_0 e, 1, c_0 e^{3c}). \quad (28.91) \quad \boxed{\text{E:kappalin}}$$

On hypothesis, $\kappa \geq \kappa_0$ and it is convenient to write

$$\kappa' = \kappa/3.$$

Let

$$T = Q^3. \quad (28.92) \quad \boxed{\text{E:Tandx}}$$

The proof divides into two cases. First we suppose that F , given by (28.89), has no zeros $\rho = \beta + i\gamma$ with $|\gamma| \leq T$ and

$$\beta > 1 - \frac{1}{\kappa_0 \log T},$$

that is, either there are no exceptional zeros, or the exceptional zero exists but satisfies $1 - \beta_1 \geq \frac{1}{\kappa_0 \log T}$. By 28.90 our sum is

$$\ll QxT^{-1}(\log x)^2 + \sum_{q \leq Q} \sum_{\chi \bmod q}^* \sum_{\rho \in \mathcal{R}(\chi)} x^\beta.$$

We have

$$x^\beta = x^{1/2} + \int_{1/2}^\beta x^u (\log x) du$$

and so the multiple sum above is

$$\leq x^{1/2} N^*(1/2, T) + \int_{1/2}^{1-1/(\kappa' \log T)} x^u N^*(u, T) (\log x) du.$$

By Theorem 28.14 this is

$$\ll x^{1/2}T^{c/2} + \int_{1/2}^{1-1/(\kappa' \log T)} x^u T^{c(1-u)} (\log x) du.$$

By (28.92) and the hypothesis on x .

$$xT^{-c} = xQ^{-3c} \geq x^{1/2} \text{ and } x^{1/2}Q^{c/2} \leq x^{3/4}.$$

Hence the sum of interest is

$$\begin{aligned} &\ll xQ^{-2}(\log x)^2 + x^{1-1/(\kappa' \log T)} T^{c/(\kappa' \log T)} \\ &= xQ^{-2}(\log x)^2 + x \exp \left(-(\log x)/(\kappa' \log Q) + c/\kappa' \right) \\ &\ll xQ^{-2}(\log x)^2 + x \exp \left(-(\log x)/(\kappa \log Q) + c/\kappa' \right). \end{aligned}$$

The remaining case is that in which there is an exceptional zero satisfying

$$\beta_1 > 1 - \frac{1}{\kappa' \log T}.$$

Now we have

$$\vartheta(x, \chi) + E_1(\chi) \frac{x^{\beta_1}}{\beta_1} \ll xT^{-1}(\log x)^2 + \sum_{\rho \in \mathcal{R}^*(\chi)} x^\beta$$

where $\mathcal{R}^*(\chi)$ denotes the set of zeros $\rho = \beta + i\gamma$ of $L(s; \chi)$, other than β_1 , with $|\gamma| \leq T$ and $\beta \geq \frac{1}{2}$.

We can proceed as above, but now the multiple sum is

$$\begin{aligned} &\ll (1 - \beta_1)(\log T)x^{1/2}T^{c/2} + \int_{1/2}^{1-\delta} (1 - \beta_1)(\log T)x^u T^{c(1-u)} du \\ &\ll (1 - \beta_1)(\log T)x^{1-\delta}T^{c\delta} \end{aligned}$$

where

$$\delta = \frac{1}{c \log T} \log \frac{1}{c_0(1 - \beta_1) \log T}.$$

Hence, by (28.62), the sum in question is

$$\begin{aligned} &\ll (1 - \beta_1)QxT^{-1}(\log x)^2 + (1 - \beta_1)(\log x)x^{1-\delta}T^{c\delta} \\ &\ll (1 - \beta_1)(\log x) \frac{x \log x}{Q \log Q} + (1 - \beta_1)(\log x)x^{1-\delta}T^{c\delta} \end{aligned}$$

where the implicit constant is absolute.

We have

$$\begin{aligned}
 x^{1-\delta} T^{c\delta} &= x \exp \left(\frac{\log(xT^{-c})}{c \log T} \log(c_0(1-\beta_1) \log T) \right) \\
 &\leq x \exp \left(-\frac{\log x - 3c \log Q}{3c \log Q} \log(\kappa'/c_0) \right) \\
 &\ll x \exp \left(-\frac{\log x - 3c \log Q}{\log Q} \right) \\
 &\ll x \exp \left(-\frac{\log x}{\log Q} \right)
 \end{aligned}$$

and that completes the proof. $\square$

The following two theorems are almost immediate

T:PXGspinap **Theorem 28.20** (Gallagher). *Suppose that $1 < q^{6c} \leq x$, $(a, q) = 1$, κ is as in Theorem 28.19 and that there is no exceptional zero β_1 with*

$$1 - \beta_1 < \frac{1}{\kappa \log q}.$$

Then

$$\vartheta(x; q, a) = \frac{x}{\phi(q)} \left(1 + O \left(\exp \left(-\frac{\log x}{\kappa \log q} \right) + \frac{\log^2 x}{q \log^2 q} \right) \right).$$

Given $q \in \mathbb{N}$ and $a \in \mathbb{Z}$ with $(q, a) = 1$ we define $p(q, a)$ to be the least prime number p such that $p \equiv a \pmod{q}$.

T:Leastpinap **Theorem 28.21** (Linnik). *There is a positive constant A such that whenever $q \in \mathbb{N}$, $a \in \mathbb{Z}$ and $(q, a) = 1$ we have $p(q, a) \leq q^A$.*

Proof We have

$$\begin{aligned}
 \vartheta(x; q, a) &= \frac{1}{\phi(q)} \sum_{\chi \bmod q} \bar{\chi}(a) \vartheta(x, \chi) \\
 &= \frac{1}{\phi(q)} \sum_{m|q} \sum_{\chi \bmod q}^* \bar{\chi}(a) \vartheta(x, \chi) + O((\log q)/\phi(q)) \quad (28.93) \quad \text{E:varther}
 \end{aligned}$$

and Theorem 28.20 follows immediately from Theorem 28.19

The proof of Theorem 28.21 divides into two cases. First, when there is no exceptional zero β_1 with

$$1 - \beta_1 < \frac{1}{\kappa \log q},$$

by Theorem 28.19 we have, for some constant c'

$$\left| \vartheta(x; q, a) - \frac{x}{\phi(q)} \right| < c' \frac{x}{\phi(q)} \left(\exp \left(-\frac{\log x}{\kappa \log q} \right) + \frac{\log^2 x}{q \log^2 q} \right)$$

and so on taking

$$x = q^A$$

with

$$A = \kappa \log(4c')$$

and $q > q_0(A)$ we have

$$\vartheta(x; q, a) > \frac{x}{2\phi(q)} > 0.$$

and this gives Theorem 28.21 in this case.

Alternatively, suppose that β_1 exists. Then, by (28.93) and Theorem 28.19,

$$\begin{aligned} \phi(q)\vartheta(x; q, a) - x + \chi_1(a) \frac{x^{\beta_1}}{\beta_1} \\ \ll (1 - \beta_1)(\log x) \left(x \exp \left(-\frac{\log x}{\log q} \right) + \frac{x \log x}{q \log q} \right) \end{aligned}$$

Since

$$1 - \beta_1 < \frac{1}{\kappa \log q} < \frac{1}{4}$$

and x is large we have

$$x - \frac{x^{\beta_1}}{\beta_1} = \int_{\beta_1}^1 \frac{u \log x - 1}{u^2} x^u du.$$

Therefore

$$\begin{aligned} x - \frac{x^{\beta_1}}{\beta_1} &\gg (1 - \beta_1)(\log x) x \exp(-(1 - \beta_1) \log x) \\ &\gg (1 - \beta_1)(\log x) x \exp(-(\log x)/(\kappa \log q)) \end{aligned}$$

where the implicit constant is absolute. Thus for positive absolute constants c' and c'' we have

$$\begin{aligned} \phi(q)\vartheta(x; q, a)x^{-1} &> \\ c'(1 - \beta_1)(\log x) \exp \left(-\frac{A}{\kappa} \right) - c''(1 - \beta_1)(\log x) &\left(\exp(-A) + \frac{\log x}{q \log q} \right). \end{aligned}$$

Thus if we choose

$$A = \frac{\kappa}{\kappa - 1} \log \frac{2c''}{c'},$$

then the above is

$$\geq (1 - \beta_1)(\log x) \frac{c'}{2} (1 - \beta_1)(\log x) \exp(-A/\kappa) - c''(1 - \beta_1)(\log x) \frac{\log x}{q \log q}$$

. Then for $q > \frac{2Ac''}{c'} e^{A/\kappa}$ we have $\vartheta(x; q, a) > 0$ and that completes the proof of Theorem 28.21. $\square$

28.8.1 Exercises

- 1 (Iwaniec ^{HI74} Iwaniec (1974)) Suppose that there is a non-negative function $f(q)$ such that $\lim_{q \rightarrow \infty} f(q) = \infty$ and

$$F(s, q) = \prod_{m|q} \prod_{\chi \bmod m}^* L(s, \chi)$$

contains no zero in the region

$$\sigma > 1 - \frac{f(q)}{\log(q(2 + |t|))},$$

then the least prime p with $p \equiv a \pmod{q}$ satisfies

$$p \ll_{\varepsilon} q^{\frac{12}{5} + \varepsilon}$$

- 2 Show that if χ is a non-principal character modulo q , then there is a constant c such that if $x > q^c$, then

$$\left| \sum_{p \leq x} \chi(p) \right| < \frac{\pi(x)}{2}.$$

- 3 (Fridlander ^{VF49} Fridlander (1949), Salié ^{HS49} Salié (1949), Montgomery ^{HM71} Montgomery (1971) Theorem 13.5.) Suppose that p is an odd prime and let $n_2(p)$ denote the least quadratic non-residue modulo p . Show that

$$n_2(p) = \Omega(\log p).$$

Assuming the generalized Riemann Hypothesis show that

$$n_2(p) = \Omega((\log p) \log \log p).$$

- 4 (Erdős ^{EP49} Erdős (1949))

- (a) Let $q = \prod_{p \leq y} p$. Show that for any $C > 0$ and any $y > y_0(c)$ there is an m such that $(m, q) = 1$ but $(m + n, q) > 1$ for $1 \leq |n| \leq cy$. (Hint: Use the method employed in proving Lemma 7.13)
- (b) Let $d_n = p_{n+1} - p_n$ where p_n denotes the n -th prime. Show that

$$\limsup_{n \rightarrow \infty} \frac{\min(d_n, d_{n+1})}{\log p_n} = \infty.$$

28.9 Maier's theorem on irregularity of primes in short intervals

S:Maier

The theorems we have established on the distribution of primes in short intervals can be used to show that the distribution of primes in very short intervals is more irregular than is predicted by the simple, classical, probabilistic model of the primes proposed by Cramér. This states that when $n \geq 2$ the probability that n is prime is taken to be

$$\frac{1}{\log n}.$$

Thus the expected number of primes not exceeding x would be

$$\sum_{2 \leq n \leq x} \frac{1}{\log n} = \text{li}(x) + O(1).$$

which fits well with what we know of the prime number theorem. It does so also with the theorems on primes in short intervals contained in section 28.5. However this model would also predict that when $\lambda > 1$ the expected number of primes p with

$$x < p \leq x + (\log x)^\lambda$$

is

$$\sum_{x < n \leq x + (\log x)^\lambda} \frac{1}{\log n} \sim (\log x)^{\lambda-1}$$

The theorem below contradicts this prediction. In retrospect this is perhaps not so surprising since probability models have some difficulty distinguishing between reduced residue classes and non-reduced residue classes.

In addition to Theorem 28.20, crucial rôles are played in the proof by Buchstab's Theorem 7.11 and the Maier matrix method. For use in the

proof we remind ourselves of the properties of the Buchstab function $\omega(u)$ defined in section 7.2, (7.37) and (7.38) by the equations

$$\omega(u) = \frac{1}{u} \quad (1 < u \leq 2),$$

$$(u\omega(u))' = \omega(u-1) \quad (u > 2)$$

and continuity at $u = 2$. Thus

$$\omega(u) = \frac{1 + \log(u-1)}{u} \quad (2 < u \leq 3).$$

Also, when $2 < u \leq 3$ we have

$$\omega'(u) = \frac{1 - (u-1)\log(u-1)}{u^2(u-1)}$$

so that $\omega'(2) = \frac{1}{4}$ and $\omega'(3) = \frac{1-2\log 2}{18} < 0$. Thus for some $u \in (2, 3)$, $\omega'(u) = 0$ and $\omega(u)$ has a local minimum. We also have for $u > 2$

$$u\omega'(u) = \omega(u-1) - \omega(u) = - \int_{u-1}^u \omega'(v) dv$$

so $\omega'(u)$ changes sign in every interval of length 1.

By considering the behaviour of the Laplace transform in Theorem 7.12 as $s \rightarrow 0+$ it follows that

$$\lim_{u \rightarrow \infty} \omega(u) = e^{-C_0}$$

where C_0 is Euler's constant.

The following lemma is particularly useful in the proof of Maier's theorem.

T:debruijn **Lemma 28.22.** *The function $\omega(u) - e^{-C_0}$ changes sign in every interval $[t-1, t]$ with $t \geq 2$.*

Proof We define the auxiliary function $\xi(u)$ for $u > -1$ by

$$\xi(u) = \int_0^\infty \exp\left(-ux - x + \int_0^x \frac{e^{-y} - 1}{y} dy\right) dx.$$

The function is differentiable for $u > -1$ and by integration by parts we have

$$u\xi'(u-1) + \xi(u) = 0. \quad (28.94) \quad \text{E:defxi}$$

We also have

$$\frac{1}{u+2} < \xi(u) < \frac{1}{u+1}.$$

For $t \geq 2$, let

$$\eta(t) = \int_{t-1}^t \omega(u)\xi(u)du + t\omega(t)\xi(t-1).$$

Then

$$\eta'(t) = 0 \quad (t > 2)$$

and

$$\lim_{t \rightarrow \infty} \eta(t) = e^{-C_0}.$$

Hence

$$\int_{t-1}^t \omega(u)\xi(u)du + t\omega(t)\xi(t-1) = e^{-C_0} \quad (t \geq 2).$$

Let

$$\nu(t) = \int_{t-1}^t \xi(u)du + t\xi(t-1).$$

Then, by (28.94), $\nu'(t) = 0$ ($t > 0$). We also have $\nu(t) \rightarrow 1$ as $t \rightarrow \infty$. Therefore

$$\int_{t-1}^t \xi(u)du + t\xi(t-1) = 1 \quad (t > 0)$$

and so

$$\int_{t-1}^t (\omega(u) - e^{-C_0})\eta(u)du + t(\omega(t) - e^{-C_0})\eta(t-1) = 0.$$

Since $\eta(t) > 0$ and $\omega(u)$ is not a constant when $2 \leq u \leq 3$, then the lemma follows. $\square$

T:Maier **Theorem 28.23** (Maier). *Let $\lambda > 1$,*

$$\Omega^-(\lambda) = \inf\{\omega(\tau) : \tau > \lambda\}, \quad \Omega^+(\lambda) = \sup\{\omega(\tau) : \tau > \lambda\}.$$

Then

$$\limsup_{x \rightarrow \infty} \frac{\pi(x + (\log x)^\lambda) - \pi(x)}{(\log x)^{\lambda-1}} \geq e^{C_0} \Omega^+(\lambda)$$

and

$$\liminf_{x \rightarrow \infty} \frac{\pi(x + (\log x)^\lambda) - \pi(x)}{(\log x)^{\lambda-1}} \leq e^{C_0} \Omega^-(\lambda).$$

Moreover $\Omega^-(\lambda) < e^{-C_0} < \Omega^+(\lambda)$ and

$$\Omega^+(\lambda) = \frac{1}{\lambda} \quad (1 < \lambda \leq 2).$$

Let

$$P(z) = \prod_{p \leq z} p.$$

T:PXGspecial

Lemma 28.24. *Suppose c and κ_0 are as in Theorem 28.20 and that $A > \max(2, 6c)$. Then there are $\kappa \in [\kappa_0, 2\kappa_0]$ and arbitrarily large $z > z_0(A, \kappa)$ such that whenever $(a, P(z)) = 1$ we have*

$$\begin{aligned} \pi(2P(z)^A, P(z), a) - \pi(P(z)^A, P(z), a) &= \frac{P(z)^A}{\phi(P(z))(A \log P(z))} \\ &\ll \frac{P(z)^A \exp(-A/\kappa)}{\phi(P(z))(A \log P(z))}. \end{aligned}$$

Proof Let p_n denote the n -th prime in order of magnitude and suppose that n is large. Then consider Theorem 28.20 with $q = P(p_n)$. If there is no exceptional zero β_1 of

$$F_n(s) = \prod_{m|P(p_n)} \prod_{\chi \bmod m}^* L(s, \chi)$$

with

$$\beta_1 > 1 - \frac{1}{\kappa_0 \log P(p_n)}, \quad (28.95) \quad \text{E:excccon}$$

then we have, provided p_n is large enough in terms of A and κ_0 .

$$\begin{aligned} \vartheta(2P(p_n)^A, P(p_n), a) - \vartheta(P(p_n)^A, P(p_n), a) &= \frac{P(p_n)^A}{\phi(P(p_n))} \\ &\ll \frac{P(p_n)^A \exp(-A/\kappa_0)}{\phi(P(p_n))}. \end{aligned}$$

Moreover, for $P(p_n)^A < p \leq 2P(p_n)^A$, we have

$$\log p = A \log P(p_n) + O(1)$$

and the desired conclusion follows with $z = p_n$.

Now suppose that there is an exceptional zero β_1 satisfying (28.95) of

$$F(s) = \prod_{m|P(p_n)} \prod_{\chi \bmod m}^* L(s, \chi)$$

and let q_1 be the corresponding conductor. Since

$$\log P(p_n) \ll \frac{1}{1 - \beta_1} \ll q_1^{1/2} \log q_1^2$$

q_1 is large in terms of n . Now choose l minimally so that $q_1 | P(p_l)$ and

consider $P(p_{l-1})$. The l will also be large in terms of n , and β_1 will satisfy (28.95) with n replaced by l , so will be exceptional for $F_l(s)$. Suppose that $F_{l-1}(s)$ has an exceptional zero β_2 , so that

$$\beta_2 > 1 - \frac{1}{\kappa_0 \log P(p_{l-1})}.$$

Then the associated conductor will divide $P(p_l)$ but by the minimality of l the exceptional conductor will differ from q_1 . But there cannot be a second exceptional zero of $F_l(s)$, so

$$\begin{aligned} \beta_2 &\leq 1 - \frac{1}{\kappa_0 \log P(p_l)} = 1 - \frac{1}{\kappa_0 (\log P(p_{l-1}) + \log p_l)} \\ &< 1 - \frac{1}{2\kappa_0 \log P(p_{l-1})}. \end{aligned}$$

Thus there are no exceptional zeros of the kind

$$\beta_1 > 1 - \frac{1}{2\kappa_0 \log P(p_{l-1})}$$

associated with $P(p_{l-1})$ and we can proceed as in the first part of the proof. $\square$

Let $\tau > \lambda$ and consider the array

$$\mathfrak{M} = (a_{uv}) \quad (P(z)^{A-1} < u \leq 2P(z)^{A-1}, 1 \leq v \leq (A \log P(z))^\tau, (v, P(z)) = 1)$$

where $a_{uv} = 1$ when $uP(z) + v$ is prime and 0 otherwise.

By Theorem 28.20, the number of non-zero entries in the v -th column is

$$\begin{aligned} &\pi(2P(z)^A + v, P(z), v) - \pi(P(z)^A + v, P(z), v) \\ &= \frac{P(z)^A}{\phi(P(z))A \log P(z)} (1 + O(\exp(-A/\kappa))) \end{aligned} \quad (28.96)$$

and so, by Theorem 7.11, the total number in the array is

$$\begin{aligned} &\frac{P(z)^A \Phi((A \log P(z))^\tau, z)}{\phi(P(z))A \log P(z)} (1 + O(\exp(-A/\kappa))) = \\ &\frac{P(z)^{A-1} (A \log(P(z)))^\tau \omega \left(\frac{\tau \log(A \log P(z))}{\log z} \right)}{A \log P(z) \prod_{p \leq z} (1 - 1/p)} (1 + O(e^{-A/\kappa})). \end{aligned}$$

Moreover, by the standard form of the prime number theorem, Theorem

6.9,

$$\begin{aligned}\log P(z) &= \vartheta(z) = z + O(z/\log z), \\ \log(A \log P(z)) &= \log z + \log A + O\left(\frac{1}{\log z}\right) \\ &= (\log z) \left(1 + O\left(\frac{\log A}{\log z}\right)\right)\end{aligned}$$

and by Merten's theorem, Theorem 2.7,

$$\prod_{p \leq z} (1 - 1/p) = \frac{e^{-C_0}}{\log z} \left(1 + O\left(\frac{1}{\log z}\right)\right).$$

Hence

$$\omega(\tau \log(A \log P(z))/\log z) = \omega(\tau)(1 + O(\log A/\log z)).$$

and the total number of non-zero entries in the array is

$$\frac{P(z)^{A-1} (A \log(P(z)))^\tau e^{C_0 \omega(\tau)}}{A \log P(z)} (1 + O(\exp(-A/\kappa))).$$

The total number of rows is $P(z)^{A-1}$. Hence there are rows with at least

$$\frac{(A \log(P(z)))^\tau e^{C_0 \omega(\tau)}}{A \log P(z)} (1 + O(\exp(-A/\kappa)))$$

non-zero entries. By dividing the primes counted in these rows into N subintervals of length

$$\frac{(A \log(P(z)))^\tau}{N}$$

where

$$N = \lceil (A \log(P(z)))^{\tau-\lambda} \rceil$$

we find that there are intervals

$$\left(X, X + \frac{(A \log(P(z)))^\tau}{N}\right]$$

containing at least

$$\frac{(A \log(P(z)))^\lambda e^{C_0 \omega(\tau)}}{A \log P(z)} (1 + O(\exp(-A/\kappa)))$$

primes where

$$P(z)^A \leq X \leq 2P(z)^A + (A \log P(z))^\tau.$$

The length of such intervals is at most

$$(A \log P(z))^\lambda \leq (\log X)^\lambda.$$

Moreover

$$A \log P(z) = \log X + O(1)$$

Thus it follows that there are arbitrarily large X such that

$$\pi(X + (\log X)^\lambda) - \pi(X) \geq e^{C_0} \omega(\tau) (\log X)^{\lambda-1} (1 + O(\exp(-A/\kappa))).$$

In the opposite direction, there are rows with at most

$$\frac{(A \log(P(z)))^\tau e^{C_0} \omega(\tau)}{A \log P(z)} (1 + O(\exp(-A/\kappa)))$$

non-zero entries. The choice

$$N = \left\lfloor \frac{(A \log(P(z)))^\tau}{(\log(2P(z)^A + (A \log P(z))^\tau))^\lambda} \right\rfloor$$

produces intervals

$$\left(X, X + \frac{(A \log(P(z)))^\tau}{N} \right]$$

of length at least

$$(\log(2P(z)^A + (A \log P(z))^\tau))^\lambda \geq (\log X)^\lambda$$

containing at most

$$e^{C_0} \omega(\tau) (\log X)^{\lambda-1} (1 + O(\exp(-A/\kappa)))$$

primes. Thus it follows that there are arbitrarily large X such that

$$\pi(X + (\log X)^\lambda) - \pi(X) \leq e^{C_0} \omega(\tau) (\log X)^{\lambda-1} (1 + O(\exp(-A/\kappa))).$$

The theorem now follows.

28.10 Notes

S:ZDT Notes

§1. The Euler product is a crucial ingredient in establishing that Dirichlet L functions have relatively few zeros, if any, off the critical line. There is an extensive literature showing that Dirichlet series without that feature can have $\asymp T$ zeros $\rho = \beta + i\gamma$ with $|\gamma| \leq T$ and $\beta > \frac{1}{2}$. See, for example, ^{BS07}Balanzario & Sanchez-Ortiz (2007), ^{BGT1}Bombieri & Ghosh

(2011), ^{BH95}Bombieri & Hejhal (1995), ^{JC61}Cassels (1961), Davenport & Heilbronn ^{HD77}Davenport (1977), ^{DH36a}Davenport & Heilbronn (1936a), ^{DH36b}Davenport & Heilbronn (1936b), ^{DG13}Dubickas, Garunkštis, Steuding, Steuding (2013), ^{KK07}Kaczorowski & Kulas (2007), ^{AK89}Karatsuba (1989), ^{AL86}Laurinćikas (1986), ^{PT35}Potter & Titchmarsh (1935), ^{SW09}Saias & Weingartner (2009), ^{RS69a, RS69b, RS94}Spira (1969a,b); ^{RV15}Spira (1994), ^{SV85}Vaughan (2015), and ^{BL14a}Voronin (1985).

§28.4. This subject was initiated by ^{JL24}Bohr & Landau (1914a) (see also ^{BL14b}Littlewood (1924)) who showed that if $\sigma > 1/2$, then $N(\sigma, T) \ll_\sigma T$, and then improved this to $N(\sigma, T) = o_\sigma(T)$, ^{FC20}Bohr & Landau (1914b). The first improvement in the exponent of T is due to ^{FC20}Carlson (1920) who obtained

$$N(\sigma, T) \ll T^{4\sigma(1-\sigma)+\varepsilon}.$$

See also ^{EL22}Landau (1922) and ^{GH30a, GH30b}Hoheisel (1930a,b) who replaced the T^ε by a powers of logarithms. ^{ET28, ET29}Titchmarsh (1928, 1929) improved the exponent to $4(1-\sigma)/(3-2\sigma)+\varepsilon$ and ^{AI37}Ingham (1937) obtained bound

$$N(\sigma, T) \ll T^{\frac{3(1-\sigma)}{2-\sigma}} (\log T)^5.$$

^{AS46}Selberg (1946) obtained a sharper bound when

$$\sigma - \frac{1}{2} \ll \frac{\log \log T}{\log T}$$

and ^{MJ82}Jutila (1982) sharpened this further and obtained

$$N(\sigma, T) \ll T^{1-(1-\delta)(\sigma-1/2)} \log T$$

for any $\delta \in (0, 3/4)$.

^{PT41}Turán in ^{PT43}Turán (1941), ^{PT51}Turán (1943), ^{PT54a}Turán (1951), ^{PT54b}Turán (1954a), ^{PT56}Turán (1954b), ^{PT58}Turán (1956), ^{GI66}Turán (1958) made a number of improvements when σ is close to 1. ^{GI66}Iglina (1966) has similar improvements near the 1-line.

Various authors, including ^{YL46a}Linnik (1946a), ^{YL46b}Linnik (1946b), ^{NC47}Chudakov (1947), ^{NC48}Chudakov (1948), ^{CH51}Haselgrove (1951) have discussed bounds for $N(\sigma, \chi, T)$, N_1 and N_2 , and ^{AR48}Rényi (1948) had shown that early forms of the large sieve already gave interesting consequences for the distribution of zeros of L -functions. ^{AV65}A. I. Vinogradov (1965), ^{AV66}Vinogradov (1966) and ^{EB65}Bombieri (1965) used the large sieve to give significant bounds for N_2 which lead to important new results on the distribution of primes in arithmetic progressions and ^{HT69}Halász and Turán (1969) introduced ideas which resulted in significant improvements when σ is close

to 1. Montgomery ^{HM71}Montgomery (1971) gives a systematic account of these developments. This led to intense activity over the next twenty years by Huxley, Jutila, Ivić and others. However, the generalization of Ingham's bound to $N_1(\sigma, q, T)$ and $N_2(\sigma, Q, T)$ when $\frac{1}{2} < \sigma < \frac{3}{4}$ has apparently not been significantly improved.

It is crucial in applications that the bounds are uniform in σ which the stated bounds are. Jutila ^{MJ77}Jutila (1977) extended the range for the density hypothesis to $\frac{11}{14} \leq \sigma \leq 1$ and Bourgain ^{JB00aa}Bourgain (2000) pushed this to $\frac{25}{39} < \sigma < 1$. This is an area which continues to be active. See Kerr ^{BK19}Kerr (2019).

The use of the mollifier M already occurs in Carlson ^{FC20}Carlson (1920), but prior to Montgomery ^{HM69}Montgomery (1969), it was used *via* variants of Littlewood's Lemma. Montgomery introduced the transform (28.39) to count zeros and this and its variants have been at the core of later developments. Montgomery obtained

$$N_1(\sigma, q, T) \ll (qT)^{\frac{2(1-\sigma)}{\sigma}} (\log qT)^{14}$$

and

$$N_2(\sigma, Q, T) \ll (Q^2T)^{\frac{2(1-\sigma)}{\sigma}} (\log Q^2T)^{14}$$

which already improves upon Ingham when $\sigma > \frac{4}{5}$. Huxley ^{MH72a}Huxley (1972a) obtained

$$N(\sigma, T) \ll T^{(5\sigma-3)(1-\sigma)/(\sigma^2+\sigma-1)} (\log T)^9$$

which gives Corollary 28.7 with $Q = 1$, and then established (28.29) and (28.31) in ^{MH72b}Huxley (1972b).

For the Riemann zeta function at least, there are numerous small improvements in the range $3/4 < \sigma < 1$. See Forti & Viola ^{FV73}Forti & Viola (1973), Heath-Brown ^{HB79}Heath-Brown (1979), Huxley ^{MH73a}Huxley (1973), ^{MH75a}Huxley (1975a), ^{MH75b}Huxley (1975b), Ivić ^{AI79b}Ivić (1979), ^{AI80b}Ivić (1980), ^{AI84b}Ivić (1984), Jutila ^{MJ72}Jutila (1972), ^{MJ77}Jutila (1977), Ramachandra ^{KR75}Ramachandra (1975). ^{AI03, Chapter 11}Ivić (2003); ? gives a comprehensive overview of the state of play in 1985.

§28.5 Results of the kind contained in this section are intimately connected with zero density estimates. The first theorem like Theorem 28.8 was established by Hoheisel ^{GH30b}Hoheisel (1930b), but with $\frac{7}{12}$ replaced by

$$1 - \frac{1}{33000}.$$

Hoheisel was constrained by only having the weaker Littlewood zero-free region available, and so his exponent depended on the constant in

that as well as the exponent in the zero-density estimate. Heilbronn, by working hard with the constants replaced that by

$$1 - \frac{1}{250}.$$

Chudakov obtained an improved zero free region for the Riemann-zeta function ^{NC36bb} from the Vinogradov mean value theorem which immediately reduced the exponent to $\frac{3}{4}$ ^{NC36a} Chudakov (1936a), and Ingham used his zero density estimate, combined with the Chudakov zero-free region, to obtain the exponent $\frac{5}{8}$ ^{AI37} Ingham (1937).

In more recent times the methods displayed here have been combined with sieve methods to show that when h is somewhat smaller than $x^{\frac{7}{12}}$ there is a prime in $(x, x+h]$. The current record, due to Baker, Harman & Pintz ^{BP01} Baker, Harman & Pintz (2001), is that this holds for $h = x^c$ with any $c > \frac{21}{40}$.

The main idea of the proof of Theorem 28.9 is taken from the proof of Lemma 6 of Saffari & Vaughan ^{SV77} Saffari & Vaughan (1977). The second part of Exercise 1 (b) is essentially due to Selberg ^{AS43} Selberg (1943). It differs in that the bound in the exercise is uniform for u close to 1, whereas Selberg apparently requires $u \ll x^{-\varepsilon}$.

§28.6 This section has its origins in the seminal paper of Hal'asz and Turán ^{HT69} Halász, Turán (1969).

§28.7 Theorem 28.14 is useful when the presence of an exception zero is a nuisance, and shows that it may not be such a big nuisance. The first results of this kind were obtained by Linnik ^{YL44a} Linnik (1944a), ^{YL44b} Linnik (1944b), ^{YL44c} Linnik (1944c) in his fundamental work on the least prime in an arithmetic progression. In addition to the original application it has also found use in work on the exceptional set in Goldbach's Problem (see Montgomery & Vaughan ^{MV75} Montgomery & Vaughan (1975)). Our exposition is inspired by Gallagher ^{PG70} Gallagher (1970) and Bombieri in §6 of ^{EB74} Bombieri (1974), the latter of which has the merit of giving the Deuring-Heilbronn, Corollary 28.15, for free. Again, stimulated by the question of the least prime in an arithmetic progression, there is a considerable history, with papers by Rodoskii ^{KR54} Rodoskii (1954), Turán ^{PT61} Turán (1961), Fogels ^{EF65} Fogels (1965), Jutila ^{MJ69} Jutila (1969), ^{MJ70} Jutila (1970), Selberg ^{AS72} Selberg (1972), Motohashi ^{YM76} Motohashi (1976). Selberg's method gives

$$N_1(\sigma, q, T) \ll (qT)^{(3+\varepsilon)(1-\sigma)}$$

and

$$N_2(\sigma, Q, T) \ll (Q^5 T^3)^{(1+\varepsilon)(1-\sigma)}.$$

§28.8 Below is a, not necessarily complete, history of bounds for the exponents A in Theorem 28.21

A	Year	
10000	1957	Pan ^{CP57} Pan (1957)
777	1965	Chen ^{JC65} Chen (1965)
550	1970	Jutila ^{MJ70} Jutila (1970)
168	1977	Chen ^{JC77} Chen (1977)
80	1977	Jutila ^{MJ77} Jutila (1977)
36	1977	Graham ^{SG77} Graham (1977)
20	1981	Graham ^{SG81} Graham (1981) (submitted before Chen's 1979 paper)
17	1979	Chen ^{JC79} Chen (1979)
16	1986	Wang ^{WW86} Wang (1986)
13.5	1989	Chen & Liu ^{CL89a} Chen & Liu (1989a), ^{CL89b} Chen & Liu (1989b), ^{CL91} Chen & Liu (1991)
8	1990	Wang ^{WW91} Wang (1991)
5.5	1992	Heath-Brown ^{HB92} Heath-Brown (1992)
5.18	2011	Xylouris ^{TX11a} Xylouris (2011a)
5	2011	Xylouris ^{TX11b} Xylouris (2011b)

§28.9 Maier's theorem ^{HM85} Maier (1985) came as something of a surprise, and caused some anxious rethinking of how we might model the primes in short intervals. Hildebrandt & Maier ^{HM89} Hildebrandt & Maier (1989) extended the method to show that even in somewhat longer intervals there are still greater biases in the distribution than had hitherto been believed. In a different direction the method has been adapted by Friedlander & Granville ^{FG89} Friedlander & Granville (1989), ^{FG91} Friedlander & Granville (1991), ^{FG92} Friedlander & Granville (1992) and by Friedlander, Granville, Hildebrandt and Maier ^{FGHM} Friedlander, Granville, Hildebrandt & Maier (1991) to show that the distribution of primes in arithmetic progressions, when the modulus is quite close to the size of the primes, is not as good as had been anticipated.

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29

Primes in Short Intervals

C:PSI

29.1 The Guth-Maynard Refinement: Initiation

S:1GMRI

Here we discuss a refinement by Maynard and Guth of the methods described in the previous chapter which have particular relevance to the distribution of primes in short intervals, but also have the potential to make relevant progress in cognate questions. The initial motivation is Theorem L.4. The case $k = 1$ occurs explicitly or implicitly in much of modern analytic number theory. The case $k = 2$ is implicit in the work of Jutila (1977) (see also Heath-Brown (1979)) but Guth and Maynard were the first to deploy it explicitly for any $k > 1$, and in particular for $k = 3$.

Fundamental ingredients in the proofs of zero density estimates are bounds for

$$\sum_{r=1}^R |D(s_r; \chi_r)|^2,$$

where

$$D(s; \chi) = \sum_{n=1}^N a_n \chi(n) n^{-s},$$

of the kind exemplified by Theorems 27.7, and 27.15. An examination of the proof of the Huxley estimate in §28.4.2 shows that we really only need to deal with dyadic ranges for n . Also, in this chapter, for simplicity of exposition, we will only concern ourselves with the case of the Riemann zeta function, so that $Q = 1$ and χ_r is identically 1. Also

$$s_r = \sigma_2 + ig_r, \quad \frac{1}{2} \leq \sigma_r < 1, \quad 0 \leq g_r \leq T \quad (29.1) \quad \text{E:valuesdef}$$

and eventually N will be quite constrained in terms of T , but for the

time being we only suppose that there are positive constants κ_1 , κ_2 and κ_3 such that

$$T^{\kappa_1} \leq N \leq T^{\kappa_2}, \quad \sum_{n=N+1}^{2N} |a_n|^2 \leq T^{\kappa_3}. \quad (29.2) \quad \boxed{\text{E:Nbound1}}$$

One further general remark needs to be made. There are many situations where it is desirable to save a large power of T . Thus statements of the kind

$$\ll_A T^{-A}$$

are to be read as holding for any fixed positive A , suitably large if necessary, and not necessarily the same on different occasions.

We concentrate on bounds for

$$\sum_{r=1}^R |D(s_r)|^2, \quad (29.3) \quad \boxed{\text{E:rRDsr1}}$$

where

$$D(s) = \sum_{n=N+1}^{2N} a_n n^{-s}. \quad (29.4) \quad \boxed{\text{E:rRDsr2}}$$

As we have already observed in the prologue to Theorem 19.4, in relation to the large sieve, by Theorem G.1 it suffices to bound the dual form

$$\sum_{n=N+1}^{2N} \left| \sum_{r=1}^R b_r n^{-s_r} \right|^2$$

and this permits, as there, the introduction of suitable weight or test functions. Here it is desirable to use test functions which are in $C^\infty(\mathbb{R})$, are 0 outside some interval such as $[N, 2N]$ and are 1 on some subinterval the interval like $[6N/5, 9N/5]$. We can do so by first constructing a function w which is in $C^\infty(\mathbb{R})$, is 0 outside the interval $[1, 2]$ and is 1 on the interval $[6/5, 9/5]$ and we will describe such a functions in detail in the next section. Hence our aim now is to bound forms of the kind

$$\sum_n \left| \sum_{r=1}^R b_r w(n/N) n^{-s_r} \right|^2,$$

and this in turn leads to the dual form

$$\sum_{r=1}^R \left| \sum_n a_n w(n/N) n^{-s_r} \right|^2$$

We will also suppose for much of this exposition that $s_r = ig_r$ where the g_r are real. This mirrors the exposition in Chapters 26 and 27, and generally the reduction to the general case is by partial integration as in the proof of Theorem 26.9.

The arguments employed here also impose some further restrictions. In counting zeros it is usual to suppose that the $g \in \mathcal{G}$ are well spaced.

D:wellsp

Definition 29.1. When there is an $H > 0$ such that

$$|g - g'| \geq H \ (g \neq g', g \in \mathcal{G}, g' \in \mathcal{G}) \quad (29.5) \quad \text{E:tspaceing}$$

then we call $\mathcal{G}$ H -spaced.

Mostly we will suppose that there is a small but fixed $\delta > 0$ such that $\mathcal{G}$ is T^δ -spaced. We will occasionally only suppose that $\mathcal{G}$ is 1-spaced. Anyway by partitioning such a $\mathcal{G}$ into T^δ -spaced subsets bounds for the latter can be applied to the former situation, typically only with the loss of some fixed power of T^δ .

Generally we will suppose that

$$\mathcal{G} \subset [T_0, T_0 + T] \quad (29.6)$$

where $T_0, T \in \mathbb{R}$ and T is sufficiently large. In fact the T_0 is spurious in the sense that the main functions we meet such as $|Q(\alpha)|$ and $E(\mathcal{G})$ are independent of T_0 , and even in the discussion of (29.11) below, if we replace m^{ig} by m^{ig+iT_0} , then the m^{iT_0} can be absorbed into the a_m . Indeed later will apply an argument based on partitioning $[0, T]$ into subintervals of equal length

For notational convenience, and to avoid confusion with the standard notation $s = \sigma + it$ we will use g_r , not t_r . Thus we suppose that

$$g_r \in \mathcal{G} \text{ and } \text{card}(\mathcal{G}) = R \quad (29.7) \quad \text{E:Tset}$$

and introduce the notation

$$\mathfrak{D}(g) = \sum_n a_n w(n/N) n^{ig}. \quad (29.8) \quad \text{E:defFrakD}$$

Then our form becomes

$$\sum_{g \in \mathcal{G}} |\mathfrak{D}(g)|^2. \quad (29.9)$$

We can also write this as

$$\sum_{m=N+1}^{2N} a_m \sum_{n=N}^{2N} \bar{a}_n \sum_{g \in \mathcal{G}} w(m/N) w(n/N) (m/n)^{ig} = \mathbf{a} \mathcal{D} \mathcal{D}^* \mathbf{a}^* \quad (29.10)$$

where $\mathbf{a}$ denotes the N dimensional vector $(a_{N+1}, \dots, a_{2N})$, $\mathcal{D}$ is the $N \times R$ matrix

$$\mathcal{D} = [w(m/N)m^{ig}] \quad (N+1 \leq m \leq 2N, g \in \mathcal{G}) \quad (29.11) \quad \boxed{\text{E: defDmatrix}}$$

and $\mathcal{D}^*$ and $\mathbf{a}^*$ denote the complex conjugate transpose respectively. As described in §L.1,

$$\mathcal{D}\mathcal{D}^* = \mathcal{H}$$

is Hermitian and its eigenvalues are non-negative, and by Theorem L.3 we have

$$\sum_{g \in \mathcal{G}} |\mathfrak{D}(g)|^2 \leq \lambda \sum_{n=N+1}^{2N} |a_n|^2 \quad (29.12) \quad \boxed{\text{E: Eigenbound}}$$

where λ is the largest eigenvalue of $\mathcal{H}$ and $\mathcal{H}^*$.

By (L.6) and the notation (L.8),

$$\tau_k = \text{tr}(\mathcal{H}^k) = \text{tr}(\mathcal{H}^{*k}), \quad (29.13) \quad \boxed{\text{E: deftauk}}$$

and by Theorem L.4 we have

$$\lambda \leq \frac{2 \text{tr}(\mathcal{H}^*)}{R} + 2 \left(\text{tr}(\mathcal{H}^{*3}) - \frac{\text{tr}(\mathcal{H}^*)^3}{R^2} \right)^{1/3}. \quad (29.14) \quad \boxed{\text{E: lambdatau}}$$

By (29.11) we have

$$\mathcal{H}^* = \left[\sum_{N+1 \leq n \leq 2N} w(m/N)^2 m^{ig_1 - ig_2} \right] \quad (\mathbf{g} \in \mathcal{G}^2).$$

Hence, by (29.13),

$$\tau_1 = \text{tr}(\mathcal{H}^*) = R \sum_{n=N+1}^{2N} w(n/N)^2. \quad (29.15) \quad \boxed{\text{E: lambdatau1}}$$

Thus the bulk of this exposition will be centered on a close examination of τ_3 . Before proceeding in that direction we need to be more specific about the weight w .

29.2 The Weight Functions

sect:weightf

It is useful to have available real functions of a real variable which are in $C^\infty(\mathbb{R})$, have compact support and majorise or minorise various in-

tervals. To this end we first define

$$\Theta(\alpha) = \begin{cases} e^{-1/\alpha} & (\alpha > 0), \\ 0 & (\alpha \leq 0) \end{cases}$$

and

$$\Upsilon(\alpha) = \frac{\Theta(\alpha)}{\Theta(\alpha) + \Theta(1-\alpha)} \quad (\alpha \in \mathbb{R})$$

Now suppose that a, b, c, d are real numbers satisfying $a < b < c < d$ and consider

$$w_0(\alpha; a, b, c, d) = \Upsilon\left(\frac{\alpha-a}{b-a}\right) \Upsilon\left(\frac{d-\alpha}{d-c}\right)$$

Then it is readily checked that Θ, Υ and $w_0 \in \mathcal{C}^\infty(\mathbb{R})$, and that $\Upsilon(\alpha)0$ is 0 when $\alpha \leq 0$ and is 1 when $\alpha \geq 1$. Thus $w_0(\alpha) = 1$ when $b \leq \alpha \leq c$, $0 < w_0(\alpha) < 1$ when $a < \alpha < b$ and $c < \alpha < d$ and $w_0(\alpha) = 0$ otherwise. The special case with $a = 1, b = 6/5, c = 9/5, d = 2$, namely

$$w(\alpha) = w_0(\alpha; 1, 6/5, 9/5, 2) = \Upsilon(5(\alpha-1)) \Upsilon(5(2-\alpha)) \quad (29.16) \quad \boxed{\text{E:weightdef}}$$

and the associated function

$$h_g(\alpha) = h_g^-(\alpha) = w(\alpha)^2 \alpha^{ig} \quad (29.17) \quad \boxed{\text{E:defh-}}$$

and its Fourier transform

$$\hat{h}_g(y) = \int_{\mathbb{R}} h_g(\alpha) e(-\alpha y) d\alpha$$

will be used extensively. Mostly for convenience we will drop the $-$. However occasionally we will need a similar function

$$h_g^+(\alpha) = w_0(\alpha; 1/2, 1, 2, 5/2) \alpha^{ig}. \quad (29.18) \quad \boxed{\text{E:defh+}}$$

where w_0 majorises rather than minorises the characteristic function of the interval $[1, 2]$ and then we will use $^\pm$ to distinguish them. Other examples we will make use of are

$$\varpi(\alpha) = w_0(\alpha; -2, -1, 1, 2) \quad (29.19) \quad \boxed{\text{E:defvarpi}}$$

which majorises the interval $[-1, 1]$ and

$$\varsigma_g(\alpha) = w_0(\alpha; 1/8, 1/4, 4, 5) \alpha^{ig}. \quad (29.20) \quad \boxed{\text{E:defvarsigma}}$$

The rapid decay of the Fourier transforms of these functions is a central feature which we will deploy on various occasions.

L:hdiff **Lemma 29.1.** *Suppose j is a positive integer. Then*

$$\hat{h}_g^\pm(y) \ll_j (1 + |g|)^j |y|^{-j} \quad (y \neq 0), \quad (29.21) \quad \text{E:hdiffpm1}$$

$$\hat{\varsigma}(y) \ll_j (1 + |g|)^j |y|^{-j} \quad (y \neq 0), \quad (29.22) \quad \text{E:hdiffvarsigma}$$

$$\hat{\varpi}_g(y) \ll_j |y|^{-j} \quad (y \neq 0), \quad (29.23) \quad \text{E:hdiffvarpi}$$

$$\int_{\mathbb{R}} w(e^{2\pi x}) e(yx) dx \ll_j |y|^{-j} \quad (y \neq 0), \quad (29.24) \quad \text{E:hdiffwE}$$

and

$$\hat{h}_g^\pm(y) \ll_j (1 + |y|)^j |g|^{-j} \quad (g \neq 0). \quad (29.25) \quad \text{E:hdiffpm2}$$

Proof We have

$$\hat{h}_g^-(y) = \int_1^2 w(\alpha)^2 \alpha^{ig} e(-\alpha y) d\alpha.$$

The stated bounds for $\hat{h}_g^-$ follow by integrating by parts with either

$$(w(\alpha)^2 \alpha^{ig}) e(-\alpha y)$$

or

$$\alpha^{ig} (w(\alpha)^2 e(-\alpha y)).$$

It is crucial that the derivatives of $\hat{h}_g^-$ are 0 at $\alpha = 1$ and 2. The bounds for $\hat{\varpi}(y)$ and

$$\hat{h}_g^+(y) = \int_{1/2}^{5/2} h_q^+(\alpha) e(-\alpha y) d\alpha$$

follow in the same way, as does (29.24). $\square$

L:averagew2 **Lemma 29.2.** *Let A be a positive real number. Then*

$$\sum_n w(n/N)^2 = N \int_{\mathbb{R}} w(\alpha)^2 d\alpha + O_A(T^{-A}).$$

Note that

$$\int_{\mathbb{R}} w(\alpha)^2 d\alpha \asymp 1.$$

Proof We have $w(n/N)^2 = h_0(n/N)$. Hence, by the Poisson summation formula

$$\sum_n w(n/N)^2 = \sum_n \int_{\mathbb{R}} h_0(\alpha/N) e(-\alpha N) d\alpha = N \sum_n \hat{h}_0(Nn).$$

The term $n = 0$ contributes

$$N \int_{\mathbb{R}} w(\alpha)^2 d\alpha$$

and by Lemma 29.1 the remaining n contribute

$$\ll_j N \sum_{n=1}^{\infty} \frac{1}{(Nn)^j}.$$

Choose $j \geq A + 2$. □

As an immediate corollary of this and (29.15) we have an approximation for the linear trace.

L:lineartrace1 **Lemma 29.3.** *Let A be a positive real number. Then*

$$\mathrm{tr}(\mathcal{H}^*) = RN \int_{\mathbb{R}} w(\alpha)^2 d\alpha + O_A(T^{-A}).$$

29.3 The Dissection of the Cubic Trace

T:cubictrace1 **Theorem 29.4.** *Suppose that $\mathcal{G} \subset [0, T]$ and is T^δ -separated. Then*

$$\mathrm{tr}(\mathcal{H}^{*3}) = RN^3 \left(\int_{\mathbb{R}} w(\alpha)^2 d\alpha \right)^3 + 3S_2 + S_3 + O_{\delta,A}(T^{-A})$$

where

$$S_2 = N \left(\int_{\mathbb{R}} w(\alpha)^2 d\alpha \right) \sum_{\substack{\mathbf{g} \in \mathcal{G}^2 \\ g_1 \neq g_2}} \left| \sum_{n \in \mathbb{N}} h_{g_1 - g_2}(n/N) \right|^2 \quad (29.26) \quad \text{E: defS2}$$

and

$$S_3 = N^3 \sum_{\substack{\mathbf{g} \in \mathcal{G}^3 \\ g_q \neq g_r}} \sum_{\mathbf{m} \in (\mathbb{Z} \setminus \{0\})^3} \hat{h}_{g_1 - g_2}(m_1 N) \hat{h}_{g_2 - g_3}(m_2 N) \hat{h}_{g_3 - g_1}(m_3 N) \quad (29.27) \quad \text{E: defS3}$$

Proof Thus $\mathrm{tr}(\mathcal{H}^{*3})$

$$\begin{aligned} &= \sum_{\mathbf{n} \in \mathbb{Z}^3} \sum_{\mathbf{g} \in \mathcal{G}^3} w\left(\frac{n_1}{N}\right)^2 w\left(\frac{n_2}{N}\right)^2 w\left(\frac{n_3}{N}\right)^2 n_1^{i(g_1 - g_2)} n_2^{i(g_2 - g_3)} n_3^{i(g_3 - g_1)} \\ &= \sum_{\mathbf{g} \in \mathcal{G}^3} \sum_{\mathbf{n} \in \mathbb{Z}^3} h_{g_1 - g_2}\left(\frac{n_1}{N}\right) h_{g_2 - g_3}\left(\frac{n_2}{N}\right) h_{g_3 - g_1}\left(\frac{n_3}{N}\right). \end{aligned}$$

We apply Poisson summation to each sum over n_j , $j = 1, 2, 3$. Thus $\text{tr}(H^3)$

$$= N^3 \sum_{\mathbf{g} \in \mathcal{G}^3} \sum_{\mathbf{m} \in \mathbb{Z}^3} \hat{h}_{g_1-g_2}(m_1 N) \hat{h}_{g_2-g_3}(m_2 N) \hat{h}_{g_3-g_1}(m_3 N)$$

We separate out the term $\mathbf{m} = \mathbf{0}$. This term contributes

$$N^3 \sum_{\mathbf{g} \in \mathcal{G}^3} \hat{h}_{g_1-g_2}(0) \hat{h}_{g_2-g_3}(0) \hat{h}_{g_3-g_1}(0)$$

We have

$$\hat{h}_g(mN) = \int_1^2 h(\alpha) \alpha^{ig} e(-\alpha mN) d\alpha \ll 1 \quad (29.28) \quad \boxed{\mathbf{E:ht(mN)}}$$

Also, since $\mathcal{G}$ is T^δ -separated, by using Lemma 29.1 on any $\hat{h}_{g_q-g_r}(0)$ with $g_q \neq g_r$ we obtain

$$RN^3 \sum_{\mathbf{g} \in \mathcal{G}^3} \hat{h}_0(0)^3 + O_j(N^3 T^{3-\delta j}).$$

In addition

$$\hat{h}_0(0) = \int_{\mathbb{R}} w(\alpha)^2 d\alpha$$

and by (29.2), on choosing j large enough, we obtain an error $\ll_{\delta,A} T^{-A}$.

Collecting together any terms with exactly two of the $m_j = 0$ and if necessary permuting the $\mathbf{g}$, $g_2 \rightarrow g_1$, $g_3 \rightarrow g_2$, $g_1 \rightarrow g_3$ we find that those terms contribute

$$3N^3 \sum_{\mathbf{g} \in \mathcal{G}^3} \sum_{m \in \mathbb{Z} \setminus \{0\}} \hat{h}_{g_1-g_2}(0) \hat{h}_{g_2-g_3}(0) \hat{h}_{g_3-g_1}(mN).$$

By (29.28) and Lemma 29.1, the contribution here from m with $|m| > T^2$ is, for sufficiently large j ,

$$\ll N^3 T^3 \sum_{m > T^2} \frac{(1+T)^j}{m^j N^j} \ll_A T^{-A}$$

and when $g_1 \neq g_2$ or $g_2 \neq g_3$ the contribution from the m with $0 < |m| \leq T^2$ is

$$\ll_j N^3 T^{5-\delta j} \ll_A T^{-A}.$$

Then the remaining terms, with $g_1 = g_2 = g_3$ and $0 < |m| \leq T^2$, contribute

$$\ll N^3 T \sum_{m=1}^{\infty} (mN)^{-j} \ll_A T^{-A}.$$

It remains to consider terms in which at most one of the $m_j = 0$. We first consider those terms in which exactly one of the $m_j = 0$. Again by permuting the g_j the contribution from these terms is

$$3N^3 \sum_{\mathbf{g} \in \mathcal{G}^3} \sum_{\mathbf{m} \in (\mathbb{Z} \setminus \{0\})^2} \hat{h}_{g_1-g_2}(m_1N) \hat{h}_{g_2-g_3}(m_2N) \hat{h}_{g_3-g_1}(0).$$

By Lemma 29.1, when $g_3 \neq g_1$,

$$\hat{h}_{g_1-g_2}(m_1N) \hat{h}_{g_2-g_3}(m_2N) \hat{h}_{g_3-g_1}(0) \ll \frac{(1+T)^4}{m_1^2 m_2^2 N^4 |g_2 - g_3|^j}.$$

Thus the total contribution from the terms with $g_3 \neq g_1$ is

$$\ll_j T^{7-\delta j} \ll_A T^{-A}.$$

The terms with $g_3 = g_1$ contribute

$$\begin{aligned} 3N^3 \hat{h}_0(0) \sum_{\mathbf{g} \in \mathcal{G}^2} \sum_{\mathbf{m} \in (\mathbb{Z} \setminus \{0\})^2} \hat{h}_{g_1-g_2}(m_1N) \hat{h}_{g_2-g_1}(m_2N) = \\ 3N^3 \hat{h}_0(0) \sum_{\mathbf{g} \in \mathcal{G}^2} \sum_{\mathbf{m} \in (\mathbb{Z} \setminus \{0\})^2} \hat{h}_{g_1-g_2}(m_1N) \hat{h}_{g_2-g_1}(-m_2N). \end{aligned}$$

We have

$$\hat{h}_{-g}(-\beta) = \int_{\mathbb{R}} w(\alpha)^2 \alpha^{-ig} e(\alpha\beta) d\alpha = \overline{\hat{h}_g(\beta)}.$$

Thus the above is

$$3N^3 \int_{\mathbb{R}} w(\alpha)^2 d\alpha \sum_{\mathbf{g} \in \mathcal{G}^2} \left| \sum_{\mathbf{m} \in \mathbb{Z} \setminus \{0\}} \hat{h}_{g_1-g_2}(mN) \right|^2. \quad (29.29) \quad \boxed{\text{E: oldhat}}$$

Again by Lemma 29.1

$$\hat{h}_0(mN) \ll_j (|m|N)^{-j}$$

and so the contribution from the terms with $g_1 = g_2$ is

$$\ll_j N^{3-2j} T \left(\sum_{m=1}^{\infty} m^{-j} \right)^2 \ll_A T^{-A}$$

by (29.2).

The purpose of having introduced the Fourier transform is to enable a novel approach later to the sum S_3 . To obtain S_2 , for the remaining terms with exactly one of the $m_j = 0$ we now revert back h_g . We observe that

$$\sum_{\mathbf{m} \in \mathbb{Z} \setminus \{0\}} N \hat{h}_{g_1-g_2}(mN) = -N \hat{h}_{g_1-g_2}(0) + \sum_n h_{g_1-g_2}(n/N)$$

and since h_g has its support in $[1, 2]$ the sum on the right will be over N . Moreover, by Lemma 29.1,

$$N\hat{h}_{g_1-g_2}(0) \ll_j NT^{-\delta j} \ll_A T^{-A}.$$

Inserting this in (29.29) gives (29.26).

We now have to deal with the terms in which none of the m_j are 0, and then in particular those in which $g_q = g_r$ for at least one pair q, r . Suppose that $g_3 = g_1$. Then, by Lemma 29.1, the contribution is

$$\ll_j N^3 T^2 \frac{(1+T)^4}{m_1^2 m_2^2 m_3^j N^j}$$

and once more by (29.2), provided that we choose j sufficiently large, the total contribution from these terms will be $\ll_A T^{-A}$.

This completes the proof of the lemma. $\square$

29.4 The Second Moment

S:4SD

Here we investigate the expression S_2 arising in Lemma 29.4. This is closely related to sums studied by ^{MJ77}Jutila (1977) and ^{HB79}Heath-Brown (1979) in connection with the density hypothesis, and can be interpreted in terms of the trace of the square of a Hermitian matrix.

We are initially motivated by one of the standard devices for obtaining an approximate functional equation for Riemann zeta function.

L:AFEht

Lemma 29.5. *Suppose that A, δ and ε are fixed positive numbers, $T^\delta \leq |g| \leq U \leq T$ and $M \geq UT^\varepsilon/N$. Then*

$$\sum_{n \in \mathbb{N}} h_g^\pm(n/N) \ll N^{1/2} \int_{|t| \leq T^\varepsilon} \left| \sum_{m \leq M} m^{-1/2+ig+it} \right| \frac{dt}{1+t^2} + \Delta$$

where

$$\Delta \ll_A T^{-A}.$$

In particular, if $N > UT^\varepsilon$, then

$$\sum_{n \in \mathbb{N}} h_g^\pm(n/N) \ll_A T^{-A}.$$

Proof As $h_g^\pm \in C^\infty(\mathbb{R})$ and has compact support in $[1, 2]$, the Mellin transform

$$H^\pm(s) = \int_{\mathbb{R}} h_0^\pm(x) x^{s-1} dx$$

is an entire function and satisfies

$$H^\pm(s) \ll_j \frac{1}{\prod_{k=1}^j |s+k|} \quad \left(\sigma \leq 3, |s+k| \geq \frac{1}{3}, 0 \leq k \leq j \right). \quad (29.30) \quad \boxed{\text{E:MellineH}}$$

As usual we will suppose that j is sufficiently large.

By the inverse Mellin transform, (5.14), we have

$$h_0^\pm(x) = \frac{1}{2\pi i} \int_{2-i\infty}^{2+i\infty} x^{-s} H^\pm(s) ds,$$

whence by absolute convergence

$$\begin{aligned} \sum_{n \in \mathbb{N}} h_g^\pm(n/N) &= \sum_{n \in \mathbb{N}} h_0^\pm(n/N) (n/N)^{ig} \\ &= \frac{1}{2\pi i} \int_{2-i\infty}^{2+i\infty} \zeta(s-ig) N^s H^\pm(s) ds. \end{aligned}$$

We now move the vertical path to the line $\operatorname{Re} s = -j + \frac{5}{2}$, picking up at $s = 1 + it$ the residue

$$N^{1+ig} H^\pm(ig) = N^{1+ig} \hat{h}_g^\pm(0)$$

by (29.30). By Lemma 29.1 and the hypothesis on g this is $\ll_A T^{-A}$.

On the path $\operatorname{Re} s = -j + \frac{5}{2}$ we apply the functional equation for $\zeta(s)$, Corollary 10.3,

$$\zeta(s-ig) = \Theta(s-ig) \zeta(1+ig-s)$$

where

$$\Theta(z) = \frac{\Gamma((1-z)/2)}{\Gamma(z/2)} \pi^{z-1/2}. \quad (29.31) \quad \boxed{\text{E:GammaU}}$$

By the uniform form of Stirling's formula, Theorem C.1, it follows that when $\operatorname{Re} s = -j + \frac{5}{2}$ we have

$$\Theta(s-ig) \ll_j (1+|t-g|)^{j-2}.$$

Also

$$\sum_{m > M} m^{3/2-j+i(t-g)} \ll M^{\frac{5}{2}-j}.$$

Thus on the line $\operatorname{Re} s = -j + \frac{5}{2}$ we have

$$\Theta(s-ig) \sum_{m > M} m^{3/2-j+i(t-g)} N^s H^\pm(s) \ll \frac{(1+|t-g|)^{j-2}}{(1+|t|)^j (MN)^{j-\frac{5}{2}}}$$

Thus the contribution from the t with $|t| \geq |g|$ to the integral

$$\frac{1}{2\pi i} \int_{\frac{5}{2}-j-i\infty}^{\frac{5}{2}-j+i\infty} \Theta(s-ig) \sum_{m>M} m^{3/2-j-i(t-g)} N^s H^\pm(s) ds$$

is

$$\ll (MN)^{\frac{5}{2}-j} \ll T^{(\frac{5}{2}-j)\varepsilon}$$

and the contribution from the t with $|t| \leq |g|$ is

$$\ll \frac{|g|^{j-2}}{(MN)^{j-\frac{5}{2}}} \ll U^{1/2} T^{(\frac{5}{2}-j)\varepsilon}.$$

Thus in either case we have a contribution $\ll_A T^{-A}$.

Note that if $M < 1$ we have nothing more to do.

When $M \geq 1$ we are left with an integral containing an integrand with the partial sum out to M . We now move this integral to the $\frac{1}{2}$ -line, which gives

$$\frac{1}{2\pi i} \int_{\frac{1}{2}-i\infty}^{\frac{1}{2}+i\infty} \Theta(s-ig) \sum_{m \leq M} m^{-1/2+i(t-g)} N^s H^\pm(s) ds.$$

The complex numbers $\Gamma((1/2 - it + ig)/2)$ and $\Gamma((1/2 + it - ig)/2)$ are complex conjugates, so, by (29.31), on this line $|\Theta(s - ig)| = 1$. Thus, by (29.30) the t with $|t| > T^\varepsilon$ contribute to the integral an amount

$$\ll \int_{T^\varepsilon}^{\infty} \frac{(MN)^{1/2}}{t^j} dt \ll_A T^{-A}.$$

which gives the first part of the lemma. If $N > UT^\varepsilon$, then we can choose $M < 1$ and the second part follows at once. $\square$

It is desirable to replace the range for m in the sum of our previous lemma by a dyadic range. To that end the following lemmas are useful.

1em:Jut1 **Lemma 29.6** (Lemma 2 of ^{MJ77}Jutila (1977)). *Suppose that*

$$|a_n| \leq Bb_n \quad (1 \leq n \leq N)$$

and $\mathcal{G}$ is a finite subset of $\mathbb{R}$. Then

$$\sum_{\mathbf{g} \in \mathcal{G}^2} \left| \sum_{n=1}^N a_n n^{ig_1 - ig_2} \right|^2 \leq B^2 \sum_{\mathbf{g} \in \mathcal{G}^2} \left| \sum_{n=1}^N b_n n^{ig_1 - ig_2} \right|^2.$$

Proof The left hand side is

$$\sum_{\mathbf{g} \in \mathcal{G}^2} \sum_{m=1}^N \sum_{n=1}^N \overline{a_m} a_n (n/m)^{ig_1 - ig_2} = \sum_{m=1}^N \sum_{n=1}^N \overline{a_m} a_n \left| \sum_{g \in \mathcal{G}} (n/m)^{ig} \right|^2.$$

We then apply the stated inequality for a_n and reverse the process. $\square$

We now introduce a useful transference principle which eases the reduction to sums over dyadic intervals.

lem:RV **Lemma 29.7.** *Suppose that $J, J', M, M', \sigma \in \mathbb{R}$ with $J' \geq J \geq 2$, $M' > M \geq 0$, and suppose further that $[J, J']$ contains at least one prime and $\mathcal{G}$ is a finite subset of $\mathbb{R}$. Then*

$$\begin{aligned} \sum_{\mathbf{g} \in \mathcal{G}^2} \left| \sum_{M < n \leq M'} a_n n^{-\sigma - i(g_1 - g_2)} \right|^2 \\ \leq \frac{1}{\sum_{J \leq p \leq J'} p^{-2\sigma}} \sum_{\mathbf{g} \in \mathcal{G}^2} \left| \sum_{JM < l \leq J'M'} A(l) l^{-\sigma - i(g_1 - g_2)} \right|^2 \end{aligned}$$

where

$$A(l) = \sum_{\substack{pn=l \\ J < p \leq J' \\ M < n \leq M'}} |a_n|.$$

Note for future reference that

$$\sum_{J \leq p \leq J'} p^{-1} = \log \frac{J'}{\log J} + O((\log J)^{-2}) \gg \frac{\log(J'/J)}{\log J} \quad (29.32) \quad \text{eq:pbound}$$

when $J' \geq 2J$ and J is sufficiently large, and

$$A(l) \leq \frac{\log J'M'}{\log J} \max_{n|l} |a_n|. \quad (29.33) \quad \text{eq:Abound}$$

Proof We evaluate

$$\int_0^1 \sum_{\mathbf{g} \in \mathcal{G}^2} \left| \sum_{M < n \leq M'} a_n n^{-\sigma - i(g_1 - g_2)} \right|^2 \left| \sum_{J \leq p \leq J'} e(\alpha p) p^{-\sigma - i(g_1 - g_2)} \right|^2 d\alpha$$

in two different ways.

First of all by Parseval's identity it is

$$\sum_{\mathbf{g} \in \mathcal{G}^2} \left| \sum_{M < n \leq M'} a_n n^{-\sigma - i(g_1 - g_2)} \right|^2 \sum_{J \leq p \leq J'} p^{-2\sigma}.$$

On the other hand the integrand is

$$\sum_{\mathbf{g} \in \mathcal{G}^2} \left| \sum_{JM < l \leq J'M'} \left(\sum_{\substack{pn=l \\ J < p \leq J' \\ M < n \leq M'}} a_n e(\alpha p) \right) l^{-\sigma-i(g_1-g_2)} \right|^2.$$

The coefficient of $l^{-\sigma-i(g_1-g_2)}$ here is bounded by $A(l)$. Hence by Lemma 29.6 the above is at most

$$\sum_{\mathbf{g} \in \mathcal{G}^2} \left| \sum_{JM < l \leq J'M'} A(l) l^{-\sigma-i(g_1-g_2)} \right|^2$$

□

It is convenient to follow ^{MJ77}Jutila (1977) and define

$$S(N) = \sum_{\mathbf{g} \in \mathcal{G}^2} \left| \sum_{N < n \leq 2N} n^{-1/2+i(g_1-g_2)} \right|^2, \quad (29.34) \quad \boxed{\text{eq: Jut2b}}$$

A trivial but useful observation is the following.

lem:trivial **Lemma 29.8.** *Suppose that $N \in \mathbb{R}$ and $k \in \mathbb{N}$. Then*

$$S(N/k) \leq kS(N)$$

Proof By following the proof of Lemma 29.6 we have

$$\begin{aligned} S(N/k) &= \sum_{N/k < m, n \leq 2N/k} (mn)^{-1/2} \left| \sum_{g \in \mathcal{G}} (m/n)^{ig} \right|^2 \\ &\leq k \sum_{N/k < m, n \leq 2N/k} (km)^{-1/2} (kn)^{-1/2} \left| \sum_{g \in \mathcal{G}} (km/kn)^{ig} \right|^2 \\ &\leq k \sum_{N < m, n \leq 2N} (mn)^{-1/2} \left| \sum_{g \in \mathcal{G}} (m/n)^{ig} \right|^2, \end{aligned}$$

and then we reverse the initial argument. □

We also make use of the following which is due to ^{HB79}Heath-Brown (1979)

lem:RV2 **Lemma 29.9.** *Suppose that $N, P \in \mathbb{R}$, $N \geq 0$, $P \geq (2N)^k$ and $\mathcal{G}$ is a finite subset of $\mathbb{R}$ with $\text{card}(\mathcal{G}) = R$. Then*

$$S(N)^k \ll 2^k (R^2)^{k-1} (\log P)^3 A_k(N)^2 S(P)$$

where

$$A_k(N) = \max_n \text{card}\{m_1, \dots, m_k : m_1 \dots m_k = n, N < m_j \leq 2N\}. \quad (29.35) \quad \boxed{\text{eq:DN}}$$

Proof By Hölder's inequality

$$\begin{aligned} S(N)^k &\leq (R^2)^{k-1} \sum_{\mathbf{g} \in \mathcal{G}^2} \left| \sum_{N < n \leq 2N} n^{-1/2-i(g_1-g_2)} \right|^{2k} \\ &= (R^2)^{k-1} \sum_{\mathbf{g} \in \mathcal{G}^2} \left| \sum_{N^k < n \leq 2^k N^k} a_n n^{-1/2-i(g_1-g_2)} \right|^2 \end{aligned}$$

where

$$a_n = \text{card}\{m_1, \dots, m_k : m_1 \dots m_k = n, N < m_j \leq 2N\}.$$

By Lemma 29.7 with M, M' replaced by $N^k, 2^k N^k$ and J, J' by $P(2N)^{-k}, 2P(2N)^{-k}$

$$\begin{aligned} \sum_{\mathbf{g} \in \mathcal{G}^2} \left| \sum_{N^k < n \leq 2^k N^k} a_n n^{-1/2-i(g_1-g_2)} \right|^2 \\ \ll (\log P) \sum_{\mathbf{g} \in \mathcal{G}^2} \left| \sum_{P2^{-k} < l \leq 2P} A(l) l^{-1/2-i(g_1-g_2)} \right|^2. \end{aligned}$$

We split the inner sum into dyadic ranges so that by Cauchy's inequality the above is

$$\ll k(\log P) \sum_{j=0}^k \sum_{\mathbf{g} \in \mathcal{G}^2} \left| \sum_{P2^{-j} < l \leq 2^{1-j} P} A(l) l^{-1/2-i(g_1-g_2)} \right|^2.$$

By Lemma 29.6 this is bounded by

$$\ll (\log P)^3 A_k(N)^2 \sum_{j=0}^k \sum_{\mathbf{g} \in \mathcal{G}^2} \left| \sum_{P2^{-j} < l \leq 2^{1-j} P} l^{-1/2-i(g_1-g_2)} \right|^2.$$

Then, by Lemma 29.8 this is

$$\ll (\log P)^3 A_k(N)^2 2^k S(P).$$

□

We now consolidate the above for $S(N)$ as follows.

lem:ult **Lemma 29.10.** *Suppose that δ and ϵ are positive real numbers, $\mathcal{G}$ is a T^δ -separated subset of $(0, T]$ and $R = \text{card } \mathcal{G}$. Then there is a positive constants C such that*

(i) *if $T^{1+\epsilon} < N \leq T^{\kappa_2}$, then*

$$S(N) \ll RN, \quad (29.36) \quad \text{eq:ult1}$$

(ii) *if $D(T)R^{\frac{1}{4}}T^{\frac{1}{2}(1+\epsilon)} \leq N \leq T^{1+\epsilon}$, then*

$$S(N) \ll RN + R^2 D(T), \quad (29.37) \quad \text{eq:ult2}$$

(iii) *and if $1 \leq N \leq D(T)R^{\frac{1}{4}}T^{\frac{1}{2}(1+\epsilon)}$, then*

$$S(N) \ll R^{\frac{5}{4}}T^{\frac{1}{2}(1+\epsilon)}D(T) + R^2 D(T) \quad (29.38) \quad \text{eq:ult3}$$

where

$$D(X) = \exp\left(\frac{C \log X}{\log \log X}\right). \quad (29.39) \quad \text{E:defDN}$$

Proof By (29.34) and Lemma 29.6 we have

$$S(N) \ll RN + N^{-1} \sum_{\mathbf{g} \in \mathcal{G}^2} \left| \sum_n h_{g_1 - g_2}^+(n/N) \right|^2.$$

When $N > T^{1+\epsilon}$ the last part of Lemma 29.5 gives (29.36) at once.

When $N \leq T^{1+\epsilon}$, the first part of that lemma with

$$M = T^{1+\epsilon}/N \quad (29.40) \quad \text{eq:M1}$$

gives

$$S(N) \ll RN + \sum_{\mathbf{g} \in \mathcal{G}^2} \left(\int_{|t| \leq T^\epsilon} \left| \sum_{m \leq M} m^{-1/2+ig_1-ig_2+it} \right| \frac{dt}{1+t^2} \right)^2.$$

and by Cauchy's inequality and Lemma 29.6 we obtain

$$S(N) \ll RN + \sum_{\mathbf{g} \in \mathcal{G}^2} \left| \sum_{m \leq M} m^{-1/2+ig_1-ig_2} \right|^2.$$

We now divide into dyadic ranges so that by Cauchy's inequality the multiple sum above is

$$\ll (\log M) \sum_{1 \leq j \ll \log M} S(M2^{-j}).$$

By Lemma 29.7 with $J = 2^{j-1}$ and $J' = 2^j$, we have

$$\begin{aligned} S(M2^{-j}) &\ll (\log M)^3 \sum_{\mathbf{g} \in \mathcal{G}^2} \left| \sum_{M/2 < n \leq 2M} n^{-\frac{1}{2} + i(g_1 - g_2)} \right|^2 \\ &\ll (\log M)^3 (S(M/2) + S(M)). \end{aligned}$$

Hence, by Lemma 29.8,

$$S(N) \ll RN + (\log M)^3 S(M). \quad (29.41) \quad \boxed{\text{eq:uni}}$$

We note that by (29.36) this also holds when $N > T^{1+\varepsilon}$.

Suppose that $N \geq 4^{1/3} T^{2(1+\varepsilon)/3}$. Then, by (29.40), $4M^2 \leq N$ and, by Lemma 29.9 with $k = 2$, N replaced by M and P replaced by N ,

$$S(M)^2 \ll R^2 (\log T)^3 A_2(M)^2 S(N).$$

Hence, by (29.41),

$$S(N) \ll RN + R(\log T)^{9/2} A_2(M)^2 S(N)^{1/2}$$

and so

$$S(N) \ll RN + R^2 (\log T)^9 A_2(M)^4$$

giving (ii) in this case.

Now suppose that $N \leq 4^{1/3} T^{2(1+\varepsilon)/3}$ but (29.40) continues to hold. Then so does (29.41). We now have

$$4M^2 = 4T^{2(1+\varepsilon)} N^{-2} \geq 4^{1/3} T^{2(1+\varepsilon)/3}.$$

Thus from above

$$S(4M^2) \ll RM^2 + R^2 (\log T)^9 A_2(4M^2)^4. \quad (29.42) \quad \boxed{\text{eq:SQ}}$$

By Lemmas 29.9

$$S(M)^2 \ll R^2 A_2(M)^2 S(4M^2),$$

whence by (29.42)

$$S(M)^2 \ll R^3 M^2 A_2(M)^2 + R^4 (\log T)^9 A_2(4M^2)^6$$

so that

$$S(M) \ll R^{\frac{3}{2}} M A_2(M) + R^2 (\log T)^5 A_2(4M^2)^3.$$

Therefore, by (29.41)

$$S(N) \ll RN + (\log T)^3 R^{\frac{3}{2}} M A_2(M) + (\log T)^8 R^2 A_2(4M^2)^3.$$

The condition $D(T)R^{\frac{1}{4}}T^{\frac{1}{2}(1+\varepsilon)} < N$ in case (ii) ensures that

$$RN \geq D(T)^2 R^{\frac{3}{2}} T^{1+\varepsilon} N^{-1} = D(T)^2 R^{\frac{3}{2}} M$$

and so the first term dominates the second and gives the desired conclusion.

Now suppose $N \leq D(T)R^{\frac{1}{4}}T^{\frac{1}{2}(1+\varepsilon)}$. Let

$$P = D(T)R^{\frac{1}{4}}T^{\frac{1}{2}(1+\varepsilon)}$$

so that $N \leq P$. By a now familiar argument based on splitting the sum and Lemma 29.6 we have $S(N) \ll (\log T)^3 S(P)$, and so by (ii)

$$S(N) \ll (\log T)^3 D(P) R^{\frac{5}{4}} T^{\frac{1}{2}(1+\varepsilon)} + R^2 D(T).$$

□

The previous lemma is less effective when N is small compared with $R^{\frac{1}{4}}T^{\frac{1}{2}}$ and in that circumstance we can obtain superior results by utilising higher powers of

$$\sum_{N < n \leq 2N} n^{-\frac{1}{2} + ig_1 - ig_2}.$$

lem:ultk

Lemma 29.11. *Suppose that δ and ε are fixed positive real numbers, $\mathcal{G}$ is a T^δ -separated subset of $(0, T]$, $R = \text{card } \mathcal{G}$ and $k \in \mathbb{N}$. Then there is a positive constant C_k such that*

$$S(N) \ll R^{2-1/k} N D(T) + R^2 \left(\frac{T^{\frac{1}{2}(1+\varepsilon)}}{R^{\frac{3}{4}}} \right)^{\frac{1}{k}} D(T) + R^2 D(T) \quad (29.43) \quad \text{eq:ultk}$$

where

$$D(T) = \exp \left(\frac{C_k \log T}{\log \log T} \right).$$

This is established by combining Lemmas 29.9 and 29.10.

We can now state a suitable bound for the sum S_2 .

T:S2bound

Theorem 29.12. *Suppose that A , δ and ε are fixed positive numbers, $\mathcal{G} \subset [0, T]$ is T^δ -separated, $k \in \mathbb{N}$, and S_2 is given by (29.26). Then*

$$S_2 \ll \left(R^{2-\frac{1}{k}} T N + R^2 N^2 (T^{1/2} R^{-3/4})^{1/k} + R^2 N^2 \right) T^\varepsilon.$$

Proof By (29.26) and Lemmas 29.5 and 29.6 we have

$$S_2 \ll N^2 S(M)$$

where $M = T^{1+\varepsilon} N^{-1}$. The conclusion then follows from 29.43 with $N = M$. □

29.4.1 Exercises

1. Prove that every closed interval of the form $[2^{k/2}, 2^{(k+1)/2}]$ ($k \in \mathbb{N}$) contains a prime.

29.5 The reduction of the third moment

S:5TDI

Now is the turn of S_3 , given by (29.27), which we rewrite as

$$S_3 = N^3 \sum_{\mathbf{m} \in (\mathbb{Z} \setminus \{0\})^3} \sum_{\substack{\mathbf{g} \in \mathcal{G}^3 \\ g_q \neq g_r}} \hat{h}_{g_1-g_2}(m_1 N) \hat{h}_{g_2-g_3}(m_2 N) h_{g_3-g_1}(m_3 N). \quad (29.44) \quad \text{E: defS3a}$$

We relate this to

$$Q(\alpha) = \sum_{g \in \mathcal{G}} \alpha^{ig}. \quad (29.45) \quad \text{E: Rsum}$$

T:S3R **Theorem 29.13.** *Suppose that A and δ are fixed positive numbers, and $\mathcal{G}$ is T^δ -separated. Then*

$$S_3 \ll \Delta + \sum_{\substack{\mathbf{m} \in (\mathbb{Z} \setminus \{0\})^3 \\ |m_1| \leq |m_2| \leq |m_3| \leq \frac{T^{1+\delta}}{N}}} N^3 \int_{\mathcal{V}(\mathbf{m})} |Q(v_1) Q\left(\frac{v_2}{v_1}\right) Q(v_2)| d\mathbf{v} \quad (29.46) \quad \text{E: S3Ra}$$

where

$$\mathcal{V}(\mathbf{m}) = \{\mathbf{v} \in [1/2, 2]^2 : |m_1 v_1 + m_2 v_2 + m_3| \leq T^\delta / N\} \quad (29.47) \quad \text{E: Vm}$$

and $\Delta \ll_A T^{-A}$. Also, when $|m_1| \leq |m_2| \leq |m_3|$, $\mathcal{V}(\mathbf{m}) = \emptyset$ unless $|m_3| \leq 5|m_2|$.

Proof An immediate observation is that if we permute m_1, m_2, m_3 , then by relabeling the g_1, g_2, g_3 to correspond we find that the general term is invariant. The number of permutations will vary according as the m_1, m_2, m_3 are distinct or not. However we are ultimately only concerned with the size of the sum. Hence

$$S_3 \leq 6N^3 \sum_{\substack{\mathbf{m} \in (\mathbb{Z} \setminus \{0\})^3 \\ |m_1| \leq |m_2| \leq |m_3|}} \left| \sum_{\substack{\mathbf{g} \in \mathcal{G}^3 \\ g_q \neq g_r}} \hat{h}_{g_1-g_2}(m_1 N) \hat{h}_{g_2-g_3}(m_2 N) \hat{h}_{g_3-g_1}(m_3 N) \right| \quad (29.48) \quad \text{eq:two2}$$

It is useful first to get rid of the large m_j . By Lemma 29.1 for any non-negative integer m ,

$$\hat{h}_g(mN) \ll_j \frac{1 + |g|^j}{|m|^j N^j}$$

Thus if $\max_k |m_k| > T^{1+\delta} N^{-1}$, then for any $j \geq 4$ we have

$$\begin{aligned} \sum_{\substack{\mathbf{m} \\ \max_k |m_k| > T^{1+\delta} N^{-1}}} \sum_{\substack{\mathbf{g} \in \mathcal{G}^3 \\ g_q \neq g_r}} N^3 |\hat{h}_{g_1-g_2}(m_1 N) \hat{h}_{g_2-g_3}(m_2 N) \hat{h}_{g_3-g_1}(m_3 N)| \\ \ll_j N^3 T^3 \sum_{\substack{0 < |m_1| \leq |m_2| \leq |m_3| \\ |m_3| > T^{1+\delta} N^{-1}}} \frac{T^j}{|m_3|^j N^j} \\ \ll_j N^3 T^3 \sum_{m_3 > T^{1+\delta} N^{-1}} \frac{m_3^2 T^j}{m_3^j N^j} \\ \ll_j T^6 T^{(3-j)\delta}. \end{aligned}$$

Thus for some j , by (29.48),

$$S_3 \ll \Delta + \sum_{\substack{\mathbf{m} \in (\mathbb{Z} \setminus \{0\})^3 \\ |m_1| \leq |m_2| \leq |m_3| \\ |m_3| \leq \frac{T^{1+\delta}}{N}}} N^3 \left| \sum_{\substack{\mathbf{g} \in \mathcal{G}^3 \\ g_q \neq g_r}} \hat{h}_{g_1-g_2}(m_1 N) \hat{h}_{g_2-g_3}(m_2 N) \hat{h}_{g_3-g_1}(m_3 N) \right| \quad (29.49) \quad \boxed{\text{eq:two5}}$$

By (29.17),

$$\begin{aligned} \hat{h}_{g_1-g_2}(m_1 N) \hat{h}_{g_2-g_3}(m_2 N) \hat{h}_{g_3-g_1}(m_3 N) = \\ \int_{\mathbb{R}^3} w_1(\mathbf{u})^2 u_1^{i(g_1-g_2)} u_2^{i(g_2-g_3)} u_3^{i(g_3-g_1)} e(-N\mathbf{m} \cdot \mathbf{u}) d\mathbf{u}. \end{aligned} \quad (29.50) \quad \boxed{\text{eq:two3}}$$

where

$$w_1(\mathbf{u}) = w(u_1)w(u_2)w(u_3). \quad (29.51) \quad \boxed{\text{eq:two4}}$$

Hence, by (29.45)

$$\begin{aligned} \sum_{\substack{\mathbf{g} \in \mathcal{G}^3 \\ g_q \neq g_r}} \hat{h}_{g_1-g_2}(m_1 N) \hat{h}_{g_2-g_3}(m_2 N) \hat{h}_{g_3-g_1}(m_3 N) \\ = \int_{\mathbb{R}^3} w_1(\mathbf{u})^2 Q(u_1/u_3) Q(u_2/u_1) Q(u_3/u_2) e(-N\mathbf{m} \cdot \mathbf{u}) d\mathbf{u} \end{aligned}$$

We make the change of variables

$$u_1, u_2 \rightarrow v_1, v_2 : u_1 = u_3 v_1, u_2 = u_3 v_2.$$

Note that the support for each u_j is in $[1, 2]$ so there are no complications with negative u_3 . Moreover the support for v_1 and v_2 will lie in $[1/u_3, 2/u_3] \subseteq [1/2, 2]$. Thus

$$\begin{aligned} & \sum_{\substack{\mathbf{g} \in \mathcal{G}^3 \\ g_q \neq g_r}} \hat{h}_{g_1-g_2}(m_1 N) \hat{h}_{g_2-g_3}(m_2 N) h_{g_3-g_1}(m_3 N) \\ &= \int_{[1/2, 2]^2} Q(v_1) Q\left(\frac{v_2}{v_1}\right) Q\left(\frac{1}{v_2}\right) J(\mathbf{v}, \mathbf{m}) d\mathbf{v}. \end{aligned} \quad (29.52) \quad \boxed{\text{eq:two6}}$$

where

$$J(\mathbf{v}, \mathbf{m}) = \int_{\mathbb{R}} w(v_1 u)^2 w(v_2 u)^2 w(u)^2 u^2 e(-Nu(m_1 v_1 + m_2 v_2 + m_3)) du$$

and the support for $w(u)$ lies in $[1, 2]$. When $|m_1 v_1 + m_2 v_2 + m_3| > 0$ we can integrate by parts in a similar manner to the proof of Lemma 29.1 and obtain

$$J(\mathbf{v}, \mathbf{m}) \ll_j N^{-j} |m_1 v_1 + m_2 v_2 + m_3|^{-j}.$$

and so

$$\begin{aligned} & \int_{\mathcal{V}(\mathbf{m})} Q(v_1) Q\left(\frac{v_2}{v_1}\right) Q\left(\frac{1}{v_2}\right) J(\mathbf{v}, \mathbf{m}) d\mathbf{v} \\ & \ll_j T^6 T^{-j\delta}. \end{aligned}$$

Hence, by (29.52)

$$\begin{aligned} & \sum_{\substack{\mathbf{g} \in \mathcal{G}^3 \\ g_q \neq g_r}} \hat{h}_{g_1-g_2}(m_1 N) \hat{h}_{g_2-g_3}(m_2 N) h_{g_3-g_1}(m_3 N) \\ &= \int_{\mathcal{V}(\mathbf{m})} Q(v_1) Q\left(\frac{v_2}{v_1}\right) Q\left(\frac{1}{v_2}\right) J(\mathbf{v}, \mathbf{m}) d\mathbf{v} + O_A(T^{-A}). \end{aligned}$$

Observing that $|Q(1/v_2)| = |Q(v_2)|$ and $J(\mathbf{v}, \mathbf{m}) \ll 1$ the main part of the theorem follows.

When $v_1, v_2 \in [1/2, 2]$ and $|m_1 v_1 + m_2 v_2 + m_3| \leq \frac{T^\delta}{N}$ we have $4|m_2| \geq |m_1 v_1 + m_2 v_2| \geq |m_3| - |m_1 v_1 + m_2 v_2 + m_3| \geq |m_3| - |m_2|$ by (29.2), on taking δ sufficiently small. Thus the integral will be 0 unless $|m_3| \leq 5|m_2|$. $\square$

We need to bound various integral means of $Q(v)$. To smooth over the cracks we introduce for an arbitrary positive number X

$$\tilde{Q}_X(u) = \left(\int_{[1/4, 4]} X \varpi(Xv - Xu) |Q(v)|^2 dv \right)^{1/2}. \quad (29.53) \quad \boxed{\text{E:Qtilde}}$$

T:T3Qtilde **Theorem 29.14.** *Suppose that A and δ are fixed positive numbers and $\mathcal{G}$ is a T^δ -separated subset of $[0, T]$. Then there are M_1, M with $0 < M_1 \leq M \leq T^{1+\delta}/N$ such that*

$$S_3 \ll \Delta + \frac{N^2 T^\delta \log^2 T}{M} \times \sum_{\substack{\frac{M_1}{2} < |m_1| \leq M_1 \\ \frac{M}{10} < |m_2| \leq |m_3| \leq M}} \int_{1/2}^2 |Q(u)| \tilde{Q}_X \left(\frac{m_1 u + m_3}{m_2 u} \right) \tilde{Q}_X \left(\frac{m_1 u + m_3}{m_2} \right) du.$$

where $\Delta \ll_A T^{-A}$ and $X = MNT^{-\delta}$.

Proof Theorem 29.13 gives

$$S_3 \ll \Delta + \sum_{\substack{\mathbf{m} \in (\mathbb{Z} \setminus \{0\})^3 \\ |m_1| \leq |m_2| \leq |m_3| \leq \frac{T^{1+\delta}}{N}}} N^3 \int_{\mathcal{V}(\mathbf{m})} |Q(v_1) Q\left(\frac{v_2}{v_1}\right) Q(v_2)| d\mathbf{v}$$

where

$$\mathcal{V}(\mathbf{m}) = \{\mathbf{v} \in [1/2, 2]^2 : |m_1 v_1 + m_2 v_2 + m_3| \leq T^\delta/N\}$$

and establishes that the integral here is 0 unless $|m_3| \leq 5|m_2|$. We divide the triple sum over m_1 and m_3 into dyadic subranges of the form

$$\begin{aligned} T^{1+\delta} N^{-1} 2^{-q} &< |m_1| \leq T^{1+\delta} N^{-1} 2^{1-q}, \\ T^{1+\delta} N^{-1} 2^{-r} &< |m_3| \leq T^{1+\delta} N^{-1} 2^{1-r} \end{aligned}$$

and observe that then

$$T^{1+\delta} N^{-1} 2^{-r} 5^{-1} < |m_2| \leq T^{1+\delta} N^{-1} 2^{1-r}.$$

Taking the choice of q and r for which the subsum is maximal we obtain for some pair M_1, M with $0 < M_1 \leq M \leq T^{1+\delta}/N$ that

$$S_3 \ll \Delta + (\log T)^2 \sum_{\substack{M_1/2 < |m_1| \leq M_1 \\ M/10 < |m_2| \leq |m_3| \leq M}} N^3 \int_{\mathcal{V}(\mathbf{m})} |Q(v_1) Q\left(\frac{v_2}{v_1}\right) Q(v_2)| d\mathbf{v}$$

Let $\mathcal{W}(\mathbf{m}; u) = \{v : 1/2 \leq v \leq 2, |m_1 u + m_2 v + m_3| \leq T^\delta/N\}$. Then

$$\int_{\mathcal{V}(\mathbf{m})} |Q(v_1)| \left| Q\left(\frac{v_2}{v_1}\right) \right| |Q(v_2)| d\mathbf{v} = \int_{1/2}^2 |Q(u)| \int_{\mathcal{W}(\mathbf{m}; u)} \left| Q\left(\frac{v}{u}\right) \right| |Q(v)| dv du$$

Hence, by the Cauchy-Schwarz inequality, a change of variable and the observation that $M < |m_2|$, the inner integral is

$$\leq \left(\int_{\mathcal{W}_1(\mathbf{m}; u)} |Q(v)|^2 dv \right)^{1/2} \left(\int_{\mathcal{W}_2(\mathbf{m}; u)} |Q(v)|^2 dv \right)^{1/2}$$

where $\mathcal{W}_1(\mathbf{m}; u) = \{v : 1/2 \leq v \leq 2, |v - (m_1 + m_3 u^{-1})/(-m_2)| \leq T^\delta/NM\}$ and $\mathcal{W}_2(\mathbf{m}; u) = \{v : 1/2 \leq v \leq 2, |v - (m_1 u + m_3)/(-m_2)| \leq T^\delta/NM\}$.

By (29.19) the function $\varpi(MNT^{-\delta}\alpha)$ majorises the interval

$$[-T^\delta(MN)^{-1}, T^\delta(MN)^{-1}].$$

Hence

$$\int_{\mathcal{W}_1(\mathbf{m}; u)} |Q(v)|^2 dv \leq \int_{1/4}^4 \varpi(MNT^{-\delta}(v - (m_1 + m_3 u^{-1})/(-m_2))) |Q(w)|^2 dv$$

and

$$\int_{\mathcal{W}_2(\mathbf{m}; u)} |Q(v)|^2 dv \leq \int_{1/4}^4 \varpi(MNT^{-\delta}(v - (m_1 u + m_3)/(-m_2))) |Q(v)|^2 dv$$

Thus, taking into account the symmetry about 0 of m_2 , by (29.53),

$$\begin{aligned} & \sum_{\substack{M_1/2 < |m_1| \leq M_1 \\ \frac{M}{10} < |m_2| \leq |m_3| \leq M}} \int_{\mathcal{V}(\mathbf{m})} |Q(v_1)| \left| Q\left(\frac{v_2}{v_1}\right) \right| |Q(v_2)| d\mathbf{v} \ll \frac{T^\delta}{MN} \times \\ & \sum_{\substack{M_1/2 < |m_1| \leq M_1 \\ \frac{M}{10} < |m_2| \leq |m_3| \leq M}} \int_{1/2}^2 |Q(u)| \tilde{Q}_X\left(\frac{m_1 u + m_3}{m_2 u}\right) \tilde{Q}_X\left(\frac{m_1 u + m_3}{m_2}\right) du. \end{aligned}$$

□

29.6 The Relationship with Additive Structure

S:AddStrut

For $\alpha \in \mathbb{R} \setminus \{0\}$, $\mathcal{G}$ a finite set of real numbers and $b_g \in \mathbb{C}$ for $g \in \mathcal{G}$ we define

$$P(\alpha) = \sum_{g \in \mathcal{G}} b_g \alpha^{ig}$$

L:L2Pmean

Lemma 29.15. *Suppose that H, K are real numbers with $H > 0$ and $K > 1$, and $\mathcal{G}$ is H -spaced. Then*

$$\int_{1/K}^K |P(\alpha)|^2 d\alpha \ll K(2 \log(K) + H^{-1}) \sum_{g \in \mathcal{G}} |b_g|^2.$$

Proof The integral is at most

$$K \int_{1/K}^K \alpha^{-1} |P(\alpha)|^2 d\alpha = K \sum_{\mathbf{g} \in \mathcal{G}^2} b_{g_1} \bar{b}_{g_2} \int_{1/K}^K \alpha^{ig_1 - ig_2 - 1} d\alpha.$$

The contribution from the terms with $g_1 = g_2$ is

$$K(2 \log K) \sum_{g \in \mathcal{G}} |b_g|^2$$

and that from the remaining terms is

$$K \sum_{\substack{\mathbf{g} \in \mathcal{G}^2 \\ g_1 \neq g_2}} b_{g_1} \bar{b}_{g_2} \frac{K^{ig_1 - ig_2} - K^{ig_2 - ig_1}}{ig_1 - ig_2}$$

and the lemma follows from Hilbert's inequality, Theorem G.14. $\square$

It is useful to have a version of this which does not necessarily depend on assumptions about the spacing of the g . First of all we recall that if $H > 0$ and

$$F_H(\alpha) = \max(0, H - |\alpha|), \quad (29.54) \quad \text{E:Fkernel}$$

then

$$\hat{F}_H(y) = \left(\frac{\sin(\pi Hy)}{\sin(\pi y)} \right)^2. \quad (29.55) \quad \text{E:Fhat}$$

L:L2Pmeana

Lemma 29.16. *Suppose that $g_0, H \in \mathbb{R}$ with $H > 0$ and define*

$$\varrho(g_0; H) = \varrho_{\mathcal{G}}(g_0; H) = \int_{\mathbb{R}} |P(e^{2\pi y})|^2 e(-yg_0) \hat{F}_H(y) dy.$$

Then

$$\varrho(g_0; H) = \sum_{\mathbf{g} \in \mathcal{G}^2} b_{g_1} \bar{b}_{g_2} \max(0, H - |g_1 - g_2 - g_0|) \quad (29.56) \quad \boxed{\text{E:L2Pa}}$$

and

$$|\varrho(g_0; H)| \leq \varrho(0; H). \quad (29.57) \quad \boxed{\text{E:L2Pb}}$$

Moreover for $k \in \mathbb{N}$ we have

$$\varrho(0; H) \leq 2k(2k+1) \sum_{\substack{\mathbf{g} \in \mathcal{G}^2 \\ |g_1 - g_2| \leq H/k}} |b_{g_1} b_{g_2}|. \quad (29.58) \quad \boxed{\text{E:L2Pc}}$$

Finally if $K > 1$ and $H = \frac{\pi}{\log K}$, then

$$\int_{1/K}^K |P(\alpha)|^2 d\alpha \ll KH^{-2} \varrho(0; H). \quad (29.59) \quad \boxed{\text{E:L2Pd}}$$

Proof The first part of the lemma follows at once by squaring out the $|P|^2$, inverting the order of summation and applying (29.55). For the second part we simply observe that

$$\left| \int_{\mathbb{R}} |P(e^{2\pi y})|^2 e(-yg_0) \hat{F}_H(y) dy \right| \leq \int_{\mathbb{R}} |P(e^{2\pi y})|^2 \hat{F}_H(y) dy.$$

For the third part we have

$$\varrho(0; H) \leq H \sum_{\substack{\mathbf{g} \in \mathcal{G}^2 \\ |g_1 - g_2| \leq H}} |b_{g_1} b_{g_2}|.$$

For each pair g_1, g_2 included in the sum there is a unique j with $|j| \leq k$ such that $jH/k - \frac{H}{2k} < g_1 - g_2 \leq jH/k + \frac{H}{2k}$. Thus the above is

$$\leq 2k \sum_{j=-k}^k \sum_{\mathbf{g} \in \mathcal{G}^2} |b_{g_1} b_{g_2}| \max(0, H/k - |g_1 - g_2 - jH/k|).$$

By (29.57) with b_g replaced by $|b_g|$ this is

$$\leq 2k(2k+1) \sum_{\mathbf{g} \in \mathcal{G}^2} |b_{g_1} b_{g_2}| \max(0, H/k - |g_1 - g_2|).$$

Finally, by the change of variable $\alpha = e^{2\pi y}$

$$\begin{aligned} \int_{1/K}^K |P(\alpha)|^2 d\alpha &= \int_{-\frac{1}{2H}}^{\frac{1}{2H}} |P(e^{2\pi y})|^2 2\pi e^{2\pi y} dy \\ &\ll KH^{-2} \int_{-\frac{1}{2H}}^{\frac{1}{2H}} |P(e^{2\pi y})|^2 \hat{F}_H(y) dy \end{aligned}$$

since $\hat{F}_H(y) \gg H^2$ when $|y| \leq \frac{1}{2H}$, and this is

$$\leq KH^{-2} \varrho(0; H).$$

as required. $\square$

As an immediate consequence we have a general bound.

L:L2kPmean

Lemma 29.17. *Suppose that $k \in \mathbb{N}$ and $K > 1$. Then*

$$\int_{1/K}^K |P(\alpha)|^{2k} d\alpha \ll K(\log(eK)) \sum_{\substack{\mathbf{g} \in \mathcal{G}^{2k} \\ |g_1 + \dots + g_k - g_{k+1} - \dots - g_{2k}| < 1}} |b_{g_1} \dots b_{g_{2k}}|.$$

Proof We have

$$P(\alpha)^k = \sum_{g \in \mathcal{G}_k} r(g) \alpha^{ig}$$

where $\mathcal{G}_k$ is the set of those g for which there exist $g_j \in \mathcal{G}$ such that $g_1 + \dots + g_k = g$ and $r(g)$ is the number of such choices for $g_1, \dots, g_k$ counted with weight $b_{g_1} \dots b_{g_k}$. Let $H = \frac{\pi}{\log K}$. Then by (29.59) our integral is

$$\ll KH^{-2} \sum_{\mathbf{g} \in \mathcal{G}_k^2} r(g_1) \bar{r}(g_2) \max(0, H - |g_1 - g_2 - g_0|).$$

Let $k = \lceil H \rceil$. Then, by (29.58), the above is

$$\ll K(\log(eK)) \sum_{\substack{\mathbf{g} \in \mathcal{G}_k^2 \\ |g_1 - g_2| < 1}} |r(g_1) r(g_2)|$$

and the lemma follows from the definition of $r(g)$ and the triangle inequality. $\square$

We now advert to Q , defined by (29.45).

L:L2kQmean

Lemma 29.18. *Suppose that $K > 1$. Then*

$$\int_{1/K}^K |Q(\alpha)|^{2k} d\alpha \ll K(\log(eK)) E_k(\mathcal{G}) \quad (29.60) \quad \text{E:Qfourth}$$

where

$$E_k(\mathcal{G}) = \text{card}\{\mathbf{g} \in \mathcal{G}^{2k} : |g_1 + \cdots g_k - g_{k+1} - \cdots - g_{2k}| < 1\}. \quad (29.61) \quad \boxed{\text{E:Ekdef}}$$

Suppose further that $\mathcal{G}$ is 1-spaced. Then

$$\int_{1/K}^K |Q(\alpha)|^2 d\alpha \ll K(\log(eK))R, \quad (29.62) \quad \boxed{\text{E:Qsquare}}$$

and

$$\int_{1/K}^K |Q(\alpha)|^4 d\alpha \ll K(\log(eK))E(\mathcal{G}) \quad (29.63) \quad \boxed{\text{E:Qfourtha}}$$

where

$$E(\mathcal{G}) = E_2(\mathcal{G}). \quad (29.64) \quad \boxed{\text{E:Edef}}$$

We also have

$$\max\left(\frac{R^4}{T}, R^2\right) \ll E(\mathcal{G}) \ll R^3. \quad (29.65) \quad \boxed{\text{E:Ebposs}}$$

We would expect that in most cases $E(\mathcal{G})$ lies between $\frac{R^4}{T}$ and R^2 .

Proof All but the lower bound in (29.65) are immediate from Lemma 29.17. To prove the lower bound observe first that in the notation of the proof of Lemma 29.17 we have

$$\begin{aligned} R^4 &= \sum_{-8T \leq k \leq 8T} \text{card}\{\mathbf{g} \in \mathcal{G}_2^2 : |g_1 - g_2 - k/2| < 1/2\} r(g_1) r(g_2) \\ &\ll \sum_{-8T \leq k \leq 8T} \sum_{\mathbf{g} \in \mathcal{G}_2^2} r(g_1) r(g_2) \max(0, 1 - |g_1 - g_2 - k/2|) \end{aligned}$$

so that by (29.57)

$$R^4 \ll T \sum_{\mathbf{g} \in \mathcal{G}_2^2} r(g_1) r(g_2) \max(0, 1 - |g_1 - g_2|) \ll TE(\mathcal{G}).$$

Also, the diagonal solutions alone contribute R^2 to $E(\mathcal{G})$. $\square$

Note that in (29.60) there is no spacing requirement on the set $\mathcal{G}$, but then it is possible that $E(\mathcal{G}) = R^4$! The upper bound (29.65) is not very helpful in our analysis here. However it is achieved when the elements of $\mathcal{G}$ are close to being in arithmetic progression. For example

consider $\mathcal{G} = \{t_r : t_r = rT/R, 0 \leq r \leq R-1\}$. Put $v = \exp\left(\frac{2\pi Ru}{T}\right)$.

$$\begin{aligned} \text{Then } \int_{[1/4, 4]} |Q(v)|^4 dv &= \frac{2\pi R}{T} \int_{|u| \leq \frac{T}{R} \log 4} \left| \sum_{r=0}^{R-1} e(ur) \right|^4 \exp\left(\frac{2\pi Ru}{T}\right) du \\ &\gg \int_0^1 \left| \sum_{r=0}^{R-1} e(ur) \right|^4 du = \int_0^1 \left| \sum_{|h| \leq R} (R - |h|) e(uh) \right|^2 du \\ &= \sum_{h=-R}^R (R - |h|)^2 = \frac{2R^3 + R}{3}. \end{aligned}$$

This may seem a problem but if $\mathcal{G}$ were like this, then

$$\sum_{g \in \mathcal{G}} |\mathfrak{D}(t)|^2 \ll (R + RN/T) \sum_{n=1}^N |b_n|^2$$

and it is plausible that the density hypothesis would follow.

The lower bound $R^4 T^{-1}$ in (29.65) is close to best possible when $\mathcal{G}$ is the set of squares or cubes in $[0, T]$. For thinner sequences other than ones close to arithmetic progressions one should expect that the lower bound R^2 is more typical behaviour.

Lemma 29.19 tells us that, at least in some regards, we can suppose that $\mathcal{G} \subset \mathbb{Z}$. There ought to be theorems in additive combinatorics that tell us in that case what happens when $E^*(\mathcal{G})$ (see Lemma 29.19 below) is large. The work in this area, such as Freiman's theorem, is concerned mostly with what happens when expressions such as $g + g'$ have excessive overlap with $\mathcal{G}$, and Freiman's theorem tells us that in that case the elements of $\mathcal{G}$ are in arithmetic progression. Is that what is happening here? Understanding this might be the key to the density hypothesis.

D:nrstint

Definition 29.2. Given a real number α we define $\llbracket \alpha \rrbracket$ to be the unique integer a with $\alpha - 1/2 < a \leq \alpha + 1/2$, and given a set of real numbers $\mathcal{G}$ define

$$\mathcal{G}^* = \{\llbracket g \rrbracket : g \in \mathcal{G}\}. \quad (29.66) \quad \text{E:defgstar}$$

L:EGEGstar

Lemma 29.19. Suppose that $\mathcal{G}$ is 1-separated. Then $\mathcal{G}^*$ is 1-separated and

$$E^*(\mathcal{G}^*) \ll E(\mathcal{G}) \ll E^*(\mathcal{G}^*)$$

where

$$E^*(\mathcal{G}^*) = \text{card}\{\mathbf{g} \in \mathcal{G}^{*4} : g_1^* + g_2^* = g_3^* + g_4^*\}.$$

Proof For $g_1, g_2 \in \mathcal{G}$ let $g_j^* = \llbracket g_j \rrbracket$, so that $g_j - \frac{1}{2} < g_j^* \leq g_j + \frac{1}{2}$. Hence if $|g_1 - g_2| \geq 1$, then $|g_1^* - g_2^*| > 0$ and since they are integers $|g_1^* - g_2^*| \geq 1$. Moreover if $g_1, g_2, g_3, g_4 \in \mathcal{G}$ and $|g_1 + g_2 - g_3 - g_4| \leq 1$, then $|g_1^* + g_2^* - g_3^* - g_4^*| \leq 2$ and so

$$\begin{aligned} E(\mathcal{G}) &\leq \int_0^1 \left| \sum_{g \in \mathcal{G}^*} e(\alpha g) \right|^4 \sum_{h=-2}^2 e(\alpha h) d\alpha \\ &\ll \int_0^1 \left| \sum_{g \in \mathcal{G}^*} e(\alpha g) \right|^4 d\alpha \end{aligned}$$

which gives $E(\mathcal{G}) \ll E^*(\mathcal{G}^*)$.

For the lower bound observe that if $g_1^* + g_2^* - g_3^* - g_4^* = 0$, then $|g_1 + g_2 - g_3 - g_4| < 2$, so that,

$$E^*(\mathcal{G}^*) \leq \sum_{\mathbf{g} \in \mathcal{G}^4} \max(4 - |g_1 + g_2 - g_3 - g_4|).$$

Thus we can use some variant of the proof of (29.58). $\square$

We also need a comcomitant bound for $\tilde{Q}$.

L:L4tildeQ **Lemma 29.20.** *Suppose that $K > 1$ and $X > 0$. Then*

$$\int_{1/K}^K \tilde{Q}_X(v)^4 \ll K(\log(eK))E(\mathcal{G}).$$

Proof By (29.53) we have

$$\begin{aligned} \int_{1/K}^K \tilde{Q}_X(v)^4 dv &= \\ \int_{1/K}^K \int_{1/K}^K \int_{1/K}^K X^2 \varpi(Xv - Xu) \varpi(Xw - Xu) |Q(v)Q(w)|^2 dw dv du \\ &= \int_{1/K}^K \int_{1/K}^K |Q(v)Q(w)|^2 \int_{X/K}^{XK} X \varpi(Xv - x) \varpi(Xw - x) dx dw dv \end{aligned}$$

By (29.19) ϖ has its support on $[-2, 2]$ and is uniformly bounded. Thus for a non-zero contribution we have $-4/X \leq v - u \leq 4/X$ and x is in

an interval of length at most 4. Thus the above is

$$\begin{aligned} &\ll \int_{1/K}^K \int_{1/K}^K (|Q(v)|^4 + |Q(w)|^4) X dw dv \\ &\ll \int_{1/K}^K |Q(v)|^4 dv \end{aligned}$$

and the lemma follows from (29.60) and (29.64). $\square$

We now take stock. A routine combination of Theorem 29.13 with the material of this section establishes the following theorem. We will later see that this can be improved significantly.

T:S3boundE **Theorem 29.21.** *Suppose that $\mathcal{G}$ is a T^δ -separated set in $[0, T]$. Then*

$$S_3 \ll T^{2+3\delta} R^{1/2} E(\mathcal{G})^{1/2}.$$

Proof By (29.62), (29.63) and Schwarz's inequality we have

$$\int_{[1/4, 4]} |Q(u)|^3 du \ll R^{1/2} E(\mathcal{G})^{1/2}.$$

Also

$$|Q(v_1)Q(v_2/v_1)Q(v_2)| \leq |Q(v_1)|^3 + |Q(v_2/v_1)|^3 + |Q(v_2)|^3.$$

Thus, by (29.47)

$$\int_{\mathcal{V}(\mathbf{m})} |Q(v_1)|^3 d\mathbf{v} \ll R^{1/2} E(\mathcal{G})^{1/2} \frac{T^\delta}{N|m_2|}.$$

Hence, summing over $\mathbf{m}$ we have

$$\begin{aligned} &\sum_{\substack{\mathbf{m} \in (\mathbb{Z} \setminus \{0\})^3 \\ |m_1| \leq |m_2| \leq |m_3| \leq T^{1+\delta} N^{-1}}} N^3 \int_{\mathcal{V}(\mathbf{m})} |Q(v_1)|^3 d\mathbf{v} \\ &\ll T^{2+2\delta} R^{1/2} E(\mathcal{G})^{1/2}. \end{aligned}$$

A similar argument holds for $|Q(v_2)|^3$. For the term $|Q(v_2/v_1)|^3$ we make the substitute $v_2 = wv_1$ and note that then $w \in [1/(2v_1), 2/v_1] \subseteq [1/4, 4]$

and the contribution becomes

$$N^3 \int_{\nu(\mathbf{m})} |Q(v_2/v_1)|^3 d\mathbf{v} =$$

$$N^3 \int_{[1/4, 4]} |Q(w)|^3 \sum_{\substack{\mathbf{m} \in (\mathbb{Z} \setminus \{0\})^3 \\ |m_1| \leq |m_2| \leq |m_3| \leq T^{1+\delta} N^{-1}}} \int_{\nu(\mathbf{m}) \cap [1/(2w), 2/w]} v dv dw$$

The condition $|v(m_1 + m_2 w) + m_3| \leq \frac{T^\delta}{N}$ together with $m_3 \neq 0$ implies that $|v(m_1 + m_2 w)| > \frac{1}{2}$ and so $|m_1 + m_2 w| > \frac{1}{16}$. Moreover the integral over v is

$$\ll \frac{T^\delta}{N|m_1 + m_2 w|}.$$

and again we can proceed as for $|Q(v_1)|^3$. Thus, by Theorem 29.13,

$$S_3 \ll T^{2+3\delta} R^{1/2} E(\mathcal{G})^{1/2}$$

as required. $\square$

29.7 Affine Transformations

To make further progress it is necessary obtain bounds for objects of the kind

$$J(f; M) = \sup_{\substack{M_1 \\ 0 < M_1 \leq M}} \int_{\mathbb{R}} \left(\sum_{\substack{M_1/2 < |m_1| \leq M_1 \\ M/10 < |m_2| \leq |m_3| \leq M}} f\left(\frac{m_1 u + m_3}{m_2}\right) \right)^2 du \quad (29.67) \quad \boxed{\text{E: def J}}$$

where f is smooth and compactly supported.

$\boxed{\text{T: Jbound}}$ **Theorem 29.22.** *Suppose that $f \in C^\infty(\mathbb{R})$ and is non-negative and supported on $[1/K, K]$ for some constant $K > 1$. Suppose also that*

$$|\hat{f}(y)| \ll_j T(T^{1+\delta}/|y|)^j \quad (y \neq 0) \quad (29.68) \quad \boxed{\text{E: hatfbound}}$$

for all $j \geq 0$. Then for every $\varepsilon > 0$ we have

$$J(f; M) \ll_{\delta, \varepsilon} T^\varepsilon M^6 \left(\int_{\mathbb{R}} f(u) du \right)^2 + T^\varepsilon M^4 \int_{\mathbb{R}} f(u)^2 du. \quad (29.69) \quad \boxed{\text{E: Jbound}}$$

Both terms on the right are necessary as the following two examples show.

A. Let $f(u) = \varepsilon^{-1}$ for $u \in [1, 1 + \varepsilon]$ and $f(u) = 0$ otherwise, and take $M = 1$. Then $J(f) \asymp \varepsilon^{-1}$, $\int_{\mathbb{R}} f(u) du = 1$, $\int_{\mathbb{R}} f(u)^2 du = \varepsilon^{-1}$.

B. Let $f(u) = 1$ for $u \in [1, 2]$ and 0 otherwise and suppose that M is large. Then $J(f) \asymp M^6$ and $\int_{\mathbb{R}} f(u) du = \int_{\mathbb{R}} f(u)^2 du = 1$.

Proof We suppose that ε is a small positive number. In the definition of J we have $M/10 < |m_3| \leq M$. Thus

$$J(f) \ll \sup_{0 < M_1 \leq M} \int_{\mathbb{R}} g(u)^2 du$$

where

$$g(u) = \sum_{\substack{M_1/2 < |m_1| \leq M_1 \\ M/10 < |m_2| \leq M}} \sum_m \varpi(m/M) f\left(\frac{m_1 u + m}{m_2}\right).$$

Hence, by Plancherel's identity

$$J(f; M) \ll \sup_{0 < M_1 \leq M} \int_{\mathbb{R}} |\hat{g}(v)|^2 dv$$

where

$$\begin{aligned} \hat{g}(v) &= \int_{\mathbb{R}} \sum_{\substack{M_1/2 < |m_1| \leq M_1 \\ M/10 < |m_2| \leq M}} \sum_m \varpi(m/M) f\left(\frac{m_1 u + m}{m_2}\right) e(-uv) du \\ &= \sum_{\substack{M_1/2 < |m_1| \leq M_1 \\ M/10 < |m_2| \leq M}} \left(\sum_m \varpi(m/M) e\left(\frac{mv}{m_1}\right) \int_{\mathbb{R}} f\left(\frac{m_1 w}{m_2}\right) e(-wv) dw \right). \end{aligned}$$

We apply the Poisson summation formula to the sum over m . Thus

$$\begin{aligned} \sum_m \varpi(m/M) e\left(\frac{mv}{m_1}\right) &= \sum_{\ell \in \mathbb{Z}} \int_{\mathbb{R}} \varpi\left(\frac{x}{M}\right) e\left(\frac{xv}{m_1} - \ell x\right) dx \\ &= \sum_{\ell \in \mathbb{Z}} M \int_{\mathbb{R}} \varpi(y) e\left(\frac{yMv}{m_1} - \ell yM\right) dy \\ &= \sum_{\ell \in \mathbb{Z}} M \hat{\varpi}\left(\frac{Mv}{m_1} - \ell M\right). \end{aligned}$$

Suppose that $|\frac{Mv}{m_1} - \ell M| > T^\varepsilon$. Then, by (29.23),

$$\hat{\varpi}\left(\frac{M_3 v}{m_1} - \ell M_3\right) \ll_{A, \varepsilon} T^{-2A}.$$

Thus the total contribution to

$$\sup_{0 < M_1 \leq M} \int_{\mathbb{R}} |\hat{g}(v)|^2 dv$$

from such terms will be $\ll_A T^{-A}$, which is certainly much smaller than the expressions on the right of (29.69) even when $A = 0$. Thus we can concentrate on

$$\sum_{\substack{M_1/2 < |m_1| \leq M_1 \\ M/10 < |m_2| \leq M}} \sum_{\substack{\ell \in \mathbb{Z} \\ |\frac{v}{m_1} - \ell| \leq \frac{T^\varepsilon}{M}}} M \hat{\omega} \left(\frac{Mv}{m_1} - \ell M \right) \int_{\mathbb{R}} f \left(\frac{m_1 w}{m_2} \right) e(-wv) dw$$

and this can be rearranged to give

$$\begin{aligned} & \sum_{\substack{\frac{M_1}{2} < |m_1| \leq M_1 \\ \frac{M}{10} < m_2 \leq M}} \sum_{\substack{\ell \in \mathbb{Z} \\ |v - m_1 \ell| \leq \frac{T^\varepsilon m_1}{M}}} M \hat{\omega} \left(\frac{Mv}{m_1} - \ell M \right) \frac{m_2}{m_1} \hat{f} \left(\frac{m_2 v}{m_1} \right) = \\ & \sum_{\substack{\frac{M_1}{2} < |m_1| \leq M_1 \\ |v - m_1 \ell| \leq \frac{T^\varepsilon m_1}{M}}} \sum_{\ell \in \mathbb{Z}} M \hat{\omega} \left(\frac{Mv}{m_1} - \ell M \right) \sum_{\frac{M}{10} < m_2 \leq M} \frac{m_2}{m_1} \hat{f} \left(\frac{m_2 v}{m_1} \right). \end{aligned}$$

and this is

$$\ll \sum_{M_1/2 < |m_1| \leq M_1} \sum_{\substack{\ell \in \mathbb{Z} \\ |v - m_1 \ell| \leq \frac{T^\varepsilon m_1}{M}}} M \left| \sum_{M/10 < m_2 \leq M} \frac{m_2}{m_1} \hat{f} \left(\frac{m_2 v}{m_1} \right) \right|.$$

Thus

$$\begin{aligned} & \int_{\mathbb{R}} |\hat{g}(v)|^2 dv \ll_A T^{-A} + \\ & \int_{\mathbb{R}} \left(\sum_{M_1/2 < |m_1| \leq M_1} \sum_{\substack{\ell \in \mathbb{Z} \\ |v - m_1 \ell| \leq \frac{T^\varepsilon m_1}{M}}} M \left| \sum_{M/10 < m_2 \leq M} \frac{m_2}{m_1} \hat{f} \left(\frac{m_2 v}{m_1} \right) \right| \right)^2 dv. \end{aligned}$$

Separating out the terms $\ell = 0$ this is

$$\begin{aligned} &\ll_A T^{-A} + \\ &\int_{-2M_1 T^\varepsilon/M}^{2M_1 T^\varepsilon/M} \left(\sum_{M_1/2 < |m_1| \leq M_1} M \left| \sum_{M/10 < m_2 \leq M} \frac{m_2}{m_1} \hat{f}\left(\frac{m_2 v}{m_1}\right) \right| \right)^2 dv + \\ &\int_{\mathbb{R}} \left(\sum_{M_1/2 < |m_1| \leq M_1} \sum_{\substack{\ell \in \mathbb{Z} \setminus \{0\} \\ |v - m_1 \ell| \leq \frac{T^\varepsilon m_1}{M}}} M \left| \sum_{M/10 < m_2 \leq M} \frac{m_2}{m_1} \hat{f}\left(\frac{m_2 v}{m_1}\right) \right| \right)^2 dv. \end{aligned}$$

Since f is non-negative we have

$$\left| \hat{f}\left(\frac{m_2 v}{m_1}\right) \right| \leq \int_{\mathbb{R}} f(u) du.$$

Hence the first integral above is

$$\ll M_1 T^\varepsilon M^5 \left(\int_{\mathbb{R}} f(u) du \right)^2 \ll T^\varepsilon M^6 \left(\int_{\mathbb{R}} f(u) du \right)^2.$$

By (29.68)

$$\hat{f}\left(\frac{m_2 v}{m_1}\right) \ll_j T \left(\frac{T^{1+\delta} M_1}{|v| M} \right)^j$$

and so the contribution when $|v| > T^{1+\delta+\varepsilon} M_1/M$ can be absorbed into the term T^{-A} . Thus it suffices to bound

$$\int_{-X}^X \left(\sum_{\frac{M_1}{2} < |m_1| \leq M_1} \sum_{\substack{\ell \in \mathbb{Z} \setminus \{0\} \\ |v - m_1 \ell| \leq \frac{T^\varepsilon M_1}{M}}} M \sum_{\frac{M}{10} < m_2 \leq M_2} \left| \frac{m_2}{m_1} \hat{f}\left(\frac{m_2 v}{m_1}\right) \right| \right)^2 dv$$

where $X = T^{1+\delta+\varepsilon}$. By Cauchy's inequality this is $\ll$

$$\begin{aligned} &\int_{-X}^X \left(\sum_{\frac{M_1}{2} < |m_1| \leq M_1} \sum_{\substack{\ell \in \mathbb{Z} \setminus \{0\} \\ |v - m_1 \ell| \leq \frac{T^\varepsilon M_1}{M}}} M^2 \sum_{\frac{M}{10} < m_2 \leq M} \frac{M}{M_1} \right) \times \\ &\sum_{M_1/2 < |m_1| \leq M_1} \sum_{\substack{\ell \in \mathbb{Z} \setminus \{0\} \\ |v - m_1 \ell| \leq \frac{T^\varepsilon m_1}{M}}} \sum_{M/10 < m_2 \leq M} \frac{m_2}{|m_1|} \left| \hat{f}\left(\frac{m_2 v}{m_1}\right) \right|^2 dv. \end{aligned}$$

In the first triple sum we now collect together the m_1 and ℓ for which for a given n we have $m_1 \ell = n$. Since $0 < |n| \ll |v| + \frac{T^\varepsilon M_1}{M} \ll T^{1+\delta+\varepsilon} M_1/M$,

the number of choices of m_1 and ℓ for a given n is $\ll T^\varepsilon$. Thus this triple sum is $\ll T^\varepsilon(1 + T^\varepsilon M_1/M)M^4 M_1^{-1}$ and so the integral is $\ll$

$$P \int_{-T^{1+\delta+\varepsilon} \frac{M_1}{M}}^{T^{1+\delta+\varepsilon} \frac{M_1}{M}} \sum_{M_1 < |m_1| \leq 2M_1} \sum_{\substack{\ell \in \mathbb{Z} \setminus \{0\} \\ |v - m_1 \ell| \leq \frac{T^\varepsilon m_1}{M}}} \sum_{\frac{M}{10} < m_2 \leq M} \frac{|m_2|}{|m_1|} \left| \hat{f}\left(\frac{m_2 v}{m_1}\right) \right|^2 dv$$

where

$$P = T^\varepsilon(M^4 M_1^{-1} + T^{2\varepsilon} M^3).$$

We move the integral inside to obtain

$$P \sum_{\frac{M_1}{2} < |m_1| \leq M_1} \sum_{\substack{\ell \in \mathbb{Z} \\ \ell \neq 0}} \sum_{\frac{M}{10} < |m_2| \leq M} \int_{I(m_1)} \frac{|m_2|}{|m_1|} \left| \hat{f}\left(\frac{m_2 v}{m_1}\right) \right|^2 dv$$

where

$$I(m) = \left\{ v : |v| \leq T^{1+\delta} M_1/M, |v - m\ell| \leq \frac{T^\delta m}{M} \right\}$$

We make the change of variables $m_2 v/m_1 = w$ to obtain

$$P \sum_{\frac{M_1}{2} < |m_1| \leq M_1} \sum_{\substack{\ell \in \mathbb{Z} \\ \ell \neq 0}} \sum_{\frac{M}{10} < |m_2| \leq M} \int_{I_1(m_2)} |\hat{f}(w)|^2 dw$$

where

$$I_1(m) = \left\{ w : |w| \leq T^{1+\delta} M_1 m_2 / (M m_1), |w - m\ell| \leq \frac{T^\delta m}{M} \right\},$$

and rearrange this to obtain the upper bound

$$P \int_{-2T^{1+\delta+\varepsilon}}^{2T^{1+\delta+\varepsilon}} \sum_{M/10 < |m_2| \leq M} \sum_{\substack{\ell \in \mathbb{Z} \setminus \{0\} \\ |w - m_2 \ell| \leq \frac{T^\varepsilon m_2}{M}}} \sum_{M_1/2 < |m_1| \leq M_1} |\hat{f}(w)|^2 dw$$

We collect together the terms with $m_2 \ell = n$ as we did previously with $m_1 \ell$. Thus the contribution from m_2 and ℓ is $\ll T^\varepsilon$ and so the triple sum here contributes $\ll T^{2\varepsilon} M_1$. Hence we obtain the bound

$$Q \int_{\mathbb{R}} |\hat{f}(w)|^2 dw$$

where

$$Q = P T^{2\varepsilon} M_1 \ll T^{4\varepsilon} M^4.$$

Hence, on choosing $\varepsilon < \delta/4$, by Plancherel's identity the above is

$$\ll T^{4\varepsilon} M^4 \int_{\mathbb{R}} f(u)^2 du.$$

□

29.8 An Improvement on Theorem 29.21

We can now state an improvement on Theorem 29.21 when $E(\mathcal{G})$ is between R^2 and R^3 . In comparing the two results it is clear that the second term on the right below is an improvement when N is small compared with T and the first term will be in those situations where we cannot show that $E(\mathcal{G}) \ll R^2$.

T:S3boundEI

Theorem 29.23. *Suppose that $\mathcal{G}$ is a T^δ -separated set contained in an interval of length T . Then*

$$S_3 \ll_\delta T^{2+2\delta+\varepsilon} R^{3/2} + T^{1+\delta+\varepsilon} N R^{1/2} E(\mathcal{G})^{1/2}.$$

Proof Theorem 29.14 states that for some M_1, M with $0 < M_1 \leq M \leq T^{1+\delta}/N$ we have

$$S_3 \ll \Delta + \frac{N^2 T^\delta \log^2 T}{M} \times \sum_{\substack{\frac{M_1}{2} < |m_1| \leq M_1 \\ \frac{M}{10} < |m_2| \leq |m_3| \leq M}} \int_{\frac{1}{2}}^2 |Q(u)| \tilde{Q}_X \left(\frac{m_1 u + m_3}{m_2 u} \right) \tilde{Q}_X \left(\frac{m_1 u + m_3}{m_2} \right) du.$$

where $\Delta \ll_A T^{-A}$ and $X = MNT^{-\delta}$. By the Cauchy-Schwarz inequality

$$\begin{aligned} & \sum_{\mathbf{m}}^* \tilde{Q}_X \left(\frac{m_1 v + m_3}{vm_2} \right) \tilde{Q}_X \left(\frac{m_1 v + m_3}{m_2} \right) \\ & \leq \left(\sum_{\mathbf{m}}^* \tilde{Q}_X \left(\frac{m_1 v + m_3}{vm_2} \right)^2 \right)^{1/2} \left(\sum_{\mathbf{m}}^* \tilde{Q}_X \left(\frac{m_1 v + m_3}{m_2} \right)^2 \right)^{1/2} \end{aligned}$$

where $\sum^*$ denotes summation over m_1, m_2, m_3 with $M_1/2 < |m_1| \leq M_1, M/10 < |m_2| \leq |m_3| \leq M$. Hence by Hölder's inequality

$$S_3 \ll \Delta + N^2 M^{-1} S_{3,1}^{1/2} S_{3,3}^{1/4} S_{3,4}^{1/4}$$

where

$$S_{3,1} = \int_{v \in [1/2, 2]} |Q(v)|^2 dv, \quad (29.70) \quad \boxed{\text{E:S31}}$$

$$S_{3,3} = \int_{v \in [1/2, 2]} \left(\sum_{\mathbf{m}}^* \tilde{Q}_X \left(\frac{m_1 v + m_3}{v m_2} \right)^2 \right)^2 dv,$$

and

$$S_{3,4} = \int_{v \in [1/2, 2]} \left(\sum_{\mathbf{m}}^* \tilde{Q}_X \left(\frac{m_1 v + m_3}{m_2} \right)^2 \right)^2 dv. \quad (29.71) \quad \boxed{\text{E:S34}}$$

By the change of variable $v \rightarrow 1/v$

$$S_{3,3} = \int_{v \in [1/2, 2]} \left(\sum_{\mathbf{m}}^* \tilde{Q}_X \left(\frac{m_1 + m_3 v}{m_2} \right)^2 \right)^2 dv. \quad (29.72) \quad \boxed{\text{E:S33}}$$

By (29.62)

$$S_{3,1} \ll R.$$

Thus

$$S_3 \ll \Delta + N^2 M^{-1} R^{1/2} S_{3,3}^{1/4} S_{3,4}^{1/4} \quad (29.73) \quad \boxed{\text{E:S3new}}$$

Let

$$f(u) = \tilde{Q}_X(u)^2. \quad (29.74) \quad \boxed{\text{E:ftildeQ}}$$

By (29.53)

$$\tilde{Q}_X(u)^2 = \int_{[1/4, 4]} MN \varpi(MN(u-w)) |Q(w)|^2 dw.$$

Hence

$$\begin{aligned} \hat{f}(y) &= \int_{\mathbb{R}} \tilde{Q}_X(u)^2 e(-uy) du \\ &= \int_{\mathbb{R}} \int_{[1/4, 4]} MN \varpi(MN(u-w)) |Q(w)|^2 dw e(-u\xi) du \\ &= \int_{[1/4, 4]} |Q(w)|^2 \int_{\mathbb{R}} MN \varpi(MN(u-w)) e(-uy) du dw. \end{aligned}$$

By the change of variable $v = MN(u-w)$ this becomes

$$\int_{[1/4, 4]} |Q(w)|^2 e(-vy) dv \hat{\varpi}(y/MN).$$

By (29.23)

$$\hat{\varpi}(y/MN) \ll_j (MN/|y|)^j \quad (y \neq 0)$$

and by (29.62)

$$\int_{[1/4,4]} |Q(w)|^2 e(-vy) dv \ll R.$$

Hence

$$\hat{f}(y) \ll_j Q(MN/|y|)^j \quad (y \neq 0).$$

Thus the hypothesis of Lemma 29.22 is satisfied by f , and we can apply it to $S_{3,3}$ and $S_{3,4}$ with $K = 2$.

By (29.53) and 29.74

$$\int_{\mathbb{R}} f(u) du = \int_{[1/4,4]} |Q(w)|^2 \int_{\mathbb{R}} MN \varpi(MN(u-w)) du dw$$

and the inner integral is

$$\int_{\mathbb{R}} \varpi(v) dv \ll 1$$

so that, by (29.62),

$$\int_{\mathbb{R}} f(u) du \ll R.$$

We also have

$$\begin{aligned} \int_{\mathbb{R}} f(u)^2 du &= \int_{[1/4,4]^2} |Q(v_1)Q(v_2)|^2 \times \\ &\quad \int_{\mathbb{R}} M^2 N^2 \tilde{\psi}(MN(u-v_1)) \tilde{\psi}(MN(u-v_2)) du |Q(v_1)Q(v_2)|^2 dv. \end{aligned}$$

We write $|Q(v_1)Q(v_2)|^2 \leq |Q(v_1)|^4 + |Q(v_2)|^4$. Then

$$\begin{aligned} \int_{\mathbb{R}} f(u)^2 du &\ll \int_{[1/4,4]} |Q(v_1)|^4 \times \\ &\quad \int_{\mathbb{R}} MN \varpi(MN(u-v_1)) \int_{\mathbb{R}} MN \varpi(MN(u-v_2)) dv_2 du dv_1 \end{aligned}$$

The innermost integral is

$$\int_{\mathbb{R}} \varpi(w) dw \ll 1.$$

We can then apply the same bound to the integral over u . Finally we can then apply Lemma 29.20 to obtain

$$\int_{\mathbb{R}} f(u)^2 du \ll E(\mathcal{G}).$$

Thus, (29.72), (29.71) and Lemma 29.22

$$\max(S_{3,3}, S_{3,4}) \ll T^\varepsilon M^6 R^2 + M^4 T^\varepsilon E(\mathcal{G}).$$

Hence, by (29.73)

$$S_3 \ll \Delta + T^\varepsilon N^2 M^2 R^{3/2} + T^\varepsilon N^2 M R^{1/2} E(\mathcal{G})^{1/2}.$$

Recall that $MN \leq T^{1+\delta}$. This completes the proof of 29.23. $\square$

29.9 Dirichlet Polynomials and Additive Structure

S:5TDIII

To make further progress we need to make some assumptions about the size of $\mathfrak{D}(g)$, and this entails restricting further both the set $\mathcal{G}$ and the sequence a_n . So before proceeding we summarize what we have established in the general cases.

Given any 1-spaced subset $\mathcal{G}$ of an interval of length T and any complex sequence $\{a_m\}$ we have shown that for some small $\delta > 0$ we have

$$\sum_{g \in \mathcal{G}} |\mathfrak{D}(g)|^2 \leq \lambda \sum_{n=1}^N |a_n|^2$$

where λ satisfies

$$\begin{aligned} \lambda \ll N + T^\delta \left(N^{\frac{2}{3}} R^{\frac{2}{3}} + N^{\frac{1}{3}} T^{\frac{1}{3}} R^{\frac{7}{12}} + N^{\frac{2}{3}} R^{\frac{2}{3}} (T^{\frac{1}{2}} R^{-\frac{3}{4}})^{\frac{1}{4}} \right. \\ \left. + T^{\frac{2}{3}} R^{\frac{1}{2}} + N^{\frac{1}{3}} T^{\frac{1}{3}} R^{\frac{1}{6}} E(\mathcal{G})^{\frac{1}{6}} \right) \end{aligned} \quad (29.75) \quad \text{eq:lambdab1}$$

To make further progress we need to ascertain some further the interrelationship between Dirichlet polynomials (29.8),

$$\mathfrak{D}(t) = \sum_n a_n w(n/N) n^{it}$$

and $E(\mathcal{G})$ when either is large. There are a number of stages in elucidating this relationship. The first is the well known observation that Dirichlet polynomials vary sufficiently slowly that they can be replaced by a smooth short average.

L:smoothingD

Lemma 29.24. *Suppose that $\varepsilon > 0$, $A > 0$. Then*

$$\mathfrak{D}(g) \ll \int_{g-T^\varepsilon}^{g+T^\varepsilon} |\mathfrak{D}(u)| du + \Delta \quad (29.76) \quad \text{E:Dave}$$

Suppose further that $\mathcal{G}$ is a finite subset of $[0, T]$ of cardinality $\ll T$ and that K is a constant with $K > 1$. Then for $1/K \leq v \leq K$ we have

$$Q(v) \ll T \int_{v-T^{\varepsilon-1}}^{v+T^{\varepsilon-1}} |Q(u)| du + \Delta \quad (29.77) \quad \text{E:Qave}$$

where $\Delta \ll_{A,\varepsilon} T^{-A}$.

Proof Let $\nu(x) = w(e^{2\pi x})$. Then

$$w(u) = \nu\left(\frac{\log u}{2\pi}\right) = \int_{\mathbb{R}} \hat{\nu}(y) e\left(-\frac{y \log u}{2\pi}\right) dy.$$

Hence

$$\begin{aligned} \mathfrak{D}(g) &= \sum_n a_n n^{ig} \int_{\mathcal{R}} \hat{\nu}(y) e\left(-\frac{y \log(n/N)}{2\pi}\right) dy \\ &= \int_{\mathbb{R}} \hat{\nu}(y) N^{iy} \mathfrak{D}(g-y) dy. \end{aligned}$$

By (29.24) $\hat{\nu}(y) \ll_j |y|^{-j}$. Hence, by (29.2),

$$\int_{|y| > T^\varepsilon} \hat{\nu}(y) N^{iy} \mathfrak{D}(g-y) dy \ll_j T^{\kappa_2 + \kappa_3} \int_{T^\varepsilon} y^{-j} dy \ll_{A,\varepsilon} T^{-A}$$

on choosing j suitably.

The second part of the lemma follows a similar pattern. Recall by (29.19) that $\varpi(\alpha) = 1$ when $-1 \leq \alpha \leq 1$. Now let

$$\nu(\alpha) = \varpi(\alpha/T)$$

Hence, by (29.45)

$$Q(\alpha) = \sum_{g \in \mathcal{G}} \alpha^{ig} \nu(g).$$

Thus

$$\begin{aligned} Q(e^{2\pi u}) &= \sum_{g \in \mathcal{G}} e(ug) \nu(g) \\ &= \int_{\mathbb{R}} \hat{\nu}(y) Q(e^{2\pi i u + 2\pi y}) dy. \end{aligned}$$

We also have $\hat{\nu}(y) = T\hat{\omega}(Ty)$ and, so by (29.23),

$$\hat{\mu}(y) \ll_j T(T|y|)^{-j}.$$

Hence

$$\int_{|y| \geq T^{\varepsilon/2-1}} \hat{\nu}(y) Q(e^{2\pi i u + 2\pi y}) dy \ll_A T^{-A}.$$

Thus, writing $v = e^{2\pi u}$ we have

$$Q(v) \ll \Delta + T \int_{|y| \geq T^{\varepsilon/2-1}} |\hat{\omega}(Ty)| |Q(v e^{2\pi y})| dy.$$

The second part of the lemma follows on making the change of variable $y = \frac{\log \frac{u}{v}}{2\pi}$. $\square$

We make use of this to connect E and Q . At this point we make the observation that hitherto we have been measuring $\mathfrak{D}(g)$ against the l^2 norm

$$\left(\sum_{n=N+1}^{2N} |a_n|^2 \right)^{1/2}.$$

However now in order to expedite the argument we need to measure $\mathfrak{D}(g)$ against the l^∞ norm, which for brevity we write as

$$\mathfrak{A} = \max_{n \in [N+1, 2N]} |a_n|. \quad (29.78) \quad \boxed{\text{eq: infnorm}}$$

L:EandQ **Lemma 29.25.** *Let V, ε be positive real numbers and suppose that $\mathcal{G}$ is a 1-separated subset of $[0, T]$ of cardinality R such that $|D(g)| \geq V\mathfrak{A}$ for $g \in \mathcal{G}$. Then*

$$E(\mathcal{G}) \ll T^\varepsilon V^{-2} \sum_{\mathbf{n} \in [N+1, 2N]^2} |Q(n_1/n_2)|^3.$$

Proof We have, by Lemma 29.24,

$$\begin{aligned} E(\mathcal{G}) &\leq \sum_{\substack{\mathbf{g} \in \mathcal{G}^4 \\ |g_1 + g_2 - g_3 - g_4| \leq 1}} \mathfrak{A}^{-2} V^{-2} |\mathfrak{D}(g_4)|^2 \\ &\ll_A T^{-A} + \mathfrak{A}^{-2} V^{-2} \sum_{\substack{\mathbf{g} \in \mathcal{G}^4 \\ |g_1 + g_2 - g_3 - g_4| \leq 1}} \int_{|u - g_4| \leq T^\varepsilon} |\mathfrak{D}(u)|^2 du \end{aligned}$$

Since the lemma is trivial when $E(\mathcal{G}) = 0$ we may suppose $E(\mathcal{G}) \geq 1$

and then suppress the T^{-A} . We make the change of variable $u = v + g_1 + g_2 - g_3$, so that

$$\begin{aligned} v &= (u - g_4) - (g_1 + g_2 - g_3 - g_4) \\ &\leq T^\varepsilon + 1 \\ &\leq 2T^\varepsilon. \end{aligned}$$

Also, since $\mathcal{G}$ is 1-separated, given g_1, g_2, g_3 there are $\ll 1$ choices for g_4 . Thus $E(\mathcal{G})$

$$\begin{aligned} &\ll \mathfrak{A}^{-2} V^{-2} \sum_{\mathbf{g} \in \mathcal{G}^3} \int_{|v| \leq 2T^\varepsilon} |\mathfrak{D}(v + g_1 + g_2 - g_3)|^2 dv \\ &= \mathfrak{A}^{-2} V^{-2} \sum_{\mathbf{n} \in [N+1, 2N]^2} a_{n_1} \overline{a_{n_2}} \int_{|v| \leq 2T^\varepsilon} \sum_{\mathbf{g} \in \mathcal{G}^3} (n_1/n_2)^{iv + ig_1 - ig_2 - ig_3} dv \\ &= \mathfrak{A}^{-2} V^{-2} \sum_{\mathbf{n} \in [N+1, 2N]^2} a_{n_1} \overline{a_{n_2}} \int_{|v| \leq 2T^\varepsilon} (n_1/n_2)^{iv} Q(n_1/n_2) Q(n_2/n_1)^2 dv \\ &\ll T^\varepsilon V^{-2} \sum_{\mathbf{n} \in [N+1, 2N]^2} |Q(n_1/n_2)|^3. \end{aligned}$$

□

A consequence of the previous lemma is Lemma 29.27 below. It is first convenient to restate essentially Lemma 29.10.

L:Qult **Lemma 29.26.** *For an $N \geq 1$ we have*

$$\sum_{\mathbf{n} \in [N+1, 2N]^2} |Q(n_1/n_2)|^2 \ll T^\varepsilon (N^2 R + T^{1/2} N R^{5/4} + N R^2)$$

Proof By squaring out and interchanging the order we have (29.34),

$$\sum_{\mathbf{n} \in [N+1, 2N]^2} |Q(n_1/n_2)|^2 \ll N S(N).$$

The lemma then follows from Lemma 29.10. □

L:EandR **Lemma 29.27.** *Let V, ε be positive real numbers and suppose that $\mathcal{G}$ is a 1-separated subset of $[0, T]$ of cardinality R such that $|D(g)| \geq \mathfrak{A}V$ for $g \in \mathcal{G}$. Then*

$$E(\mathcal{G}) \ll T^\varepsilon (R^2 N^2 V^{-2} + R^3 N V^{-2} + R^{9/4} T^{1/2} N V^{-2}).$$

In particular, if $N \gg T^{2/3}$, then

$$E(\mathcal{G}) \ll T^\varepsilon (R^2 N^2 V^{-2} + R^3 N V^{-2}).$$

Proof We have

$$N^{-1} \sum_{\mathbf{n} \in [N+1, 2N]^2} |Q(n_1/n_2)|^2 \ll \sum_{\mathbf{n} \in [N+1, 2N]^2} n_1^{-1/2} n_2^{-1/2} |Q(n_1/n_2)|^2.$$

The crude bound $|Q(n_1/n_2)| \leq R$ gives the first part of the lemma. The second part follows on observing that

$$R^{9/4} T^{1/2} N = (T/N^{3/2})^{1/2} (R^2 N^2)^{3/4} (R^3 N)^{1/4}.$$

□

The above uses the crude estimate $|Q(n_1/n_2)| \leq R$. To avoid this one needs to invoke the fourth moment.

L:EandR4 **Lemma 29.28.** *For any $N \geq 1$ we have*

$$\sum_{\mathbf{n} \in [N, 2N]^2} |Q(n_1/n_2)|^4 \ll T^\varepsilon (N^2 E(\mathcal{G}) + NT^{1/2} E(\mathcal{G})^{3/4} R + NR^4).$$

Proof By multiply out and inverting the order we have

$$\sum_{\mathbf{g} \in \mathcal{G}^4} \left| \sum_{n \in [N+1, 2N]} n^{ig_1 + ig_2 - ig_3 - ig_4} \right|^2.$$

By Lemma 29.24, Cauchy's inequality and Definition 29.2, this is

$$\begin{aligned} & \sum_{\mathbf{n} \in [N, 2N]^2} |Q(n_1/n_2)|^4 \\ & \ll \Delta + T^\varepsilon \sum_{\mathbf{g} \in \mathcal{G}^4} \int_{g_1 + g_2 - g_3 - g_4 - T^\varepsilon}^{g_1 + g_2 - g_3 - g_4 + T^\varepsilon} \left| \sum_{n \in [N+1, 2N]} n^{iu} \right|^2 du \\ & \leq \Delta + T^\varepsilon \sum_{\mathbf{g} \in \mathcal{G}^4} \int_{g_1^* + g_2^* - g_3^* - g_4^* - T^\varepsilon - 2}^{g_1^* + g_2^* - g_3^* - g_4^* + T^\varepsilon + 2} \left| \sum_{n \in [N+1, 2N]} n^{iu} \right|^2 du \\ & = \Delta + T^\varepsilon \sum_{\mathbf{g} \in \mathcal{G}^4} \int_{-T^\varepsilon - 2}^{T^\varepsilon + 2} \left| \sum_{n \in [N+1, 2N]} n^{iv + ig_1^* + ig_2^* - ig_3^* - ig_4^*} \right|^2 dv \\ & \leq \Delta + T^\varepsilon \int_{3T^\varepsilon}^{3T^\varepsilon} \sum_{\mathbf{g} \in \mathcal{G}^4} \left| \sum_{n \in [N+1, 2N]} n^{iv + ig_1^* + ig_2^* - ig_3^* - ig_4^*} \right|^2 dv \end{aligned}$$

For $m \in \mathbb{Z}$ let $r(m)$ denote the number of pairs $\mathbf{g} \in \mathcal{G}^2$ such that $g_1^* - g_2^* = m$ and then let $\mathcal{M}_j$ denote the set of $m \in \mathbb{Z}$ such that $2^j \leq r(m) \leq 2^{j+1}$. The set $\mathcal{M}_j$ will be non-empty if and only if there are m

such that $2^j \leq r(m) \leq 2^{j+1}$, and every m for which $r(m) \neq 0$ will be in some $\mathcal{M}_j$ with $j \geq 0$. Thus

$$\begin{aligned} \sum_j 2^j \text{card}(\mathcal{M}_j) &\leq \sum_m r(m) \leq R^2 \text{ and} \\ \sum_j 2^{2j} \text{card}(\mathcal{M}_j) &\leq \sum_m r(m)^2 \leq E(\mathcal{G}). \end{aligned} \quad (29.79) \quad \boxed{\text{E:Mjbounds}}$$

We also have

$$\begin{aligned} T^\varepsilon \int_{-3T^\varepsilon}^{3T^\varepsilon} \sum_{\mathbf{g} \in \mathcal{G}^4} \left| \sum_{n \in [N+1, 2N]} n^{iv+ig_1^*+ig_2^*-ig_3^*-ig_4^*} \right|^2 dv = \\ T^\varepsilon \int_{-3T^\varepsilon}^{3T^\varepsilon} \sum_{\mathbf{m}} r(m_1)r(m_2) \left| \sum_{n \in [N+1, 2N]} n^{iv+im_1-im_2} \right|^2 dv \leq \\ T^\varepsilon \int_{-3T^\varepsilon}^{3T^\varepsilon} \sum_{\mathbf{j}} \sum_{\mathbf{m} \in \mathcal{M}_{j_1} \times \mathcal{M}_{j_2}} r(m_1)r(m_2) \left| \sum_{n \in [N+1, 2N]} n^{iv+im_1-im_2} \right|^2 dv \end{aligned}$$

Expanding the square and interchanging the order of summation we obtain

$$T^\varepsilon \int_{-3T^\varepsilon}^{3T^\varepsilon} \sum_{\mathbf{n} \in [N+1, 2N]^2} (n_1/n_2)^{iv} \sum_{\mathbf{j}} F_{j_1}(n_1/n_2) \bar{F}_{j_2}(n_1/n_2) dv$$

where

$$F_j(\alpha) = \sum_{m \in \mathcal{M}_j} r(m) \alpha^{im}.$$

By Cauchy's inequality and (29.34) with $\mathcal{G} = \mathcal{M}_j$ this is

$$\begin{aligned} &\ll (\log T) \sum_{\mathbf{j}} T^{2\varepsilon} N \sum_{\mathbf{n} \in [N+1, 2N]^2} n_1^{-1/2} n_2^{-1/2} |F_{j_1}(n_1/n_2)|^2 \\ &= (\log T) \sum_{\mathbf{j}} T^{2\varepsilon} N \sum_{\mathbf{m} \in \mathcal{M}_j^2} r(m_1)r(m_2) \left| \sum_{n \in [N+1, 2N]} n^{im_1-im_2-\frac{1}{2}} \right|^2 \\ &\ll (\log T) \sum_{\mathbf{j}} 2^{2j} T^{2\varepsilon} N S(N). \end{aligned}$$

By Lemma 29.11 with $\mathcal{G} = \mathcal{M}_j$ and $k = 1$ this is

$$\ll T^{3\varepsilon} \sum_j 2^{2j} (N^2 \text{card}(\mathcal{M}_j) + NT^{1/2} \text{card}(\mathcal{M}_j)^{5/4} + N \text{card}(\mathcal{M}_j)^2).$$

Now, by (29.79),

$$\begin{aligned} & \sum_j 2^{2j} \text{card}(\mathcal{M}_j)^{5/4} \\ & \leq \left(\sum_j 2^{2j} \text{card}(\mathcal{M}_j)^{3/4} \right) \left(\sum_j 2^{2j} \text{card}(\mathcal{M}_j) \right)^{1/2} \\ & \ll E(\mathcal{G})^{5/4} R \end{aligned}$$

so that the sum in question is

$$T^\varepsilon (N^2 E(\mathcal{G}) + NT^{1/2} E(\mathcal{G})^{3/4} R + NR^4)$$

as required. $\square$

There is already a well established technology for handling expressions of the form

$$\sum_{q \leq Q} \sum_{\substack{a=1 \\ (a,q)=1}} F(a/q)$$

and this gives us an additional way of attacking the expressions considered here.

L:smallgcd **Lemma 29.29.** *We have*

$$\sum_{\substack{\mathbf{n} \in [N+1, 2N]^2 \\ \gcd(n_1, n_2) \leq K}} |Q(n_1/n_2)|^3 \ll T^\varepsilon (KT + N^2) R^{1/2} E(\mathcal{G})^{1/2}$$

Proof By (29.77) and Hölder's inequality we have

$$Q(v)^3 \ll T^{1+2\varepsilon} \int_{v-T^{\varepsilon-1}}^{v+T^{\varepsilon-1}} |Q(u)|^3 du + T^{-2}.$$

Given $X \geq 1$, when $X \leq a, a'q, q' \leq 2X$ and $\frac{a}{q} \neq \frac{a'}{q'}$ we have

$$\left| \frac{a}{q} - \frac{a'}{q'} \right| = \frac{|aq' - a'q|}{qq'} \geq X^{-2}.$$

Thus an interval of length ℓ can contain at most $1 + \ell X^2$ of the fractions

a/q . Hence

$$\begin{aligned} & \sum_{X \leq a \leq 2X} \sum_{\substack{X \leq q \leq 2X \\ (q,a)=1}} |Q(a/q)|^3 \\ & \ll T^{1+2\varepsilon} \int_{\frac{1}{2}-T^{\varepsilon-1}}^{2+T^{\varepsilon-1}} (1 + X^2 T^{\varepsilon-1}) |Q(u)|^3 du + X^2 T^{-2} \\ & \ll T^{3\varepsilon} (T + X^2) \int_{\frac{1}{2}-T^{\varepsilon-1}}^{2+T^{\varepsilon-1}} |Q(u)|^3 du + X^2 T^{-2}. \end{aligned}$$

Now in the sum in the lemma we sort the terms according to $d = (n_1, n_2)$ and put $n_1 = ad$, $n_2 = qd$, $X = N/d$ so that for given d we have the upper bound

$$T^{3\varepsilon} (T + N^2 d^{-2}) \int_{\frac{1}{2}-T^{\varepsilon-1}}^{2+T^{\varepsilon-1}} |Q(u)|^3 du + N^2 d^{-2} T^{-2}.$$

summing over the $d \leq K$ and combining the Cauchy-Schwarz inequality with (29.64) and (29.65) gives the desired conclusion, since the result is trivial when $\mathcal{G}$ is empty. $\square$

We also need a bound for the contribution from the large values (n_1, n_2) .

L:largegcd **Lemma 29.30.** *Suppose that $N \geq T^{3/4}$. Then*

$$\begin{aligned} & \sum_{\substack{\mathbf{n} \in [N, 2N]^2 \\ \gcd(n_1, n_2) \geq N^2/T}} |Q(n_1/n_2)|^3 \ll \\ & T^\varepsilon (NR^3 + NT^{1/4} R^{21/8} + N^2 E(\mathcal{G})^{1/2} R^{1/2}). \end{aligned}$$

Note that the proof actually gives $NE(\mathcal{G})^{3/8} T^{1/2} R^{9/8}$ in place of the last term. However the last term here also occurs in Lemma 29.29, and there it seems hard to improve the term. This is not surprising since one expects that our sum is dominated by the small greatest common divisors, and indeed by those terms with $(n_1, n_2) = 1$.

Proof By Cauchy's inequality

$$\begin{aligned} & \sum_{X < q \leq 2X} \sum_{\substack{X < a \leq 2X \\ (q,a)=1}} |Q(a/q)|^3 \\ & \leq \left(\sum_{X < a, q \leq 2X} |Q(a/q)|^2 \right)^{1/2} \left(\sum_{X < a, q \leq 2X} |Q(a/q)|^4 \right)^{1/2} \end{aligned}$$

By Lemmas 29.26 and 29.28 this is

$$\ll T^\varepsilon (X^2 R + T^{\frac{1}{2}} X R^{\frac{5}{4}} + X R^2)^{\frac{1}{2}} (X^2 E(\mathcal{G}) + X T^{\frac{1}{2}} E(\mathcal{G})^{\frac{3}{4}} R + X R^4)^{\frac{1}{2}}$$

We also have

$$\begin{aligned} & \sum_{\substack{\mathbf{n} \in [N, 2N]^2 \\ \gcd(n_1, n_2) \geq N^2/T}} |Q(n_1/n_2)|^3 \\ & \leq \sum_{N^2/T < d \leq N} \sum_{N/d < q \leq 2N/d} \sum_{\substack{N/d < a \leq 2N/d \\ (q,a)=1}} |Q(a/q)|^3. \end{aligned}$$

We take $X = N/d$ and perform the summation. Thus

$$\begin{aligned} & \sum_{\substack{\mathbf{n} \in [N, 2N]^2 \\ \gcd(n_1, n_2) \geq N^2/T}} |Q(n_1/n_2)|^3 \leq \\ & T^{2\varepsilon} (TR + T^{1/2} NR^{5/4} + NR^2)^{1/2} (TE(\mathcal{G}) + NT^{1/2} E(\mathcal{G})^{3/4} R + NR^4)^{1/2}. \end{aligned}$$

The condition $N \geq T^{3/4}$ ensures that $TR \leq T^{1/2} N^{2/3} R \leq T^{1/2} NR^{5/4}$ and in addition the upper bound in (29.65) ensures that $TE(\mathcal{G}) \ll N^{2/3} T^{1/2} E(\mathcal{G})^{3/4} R^{3/4}$. Thus

$$\begin{aligned} & \sum_{\substack{\mathbf{n} \in [N, 2N]^2 \\ \gcd(n_1, n_2) \geq N^2/T}} |Q(n_1/n_2)|^3 \\ & \ll T^{2\varepsilon} N (T^{1/2} R^{5/4} + R^2)^{1/2} (T^{1/2} E(\mathcal{G})^{3/4} R + R^4)^{1/2} \\ & \ll T^{2\varepsilon} N (NR^3 + NT^{1/4} R^{21/8} + NE(\mathcal{G})^{3/8} T^{1/2} R^{9/8} \\ & \quad + NE(\mathcal{G})^{3/8} T^{1/4} R^{3/2}). \end{aligned}$$

If $R \leq T^{2/3}$, then $NE(\mathcal{G})^{3/8} T^{1/2} R^{9/8} + NE(\mathcal{G})^{3/8} T^{1/4} R^{3/2}$. On the other hand, if $R > T^{2/3}$, then by the upper bound in (29.65) we have $NE(\mathcal{G})^{3/8} T^{1/4} R^{3/2} \ll NT^{1/4} R^{21/8} \ll NR^3$. By the lower bound in

(29.65)

$$\begin{aligned}
NE(\mathcal{G})^{3/8}T^{1/2}R^{9/8} &\ll N^2E(\mathcal{G})^{1/2}R^{1/2}(TR^{-4})^{1/8}T^{1/2}R^{5/8}N^{-1} \\
&\ll N^2E(\mathcal{G})^{1/2}R^{1/2}(T^{5/8}R^{1/8}/N) \\
&\ll N^2E(\mathcal{G})^{1/2}R^{1/2}(T^{3/4}/N) \\
&\ll N^2E(\mathcal{G})^{1/2}R^{1/2}.
\end{aligned}$$

and this completes the proof of the lemma $\square$

We can now wrap up our analysis of S_3 .

T:FinalE **Theorem 29.31.** *Suppose that $V > 0$, $\varepsilon > 0$, $\mathcal{G}$ is a 1-separated set contained in an interval of length T , that $T^{3/4} \leq N \leq T$ and that $\mathfrak{D}(t) \geq \mathfrak{A}V$ for $g \in \mathcal{G}$. Let $R = \text{card } \mathcal{G}$. Then*

$$E(\mathcal{G}) \ll T^\varepsilon RN^4V^{-4} + T^{\frac{1}{4}+\varepsilon}R^{21/8}NV^{-2} + T^\varepsilon R^3NV^{-2}.$$

Proof By Lemma 29.25

$$E(\mathcal{G}) \ll T^\varepsilon \mathfrak{A}(N)^2V^{-2} \sum_{\mathbf{n} \in [N, 2N]^2} |Q(n_1/n_2)|^3.$$

where $\mathfrak{A}(N) = \max_{n \in [N+1, 2N]} |a_n|$. Then by Lemmas 29.29 and 29.30 this is

$$\ll T^\varepsilon \mathfrak{A}(N)^2V^{-2} (NR^3 + NT^{1/4}R^{21/8} + N^2E(\mathcal{G})^{1/2}R^{1/2})$$

Solving for $E(\mathcal{G})$ establishes the theorem. $\square$

We now insert this in Theorem 29.23.

T:S3another **Theorem 29.32.** *Suppose that $V, \varepsilon > 0$ and $\mathcal{G}$ is a 1-separated set contained in an interval of length T , that $N \geq T^{3/4}$ and that $\mathfrak{D}(g) \geq \mathfrak{A}V$ for $g \in \mathcal{G}$. Let $R = \text{card}(\mathcal{G})$. Then*

$$S_3 \ll T^{2+\varepsilon}R^{\frac{3}{2}} + T^{1+\varepsilon}N^3RV^{-2} + T^{1+\varepsilon}N^{\frac{3}{2}}R^2V^{-1} + T^{\frac{9}{8}+\varepsilon}N^{\frac{3}{2}}R^{\frac{29}{16}}V^{-1}.$$

29.10 A Bound for the Bilinear Form

We now establish a bound for the bilinear form at the centre of our study.

L:newform **Lemma 29.33.** *Let a_n ($N+1 \leq n \leq 2N$) and s_r ($1 \leq r \leq R$) be*

complex numbers, with $\text{Im } s_r - 1$ spaced and lying in an interval of length T , and $\text{Re } s_r \geq \theta$, and let D be as in (29.4). Then

$$\begin{aligned} \sum_{r=1}^R |D(s_r)|^2 &\ll_{\varepsilon} \mathfrak{A}^2 T^{\varepsilon} N^{-2\theta} (N^2 + N^{5/3} T^{1/24} R^{29/48} + N^{5/3} R^{2/3} \\ &\quad + N T^{2/3} R^{1/2} + N^{3/2} T^{1/4} R^{1/2} + N^{9/7} T^{2/7} R^{5/7} \\ &\quad + N^{9/7} T^{9/28} R^{37/56}) \end{aligned}$$

where $\mathfrak{A} = \max_n |a_n|$.

Proof We return to the inequalities (29.12), (29.14) and (29.15). The set $\mathcal{G}$ is a set of R points g in an interval of length T . Let $j \in \mathbb{N}$ and suppose that $V_j = 2^j$ and $\mathcal{G}_j$ is the subset of G such that $V_j/2 \leq |\mathfrak{D}(g)|/\mathfrak{A} \leq V_j$. Let $R_j = \text{card}(\mathcal{G}_j)$. For the time being, for simplicity, we drop the j . When combined with Theorem 29.4 this gives

$$\lambda \ll N + (S_2 + |S_3|)^{1/3}.$$

We write

$$\mathfrak{N} = \mathfrak{A}^{-2} \sum_{n=N+1}^{2N} |a_n|^2.$$

Then by Theorems 29.12 (with $k = 4$), and 29.32, we have

$$\begin{aligned} (RV^2)^3 &\ll \left(\sum_{g \in \mathcal{G}_j} \mathfrak{A}^{-2} |\mathfrak{D}(g)|^2 \right)^3 \ll T^{\varepsilon} \mathfrak{N}^3 (N^3 + R^{7/4} TN + R^{29/16} N^2 T^{1/8} \\ &\quad + R^2 N^2 + T^2 R^{3/2} + TN^3 RV^{-2} + TN^{3/2} R^2 V^{-1} + T^{9/8} N^{3/2} R^{29/16} V^{-1}). \end{aligned}$$

We have $R^{7/4} TN = (R^2 N^2)^{1/2} (R^{3/2} T^2)^{1/2} \leq R^2 N^2 + R^{3/2} T^2$ and so we can suppress this term and obtain

$$\begin{aligned} R^3 V^6 &\ll T^{\varepsilon} \mathfrak{N}^3 \mathfrak{A}^{-6} (N^3 + R^{29/16} N^2 T^{1/8} + R^2 N^2 + T^2 R^{3/2} + TN^3 RV^{-2} \\ &\quad + TN^{3/2} R^2 V^{-1} + T^{9/8} N^{3/2} R^{29/16} V^{-1}). \end{aligned}$$

Solving for V^2 now gives

$$\begin{aligned} \sum_{g \in \mathcal{G}_j} \mathfrak{A}^{-2} |\mathfrak{D}(g)|^2 &\ll R_j V_j^2 \\ &\ll T^{\varepsilon} (\mathfrak{N} N + \mathfrak{N} N^{2/3} T^{1/24} R_j^{29/48} + \mathfrak{N} N^{2/3} R_j^{2/3} \\ &\quad + \mathfrak{N} T^{2/3} R_j^{1/2} + \mathfrak{N}^{3/4} N^{3/4} T^{1/4} R_j^{1/2} \\ &\quad + \mathfrak{N}^{6/7} N^{3/7} T^{2/7} R_j^{5/7} + \mathfrak{N}^{6/7} N^{3/7} T^{9/28} R_j^{37/56}). \end{aligned}$$

Summing over j for which $\mathcal{G}_j \neq \emptyset$ and noting that the contribution from the $|\mathfrak{D}(g)| \leq \mathfrak{A}$ is at most $R \leq T$ we have

$$\begin{aligned} \sum_{g \in \mathcal{G}} |\mathfrak{D}(g)|^2 &\ll_{\varepsilon} \mathfrak{A}^2 T^{\varepsilon} (N^2 + N^{5/3} T^{1/24} R^{29/48} + N^{5/3} R^{2/3} \\ &\quad + NT^{2/3} R^{1/2} + N^{3/2} T^{1/4} R^{1/2} + N^{9/7} T^{2/7} R^{5/7} \\ &\quad + N^{9/7} T^{9/28} R^{37/56}) \end{aligned}$$

We can now apply the usual partial summation technique to bound the sum

$$\sum_{r=1}^R |D(s_r)|^2.$$

□

29.11 Counting Zeros

S:zerocount

We first need a consequence of the previous lemma.

L:newcount

Lemma 29.34. *Let a_n ($N+1 \leq n \leq 2N$), let D be as in (29.4) and let s_r ($1 \leq r \leq R$) be complex numbers, with $\text{Im } s_r$ 1-spaced and lying in an interval of length T , and such that $\text{Re } s_r \geq \theta$ and $|D(s_r)| \geq 1$. Suppose further that $\frac{7}{10} \leq \theta \leq \frac{8}{10}$. Then*

$$R \ll_{\varepsilon} \mathfrak{A}^2 T^{\varepsilon} (T + N^{6/5}) N^{\frac{12-20\theta}{5}}.$$

where $\mathfrak{A} = \max_n |a_n|$.

Proof At once by Lemma 29.33 we have

$$\begin{aligned} R &\ll_{\varepsilon} \mathfrak{A}^2 T^{\varepsilon} N^{-2\theta} (N^2 + N^{5/3} T^{1/24} R^{29/48} + N^{5/3} R^{2/3} \\ &\quad + NT^{2/3} R^{1/2} + N^{3/2} T^{1/4} R^{1/2} + N^{9/7} T^{2/7} R^{5/7} \\ &\quad + N^{9/7} T^{9/28} R^{37/56}). \end{aligned}$$

Solving for R ,

$$\begin{aligned} R &\ll_{\varepsilon} \mathfrak{A}^2 T^{\varepsilon} (N^{2-2\theta} + N^{\frac{80-96\theta}{19}} T^{2/19} + N^{5-6\theta} \\ &\quad + N^{2-4\theta} T^{4/3} + N^{3-4\theta} T^{1/2} + N^{\frac{9-14\theta}{2}} T + N^{\frac{72-112\theta}{19}} T^{18/19} \\ &\quad). \end{aligned}$$

Now we adopt the Huxley manoeuvre. We divide the interval $[0, T]$

into J subintervals $[(j-1)T/J, jT/J]$ ($1 \leq j \leq J$) of length $T/J \ll \min(T, N^{6/5})$ where $J = \lfloor TN^{-6/5} \rfloor + 1 \leq (T + N^{6/5})N^{-6/5}$, and sum over j . Thus

$$R \ll_{\varepsilon} \mathfrak{A}^2 T^{\varepsilon} (T + N^{6/5}) \left(N^{4-10\theta} + N^{\frac{298-480\theta}{95}} + N^{\frac{12-20\theta}{5}} + N^{\frac{9-14\theta}{2}} + N^{\frac{354-560\theta}{95}} \right).$$

As $\frac{7}{10} \leq \theta \leq \frac{8}{10}$ the other terms are bounded by the third. $\square$

It is also useful to state the following consequences of Theorems 26.9 and 27.4, *viz.*

$$R \ll \mathfrak{A}^2 T^{\varepsilon} (N^{2-2\theta} + T \min(N^{1-2\theta}, N^{4-6\theta})). \quad (29.80) \quad \boxed{\text{eq:combin}}$$

which holds under the same initial hypothesis as the above lemma, but with the more relaxed constraint $0 \leq \theta \leq 1$.

We can now state the refined zero density theorem for the Riemann zeta function.

T:GMzerod **Theorem 29.35.** *Suppose that $\frac{7}{10} \leq \theta \leq \frac{8}{10}$. Then*

$$N(\theta, T) \ll T^{\frac{15(1-\theta)}{3+5\theta} + \varepsilon}.$$

Proof We now largely follow the section §28.4.2, but as applied solely to the Riemann zeta function. There we found that given $\eta > 0$ the zeros ρ of $\zeta(s)$ with $0 < \text{Im } \rho \leq T$ and $\text{Re } \rho \geq \theta$ can be divided into $\ll \log T$ subsets $\mathcal{P}_j$ of cardinality N_j and that there is K_j with

$$T^{\eta/5} < K_j \ll T^{1/2} (\log T)^3 \quad (29.81) \quad \boxed{\text{eq:Kayjay}}$$

such that for any fixed k and $\rho \in \mathcal{P}_j(\chi)$,

$$1 \ll (\log T)^{3k} \left(\left| \sum_{n \in J_j} c_1(n) \chi(n) n^{-\rho} \right|^2 + \int_{-\infty}^{\infty} \left| \sum_{n \in J_j} c_2(n; v) \chi(n) n^{-\rho} \right|^2 e^{-|v|} dv \right).$$

where

$$J_j \subset (K_j^k, K_j'^k], \quad |c_1(n)| \ll d_{2k}(n), \quad |c_2(n; v)| \ll d_{2k}(n).$$

We introduce parameters X and Y with

$$T^{\frac{5}{14}} \leq X = T^{\frac{5}{6+10\theta}} \leq T^{\frac{5}{13}} \quad \text{and} \quad T^{\frac{45}{104}} \leq Y = T^{\frac{75(1-\theta)}{(9-10\theta)(12+20\theta)}} \leq T^{\frac{15}{28}}. \quad (29.82) \quad \boxed{\text{eq:Exe}}$$

Then, much as in 28.4.2, our argument depends on the size of K_j .

1. If $Y < K_j$, since also the upper bound in (29.81) holds, we may take $N = K_j^2$ and apply the first minimum in (29.80) to obtain

$$R \ll T^{1+\eta} Y^{2-4\theta} = T^{\eta + \frac{75(1-2\theta)(1-\theta) + 2(9-10\theta)(3+5\theta)}{2(9-10\theta)(3+5\theta)}}.$$

We need to show that the function of θ in the exponent does not exceed

$$\frac{15(1-\theta)}{3+5\theta}.$$

The quadratic polynomial $250\theta^2 - 3750\theta + 141$ has its minimum at $\theta = \frac{3}{4}$ and its value there is

$$0.375.$$

Hence it is non-negative. Therefore

$$2(9-10\theta)(3+5\theta) \leq 15(1-\theta)(13-10\theta).$$

Adding $75(1-\theta)(1-2\theta)$ to both sides gives

$$75(1-\theta)(1-2\theta) + 2(9-10\theta)(3+5\theta) \leq 30(1-\theta)(9-10\theta)$$

which gives the desired inequality.

2. If $X < K_j \leq Y$, then we again take $N = K_j^2$ but now apply Lemma 29.34. This gives at once

$$R \ll T^{\varepsilon + \frac{15(1-\theta)}{3+5\theta}}.$$

3. If $T^{1/3} < K_j \leq X$, then we take $N = K_j^3$, and apply (29.80) once more, but now with the second part of the minimum, which gives,

$$R \ll T^{\varepsilon} (X^{6-6\theta} + T^{2-2\theta})$$

and the result follows.

4. If $Y^{2/3} < K_j \leq T^{1/3}$, then let $N = K_j^3$ and argue as in 1.

5. If $X^{2/3} < K_j \leq Y^{2/3}$, then let $N = K_j^3$ and argue as in 2.

6. If $X^{2/k} < K_j \leq X^{3/k}$, then let $N = K_j^k$, so that $X^2 < K_j^k \leq X^3$, and argue as in 3., 4., or 5.

7. Finally observe that for $k \in \mathbb{N}$ we have $2/k \leq 3/(k+1)$. $\square$

29.12 Primes in Short Intervals II

S:GMShorts

We now reach the main aim of the material in this chapter, namely an improved approximation for the distribution of primes in short intervals.

GMinshorts **Theorem 29.36.** *Let $\nu \in (\frac{17}{30}, 1]$. There is a positive number c such that if the function $G(x)$ satisfies $G(x) \geq 1$ and $x^\nu (\log x)^c G(x)^{7/2} \leq h \leq x$, then*

$$\psi(x+h) - \psi(x) = h + O\left(\frac{h}{G(x)} + h \exp\left(-\frac{(\log x)^{\frac{1}{3}}}{(c \log \log x)^{\frac{1}{3}}}\right)\right) \quad (29.83) \quad \text{eq:GMpsi}$$

and

$$\pi(x+h) - \pi(x) - \int_x^{x+h} \frac{du}{\log u} \ll \frac{h}{G(x)} + h \exp\left(-\frac{(\log x)^{\frac{1}{3}}}{(c \log \log x)^{\frac{1}{3}}}\right). \quad (29.84) \quad \text{eq:GMpi}$$

Proof As presaged in the comment after Theorem 28.8 this follows the proof of that theorem with obvious modifications to accept Theorem 29.35. With the same notation as there we obtain

$$\psi(x+h) - \psi(x) - h \ll hx^{-1} \sum_{\substack{|\gamma| \leq T \\ \beta \geq 1/2}} x^\beta + xT^{-1}(\log x)^2 \quad (29.85) \quad \text{eq:GMesta}$$

with

$$\sum_{\substack{|\gamma| \leq T \\ \beta \geq 1/2}} x^\beta = x^{1/2} N(1/2, T) + \int_{1/2}^{1-\delta} x^u N(u, T) (\log x) du. \quad (29.86) \quad \text{eq:GMestb}$$

By Theorem 28.4 and Corollary 28.5

$$\begin{aligned} x^{1/2} N(1/2, T) + \int_{1/2}^{7/10} x^u N(u, T) (\log x) du \\ \ll (x^{1/2} T + x^{7/10} T^{9/13} + x^{4/5} T^{3/7}) (\log x)^{c_2} \end{aligned}$$

and

$$\int_{8/10}^{1-\delta} x^u N(u, T) (\log x) du \ll (x^{8/10} T^{18/25} + x(T^{15/7}/x)^\delta) (\log x)^{c_2}.$$

Also, by Theorem 29.35

$$\begin{aligned} \int_{7/10}^{4/5} x^u N(u, T) (\log x) du &\ll \int_{7/10}^{4/5} x^u T^{30(1-u)/13+\varepsilon} du \\ &\ll x^{7/10} T^{9/13+\varepsilon} + x^{4/5} T^{6/13+\varepsilon}. \end{aligned}$$

We now make the choice

$$T = xh^{-1}(\log x)^{-2}G(x)$$

so that

$$T \leq x^{1-\nu}(\log x)^{-c}G(x)^{-7/2}.$$

Then, by (29.85) and (29.86)

$$\psi(x+h) - \psi(x) - h \ll hG(x)^{-1} + h(\log x)^{c_2}(T^{15/7}/x)^\delta.$$

By (28.52),

$$(T^{15/7}/x)^\delta \leq \exp\left(-\frac{\log x}{c_3(\log x)^{2/3}(\log \log x)^{1/3}}\right),$$

which gives (29.83).

The translation from ψ to π can be accomplished as in the proof of Theorem 29.35. $\square$

29.13 Notes

S:NotesPISI

§S:1GMRI Most of the material of this chapter is from ^{GM24}Guth & Maynard (2024). It is often asserted that to make progress on understanding the zeros of the Riemann zeta function we should make more use of the additive structure of numbers. One glimpse of this is the use in Chapter 24 of the Vinogradov Mean Value Theorem to improve the known zero-free region. A notable development of the Guth-Maynard work is their discovery of another connection with additive structure. In particular they harnesses for multiplicative number theory the power of classical harmonic analysis, that is, Fourier analysis on the additive structure of $\mathbb{R}$. It also makes better use than hitherto of material such as that in Appendix L. In particular the inequality (??) is central to the work.

§29.2 Weights of this kind are something of a novelty in analytic number theory.

§29.4 ^{HB79}Lemma 29.10 is a slightly refined version of Theorem 1 of Heath-Brown ^{HB79}Heath-Brown (1979).

29.14 References

- ^{GM24}L. Guth & J. A. Maynard (2024), New large value estimates for Dirichlet polynomials, arXiv:2405.20552 [math.NT]
- ^{HB79}D. R. Heath-Brown (1979), A large values estimate for Dirichlet polynomials, J. London Math. Soc. (2), 20(1979), 8–18.
- ^{MJ77}M. Jutila (1977), Zero-density estimates for L -functions, Acta Arith., 32(1977), 55–62.

30

The pair correlation of zeta zeros

C:PairCorr

As usual, we let $N(T)$ denote the number of zeros $\rho = \beta + i\gamma$ of the Riemann zeta function with $0 < \gamma \leq T$. Multiple zeros are counted according to their multiplicity. From Theorem 10.13 we know that

$$N(T+1) - N(T) \ll \log T \quad (30.1) \quad \text{E:DiffN(T)}$$

for $T \geq 2$, and in Corollary 14.3 we found that

$$N(T) = \frac{T}{2\pi} \log \frac{T}{2\pi} - \frac{T}{2\pi} + O(\log T). \quad (30.2) \quad \text{E:N(T)Est}$$

Thus we see that the average spacing between the consecutive γ at height T is approximately $2\pi/\log T$. Our object in this Chapter (which will only be partially achieved) is to determine the distribution of the differences $\gamma - \gamma'$ between the ordinates on the scale of $2\pi/\log T$. We shall assume the Riemann Hypothesis throughout this Chapter.

Our approach to the distribution of the numbers $\gamma - \gamma'$ is to try to determine the Fourier Transform of this distribution, which is to say the asymptotic size of sums of $X^{i(\gamma-\gamma')}$. In summing this quantity we introduce a weighting $w(\gamma - \gamma')$, so that pairs of zeros that are far apart receive little weight. Specifically, for $X > 0$ and $T \geq 2$ we set

$$F(X, T) = \sum_{\substack{0 < \gamma \leq T \\ 0 < \gamma' \leq T}} X^{i(\gamma-\gamma')} w(\gamma - \gamma') \quad (30.3) \quad \text{E:DefF}$$

where $w(u) = 4/(4+u^2)$. Since w is an even function, on exchanging the roles of γ and γ' in the above it follows that $F(X, T)$ is real for all X and T . The weight $w(u)$ arises naturally in our analysis in the equation

(30.9), from which it follows that

$$\frac{4}{T \log \frac{T}{2\pi}} \int_{-\infty}^{\infty} \left| \sum_{0 < \gamma \leq T} \frac{X^{i\gamma}}{1 + (t - \gamma)^2} \right|^2 dt. \quad (30.4) \quad \boxed{\text{E:Fform1}}$$

From this formula we see that $F(X, T) \geq 0$ for all X and T .

To derive an asymptotic formula for $F(X, T)$ we start from a convenient explicit formula (proved below) which asserts that if $X \geq 1$, then

$$\begin{aligned} & 2 \sum_{\gamma} \frac{X^{i\gamma}}{1 + (t - \gamma)^2} \\ &= -X^{-1/2} \left(\sum_{n \leq X} \Lambda(n) (X/n)^{-1/2+it} + \sum_{n > X} \Lambda(n) (X/n)^{3/2+it} \right) \\ & \quad + X^{-1+it} \log \tau + O(X^{-1}) + O(X^{1/2} \tau^{-2}) \end{aligned} \quad (30.5) \quad \boxed{\text{E:ExpEq1}}$$

where $\tau = |t| + 4$. We write the above briefly as $L(X, T) = R(X, T)$. In Section 30.1 we shall show that $\int_0^T |L(X, T)|^2 dt$ is approximately $F(X, T)$, while the mean value theorems of Chapter ?? will be used to estimate $\int |R(X, T)|^2 dt$. Unfortunately, in this latter endeavour the main term is larger than the error term only when $X = o(T(\log T)/\log \log T)$.

To see how we might derive useful information concerning the zeros from asymptotic information concerning $F(X, T)$, we observe that if $R(\alpha) \in L^1(\mathbb{R})$, then

$$\int_{-\infty}^{\infty} R(\alpha) F(T^\alpha, T) d\alpha = \sum_{\substack{0 < \gamma \leq T \\ 0 < \gamma' \leq T}} w(\gamma - \gamma') \hat{R}\left(\frac{(\gamma - \gamma') \log T}{2\pi}\right). \quad (30.6) \quad \boxed{\text{E:intRF}}$$

In view of the remarks we have already made, we see that we can determine the asymptotic size of left hand side of the above only when $\text{supp } R \subseteq [-1, 1]$, but it is still the case that

$$\int_{-\infty}^{\infty} R(\alpha) F(T^\alpha, T) d\alpha \geq \int_{-1}^1 R(\alpha) F(T^\alpha, T) d\alpha \quad (30.7) \quad \boxed{\text{E:intRFineq}}$$

for R with arbitrary support, provided that $R(\alpha) \geq 0$ whenever $|\alpha| \geq 1$. (The above inequality is reversed in case $R(\alpha) \leq 0$ when $|\alpha| \geq 1$.)

30.1 The basic asymptotic estimate and conjectures

S:PCEst

We now prove the approximate identity (30.5). To this end, let $K(w) = \frac{2}{1-w^2} = \frac{1}{1-w} + \frac{1}{1+w}$. Thus $K(w)$ has simple poles at ± 1 , and $K(iv) =$

$\frac{1}{1+v^2}$. By the calculus of residues we see that

$$\frac{1}{2\pi i} \int_{a-i\infty}^{a+i\infty} K(w)x^w dw = \begin{cases} \frac{1}{x} - x & (x \geq 1), \\ 0 & (0 \leq x < 1). \end{cases}$$

Hence

$$\frac{1}{2\pi i} \int_{a-i\infty}^{a+i\infty} \frac{\zeta'}{\zeta} \left(w + \frac{1}{2} + it\right) K(w) X^w dw = \sum_{n \leq X} \frac{\Lambda(n)}{n^{\frac{1}{2}+it}} \left(\frac{X}{n} - \frac{n}{X}\right)$$

for $a > 1$. Now $\frac{\zeta'}{\zeta} \left(w + \frac{1}{2} + it\right)$ has a simple pole at $\frac{1}{2} - it$ with residue -1 , simple poles at $-2n - \frac{1}{2} - it$ with residue 1 for $n = 1, 2, \dots$, and simple poles at $i(\gamma - t)$ with residue 1 . The convention here is that if a zero $\rho = \frac{1}{2} + i\gamma$ has multiplicity m_ρ , then in a sum over zeros a summand corresponding to ρ is repeated m_ρ times. Write $w = u + iv$. On moving the contour from the abscissa $u = a > 1$ to $u = -\infty$, we see that the above is

$$\begin{aligned} &= -\frac{\zeta'}{\zeta} \left(\frac{3}{2} + it\right) X + \frac{\zeta'}{\zeta} \left(-\frac{1}{2} + it\right) X^{-1} - K\left(\frac{1}{2} - it\right) X^{\frac{1}{2}-it} \\ &\quad + 2 \sum_{\gamma} \frac{X^{i(\gamma-t)}}{1 + (\gamma - t)^2} + \sum_{n=1}^{\infty} K\left(-2n - \frac{1}{2} - it\right) X^{-2n-\frac{1}{2}-it}. \end{aligned}$$

That the contour may be moved with this result is justified in the same way that it was justified in §12.1 when we discussed the classical explicit formulæ. We note that $-\frac{\zeta'}{\zeta} \left(\frac{3}{2} + it\right) = \sum_n \Lambda(n) n^{-\frac{3}{2}-it}$, we multiply both sides of the above equation by X^{it} , and rearrange to see that

$$\begin{aligned} 2 \sum_{\gamma} \frac{X^{i\gamma}}{1 + (\gamma - t)^2} &= \frac{-1}{X^{1/2}} \left(\sum_{n \leq X} \Lambda(n) \left(\frac{X}{n}\right)^{-\frac{1}{2}+it} + \sum_{n > X} \Lambda(n) \left(\frac{X}{n}\right)^{\frac{3}{2}+it} \right) \\ &\quad - \frac{\zeta'}{\zeta} \left(-\frac{1}{2} + it\right) X^{-1+it} + K\left(\frac{1}{2} - it\right) X^{\frac{1}{2}} \\ &\quad - \sum_{n=1}^{\infty} K\left(-2n - \frac{1}{2} - it\right) X^{-2n-\frac{1}{2}}. \end{aligned}$$

This is an exact equation; to complete the proof of (30.5) we now estimate the last three terms above. In §10.2 we took the logarithmic derivative of the asymmetric form of the functional equation to show that

$$\frac{\zeta'}{\zeta}(s) = -\frac{\zeta'}{\zeta}(1-s) + \log 2\pi - \frac{\Gamma'}{\Gamma}(1-s) + \frac{\pi}{2} \cot \frac{\pi s}{2}.$$

Also, Theorem C.1 asserts that

$$\frac{\Gamma'}{\Gamma}(s) = \log s + O(1/|s|)$$

provided that $|s| \geq \delta$ and $|\arg s| < \pi - \delta$. Hence

$$\frac{\zeta'}{\zeta}\left(-\frac{1}{2} + it\right) = -\frac{\zeta'}{\zeta}\left(\frac{3}{2} - it\right) - \frac{\Gamma'}{\Gamma}\left(\frac{3}{2} - it\right) + O(1) = -\log \tau + O(1).$$

For the second term it suffices to observe that $K\left(\frac{1}{2} - it\right) \ll \tau^{-2}$. Since $X \geq 1$, the sum over n is $\ll X^{-\frac{5}{2}} \sum_n (n^2 + t^2)^{-1} \ll X^{-\frac{5}{2}} \tau^{-1} \ll X^{-1}$. Thus the proof of (30.5) is complete.

We now estimate the integral of the square of the modulus of the respective sides of (30.5) to establish

T:PC1 **Theorem 30.1.** (Assume RH.) Let $F(X, T)$ be defined as in (30.3). Then $F(1/X, T) = F(X, T)$ for $X > 0$, and

$$\begin{aligned} F(X, T) &= \frac{\log \frac{T}{2\pi} - 2}{X^2} + \frac{\log X}{\log \frac{T}{2\pi}} \\ &\quad + O\left(\frac{(\log 2X)^3}{X^{1/2} \log T}\right) + O\left(\frac{X \log \log 4X}{T \log T}\right) \end{aligned} \quad (30.8) \quad \text{E:PC1}$$

for $1 \leq X \leq T$.

In the notation $X = \left(\frac{T}{2\pi}\right)^\alpha$, the main term above takes the shape

$$\left(\frac{T}{2\pi}\right)^{2\alpha} \left(\log \frac{T}{2\pi} - 2\right) + \alpha.$$

Here the first term behaves in the limit as a Dirac delta with mass 1 at the origin, since

$$\int_{-\infty}^{\infty} \left(\frac{T}{2\pi}\right)^{2\alpha} \log \frac{T}{2\pi} d\alpha = 1.$$

Proof Let $L(X, t)$ denote the left hand side of (30.5). Then

$$\int_0^T |L(X, t)|^2 dt = 4 \sum_{\gamma, \gamma'} X^{i(\gamma - \gamma')} \int_0^T \frac{dt}{(1 + (t - \gamma)^2)(1 + (t - \gamma')^2)}.$$

From (30.1) we see that if $0 \leq t \leq T$, then

$$\sum_{\gamma \notin [0, T]} \frac{1}{1 + (t - \gamma)^2} \ll \left(\frac{1}{t+1} + \frac{1}{T-t+1}\right) \log T,$$

and

$$\sum_{\gamma'} \frac{1}{1 + (t - \gamma')^2} \ll \log T.$$

Hence

$$\sum_{\gamma \notin [0, T]} \sum_{\gamma'} \int_0^T \frac{dt}{(1 + (t - \gamma)^2)(1 + (t - \gamma')^2)} \ll (\log T)^3,$$

and so

$$\begin{aligned} \int_0^T |L(X, t)|^2 dt &= 4 \sum_{\substack{0 < \gamma \leq T \\ 0 < \gamma' \leq T}} X^{i(\gamma - \gamma')} \int_0^T \frac{dt}{(1 + (t - \gamma)^2)(1 + (t - \gamma')^2)} \\ &\quad + O((\log T)^3). \end{aligned}$$

If $t \geq T$, then

$$\sum_{0 < \gamma \leq T} \frac{1}{1 + (t - \gamma)^2} \ll \frac{\log T}{t - T + 1},$$

and similarly for γ' , so

$$\begin{aligned} \sum_{\substack{0 < \gamma \leq T \\ 0 < \gamma' \leq T}} \int_T^\infty \frac{dt}{(1 + (t - \gamma)^2)(1 + (t - \gamma')^2)} &\ll (\log T)^2 \int_T^\infty \frac{dt}{(t - T + 1)^2} \\ &\ll (\log T)^3. \end{aligned}$$

The estimate is the same if we integrate instead over the interval $-\infty < t \leq 0$. Finally, it is easy to see that by the calculus of residues that

$$\int_{-\infty}^\infty \frac{dt}{(1 + (t - a)^2)(1 + (t - b)^2)} = \frac{2\pi}{(a - b)^2 + 4} = \frac{\pi}{2} w(a - b). \quad (30.9) \quad \boxed{\text{E:Def2w}}$$

Thus we conclude that

$$\int_0^T |L(X, t)|^2 dt = F(X, T) T \log \frac{T}{2\pi} + O((\log T)^3)$$

uniformly for $X > 0$.

The error terms on the right hand side of (30.5) are due mainly to the pole and trivial zeros of the zeta function, but they are troublesome only when t is small. To avoid considering them for small t we let V be a parameter to be chosen laeter, and employ a crude method for the range $0 \leq t \leq V$. From (30.1) we see that

$$|R(X, t)| = |L(X, t)| \leq L(1, t) \ll \log \tau$$

for all X . Hence

$$\int_0^V |R(X, t)|^2 dt \ll V(\log V)^2$$

for $V \geq 2$.

We now consider the contributions of the main terms in $R(X, t)$ when $V \leq t \leq T$. We first observe that from Corollaries 26.5 and 26.6 it follows that

$$\begin{aligned} \int_V^T \left| \sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^{1/2}} W(n/X) n^{-it} + \frac{1}{X} \sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^{1/2}} n^{it} \right|^2 dt & \quad (30.10) \quad \boxed{\text{E:Term1}} \\ &= (T - V) \left(\sum_{n=1}^{\infty} \frac{\Lambda(n)^2}{n} W(n/X)^2 + \frac{1}{X^2} \sum_{n=1}^{\infty} \frac{\Lambda(n)^2}{n^3} \right) \\ &\quad + O \left(\sum_{n=1}^{\infty} \frac{\Lambda(n)^2}{d_n} W(n/X)^2 + \frac{1}{X^2} \sum_{n=1}^{\infty} \frac{\Lambda(n)^2}{n^2} \right) \end{aligned}$$

where d_n denotes the distance from n to the nearest other primepower. From the estimate $\psi(x) = x + O(x^{1/2}(\log 2x)^2)$ of Theorem 13.1 we see by integration by parts that

$$\begin{aligned} \sum_{n \leq x} \Lambda(n)^2 &= \sum_{n \leq x} \Lambda(n) \log n + O((\psi(x) - \vartheta(x)) \log x) \\ &= x \log x - x + O(x^{1/2}(\log 2x)^3). \end{aligned}$$

Write this as $S(x) = x \log x - x + R(x)$. Then by a further integration by parts we see that

$$\begin{aligned} \sum_{n=1}^{\infty} \frac{\Lambda(n)^2}{n} W(n/X)^2 &= \int_0^{\infty} \frac{1}{u} W(u^2/X^2) dS(u) \\ &= \int_0^{\infty} \frac{\log u}{u} W(u^2/X^2) du + \left[\frac{R(u)}{u} W(u^2/X^2) \right]_0^{\infty} \\ &\quad - \frac{1}{X^2} \int_0^X R(u) du + 3X^2 \int_X^{\infty} \frac{R(u)}{u^4} du \\ &= \log X + O(X^{-1/2}(\log X)^3). \end{aligned}$$

By Theorem 26.7, the error terms in (30.10) are $\ll X \log \log 4X$. Thus the expression in (30.10) is

$$(T - V) \left(\log X + O(X^{-1/2}(\log X)^3) \right) + O(X \log \log 4X). \quad (30.11) \quad \boxed{\text{E:Term1again}}$$

The above accounts for the contributions of two of the main terms on

the right hand side of (30.5). The mean square of the third main term is

$$\begin{aligned} X^{-2} \int_V^T \left(\log \frac{\tau}{2\pi} \right)^2 dt &= X^{-2} T \left(\log \frac{T}{2\pi} \right)^2 - 2X^{-2} T \log \frac{T}{2\pi} \\ &= O(X^{-2} T) + O(X^{-2} V (\log V)^2). \end{aligned} \quad (30.12) \quad \boxed{\text{E:Term2}}$$

The mean square size of the main terms in (30.5) are determined in (30.11) and (30.12), but it remains to show that the correlations between these two main terms is small. The first main term, times the complex conjugate of the second one, integrated, is

$$\frac{1}{X} \int_V^T \left(\sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^{1/2}} W(n/X) n^{-it} + \frac{1}{X} \sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^{3/2}} n^{it} \right) \log \frac{\tau}{2\pi} dt.$$

If $t > 0$, then $\tau = t + 4$, by definition. Thus if $0 < a < b$ and c is a nonzero real number, then $\int_a^b e^{ict} \log \tau dt \ll \frac{\log(b+4)}{|c|}$. This can be shown by integration by parts, or by appealing to Theorem ???. Thus the above is

$$\begin{aligned} &\ll \frac{\log T}{X} \left(\sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^{1/2} \log n} W(n/X) + \frac{1}{X} \sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^{3/2} \log n} \right) \\ &\ll \frac{\log T}{X^{1/2} \log 2X}. \end{aligned} \quad (30.13) \quad \boxed{\text{E:Term12}}$$

On combining these estimates we deduce that

$$\begin{aligned} &\int_V^T \left| \sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^{1/2}} W(n/X) n^{-it} - \frac{1}{X} \left(\log \frac{\tau}{2\pi} + \frac{\zeta'}{\zeta} \left(\frac{3}{2} - it \right) \right) \right|^2 dt \\ &= T \log X + T X^{-2} \left(\log \frac{T}{2\pi} \right)^2 - 2T \log \frac{T}{2\pi} \\ &\quad + O(T X^{-1/2} (\log X)^3) + O(X \log \log 4X) \\ &\quad + O(V \log X) + O(V X^{-2} (\log V)^2). \end{aligned} \quad (30.14) \quad \boxed{\text{E:M1Est}}$$

The integral of the square of the sum of the error terms in (30.5) is

$$\int_V^T |X^{-1} \tau^{-1} + X^{1/2} \tau^{-2}|^2 dt \ll X^{-2} V^{-1} + X V^{-3}. \quad (30.15) \quad \boxed{\text{E:M2Est}}$$

To ensure that the last two error terms in (30.14) are majorized by the other error terms in that formula, we take $V = T X^{-1/2} (\log T)^{-2}$. Suppose that f_1 and f_2 are measurable functions, that $M_i = \int_a^b |f_i(t)|^2 dt$, and that $M_1 \geq M_2$. Then

$$\int_a^b |f_1(t) + f_2(t)|^2 dt = M_1 + O(\sqrt{M_1 M_2})$$

by the Cauchy–Schwarz inequality. Let M_1 Denote the integral in (30.14) and M_2 the integral in (30.15). In order that M_2 should make no contribution to our final result, we need to know that not only is M_2 majorized by the first two error terms in (30.14), but also that the larger quantity $\sqrt{M_1 M_2}$ is also majorized by these error terms. Since $M_1 \ll T(\log T)^2$ in all cases, this will be the case provided that $M_2 \ll TX^{-1}(\log T)^{-2}$. It is now easy to verify that this is the case (by a wide margin) when our choice of V is substituted into (30.15). $\square$

If we could derive (assuming RH) an asymptotic formula for $F(X, T)$ when $T < X < T^A$ for any fixed $A > 1$, then by Fourier inversion we could determine (still assuming RH) the distribution of the differences $\gamma - \gamma'$ relative to the average spacing. The main issue would be to derive an asymptotic estimate for

$$\int_V^T \left| \sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^{1/2}} W(n/X) n^{-it} \right|^2 dt.$$

To assess the difficulty here, consider the simple formula

$$\int_{-T}^T \left| \sum_{n \leq N} a_n n^{-it} \right|^2 dt = 2T \sum_{n \leq N} |a_n|^2 + 2T \sum_{\substack{m, n \leq N \\ m \neq n}} a_m \overline{a_n} \frac{\sin(T(\log m/n))}{T(\log m/n)}.$$

Let $h = n - m$. If $0 < |h| \leq m/T$, then

$$\frac{\sin(T(\log m/n))}{T(\log m/n)} \asymp 1,$$

so such nondiagonal terms carry a weight comparable to that of a diagonal term. Of course, such nondiagonal terms exist only when N is larger than T . The function $\frac{\sin u}{u}$ is not in $L^1(\mathbb{R})$, nor is it of bounded variation. Thus it is not easy to determine the contribution of nondiagonal terms directly from the above. The situation is improved if we average. For example,

$$\begin{aligned} \int_{-T}^T \left(1 - \frac{|t|}{T}\right) \left| \sum_{n \leq N} a_n n^{-it} \right|^2 dt &= T \sum_{n \leq N} |a_n|^2 \\ &\quad + T \sum_{\substack{m, n \leq N \\ m \neq n}} a_m \overline{a_n} \left(\frac{\sin \frac{1}{2} T(\log n/m)}{\frac{1}{2} T(\log n/m)} \right)^2. \end{aligned}$$

The function $\left(\frac{\sin u}{u}\right)^2$ is in $L^1(\mathbb{R})$ and is also of bounded variation, and we

could proceed from the above, but our work is made easier if we instead employ the smoothing found in Corollary 26.11.

22.1.1 Exercises

1. From (30.1) it follows that $\sum_{\gamma} (1 + (t - \gamma)^2)^{-1} \ll \log t$. Use (30.5) to show that

$$\sum_{\gamma} \frac{1}{1 + (t - \gamma)^2} = \log \frac{t}{2\pi} + 2 \operatorname{Re} \frac{\zeta'}{\zeta} \left(\frac{3}{2} + it \right) + O(1/t).$$

Exer: Lambda^2 2. For $\sigma > 1$ set

$$f(s) = \sum_{n=1}^{\infty} \frac{\Lambda(n)^2}{n^s} = \sum_p \sum_{k=1}^{\infty} \frac{(\log p)^2}{p^{ks}},$$

$$g(s) = \sum_{n=1}^{\infty} \frac{\Lambda(n) \log n}{n^s} = \sum_p \sum_{k=1}^{\infty} \frac{k(\log p)^2}{p^{ks}} = \left(\frac{\zeta'}{\zeta}(s) \right)',$$

and put $F(x) = \sum_{n \leq x} \Lambda(n)^2$, $G(x) = \sum_{n \leq x} \Lambda(n) \log n$.

- (a) Show that $\sum_{r=1}^{\infty} \varphi(r) f(rs) = g(s)$.
 (b) Show that

$$\sum_{r=1}^{\infty} a_r g(rs) = f(s) \quad (30.16) \quad \text{E:fginverse}$$

for $\sigma > 1$ if and only if

$$\sum_{\substack{r,k \\ rk=m}} a_r \frac{m}{r} = 1$$

for all m .

- (c) Show that the identity immediately above is equivalent to asserting that

$$\zeta(s-1) \sum_{r=1}^{\infty} \frac{a_r}{r^s} = \zeta(s)$$

for $\sigma > 2$.

- (d) Deduce that for $\sigma > 2$,

$$\sum_{r=1}^{\infty} \frac{a_r}{r^s} = \prod_p \left(1 + \frac{1-p}{p^s} + \frac{1-p}{p^{2s}} + \cdots \right).$$

(e) Conclude that (30.16) holds if

$$a_r = \prod_{p|r} (1-p).$$

3. Let $U = T/(2\pi)$. Determine the size of $U^{-2\alpha} \log U$ for the following values of α :

- (a) $\alpha = 1/\log U$;
- (b) $\alpha = (\log \log U)/\log U$;
- (c) $\alpha = (\log \log U - \frac{1}{2} \log \log \log U)/\log U$.

Exer:w,whatinvert

4. For $a > 0$ let

$$I(a, u) = 2\pi \int_{-\infty}^{\infty} e^{-2\pi a|x|} e(ux) dx, \quad J(a, x) = \int_{-\infty}^{\infty} \frac{2a}{a^2 + u^2} e(xu) du.$$

(a) Show that

$$I(a, u) = 2\pi \int_0^{\infty} e^{-2\pi ax} (e(ux) + e(-ux)) dx.$$

(b) Show that the above is

$$= \left[\frac{e^{-2\pi(a+iu)x}}{a+iu} \right]_0^{\infty} + \left[\frac{e^{-2\pi(a-iu)x}}{a-iu} \right]_0^{\infty}.$$

(c) Deduce that

$$I(a, u) = \frac{2a}{a^2 + u^2}.$$

(d) Deduce that

$$w(u) = 2\pi \int_{-\infty}^{\infty} e^{-4\pi|x|} e(ux) dx. \quad (30.17) \quad \text{E:wexpand}$$

(e) Show that $J(a, -x) = J(a, x)$ for all x .

(f) Replace the real variable u in the definition of $J(a, x)$ by the complex variable $w = u + iv$, consider the integral to be a contour integral in the complex plane, assume that $x \geq 0$, form a semi-circular path in the upper halfplane, and calculate the residue at the pole at $w = ia$ to show that

$$J(a, x) = 2\pi e^{-2\pi a|x|}.$$

(g) Deduce that

$$\widehat{w}(x) = \int_{-\infty}^{\infty} w(u)e(-xu) du = 2\pi e^{-4\pi|x|}. \quad (30.18) \quad \boxed{\text{E:whatexpand}}$$

Thus (30.17) asserts that $w(u) = \int_{\mathbb{R}} \widehat{w}(x)e(ux) dx$ is a valid Fourier expansion.

5. With $\widehat{w}(x)$ defined as above, show that

$$F(X, T2\pi \log T)^{-1} \int_{-\infty}^{\infty} \widehat{w}(x) \left| \sum_{0 < \gamma \leq T} X^{i\gamma} e(\gamma x) \right|^2 dx. \quad (30.19) \quad \boxed{\text{E:Fform2}}$$

6. Let $a_1, a_2, a_3, \dots$ be real numbers.

(a) Show that

$$\begin{aligned} \int_{-\infty}^{\infty} \left(\sum_{i=1}^n \frac{1}{1 + (x - a_i)^2} \right)^2 dx &= 2\pi \sum_{1 \leq i, j \leq n} \frac{1}{4 + (a_i - a_j)^2} \\ &\asymp \sum_{1 \leq i, j \leq n} \frac{1}{1 + (a_i - a_j)^2}. \end{aligned}$$

(b) By Cauchy's inequality, or otherwise, show that

$$n^4 \leq \left(\sum_{1 \leq i, j \leq n} \frac{1}{1 + (a_i - a_j)^2} \right) \left(\sum_{1 \leq i, j \leq n} (1 + (a_i - a_j)^2) \right).$$

(c) (Putnam 2011 B5) Show that if there is a constant $A > 0$ such that

$$\int_{-\infty}^{\infty} \left(\sum_{i=1}^n \frac{1}{1 + (x - a_i)^2} \right)^2 dx \leq An$$

for all n , then there is a constant B such that

$$\sum_{1 \leq i, j \leq n} (1 + (a_i - a_j)^2) \geq Bn^3$$

for all n .

30.2 Applications

S:PCapps

22.3.1 Exercises

30.3 An arithmetic equivalent of the Pair Correlation Conjecture

S:PCArithEquiv

22.4.1 Exercises

30.4 Notes

S:NotesPairCorr

^{MVB88}Berry, (1988) used physical reasoning to conjecture that

$$\int_0^T (S(t + \delta) - S(t))^2 dt = f(T)T + o(T)$$

where $f(T) =$ ^{AF90}Fujii (1990) used a formula for $S(t)$ due to ^{DAG87}Goldston (1987) that is similar to, but more useful than a similar formula of ^{AS46}Selberg (1946) to show that ... assuming RH. If, in addition the Strong Pair Correlation Conjecture is assumed, then Berry's Conjecture follows.

30.5 References

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Appendix I

The Weak Distribution of Measures

C:LimDist

I.1 Basic theory

S:BasicThy

We now develop an analogue for the real line $\mathbb{R}$ of uniform distribution in the circle group $\mathbb{T}$. This is useful in discussing the limiting distribution of the error term in the Prime Number Theorem, and in Chapter 23 the limiting distribution of additive functions.

Let μ be a measure on the real line. We say that μ is a *probability measure* if it is nonnegative and if its total mass is 1. That is, $\mu(\mathcal{S}) \geq 0$ for all measurable sets $\mathcal{S}$ and $\mu(\mathbb{R}) = 1$. The *distribution function* of μ is the function $F(x) = \mu((-\infty, x])$. Clearly μ is a probability measure if and only if F is increasing and

$$\lim_{x \rightarrow -\infty} F(x) = 0, \quad \lim_{x \rightarrow +\infty} F(x) = 1. \quad (\text{I.1}) \quad \text{E:limF(x)}$$

Conversely, if F is increasing, right-continuous (i.e., $F(x^+) = F(x)$ for all real x), and satisfies (I.1), then there is a probability measure μ of which F is the distribution function. If μ is a measure, we say that a sequence of probability measures μ_N *tends weakly* to μ if $\lim_{N \rightarrow \infty} F_N(x) = F(x)$ whenever x is a point of continuity of F . We note that the distribution function F of a nonnegative measure can have at most countably many jump discontinuities. As a first observation about sequences of measures, we have

T:HellyThm

Theorem I.1 (Helly's theorem). *Let $\mu_1, \mu_2, \mu_3, \dots$ be a sequence of probability measures. Then there is a strictly increasing sequence n_k of positive integers such that the subsequence μ_{n_k} is weakly convergent.*

Proof We proceed by a Cantor diagonal process. Let $x_1, x_2, \dots$ be distinct and dense in $\mathbb{R}$. The numbers $F_N(x_1)$ have a limit point. Thus we may choose indices $N(j, 1)$ so that $F_{N(j, 1)}(x_1)$ converges. From the

indices $N(j, 1)$ we choose a subsequence $N(j, 2)$ so that $F_{N(j, 2)}(x_2)$ converges. We continue in this manner, and find that the sequence $F_{N(j, j)}(x_k)$ converges for all k . Hence there is a unique right-continuous function F such that $F_{N(j, j)}(x)$ tends to $F(x)$ for all points of continuity x of F . $\square$

Consider now the particular case in which μ is the probability measure that attaches weight 1 to the integer N . The sequence μ_N converges to the measure that is identically 0. Thus a limit of probability measures need not be a probability measure. This can be explained by noting that (I.1) need not hold uniformly for all measures μ_N in a sequence. We say, however, that a collection of measures is *tight* if (I.1) holds uniformly for all measures μ of the family. Our first observation concerning tight families of measures is obvious.

T:ConvTightFam

Theorem I.2. *If $\mu_1, \mu_2, \dots$ is a sequence of probability measures converging weakly to a measure μ , and if the sequence is tight, then μ is a probability measure.*

If μ is a probability measure and f is bounded and continuous, then the integral $\int_{-\infty}^{+\infty} f(x) d\mu(x)$ is well-defined. In particular, for real t we let $\hat{\mu}(t)$ denote the Fourier transform of the measure,

$$\hat{\mu}(t) = \int_{-\infty}^{\infty} e(-tx) d\mu(x).$$

We note that $\hat{\mu}(0)$ is the total mass of a measure, so that $\hat{\mu}(0) = 1$ for a probability measure.

L:PropMuHat

Lemma I.3. *Let μ be a probability measure. Then $|\hat{\mu}(t)| \leq 1$ for all t , and $\hat{\mu}(t)$ is continuous. Let $\mathcal{S}$ be a tight family of probability measures. Then $\hat{\mu}(t)$ is uniformly continuous in t , and uniformly so for $\mu \in \mathcal{S}$.*

Proof By the triangle inequality, $|\hat{\mu}(t)| \leq \int_{-\infty}^{\infty} 1 d\mu = 1$. Suppose that $\varepsilon > 0$ is given, and that A is so large that $\mu([-A, A]) > 1 - \varepsilon$. Then for any real t ,

$$\left| \int_{-A}^A e(-tx) d\mu(x) - \hat{\mu}(t) \right| < \varepsilon.$$

Hence by the triangle inequality,

$$|\hat{\mu}(t_1) - \hat{\mu}(t_2)| < 2\varepsilon + \int_{-A}^A |e(-t_1x) - e(-t_2x)| d\mu(x).$$

Here the integrand is $2|\sin \pi(t_1 - t_2)x| \leq 2\pi|t_1 - t_2|A$. Thus if $|t_1 - t_2| <$

ε/A , then the integrand is uniformly $< 2\pi\varepsilon$. Hence the integral is $< 2\pi\varepsilon$, so that $|\hat{\mu}(t_1) - \hat{\mu}(t_2)| < 9\varepsilon$. Thus $\hat{\mu}$ is uniformly continuous. If $\mathcal{S}$ is a tight family of measures, then the choice of A depends on ε and $\mathcal{S}$, but is independent of the individual $\mu \in \mathcal{S}$. $\square$

L: Intfdmu

Lemma I.4. *If $f(t) \in L^1(\mathbb{R})$ and μ is a probability measure, then*

$$\int_{-\infty}^{\infty} \hat{f}(x) d\mu(x) = \int_{-\infty}^{\infty} f(t) \hat{\mu}(t) dt.$$

We note that $\hat{f}$ is bounded and continuous, so that the integral on the left can be considered to be a Riemann–Stieltjes integral $\int \hat{f} dF$. As $|f\hat{\mu}| \leq |f|$, the integral on the right is a Lebesgue integral for $f \in L^1(\mathbb{R})$.

Proof By the definitions of $\hat{\mu}$ and $\hat{f}$, the above asserts that

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(t) e(-tx) dt d\mu(x) = \int_{-\infty}^{\infty} f(t) \int_{-\infty}^{\infty} e(-tx) d\mu(x) dt.$$

The interchange of integrals is justified by Fubini's theorem, in view of the joint integrability of $f(t)e(-tx)$. $\square$

Co: FourierTransUnique

Corollary I.5. *If μ_1 and μ_2 are probability measures such that $\hat{\mu}_1(t) = \hat{\mu}_2(t)$ for all real t , then $\mu_1 = \mu_2$.*

Proof Let $I = [a, b]$ be a given interval, and let $f(x)$ be determined so that $\hat{f}(t)$ is the piecewise linear function whose graph passes through the points $(-\infty, 0)$, $(a - \delta, 0)$, $(a, 1)$, $(b, 1)$, $(b + \delta, 0)$, $(\infty, 0)$. Then $f(x) \ll_{\delta} x^{-2}$, and hence $f \in L^1(\mathbb{R})$. By Lemma I.4 we see that

$$\int_{-\infty}^{\infty} \hat{f} d\mu_1 = \int_{-\infty}^{\infty} \hat{f} d\mu_2.$$

On the other hand, as δ tends to 0, the respective sides of this decrease to $\mu_i([a, b])$. Hence $\mu_1(I) = \mu_2(I)$ for all closed bounded intervals I , and it follows that $\mu_2 = \mu_1$. $\square$

We now characterize weak convergence in a number of useful ways.

T: CharactWeakConv

Theorem I.6. *If $\mu, \mu_1, \mu_2, \dots$ are probability measures, then the following are equivalent:*

- (a) *The μ_N tend weakly to μ ;*
- (b) *For every bounded continuous function f , the integral $\int_{-\infty}^{\infty} f d\mu_N$ tends to $\int_{-\infty}^{\infty} f d\mu$;*
- (c) *For each real number t , $\hat{\mu}_N(t)$ tends to $\hat{\mu}(t)$;*

(d) For each real number $T > 0$, $\hat{\mu}_N(t)$ tends to $\hat{\mu}(t)$ uniformly for $|t| \leq T$.

Proof We first demonstrate that any one of the above conditions implies that the μ_N are tight. Suppose that (a) holds. There is an A such that $\mu([-A, A]) > 1 - \varepsilon$. Hence $\mu_N([-A, A]) > 1 - 2\varepsilon$ for all $N > N_0$, and consequently there is a B such that $\mu_N([-B, B]) > 1 - 2\varepsilon$ for all N . Since both (b) and (d) include (c), we now suppose that (c) holds. Take $f(t) = \chi_{[-\delta, \delta]}(t)/(2\delta)$ in Lemma I.4. Then

$$\frac{1}{2\delta} \int_{-\delta}^{\delta} \hat{\mu}_N(t) dt = \int_{-\infty}^{\infty} \frac{\sin 2\pi\delta x}{2\pi\delta x} d\mu_N(x). \quad (\text{I.2}) \quad \boxed{\text{E: IntmuN}}$$

By the principle of dominated convergence, the left hand side tends to $\frac{1}{2\delta} \int_{-\delta}^{\delta} \hat{\mu}(t) dt$ as N tends to infinity. But $\hat{\mu}(0) = 1$ and $\hat{\mu}$ is continuous at 0, so if δ is small, then

$$\left| \frac{1}{2\delta} \int_{-\delta}^{\delta} \hat{\mu}(t) dt - 1 \right| < \varepsilon.$$

Hence

$$\int_{-\infty}^{\infty} \frac{\sin 2\pi\delta x}{2\pi\delta x} d\mu_N(x) > 1 - 2\varepsilon$$

for all $N > N_0$. Here the integrand is ≤ 1 for all real x , and is $\leq 1/2$ when $|x| \geq B = 1/\delta$, so that this integral is

$$\leq \mu_N([-B, B]) + \frac{1}{2} \mu_N\{|x| > B\} = \frac{1}{2} + \frac{1}{2} \mu_N([-B, B]).$$

Hence $\mu_N([-B, B]) > 1 - \varepsilon$ for $N > N_0$, and therefore the μ_N are tight.

We now derive (b) from (a). Suppose that $|f(x)| \leq M$ for all x , that $F(x)$ is the distribution function of μ , and that F_N is the distribution function of μ_N . Since F is increasing, the set of points of discontinuity of F is at most countable. The same is true of each F_N , and hence there a set $\mathcal{D}$ that is at most countably infinite, such that if $x \notin \mathcal{D}$, then F and each of the functions F_N is continuous at x . Suppose that $\varepsilon > 0$ is given. Choose $A \notin \mathcal{C}$ so large that $\mu\{|x| > A\} < \varepsilon$ and $\mu_N\{|x| > A\} < \varepsilon$ for all N . Such an A exists because the family is tight. Hence

$$\left| \int_{-\infty}^{\infty} f(x) d\mu_N - \int_{-\infty}^{\infty} f(x) d\mu \right| \leq \left| \int_{-A}^A f(x) d\mu_N - \int_{-A}^A f(x) d\mu \right| + 2M\varepsilon.$$

Since $f(x)$ is uniformly continuous on $[-A, A]$, we may choose numbers x_k , $-A = x_0 < x_1 < \dots < x_k = A$, so that $|f(x) - f(x')| < \varepsilon$ if

$x_{k-1} \leq x \leq x' \leq x_k$ for some k . Moreover, these x_k may be chosen so that none of them is a member of $\mathcal{D}$. Then

$$\left| \int_{x_{k-1}}^{x_k} f(x) d\mu - f(x_k)(F(x_k) - F(x_{k-1})) \right| \leq \varepsilon (F(x_k) - F(x_{k-1})).$$

This also holds with μ replaced by μ_N . On summing over k and using the triangle inequality, we find that the difference in question has absolute value not exceeding

$$M \sum_{k=1}^K |\mu_N([x_{k-1}, x_k]) - \mu([x_{k-1}, x_k])| + 2(M+1)\varepsilon.$$

From the hypothesis (a) it follows that this first term tends to 0 as N tends to infinity. Since ε is arbitrarily small, this gives (b).

We note that (c) is the special case $f(x) = e(-tx)$ of (b), and hence (b) implies (c).

Suppose that (c) holds. Since the μ_N are tight, by Lemma I.3 it follows that the $\mu_N(t)$ are uniformly continuous both in t and in N . Hence (d) follows from (c).

To complete the proof it remains to show that (d) implies (a). Suppose that $f \in L^1(\mathbb{R})$. Then by the principle of dominated convergence

$$\lim_{N \rightarrow \infty} \int_{-\infty}^{\infty} f(t) \hat{\mu}_N(t) dt = \int_{-\infty}^{\infty} f(t) \hat{\mu}(t) dt,$$

and hence by Lemma I.4,

$$\lim_{N \rightarrow \infty} \int_{-\infty}^{\infty} \hat{f}(x) d\mu_N(x) = \int_{-\infty}^{\infty} \hat{f}(x) d\mu(x).$$

Let f be chosen as in the proof of Corollary I.5, so that $f \in L^1(\mathbb{R})$ and f is a piecewise-linear majorant of $\chi_{[a,b]}$. Then

$$\mu_N([a, b]) \leq \int_{-\infty}^{\infty} \hat{f}(x) d\mu_N(x) \rightarrow \int_{-\infty}^{\infty} \hat{f}(x) d\mu(x) \leq \mu([a - \delta, b + \delta])$$

as $N \rightarrow \infty$. Here the last member tends to $\mu([a, b])$ as δ tends to 0, and thus we conclude that $\limsup_{N \rightarrow \infty} \mu_N([a, b]) \leq \mu([a, b])$. Supposing that $a < b$, we may construct $f \in L^1(\mathbb{R})$ so that $\hat{f}$ forms a piecewise-linear minorant of $\chi_{[a,b]}$, and thus similarly show that $\liminf_{N \rightarrow \infty} \mu_N([a, b]) \geq \lim_{\delta \rightarrow 0+} \mu([a + \delta, b - \delta])$. This last expression is equal to $\mu([a, b])$ provided that $\mu(\{a\}) = \mu(\{b\}) = 0$, i.e. if the distribution function F is continuous at a and at b . Thus we have shown that

$$\lim_{N \rightarrow \infty} F_N(b) - F_N(a) = F(b) - F(a)$$

whenever a and b are points of continuity of F . Since the family μ_N is tight, the numbers $F_N(a)$ and $F(a)$ are near 0 if a is large and negative. Hence $\lim_{N \rightarrow \infty} F_N(b) = F(b)$, which is (a). $\square$

We now sharpen Theorem I.6 by showing that the limiting measure need not be given in advance.

T:muNConv

Theorem I.7. *Let $\mu_1, \mu_2, \dots$ be probability measures, and suppose that for each t the sequence $\hat{\mu}_1(t), \hat{\mu}_2(t), \dots$ converges. Call the limit $r(t)$. If r is continuous at 0, then the μ_N are weakly convergent to a probability measure μ , and $r(t) = \hat{\mu}(t)$.*

Proof We note that $\hat{\mu}_N(0) = 1$ for all N , so that $r(0) = 1$. In proving Theorem I.6 we showed that condition (c) implies that the μ_N are tight. The only properties of $\hat{\mu}(t)$ used in that argument were that $\hat{\mu}(0) = 1$ and that $\hat{\mu}(t)$ is continuous at 0. Hence by the same method we may show that the μ_N are tight. By Theorem I.1 there is a subsequence $\mu_{N(k)}$ that is weakly convergent to a measure μ . By Theorem I.2 we see that μ is a probability measure. By Theorem I.6 applied to the subsequence $\mu_{N(k)}$, we deduce that $\hat{\mu}_{N(k)}(t) \rightarrow \hat{\mu}(t)$ as $k \rightarrow \infty$, for all real t . Hence $\hat{\mu} = r$, and $\hat{\mu}_N(t) \rightarrow \hat{\mu}(t)$. Then by Theorem I.6 again we conclude that the μ_N tend weakly to μ . $\square$

We now characterize continuous measures in terms of the behaviour of $\hat{\mu}$.

T:CharContMeas

Theorem I.8. *Let μ be a probability measure. Then μ is continuous if and only if*

$$\int_{-T}^T |\hat{\mu}(t)|^2 dt = o(T)$$

as T tends to ∞ .

Proof The left hand side is

$$\begin{aligned} \int_{-T}^T \int_{-\infty}^{\infty} e(-tx) d\mu(x) \int_{-\infty}^{\infty} e(ty) d\mu(y) dt \\ = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-T}^T e(t(y-x)) dt d\mu(x) d\mu(y) \\ = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \frac{\sin 2\pi T(y-x)}{\pi(y-x)} d\mu(x) d\mu(y). \end{aligned}$$

If we divide by $2T$, then by the principle of dominated convergence this

latter integral tends to

$$\iint_{x=y} d\mu(x) d\mu(y) = \sum_i d_i^2$$

as T tends to infinity, where the d_i are the heights of the jump discontinuities of μ . But μ is continuous if and only if $\sum d_i^2 = 0$, so we have the result. $\square$

??.1 Exercises

1. Suppose that $\mu_1, \mu_2, \dots$ are probability measures such that for each convergent subsequence the limit is a probability measure. Show that the sequence is tight.
2. Construct an example to show that in the situation of Theorem I.6, the convergence of the $\hat{\mu}_N(t)$ to $\hat{\mu}(t)$ need not be uniform in t .
3. Suppose that μ_N is a sequence of probability measures on $\mathbb{T}$, and that μ is also a probability measure on $\mathbb{T}$. We say that the μ_N tend weakly to μ if $\mu_N([\alpha, \beta]) \rightarrow \mu([\alpha, \beta])$ whenever $0 \leq \beta - \alpha \leq 1$ and $\mu(\{\alpha\}) = \mu(\{\beta\}) = 0$. Show that the following are equivalent:
 - (a) The μ_N tend weakly to μ ;
 - (b) For each integer k , $\hat{\mu}_N(k)$ tends to $\hat{\mu}(k)$ as N tends to infinity;
 - (c) If f is a continuous function defined on $\mathbb{T}$, then $\int_{\mathbb{T}} f(x) d\mu_N(x)$ tends to $\int_{\mathbb{T}} f(x) d\mu$ as N tends to infinity.
4. Let μ be the measure on $\mathbb{R}$ defined by $d\mu = (2\pi)^{-1/2} e^{-x^2/2} d\lambda$ where λ denotes Lebesgue measure. Show that μ is a probability measure. (This is called emphnormal distribution with mean 0 and variance 1.) Show that $\int_{\mathbb{R}} x d\mu(x) = 0$, that $\int x^2 d\mu(x) = 1$, and that $\hat{\mu}(t) = e^{-2\pi^2 t^2}$.
5. Let p be a fixed real number, $0 < p < 1$. For each positive integer N let μ_N be the discrete measure that has a point mass of weight $\binom{N,n}{p} (1-p)^{N-n}$ at $(n - Np)/\sigma$ for $n = 0, 1, 2, \dots, N$ where $\sigma = \sqrt{np(1-p)}$. Show that μ_N is a probability measure. Show that $\hat{\mu}_N(t) = (pe^{-t(1-p)/\sigma} + (1-p)e^{-tp/\sigma})^N$. Show that if T is given, then $\hat{\mu}_N(t) = e^{-2\pi^2 t^2} + O(n^{-1/2})$ as N tends to infinity, uniformly for $|t| \leq T$. Conclude that the measures μ tend weakly to the measure μ of the preceding exercise. (This is a special case of the Central Limit Theorem of probability theory.)

I.2 Notes

S:NotesLimDist

Section ?? . The result of Exercise ?? is due to ^{Bohl}Bohl (1906) (see p. 279 of his paper). Later, ^{BohrAPF}Bohr (1925) created an extensive theory of almost periodic functions, and in the course of this demonstrated (cf pp. 119–121) that Bohl’s Theorem is equivalent to the localized form of Kronecker’s Theorem, i.e., to our Corollary ??.

I.3 References

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- ^{Franel}Franel, J. 1924. Les suites de Farey et le problème des nombres premiers. *Nachr. Akad. Wiss. Göttingen*, 198–201.
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Appendix J

Topics in harmonic analysis III

C:AppJHarmAnalIII

J.1 A majorant inequality

S:majorant
T:L2Majorant

Theorem J.1. *For positive integers n let λ_n be real, suppose that $|a_n| \leq b_n$ for all n , and that $\sum_{n=1}^{\infty} b_n < \infty$. Then for any real T_0 and any $T > 0$,*

$$\int_{T_0-T}^{T_0+T} \left| \sum_{n=1}^{\infty} a_n e(\lambda_n t) \right|^2 dt \leq 3 \int_{-T}^T \left| \sum_{n=1}^{\infty} b_n e(\lambda_n t) \right|^2 dt.$$

Proof It suffices to prove the inequality when $T_0 = 0$, for once this is done, the general case follows by replacing a_n by $a_n e(\lambda_n T_0)$. Let $K(t) = \max(1 - |t|/T, 0)$. Then

$$\widehat{K}(u) = \frac{1}{T} \left(\frac{\sin \pi T u}{\pi u} \right)^2 \geq 0,$$

so

$$\begin{aligned} \int_{-\infty}^{\infty} K(t) \left| \sum_{n=1}^{\infty} a_n e(\lambda_n t) \right|^2 dt &= \sum_{m,n} a_m \overline{a_n} \widehat{K}(\lambda_n - \lambda_m) \\ &\leq \sum_{m,n} b_m b_n \widehat{K}(\lambda_n - \lambda_m) \\ &= \int_{-\infty}^{\infty} K(t) \left| \sum_{n=1}^{\infty} b_n e(\lambda_n t) \right|^2 dt. \end{aligned}$$

By replacing a_n by $a_n e(\lambda_n T_0)$, we see more generally that

$$\int_{-\infty}^{\infty} K(t - T_0) \left| \sum_{n=1}^{\infty} a_n e(\lambda_n t) \right|^2 dt \leq \int_{-\infty}^{\infty} K(t) \left| \sum_{n=1}^{\infty} b_n e(\lambda_n t) \right|^2 dt. \quad (\text{J.1})$$

E:WeightedL2Est

But $\chi_{[-T,T]}(t) \leq K(t+T) + K(t) + K(t-T)$, so we apply (J.1) three

times, with $T_0 = -T$, $T_0 = 0$, and $T_0 = T$, and sum to find that

$$\int_{-T}^T \left| \sum_{n=1}^{\infty} a_n e(\lambda_n t) \right|^2 dt \leq 3 \int_{-\infty}^{\infty} K(t) \left| \sum_{n=1}^{\infty} b_n e(\lambda_n t) \right|^2 dt.$$

This gives the stated result, since $K(t) \leq \chi_{[-T, T]}(t)$. □

Appendix K

Notes on Turán's Power sum method

C:Turán

K.1 Turán's First Main Theorem

S:Turán1

Let $a_n, z_n \in \mathbb{C}$ and consider the power sum

$$S_h = \sum_{n=1}^N a_n z_n^h \quad (h = 0, 1, 2, \dots). \quad (\text{K.1}) \quad \text{E:PowSum}$$

It is sometimes useful to know that h can be chosen so that the majority of the terms $a_n z_n^h$ point in approximately the same direction, And to have some control over the size or range of h for which this occurs. If the numbers $\arg z_n$ are linearly independent over $\mathbb{Q}$, then it follows by Kronecker's theorem that given $\varepsilon > 0$ there are h such that

$$|S_h| \geq (1 - \varepsilon) \sum_{n=1}^N |a_n z_n^h|. \quad (\text{K.2}) \quad \text{E:PowKron}$$

The difficulty with this in applications is that we do not have any control over the size of h in terms of ε .

Any lower bound we obtain for $|S_h|$ will depend on the nature of the a_n . The simplest general comparison, therefore is with S_0 .

T:Tur1

Theorem K.1 (Turán's First Main Theorem). *Suppose that $|z_n| \geq 1$ ($n \in \mathbb{N}$). Then for any $M \geq 0$ and $N \geq 1$ there is an integer $h \in [M + 1, M + N]$ such that*

$$|S_h| \geq \lambda(M, N) |S_0| \quad (\text{K.3}) \quad \text{E:Tur1b1}$$

where

$$\lambda(M, N) = \frac{1}{\sum_{k=0}^{N-1} \binom{M+k}{k} 2^k}. \quad (\text{K.4}) \quad \text{E:Tur1lam}$$

Moreover

$$\lambda(M, N) \geq \left(\frac{N}{2e(M+N)} \right)^{N-1} \quad (\text{K.5}) \quad \boxed{\text{E:Tur2lam}}$$

T:Tur1Cor **Corollary K.2.** For any $M \geq 0$ there is an integer $h \in [M+1, M+N]$ such that

$$|S_h| \geq \left(\min_{1 \leq n \leq N} |z_n| \right)^h \left(\frac{N}{2e(M+N)} \right)^{N-1} |S_0|. \quad (\text{K.6}) \quad \boxed{\text{E:Tur1b2}}$$

Proof It suffices to prove the theorem when the z_n are distinct, for then the case when two or more of the z_n are identical follows by combining them and appealing to the case of smaller N . Note that, for a given M , $\lambda(M, N)$ is an increasing function of N .

Let $b_0, \dots, b_{N-1}$ be complex numbers at our disposal. Then

$$\left| \sum_{j=0}^{N-1} b_j S_{M+1+j} \right| \leq \sum_{j=0}^{N-1} |b_j| \max_{M+1 \leq h \leq M+N} |S_h|. \quad (\text{K.7}) \quad \boxed{\text{E:blincom}}$$

Inserting the definition (K.1) of S_h into the left hand side and interchanging the order of summation gives

$$\sum_{n=1}^N a_n z_n^{M+1} \sum_{j=0}^{N-1} b_j z_n^j = \sum_{n=1}^N a_n z_n^{M+1} \mathcal{P}(z_n) \quad (\text{K.8}) \quad \boxed{\text{E:blinpol}}$$

where

$$\mathcal{P}(z) = \sum_{j=0}^{N-1} b_j z^j.$$

Consider the system of N linear equations in N variables $b_0, \dots, b_{N-1}$

$$\mathcal{P}(z_n) = z_n^{-M-1} \quad (n = 1, \dots, N). \quad (\text{K.9}) \quad \boxed{\text{E:blineq}}$$

The coefficient matrix is a Vandermonde matrix whose determinant

$$\prod_{1 \leq i < j \leq N} (z_j - z_i)$$

is non-zero. Thus (K.9) determines the b_j uniquely and, by (K.7) and (K.8),

$$|S_0| \leq \sum_{j=0}^{N-1} |b_j| \max_{M+1 \leq h \leq M+N} |S_h|. \quad (\text{K.10}) \quad \boxed{\text{E:blub}}$$

Let $Q_0(z) = 1$ and

$$Q_k(z) = \prod_{n=1}^k (z - z_n).$$

Then there are c_k such that

$$\mathcal{P}(z) = \sum_{k=0}^{N-1} c_k Q_k(z),$$

since the system of linear equations connecting the c_j and the b_j is triangular with 1s on diagonal.

Now consider for $k \geq 0$ the integral

$$\frac{1}{2\pi i} \int_{C_R} \frac{\mathcal{P}(z)}{Q_{k+1}(z)} dz$$

where C_R denotes the circle centered at 0 and of radius R , described in the positive sense, and such that

$$R > |z_n| \quad (n = 1, \dots, N).$$

The function

$$Q_{k+1}^{-1}(z) \sum_{j=k+1}^{N-1} c_j Q_j(z)$$

has only removable singularities. Hence

$$\frac{1}{2\pi i} \int_{C_R} \frac{\sum_{j=k+1}^{N-1} c_j Q_j(z)}{Q_{k+1}(z)} dz = 0$$

Moreover when $j \leq k-1$

$$\frac{Q_j(z)}{Q_{k+1}} \ll R^{-2}$$

so, by Cauchy's theorem, letting $R \rightarrow \infty$ shows that

$$\frac{1}{2\pi i} \int_{C_R} \frac{\sum_{j=0}^{k-1} c_j Q_j(z)}{Q_{k+1}(z)} dz = 0.$$

Therefore

$$\frac{1}{2\pi i} \int_{C_R} \frac{\mathcal{P}(z)}{Q_{k+1}(z)} dz = c_k.$$

We also have

$$\frac{z^{-M-1}}{Q_{k+1}(z)} \ll R^{-2}.$$

Hence

$$c_k = \frac{1}{2\pi i} \int_{C_R} \frac{\mathcal{P}(z) - z^{-M-1}}{Q_{k+1}(z)} dz$$

By (K.9), $z^{M+1}\mathcal{P}(z) - 1$ has a zero at z_n ($n = 1, \dots, N$). Thus the integrand is analytic for $z \neq 0$ and we can replace C_R by C_r where $0 < r < 1$. But now $\mathcal{P}(z)/Q_{k+1}(z)$ is analytic for $|z| \leq r$. Therefore

$$\begin{aligned} c_k &= \frac{1}{2\pi i} \int_{C_r} \frac{-z^{-M-1}}{Q_{k+1}(z)} dz \\ &= \frac{(-1)^k \lambda_k(M)}{z_1 \dots z_{k+1}} \end{aligned}$$

where

$$\lambda_k(M) = \frac{1}{2\pi i} \int_{C_r} z^{-M-1} \prod_{n=1}^{k+1} (1 - z/z_n)^{-1} dz.$$

When $|w| < r$ we have

$$\begin{aligned} \sum_{M=0}^{\infty} \lambda_k(M) w^M &= \frac{1}{2\pi i} \int_{C_r} \frac{\prod_{n=1}^{k+1} (1 - z/z_n)^{-1}}{z - w} dz. \\ &= \prod_{n=1}^{k+1} (1 - w/z_n)^{-1}. \end{aligned}$$

Since

$$(1 - w/z_n)^{-1} = \sum_{M=0}^{\infty} z_n^{-M} w^M$$

has coefficients of modulus $|z_n|^{-M} \leq 1$ it follows that $|\lambda_k(M)|$ does not exceed the coefficient of w^M in $(1 - w)^{-k-1}$ and this is

$$(-1)^M \binom{-k-1}{M} = \binom{M+k}{k}.$$

Thus

$$|c_k| \leq \binom{M+k}{k} / |z_1 \dots z_{k+1}|.$$

Now we compare

$$\sum_{k=0}^{N-1} c_k Q_k(z) = \mathcal{P}(z) = \sum_{h=0}^{N-1} b_h z^h$$

with

$$\sum_{k=0}^{N-1} \binom{M+k}{k} (z+1)^k = Q(z) = \sum_{h=0}^{N-1} B_h z^h.$$

The coefficients of the polynomial $Q(z)$ majorise those of $\mathcal{P}(z)$. Thus

$$\sum_{h=0}^{N-1} |b_h| \leq \sum_{h=0}^{N-1} B_h = Q(1) = \sum_{k=0}^{N-1} \binom{M+k}{k} 2^k,$$

and, by (K.10), gives (K.3) with (K.4). To obtain (K.5) observe that

$$\sum_{k=0}^{N-1} \binom{M+k}{k} 2^k < 2^{N-1} \sum_{k=0}^{N-1} \binom{M+k}{k}$$

and here the sum is the coefficient of z^M in

$$\sum_{k=0}^{N-1} (1+z)^{M+k} = z^{-1} (1+z)^M ((1+z)^N - 1)$$

and this is

$$\binom{N+M}{M+1} \leq \frac{(M+N)^{N-1}}{(N-1)!} \leq \left(\frac{eM + eN}{N} \right)^{N-1}$$

which completes the proof of the theorem. $\square$

I.1 Exercises

1. Prove that the constant $\lambda(M, N)$ in Theorem K.1 cannot be made any smaller.
2. Prove that for any $d \in \mathbb{N}$ there is a polynomial $Q(z)$ of degree d such that $Q(0) = Q(1) = 1$ and

$$\max_{|z| \leq 1} |Q(z)| \leq 1 + 2/d.$$

3. Suppose that $|z_n| \geq 1$ ($1 \leq n \leq N$).

(a) Prove that if $N \leq H \leq N^2$, then

$$\sum_{h=1}^H |S_h|^2 \geq e^{-8N^2/H} |S_0|^2.$$

(b) Prove that if $N \leq H \leq N^2$, then

$$\sum_{h=1}^H |S_h|^2 \geq e^{-8N^2/H} |S_0|^2.$$

(c) Prove that if $H \geq N^2$, then

$$\sum_{h=1}^H |S_h|^2 \gg HN^{-2} |S_0|^2.$$

K.2 Turán's Second main Theorem

S:Turan2

In §K.1 lower bounds were obtained for S_h in terms of S_0 and $(\min_n |z_n|)^h$. For some questions it is necessary to have bounds in term of S_0 and $(\max_n |z_n|)^h$. However if we take ε to be a small positive number and define

$$z_1 = 1, z_n = \varepsilon (n > 1), b_1 = \varepsilon, b_n = \frac{1}{N-1} (n > 1)$$

it follows that

$$S_0 = 1 + \varepsilon, |S_h| = \varepsilon^h + \varepsilon \leq 2\varepsilon |S_0|.$$

Thus it is necessary to have extra constraints in order to deal with the more demanding requirements.

T:Tur2

Theorem K.3 (Turán's Second Main Theorem). *Suppose that*

$$1 = |z_1| \geq |z_2| \geq \dots \geq |z_N|.$$

Then for any $M \geq 0$ and $N \geq 1$ there is an h such that $M+1 \leq h \leq M+N$ and

$$|S_h| \geq 2 \left(\frac{N}{8e(M+N)} \right)^N \min_{1 \leq J \leq N} \left| \sum_{n=1}^J a_n \right|.$$

T:Tur2cor

Corollary K.4. *Suppose that $|z_1| \geq |z_2| \geq \dots \geq |z_N|$. Then for any $M \geq 0$ and $N \geq 1$ there is an h such that $M+1 \leq h \leq M+N$ and*

$$|S_h| \geq 2 \left(\max_{1 \leq n \leq N} |z_n| \right)^h \left(\frac{N}{8e(M+N)} \right)^N \min_{1 \leq J \leq N} \left| \sum_{n=1}^J a_n \right|.$$

In applications it is useful to have a more relaxed range for h .

T:Tur2corcor

Corollary K.5. Suppose that $|z_1| \geq |z_2| \geq \dots \geq |z_N|$. Then for any $L \geq N$ and $N \geq 1$ there is an h such that $L+1 \leq h \leq 2L$ and

$$|S_h| \geq 2(16e)^{-L} \left(\max_{1 \leq n \leq N} |z_n| \right)^h \min_{1 \leq J \leq N} \left| \sum_{n=1}^J a_n \right|.$$

To deduce this corollary let $M = L$ in Corollary K.4 and observe that the inequality

$$1 + y/2 \leq (16e)^y$$

holds for all $y \geq 0$. Then with $y = -1 + L/N$ it follows that

$$8e(N+L)/N \leq (16e)^{L/N}, \left(\frac{N}{8e(L+N)} \right)^N \geq (16e)^{-L}.$$

Proof To prove Theorem K.3 we may suppose as in the proof of Theorem K.1 that the z_n are distinct.

A theorem of Chebyshev (see Exercise K.2.1) states that if $f(x)$ is a monic polynomial of degree d and I is an interval of length l , then

$$\max_{x \in I} |f(x)| \geq 2(l/4)^d.$$

We apply this to

$$f(x) = \prod_{n=1}^N (x - |z_n|)$$

and $I = [a, 1]$ where

$$a = \frac{M}{N+M}.$$

Then there is an $r \in [a, 1]$ such that

$$|f(r)| \geq 2 \left(\frac{1-a}{4} \right)^N$$

and so

$$\begin{aligned} r^M \prod_{n=1}^N |r - |z_n|| &\geq 2(1 + N/M)^{-M} \left(\frac{N}{4M + 4N} \right)^N \\ &> 2 \left(\frac{N}{4e(M+N)} \right)^N. \end{aligned} \quad (\text{K.11}) \quad \text{E:auxbd}$$

Note that $r \neq |z_n|$ ($1 \leq n \leq N$). Now we choose $J \geq 1$ maximally so that $|z_J| > r$, and then for $n > J$ we have $|z_n| < r$.

We follow essentially the same idea as in Theorem K.1. We now choose the coefficients b_j of $\mathcal{P}$ so that

$$\mathcal{P}(z_n) = \begin{cases} z_n^{-M-1} & (1 \leq n \leq J), \\ 0 & (J < n \leq N). \end{cases}$$

and recall that they are uniquely determined. Then in place of (K.10) we obtain

$$\left| \sum_{n=1}^J a_n \right| \leq \sum_{j=0}^{N-1} |b_j| \max_{M+1 \leq h \leq M+N} |S_h|.$$

Again, as before,

$$\frac{1}{2\pi i} \int_{C_R} \frac{\mathcal{P}(z) - z^{-M-1}}{Q_{k+1}(z)} dz = c_k.$$

Also $z^{M+1}\mathcal{P}(z) - 1$ has a zero at z_n when $1 \leq n \leq J$. Hence the integrand has only removeable singularities for $|z| \geq r$ and so

$$c_k = \frac{1}{2\pi i} \int_{C_r} \frac{\mathcal{P}(z) - z^{-M-1}}{Q_{k+1}(z)} dz.$$

The zeros of $Q_{k+1}(z)$ with $|z| < r$ are of the form $z = z_n$ with $n > J$. But such z_n are also zeros of $\mathcal{P}(z)$. Hence $\mathcal{P}(z)/Q_{k+1}(z)$ only has removeable singularities on and inside C_r . Therefore

$$c_k = \frac{1}{2\pi i} \int_{C_r} \frac{-1}{z^{M+1} Q_{k+1}(z)} dz.$$

and so

$$|c_k| \leq \frac{1}{r^M \prod_{n=1}^{k+1} |r - |z_n||}.$$

Therefore, by (K.11),

$$|c_k| \leq \lambda$$

where

$$\lambda = \frac{1}{2} \left(\frac{4e(M+N)}{N} \right)^N.$$

Now we compare

$$\sum_{k=0}^{N-1} c_k Q_k(z) = \mathcal{P}(z) = \sum_{h=0}^{N-1} b_h z^h$$

with

$$\sum_{k=0}^{n-1} \lambda(z+1)^k = Q(z) = \sum_{h=0}^{N-1} B_h z^h.$$

The coefficients of the polynomial $Q(z)$ majorise those of $\mathcal{P}(z)$. Thus

$$\sum_{h=0}^{N-1} |b_h| \leq \sum_{h=0}^{N-1} B_h = Q(1) = \sum_{k=0}^{N-1} \lambda 2^k < \lambda 2^N,$$

The theorem now follows. $\square$

I.2 Exercises

1. The n -th Chebyshev polynomial $T_n(x)$ ($x \in \mathbb{R}$) is defined for $x \in [-1, 1]$ by $T_n(x) = \cos(n \cos^{-1} x)$.

- (a) Prove that $T_0(x) = 1$, $T_1(x) = x$, $T_2(x) = 2x^2 - 1$, $T_3(x) = 4x^3 - 3x$ and

$$T_{n+1}(x) = 2xT_n(x) - T_{n-1}(x).$$

- (b) Prove that

$$\max_{x \in [-1, 1]} |T_n(x)| = 1$$

and that the maximum is attained precisely when $x = x_k = \cos \frac{\pi k}{n}$ ($0 \leq k \leq n$) and then $T_n(x_k) = (-1)^k$.

- (c) Prove that if $f(x)$ is a monic polynomial of degree n , then

$$\max_{x \in [-1, 1]} |f(x)| \geq \frac{1}{2^{n-1}}.$$

Hint: Show that if the maximum is smaller than 2^{n-1} , then $f(x) - 2^{1-n}T_n(x)$ changes sign at least n times.

- (d) Prove that if $f(x)$ is a monic polynomial of degree n , and I is an arbitrary interval of length l , then

$$\max_{x \in I} |f(x)| \geq \frac{1}{2} \left(\frac{l}{4} \right)^n.$$

2. Suppose that $b_n = 1$ ($n = 1, \dots, N$)

- (a) Prove that if $|z_n| \geq 1$ ($n = 1, \dots, N$), then there is an h with $1 \leq h \leq N$ such that $|S_h| \geq 1$, and show that this is best possible.
- (b) Prove that if S_h is real for all positive integers h then there is an h with $1 \leq h \leq N + 1$ such that $S_h \geq 0$.
- (c) Suppose that $\max_{1 \leq n \leq N} |z_n| = 1$. Prove that, for some h with $1 \leq h \leq 2N - 1$, $|S_h| \geq 1$.

Appendix L

Bounds for Bilinear Forms II

C:BilinFormsII

L.1 Hermitian matrices

S:HermMat

In many parts of this work, including nearly all the chapters of volume II and Chapters 26, 27, 28 and 30 of this volume, we encounter expressions of the kind

$$\sum_{r=1}^R \left| \sum_{n=1}^N a_n c_{nr} \right|^2 = \sum_{m=1}^N a_m \sum_{n=1}^N \overline{a_n} \sum_{r=1}^R c_{mr} \overline{c_{nr}} \quad (\text{L.1}) \quad \boxed{\text{eq:L1}}$$

where the c_{rn} are complex numbers we have some control over, and the a_n are complex numbers we have less control over. It is often convenient to write this as

$$\mathbf{a} \mathcal{C} \mathcal{C}^* \mathbf{a}^* \quad (\text{L.2}) \quad \boxed{\text{eq:L2}}$$

where $\mathcal{C}$ is the $N \times R$ matrix

$$\mathcal{C} = (c_{nr}) \quad (\text{L.3}) \quad \boxed{\text{eq:L3}}$$

and the $\mathcal{X}^*$ denotes the complex conjugate transpose. The $N \times N$ matrix

$$\mathcal{H} = \mathcal{C} \mathcal{C}^* \quad (\text{L.4}) \quad \boxed{\text{eq:L4}}$$

is Hermitian and the diagonal terms

$$\sum_{r=1}^R c_{mr} \overline{c_{mr}} = \sum_{r=1}^R |c_{mr}|^2 \quad (\text{L.5}) \quad \boxed{\text{eq:L5}}$$

are non-negative. Note also that

$$\text{tr}(\mathcal{H}) = \sum_{m=1}^N \sum_{r=1}^R |c_{mr}|^2 = \text{tr}(\mathcal{H}^*). \quad (\text{L.6}) \quad \boxed{\text{E:trHHstar}}$$

If λ is an eigenvalue associated with $\mathcal{H}$ and $\mathbf{a}$ is a non-zero eigenvector associated with λ , then we have

$$\lambda \mathbf{a} \cdot \bar{\mathbf{a}} = \mathbf{a} \mathcal{H} \bar{\mathbf{a}} = \mathbf{a} \mathcal{C} \mathcal{C}^* \mathbf{a}^* \geq 0. \quad (\text{L.7}) \quad \boxed{\text{eq:L6}}$$

Thus each eigenvalue λ of $\mathcal{H}$ is a non-negative number. In many applications we are, in fact, simply obtaining a bound for the largest eigenvalue of such a matrix. Thus it is useful to bring together here a few additional items not included in Appendix G. Unless stated otherwise the entries in subsequent matrices are assumed to be complex numbers.

T:trace **Theorem L.1.** *The trace of a square matrix is both the sum of the diagonal terms and the sum of the eigenvalues.*

Proof The first part is immediate by the definition of trace. Let $\mathcal{A}$ denote an $N \times N$ matrix. Then, by definition of eigenvalue, we have

$$P(x) = \prod_{j=1}^N (x - \lambda_j) = \det(xI - \mathcal{A})$$

where in the product the λ_j are an ordering of the eigenvalues, counted with multiplicity. The coefficient of x^{N-1} in $P(x)$ is

$$- \sum_{j=1}^N \lambda_j.$$

In the usual definition of the determinant as the sum over products of entries in the matrix weighted by signs, the only factors that can contribute an x are the diagonal terms $(x - a_{nn})$, so at least $N - 1$ of these must occur in the product. But this then fixes the final factor as also being a diagonal term. Thus the coefficient of x^{N-1} in $P(x)$ is also the coefficient of x^{N-1} in

$$\prod_{n=1}^N (x - a_{nn})$$

and this coefficient is $-\text{tr } \mathcal{A}$ □

There is also a useful relationship between the eigenvalues of a square matrix and those of its k -th power.

T:L2 **Theorem L.2.** *The eigenvalues of a the k -th power of a square matrix are the k -th powers of the eigenvalues of the original matrix, with the same multiplicity.*

Proof Let $P_k(x)$ denote the characteristic polynomial of the k -th power of the $N \times N$ matrix A . Then

$$\begin{aligned} P_k(x^k) &= \det(x^k I^k - \mathcal{A}^k) \\ &= \det \prod_{\rho^k=1}^{\rho} (xI - \rho \mathcal{A}) \\ &= \prod_{\rho^k=1}^{\rho} P_1(x\bar{\rho}) \\ &= \prod_{\rho^k=1}^{\rho} \prod_{n=1}^N (x\bar{\rho} - \lambda_n) \end{aligned}$$

where the λ_n are an ordering of the eigenvalues of $\mathcal{A}$ counted with multiplicity. Thus

$$P_k(x^k) = \prod_{n=1}^N \prod_{\rho^k=1}^{\rho} (x\bar{\rho} - \lambda_n) = \prod_{n=1}^N (x^k - \lambda_n^k).$$

Hence

$$P_k(y) = \prod_{n=1}^N (y - \lambda_n^k).$$

□

We can combine (L.7) with Corollary G.11, and then appeal to duality (Theorem G.1) to obtain the next theorem.

T:L3 **Theorem L.3.** *Suppose that $\mathcal{H}$ is an $N \times N$ Hermitian matrix satisfying (L.4) with (L.3). Then every eigenvalue of $\mathcal{H}$ is non-negative and for any set of N complex numbers a_n we have*

$$\sum_{r=1}^R \left| \sum_{n=1}^N a_n c_{nr} \right|^2 \leq \lambda \mathbf{a} \cdot \mathbf{a}^*$$

where λ is the largest eigenvalue of $\mathcal{H}$. Moreover λ is also the largest eigenvalue of the $R \times R$ matrix $\mathcal{H}^* \mathcal{H}$.

Given $\mathcal{H}$ as above, let τ_k denote

$$\tau_k = \text{tr}(\mathcal{H}^k). \quad (\text{L.8}) \quad \boxed{\text{eq:L7}}$$

T:lambdabound

Theorem L.4. Suppose that $\mathcal{H}$ is an $N \times N$ Hermitian matrix as above, with $N \geq 2$, and λ is an eigenvalue of $\mathcal{H}$. Then, for $k \in \mathbb{N}$ and $k \geq 2$,

$$\lambda \leq \frac{2\tau_1}{N} + 2 \left(\tau_k - \frac{\tau_1^k}{N^{k-1}} \right)^{1/k}.$$

This just follows from Hölder's inequality and really does not have anything to do with eigenvalues. Note that it is essentially best possible when the eigenvalues are the same, or when there is one dominant eigenvalue and N is large.

Proof Suppose that $\lambda_1, \dots, \lambda_N$ are N non-negative real numbers. Then, by Hölder's inequality,

$$(\lambda_2 + \dots + \lambda_N)^k \leq (N-1)^{k-1} (\lambda_2^k + \dots + \lambda_N^k).$$

Hence, by (L.8) and Theorem L.2

$$(\tau_1 - \lambda_1)^k \leq (N-1)^{k-1} (\tau_k - \lambda_1^k). \quad (\text{L.9}) \quad \boxed{\text{eq:L8}}$$

If $\lambda = \lambda_1$ and $\lambda_1 \leq \frac{2\tau_1}{N}$, then we are done. Hence we may suppose that

$$\lambda_1 \leq \tau_1 \leq N\lambda_1/2.$$

By (L.9) we have

$$(N-1)^{1-k} (\tau_1 - \lambda_1)^k + \lambda_1^k - N^{1-k} \tau_1^k \leq \tau_k - \tau_1^k N^{1-k}. \quad (\text{L.10}) \quad \boxed{\text{eq:L9}}$$

Viewing the left hand side as a function of τ_1 but keeping λ_1 fixed we see that it is decreasing for $\tau_1 \leq N\lambda_1$ and 0 when $\tau_1 = N\lambda_1$. Hence

$$\begin{aligned} & (N-1)^{1-k} (\tau_1 - \lambda_1)^k + \lambda_1^k - N^{1-k} \tau_1^k \\ & \geq \lambda_1^k ((N-1)^{1-k} (N/2 - 1)^k + 1 - N^{1-k} 2^{-k}) \\ & = \lambda_1^k ((N-1) (1 - \frac{N}{2(N-1)})^k + 1 - N^{1-k} 2^{-k}) \\ & = \lambda_1^k \left(1 - \int_{N-1}^N \left(1 - \frac{N}{2u} \right)^k + k \left(1 - \frac{N}{2u} \right)^{k-1} \frac{N}{2u} du \right) \\ & = \lambda_1^k \left(1 - \int_{N-1}^N \left(1 - \frac{N}{2u} \right)^{k-1} \left(1 + \frac{(k-1)N}{2u} \right) du \right). \end{aligned}$$

By the binomial inequality this is

$$\begin{aligned}
 &\geq \lambda_1^k \left(1 - \int_{N-1}^N \left(1 - \frac{N}{2u} \right)^{k-1} \left(1 + \frac{N}{2u} \right)^{k-1} du \right) \\
 &= \lambda_1^k \left(1 - \int_{N-1}^N \left(1 - \frac{N^2}{4u^2} \right)^{k-1} du \right) \\
 &\geq \lambda_1^k \left(1 - \int_{N-1}^N \left(1 - \frac{N^2}{4u^2} \right) du \right) \\
 &= \lambda_1^k \frac{N}{4(N-1)}.
 \end{aligned}$$

Hence, by (L.10),

$$\lambda_1^k \frac{N}{4(N-1)} \leq \tau_k - \tau_1^k N^{1-k}$$

and so

$$\lambda_1 \leq \left(\frac{4(N-1)}{N} \right)^{1/k} (\tau_k - \tau_1^k N^{1-k})^{1/k}.$$

□